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Cellular reprogramming to reverse aging and promote organ and tissue regeneration

Abstract

Provided herein are engineered nucleic acids (e.g., expression vectors, including viral vectors, such as lentiviral vectors, adenoviral vectors, AAV vectors, herpes viral vectors, and retroviral vectors) that encode OCT4; KLF4; SOX2; or any combination thereof that are useful, for example, in inducing cellular reprogramming, tissue repair, tissue regeneration, organ regeneration, reversing aging, or any combination thereof. Also provided herein are recombinant viruses (e.g., lentiviruses, alphaviruses, vaccinia viruses, adenoviruses, herpes viruses, retroviruses, or AAVs) comprising the engineered nucleic acids (e.g., engineered nucleic acids), engineered cells, compositions comprising the engineered nucleic acids, the recombinant viruses, engineered cells, engineered proteins, chemical agents that are capable of activating expression of OCT4; KLF4; SOX2; or any combination thereof, an engineered protein selected from the group consisting of OCT4; KLF4; SOX2; or any combination thereof, an antibody capable of activating expression of OCT4; KLF4; SOX2; or any combination thereof, and methods of treating a (e.g., ocular disease), preventing a disease (e.g., ocular disease), regulating (e.g., inducing or inducing and then stopping) cellular reprogramming, regulating tissue repair, regulating tissue regeneration, or any combination thereof).

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of and claims priority under 35 U.S.C. § 120 to U.S. patent application U.S. Ser. No. 18/318,566, filed May 16, 2023, which is a continuation of and claims priority under 35 U.S.C. § 120 to U.S. patent application U.S. Ser. No. 17/280,384, filed Mar. 26, 2021, which is a national stage filing under 35 U.S.C. § 371 of international PCT application number PCT/US2019/053545, filed Sep. 27, 2019, which claims priority under 35 U.S.C. § 119(e) of U.S. provisional application No. 62/738,922, filed Sep. 28, 2018, U.S. provisional application No. 62/792,283, filed Jan. 14, 2019, U.S. provisional application No. 62/865,877, filed Jun. 24, 2019, and U.S. provisional application No. 62/880,488, filed Jul. 30, 2019, each of which is incorporated by reference herein in its entirety.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING
(1) The contents of the electronic sequence listing (H082470296US07-SEQ-FL.xml; Size: 251,743 bytes; and Date of Creation: Feb. 12, 2024) is herein incorporated by reference in its entirety.
BACKGROUND OF THE INVENTION
(2) In many animals, including vertebrates, vital organs have a limited intrinsic capacity for regeneration and repair. Acute injury and chronic disorders can damage vital organs and tissues, including the heart, pancreas, brain, kidney, muscles, skin and neuronal tissue, among others. Mature somatic cells, however, often cannot survive these insults, and even if they do, they are unable to self-renew and transdifferentiate to replace damaged cells. Furthermore, cells that are capable of self-renewal can be limited in quantity, have limited capacity and are susceptible to damage, especially with age. In contrast to somatic cells from adults, cells from individuals that are chronologically closer to fertilization, such as those from embryos and infants, display cellular youthfulness and have a greater capacity to resist injury and stress, to heal, renew, and regenerate organs and tissues. Thus, compositions and methods directed at rejuvenating cells, thereby restoring them from an aged, mature state to a younger, more vital state, have long been sought to treat certain injuries and diseases, as well as generally reverse and prevent aging in entire organisms.
(3) There are two types of information in the body: digital and analog. DNA is digital information and the epigenome is analog information. Analog information never lasts as long as digital, nor can analog information be copied with high fidelity compared to digital information. This has consequences for how long organisms live and thrive. Aging was once thought of as a process driven by mutations in the genetic material of a cell. This has largely been abandoned as an explanation. A major cause of aging is now thought to be due to epigenetic changes that cause cells to transcribe the wrong genes at the wrong time for optimal function, a process that becomes more dysfunctional over time, leading to diseases, an inability to heal and eventually to death. The Yamanaka factors (OCT4, SOX2, c-Myc, and KLF4) have previously been shown to induce pluripotency in vitro (Takahashi et al., Cell. 2006 Aug. 25; 126(4):663-76) and reverse the DNA methlylation clock of aging (Horvath, Genome Biol. 2013). Nanog and Lin28 can help induce pluoripotency together with Yamanaka factors. And Tet1, NR5A-2, Sall4, NKX3-1 can replace Oct4 (Gao et al., Cell Stem Cell 12, 1-17, Apr. 4, 2013 and Mai et al., Nature Cell Biology 20, 900-908, 2018). Expression of original four transcription factors in transgenic mice, however, induce teratomas in vivo, along with other acute toxicities like dysplasia in the intestinal epithelium, that can kill an animal in a few days (Abad et al., Nature. 2013 Oct. 17; 502(7471):340-5). Therefore, non-toxic and efficient methods of cellular reprogramming are needed.
SUMMARY OF THE INVENTION

(4) The cellular aging process has been postulated to be caused by the loss of both genetic and epigenetic information. While previous studies have hypothesized that aging is caused primarily by the loss of genetic information (most commonly in the form of genetic mutations such as substitutions, and deletions in an organism's genome), the compositions and methods of the present disclosure are informed by the unexpected finding that aging is primarily driven by a loss in the particular epigenetic information that is established closer to fertilization and final differentiation of particular cells. Epigenetic information, which commonly takes the form of covalent modifications to DNA, such as 5-methylcytosine(5mC), hydroxymethylcytosine (5hmeC), 5-formylcytosine (fC), and 5-carboxylcytosine (caC) and adenine methylation, and to certain proteins, such as lysine acetylation, lysine and arginine methylation, serine and threonine phosphorylation, and lysine ubiquitination and sumoylation of histone proteins, is sometimes referred to as the "analog" information of the cell. The loss of this analog information can result in dysregulation of vital cellular processes, such as the processes that maintain cell identity, causing cells to exhibit traits that are typically associated with aging such as senescence.
(5) The methods, compositions, and kits of the present disclosure rejuvenate cells by preventing and reversing the cellular causes of aging. Without being bound by a particular theory, more specifically, the methods, compositions and kits of the present disclosure rejuvenate cells by restoring epigenetic information that has been lost due to the aging process, injury or disease. The methods compositions and kits of the present disclosure comprise the transcription factors OCT4, SOX2 and KLF4. OCT4, SOX2 and KLF4 are three of the four "Yamanaka Factors", with the fourth being c-Myc. The Yamanaka Factors have traditionally been used to reprogram cells to a pluripotent state. However, the induction of expression of the four transcription factors in transgenic mice resulted in the formation of teratomas in vivo, along with other acute toxicities like dysplasia in the intestinal epithelium, which can kill the animal in a few days. Moreover, the fact that the four Yamanaka Factors are typically used to reprogram cells to a completely pluripotent state, wherein the cell loses its pre-established cellular identity, can be dangerous for in vivo applications where the cellular identity of target cells must be maintained for tissue and/or organ integrity. In contrast, in some embodiments, the methods described herein, allow incomplete reprogramming and do not result in global changes in demethylation. In some embodiments, the methods described herein do not require complete de-differentiation of cells. For example, while expression of OCT4, SOX2, and KLF4 promoted regeneration following injury in young and old mice and following vincristine-induced injury in human neurons, expression of OCT4, SOX2, and KLF4 did not induce a global reduction of DNA methylation (see e.g., FIGS. **45**B-**45**C).
(6) In some embodiments, the results disclosed herein suggest that expression of OCT4, SOX2, and KLF4 can allow diseased cells to revert to a healthier state without inducing complete reprogramming. Without being bound by a particular theory, the results disclosed herein suggest that cells maintain a backup epigenome that can be restored using the methods described herein.
(7) The methods, compositions and kits of the present disclosure are in part informed by the surprising and unexpected discovery that the spatially and temporally specific induction of OCT4, SOX2, and KLF4 expression in the absence of the induction of c-Myc expression can rejuvenate a cell without reprogramming the cell to a pluripotent state. Using inducible promoters, the expression of OCT4, SOX2 and KLF4 can be carefully controlled to decrease and reverse epigenetic marks associated with aging, increase the epigenetic marks associated with cellular youthfulness, decrease the expression of aging related proteins, increase the expression of proteins associated with a youthful cellular state, restore the balance between euchromatin and heterochromatin, prevent loss of cellular identity, restore cellular identity, reversing the aging related changes in DNA methylation, thereby rejuvenating the cell without reprogramming the cell to a pluripotent state.
(8) Thus, in various embodiments the methods of the invention rejuvenates a cell by restoring the cellular identity of the cell by reversing the effects of or preventing of one or more dysregulated

developmental pathways. For example, the methods: (i) increase the abundance of at least one of histone H2A, histone H2B, histone H3, histone H4, or any combination thereof in the cell; (ii) increase the abundance of at least one of CHAF1a, CHAF1b, HP1α, NuRD or any combination thereof in the cell; (iii) increase at least one heterochromatin mark in the cell such as for example H3K9me3, H3K27me3 or any combination thereof; or decrease one heterochromatin mark such as H4K20me3 or euchromatin mark H3K4me3; (iv) increase/decrease DNA methylation of at least one age-related CpG site in the cell towards young level; (v) increase the abundance of lamin B1 in the cell; (vi) increase acetylation of histone H3 at lysine 27 (H3K27ac), increase acetylation of histone H3 at lysine 56 (H3K56ac) or any combination thereof in the cell; (vii) decrease acetylation of histone H3 at lysine 122 (H3K122Ac) or histone H4 at lysine 16 (H4K16ac), or any combination thereof in the cell (viii) decrease the abundance of IL6, Ccl2, Ccl20, Apob, p16, LINE-1 repeats, Sat III repeats, Alu elements, IAP or any combination thereof; (ix) restores the balance between euchromatin epigenetic marks such as H3K4me3 and heterochromatin epigenetic marks such as for example H3K9me3 or H3K27me3 (x) induces the formation of euchromatin; (xi) restores youthful levels of at least one repressive heterochromatin epigenetic mark; and/or (xii) restores the expression of at least one of the genes recited in Table 5 to youthful levels.

(9) The present disclosure stems from the unexpected discovery that, in some embodiments, precise expression of OCT4, SOX2, and KLF4 in the absence of exogenous c-Myc expression can be used to promote reprogramming and tissue regeneration in vivo without acute toxicity. The expression vectors provided herein, in certain embodiments, allow for precise control of OCT4, SOX2, and KLF4 (OSK) expression, incorporation into viruses (e.g., adeno-associated virus (AAV) at a high viral titer (e.g., more than 2×10.sup.12 particles per preparation, 1×10.sup.13 particles per mL), reversing aging, treating diseases, including ocular diseases, and/or tissue regeneration (e.g. optic nerve regeneration) in vivo following damage.

(10) As shown in FIG. **14**, mice with inducible transgene expression of OCT4, SOX2, and KLF4 (OSK) died two days after induction of OSK expression, due to generalized cytological and architectural dysplasia in the intestinal epithelium. A similar finding has been reported in mice with transgene of OCT4, SOX2, and KLF4 μlus c-Myc (Abad et al., Nature. 2013 Oct. 17; 502(7471):340-5; Ocampo et al., Cell. 2016 Dec. 16; 167:1719-33). Surprisingly, in some embodiments as shown in FIG. **14**, expression of OCT4, SOX2, and KLF4 did not cause toxicity or cancer in vivo. Continuous expression (e.g., induction by doxycycline administration) of OCT4, SOX2, and KLF4 through AAV9 delivery (TRE−OSK with UBC−rtTA4) did not result in teratoma formation in vivo. No teratoma or body weight loss was detected for three months when AAV9 viruses encoding these three transcription factors were delivered to the entire body of mice (FIG. **14**).

(11) Accordingly, provided herein, in certain embodiments, are nucleic acids (e.g., engineered nucleic acid) capable of inducing expression of OCT4, KLF4, inducing agent, and/or SOX2 and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) comprising the same. The nucleic acids may encode OCT4, KLF4, and/or SOX2. The nucleic acids may encode a transcription factor selected from the group consisting of OCT4; KLF4; SOX2; and any combinations thereof. In certain embodiments, a nucleic encodes two or more transcription factors selected from the group consisting of OCT4, KLF4, and SOX2. In certain embodiments, a nucleic acid encodes OCT4 and SOX2, OCT4 and KLF4. In certain embodiments, a nucleic acid encodes SOX2 and KLF4. In certain embodiments, a nucleic encodes OCT4, KLF4, and SOX2. In certain embodiments, a nucleic acid encodes four or more transcription factors (e.g., OCT4, SOX2, KLF4, and another transcription factor). In some embodiments, the present disclosure provides nucleic acids encoding an inducing agent (e.g., an inducing agent that is capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof). In some embodiments, the nucleic acids encode a Cas9 fusion protein (CRISPR activator) and a guide RNA sequence targeting a promoter or enhancer at the endogenous locus of OCT4,

KLF4, and/or SOX2. In some embodiments, the nucleic acids encode a Cas9 fusion protein (CRISPR activator) and a guide RNA sequence targeting a promoter or enhancer at the endogenous locus of OCT4, SOX2, KLF4, or any combination thereof.

(12) Aspects of the present disclosure also provide methods of regulating cellular reprogramming, promoting tissue repair, promoting tissue survival, promoting tissue regeneration, promoting tissue growth, regulating tissue function, promoting organ regeneration, promoting organ survival, regulating organ function, treating and/or preventing disease, or any combination thereof. Regulating may comprise inducing cellular reprogramming, reversing aging, improving tissue function, improving organ function, tissue repair, tissue survival, tissue regeneration, tissue growth, promoting angiogenesis, reducing scar formation, reducing the appearance of aging, promoting organ regeneration, promoting organ survival, altering the taste and quality of agricultural products derived from animals, treating a disease, or any combination thereof, in vivo or in vitro. The methods may comprise administering any of the nucleic acids described herein (e.g., DNA and/or RNA), any of the engineered proteins encoding KLF4, OCT4, an inducing agent, and/or SOX2, any of the chemical agents activating (e.g., inducing expression of) OCT4, KLF4, an inducing agent, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, an inducing agent, and/or SOX2, and/or any of the recombinant viruses described herein. The methods may comprise administering any of the nucleic acids described herein (e.g., DNA and/or RNA), any of the engineered proteins encoding KLF4, SOX2, OCT4, or any combination thereof, any of the chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof and/or any of the recombinant viruses described herein. In certain embodiments, the engineered nucleic acids comprise DNA and/or RNA. The engineered nucleic acid may be an expression vector or not an expression vector. For example, the engineered nucleic acid may be mRNA or plasmid DNA. In certain embodiments, the method further comprises administering a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent, an engineered protein encoding an inducing agent, a chemical agent capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or a recombinant virus encoding an inducing agent. For example, the engineered nucleic acid may be mRNA or plasmid DNA.

(13) One aspect of the present disclosure provide vectors (e.g., expression vectors) comprising a first nucleic acid (e.g., engineered nucleic acid) encoding OCT4, a second nucleic acid (e.g., engineered nucleic acid) encoding SOX2, a third nucleic acid (e.g., engineered nucleic acid) encoding KLF4, alone or in combination and in the absence of an exogenous nucleic acid (e.g., engineered nucleic acid) capable of expressing c-Myc. In certain embodiments, a vector (e.g., expression vector) comprising a first nucleic acid (e.g., engineered nucleic acid) encoding OCT4, a second nucleic acid (e.g., engineered nucleic acid) encoding SOX2, a third nucleic acid (e.g., engineered nucleic acid) encoding KLF4, or any combination thereof. In certain embodiments, the first, second, and third nucleic acids (e.g., engineered nucleic acids) are present on separate expression vectors. In certain embodiments, two of the first, second, and third nucleic acids (e.g., engineered nucleic acids) are present on the same expression vector. In some embodiments, all three nucleic acids (e.g., engineered nucleic acids) are present on the same expression vector. In certain embodiments, the sequence encoding OCT4 is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 2 or 41. In certain embodiments, the sequence encoding SOX2 is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 4 or 43. In certain embodiments, the sequence encoding KLF4 is at least 70% identical to SEQ ID NO: 6 or 45. In certain embodiments, OCT4, SOX2, KLF4, or any combination thereof is a human protein. In certain embodiments, OCT4, SOX2, KLF4, or any combination thereof is a non-human protein (for example, mammals (e.g., primates (e.g., cynomolgus monkeys, rhesus monkeys); commercially relevant mammals, such as cattle, pigs, horses, sheep, goats, cats, and/or dogs) and birds (e.g., commercially relevant birds, such as

chickens, ducks, geese, and/or turkeys). If two or more of OCT4, SOX2, and KLF4 are on one vector, they may be in any order. The words "first," "second," and "third" are not meant to imply an order of the genes on the vector.

(14) An expression vector of the present disclosure may further comprise an inducible promoter. An expression vector may only have one inducible promoter. In such instances, the expression of OCT4, SOX2, and KLF4 are under the control of the same inducible promoter. In some instances, an expression vector comprises more than one inducible promoter. The inducible promoter may comprise a tetracycline-responsive element (TRE) (e.g., a TRE3G promoter, a TRE2 promoter, or a P tight promoter), mifepristone-responsive promoters (e.g., GAL4-Elb promoter), or a coumermycin-responsive). As an example, a TRE (e.g., TRE3G) promoter may comprise a nucleic acid (e.g., engineered nucleic acid) sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 7. As an example, a TRE (e.g., TRE2) promoter may comprise a nucleic acid (e.g., engineered nucleic acid) sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 23. As an example, a TRE (e.g., P tight) promoter may comprise a nucleic acid (e.g., engineered nucleic acid) sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 24. See, e.g., U.S. Provisional Application, U.S. Ser. No. 62/738,894, entitled MUTANT REVERSE TETRACYCLINE TRANSACTIVATORS FOR EXPRESSION OF GENES, which was filed on Sep. 28, 2018, and the International Patent Application, PCT/US2019/053492, titled MUTANT REVERSE TETRACYCLINE TRANSACTIVATORS FOR EXPRESSION OF GENES, which was filed on the same day as the instant application, each of which is herein incorporated by reference in its entirety.

(15) In certain embodiments, an inducing agent is capable of inducing expression of the first (e.g., OCT4), second (e.g., SOX2), third (e.g., KLF4) nucleic acids (e.g., engineered nucleic acids), or any combination thereof from the inducible promoter in the presence of a tetracycline (e.g., doxycycline). In certain embodiments, the inducing agent is reverse tetracycline-controlled transactivator (rtTA) (e.g., M2-rtTA, rtTA3 or rtTA4). In certain embodiments, the rtTA is rtTA3 comprising an amino acid sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 11. In certain embodiments, the rtTA is rtTA4 and comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 13. In certain embodiments, the rtTA is M2-rtTA and comprises a sequence that is at least 70% % (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 15.

(16) In certain embodiments, an inducing agent is capable of inducing expression of expression of the first nucleic acid (e.g., engineered nucleic acid) (e.g., OCT4), second nucleic acid (e.g., engineered nucleic acid) (e.g., SOX2), third nucleic (e.g., KLF4), or any combination thereof from the inducible promoter in the absence of tetracycline (e.g., doxycycline). In certain embodiments, the inducing agent is tetracycline-controlled transactivator (tTA).

(17) In certain embodiments, an expression vector of the present disclosure comprises a constitutive promoter (e.g., CP1, CMV, EF1 alpha, SV40, PGK1, Ubc, human beta actin, CAG, Ac5, polyhedrin, TEF1, GDS, CaM3 5S, Ubi, H1, and U6 promoter). A constitutive promoter may be operably linked to nucleic acid (e.g., engineered nucleic acid) sequences encoding OCT4, KLF4, SOX2, an inducing agent, or a combination thereof. In some embodiments, an expression vector comprises one constitutive promoter. In some embodiments, an expression vector comprises more than one constitutive promoter.

(18) In certain embodiments, an expression vector of the present disclosure comprises an SV40-derived terminator sequence. In certain embodiments, the SV40-derived sequence is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 8.

(19) In certain embodiments, an expression vector of the present disclosure comprises a separator sequence, which may be useful in producing two separate amino acid sequences from one

transcript. The separator sequence may encode a self-cleaving peptide (e.g., 2A peptide, including a 2A peptide sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 9). In certain embodiments, the separator sequence is an Internal Ribosome Entry Site (IRES).

(20) In certain embodiments, the expression vector is a viral vector (e.g., a lentiviral, a retroviral, or an adeno-associated virus (AAV) vector) (e.g., FIGS. **2-3**). An AAV vector of the present disclosure generally comprises inverted terminal repeats (ITRs) flanking a transgene of interest (e.g., a nucleic acid (e.g., engineered nucleic acid) encoding OCT4, SOX2, KLF4, an inducing agent, or a combination thereof). In some embodiments, the distance between two inverted terminal repeats is less than 5.0 kilobases (kb) (e.g., less than 4.9 kb, less than 4.8 kb, less than 4.7 kb, less than 4.6 kb, less than 4.5 kb, less than 4.4 kb, less than 4.3 kb, less than 4.2 kb, less than 4.1 kb, less than 4 kb, less than 3.5 kb, less than 3 kb, less than 2.5 kb, less than 2 kb, less than 1.5 kb, less than 1 kb, or less than 0.5 kb).

(21) In certain embodiments, an expression vector (e.g., an expression vector encoding OCT4, KLF4, SOX2, an inducing agent, or a combination thereof) of the present disclosure may further comprise a selection agent (e.g., an antibiotic, including blasticidin, geneticin, hygromycin B, mycophenolic acid, puromycin, zeocin, actinomycin D, ampicillin, carbenicillin, kanamycin, and neomycin) and/or detectable marker (e.g., GFP, RFP, luciferase, CFP, mCherry, DsRed2FP, mKate, biotin, FLAG-tag, HA-tag, His-tag, Myc-tag, V5-tag, etc.).

(22) In some embodiments, the expression vector (e.g., viral vector) encoding OCT4, KLF4, and SOX2 comprises the sequence provided in SEQ ID NO: 16, SEQ ID NO: 105, or SEQ ID NO: 121. In some embodiments, the expression vector encoding OCT4, KLF4, and SOX2 comprise the elements depicted in FIG. **2**, FIG. **3**, FIGS. **4**A-**4**AL, FIGS. **5**A-**5**D, or a combination thereof. Viral vectors include adeno-associated virus (AAV) vectors, retroviral vectors, lentiviral vectors, and herpes viral vectors.

(23) In another aspect, the present disclosure provides recombinant viruses (e.g., lentivirus, adenovirus, retrovirus, herpes virus, alphavirus, vaccinia virus or adeno-associated virus (AAV)) comprising any of the expression vectors described herein. In certain embodiments, a recombinant virus encodes a transcription factor selected from OCT4; KLF4; SOX2; and any combinations thereof. In certain embodiments, a recombinant virus encodes two or more transcription factors selected from the group consisting of OCT4, KLF4, and SOX2. In certain embodiments, a recombinant virus encodes OCT4 and SOX2, OCT4 and KLF4, OCT4, KLF4, and SOX2, or SOX2 and KLF4. In certain embodiments, a recombinant virus encodes OCT4, KLF4, and SOX2. In certain embodiments, a four or more transcription factors encodes four or more transcription factors (e.g., OCT4, SOX2, KLF4, and another transcription factor).

(24) In yet another aspect, the present disclosure provides methods of regulating (e.g., inducing) cellular reprogramming, tissue repair, tissue regeneration, organ regeneration, reversing aging, or any combination thereof comprising administering to a cell a first nucleic acid (e.g., engineered nucleic acid) encoding OCT4, a second nucleic acid (e.g., engineered nucleic acid) encoding SOX2, and a third nucleic acid (e.g., engineered nucleic acid) encoding KLF4 in the absence of an exogenous nucleic acid (e.g., engineered nucleic acid) capable of expressing c-Myc. In certain embodiments, the first nucleic acid (e.g., engineered nucleic acid) encoding OCT4, the second nucleic acid (e.g., engineered nucleic acid) encoding SOX2, and the third nucleic acid (e.g., engineered nucleic acid) encoding KLF4 is administered to a subject. The subject may be human or non-human. Non-human subjects include, for example, mammals (e.g., primates (e.g., cynomolgus monkeys, rhesus monkeys); commercially relevant mammals, such as cattle, pigs, horses, sheep, goats, cats, and/or dogs) and birds (e.g., commercially relevant birds, such as chickens, ducks, geese, and/or turkeys). In certain embodiments, the three nucleic acids (e.g., engineered nucleic acids) are administered simultaneously. In certain embodiments, the three nucleic acids (e.g., engineered nucleic acids) are administered simultaneously on the same vector.

(25) In yet another aspect, the present disclosure provides methods of regulating (e.g., inducing) cellular reprogramming, tissue repair, tissue regeneration, organ regeneration, reversing aging, or any combination thereof comprising administering to a cell a first nucleic acid (e.g., engineered nucleic acid) encoding OCT4, a second nucleic acid (e.g., engineered nucleic acid) encoding SOX2, a third nucleic acid (e.g., engineered nucleic acid) encoding KLF4, or any combination thereof. In certain embodiments, the first nucleic acid (e.g., engineered nucleic acid) encoding OCT4, the second nucleic acid (e.g., engineered nucleic acid) encoding SOX2, the third nucleic acid (e.g., engineered nucleic acid) encoding KLF4, or any combination thereof is administered to a subject.
(26) The expression vector comprising one or more of the first, second, and third nucleic acids (e.g., engineered nucleic acids) may be any of the expression vectors described above and herein. In some embodiments, the first nucleic acid, the second nucleic acid, the third nucleic acid, or any combination thereof are present on separate expression vectors. In certain embodiments, two of the first nucleic acid, the second nucleic acid, the third nucleic acid, or any combination thereof are present on the same expression vector. In certain embodiments, all three nucleic acids (e.g., engineered nucleic acids) are present on the same expression vector. In certain embodiments, at least two of the first, second, or third nucleic acids (e.g., engineered nucleic acids) are operably linked to the same promoter. In certain embodiments, all three of the first, second, and third nucleic acids (e.g., engineered nucleic acids) are operably linked to the same promoter.
(27) In some embodiments, the expression vector (e.g., viral expression vector, including lentiviral, retroviral, adeno-associated viral vectors) comprises an inducible promoter (e.g., a promoter comprising a tetracycline-responsive element (TRE) including a TRE3G sequence, a TRE2 sequence, or a P tight sequence), and the method further comprises administering an inducing agent (e.g., a chemical agent, a nucleic acid (e.g., engineered nucleic acid) (e.g., nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent), a protein, light, or temperature). In some embodiments, a pH is used to induce expression of a nucleic acid operably linked to a promoter. In certain embodiments, a chemical agent capable of modulating the activity of an inducing agent is tetracycline (e.g., doxycycline). As a non-limiting example, tetracycline-controlled transactivator (tTA) is an inducing agent whose activity is inhibited by tetracycline. As a non-limiting example, reverse tetracycline-controlled transactivator (rtTA) is an inducing agent whose activity is activated by tetracycline. The inducing agent (e.g., rtTA or tTA) may be encoded by a fourth nucleic acid (e.g., engineered nucleic acid) that is administered nucleic acid. In certain embodiments, the inducing agent (e.g., a chemical agent, a nucleic acid (e.g., engineered nucleic acid) (e.g., a nucleic acid comprising RNA and/or DNA encoding an inducing agent), a protein, light, a particular pH, or temperature) is introduced simultaneously with the nucleic acids (e.g., engineered nucleic acids) encoding OCT4, SOX2, and KLF4. In certain embodiments, the inducing agent (e.g., a chemical agent, a nucleic acid (e.g., engineered nucleic acid) (e.g., a nucleic acid comprising RNA and/or DNA encoding an inducing agent), a protein, light, a particular pH, or temperature) is introduced simultaneously with the nucleic acids (e.g., engineered nucleic acids) encoding one or more (e.g., two or more or three or more) transcription factors selected from OCT4; SOX2; KLF4; and any combinations thereof. A promoter (e.g., constitutive promoter, including CAG and Ubc, or an inducible promoter) may be operably linked to the nucleic acid (e.g., engineered nucleic acid) encoding the inducing agent. In certain embodiments, the promoter operably linked to the nucleic acid (e.g., engineered nucleic acid) encoding the inducing agent is a tissue-specific promoter.
(28) In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding the inducing agent is present on the same expression vector as at least one of the nucleic acids (e.g., engineered nucleic acids) encoding OCT4, SOX2, KLF4, or a combination thereof. In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding the inducing agent is present on a separate expression vector from the nucleic acid (e.g., engineered nucleic acid) encoding OCT4, the nucleic acid (e.g., engineered nucleic acid) SOX2, and the nucleic acid (e.g., engineered nucleic acid)

encoding KLF4. In certain embodiments, the nucleic acids (e.g., engineered nucleic acids) encoding OCT4, SOX2, and KLF4 are present on a first expression vector, and the fourth nucleic acid (e.g., engineered nucleic acid) is present on a second expression vector.

(29) In some embodiments, a nucleic acid encoding OCT4, SOX2, KLF4, and/or an inducing agent is not present on a viral vector. In some embodiments, a nucleic acid encoding one or more (e.g., two or more or three or more) transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof is not present on a viral vector. In some embodiments, the nucleic acid is delivered without a viral vector. In some embodiments, delivery of the nucleic acid that is not on a viral vector comprises administration of a naked nucleic acid, electroporation, use of a nanoparticle, and/or use of a liposome.

(30) The expression vectors may be viral vectors (e.g., lentivirus vectors, adenovirus vectors, retrovirus vectors, herpes virus vectors, alphavirus, vaccinia virus, or AAV vectors). For example, the first expression vector encoding OCT4, SOX2, and KLF4 may comprise the nucleic acid (e.g., engineered nucleic acid) sequence set forth in SEQ ID NO: 16. In some embodiments, the expression vector encoding an inducing agent comprises the sequence provided in SEQ ID NO: 17 (e.g., FIG. **12**), SEQ ID NO: 31 (e.g., FIG. **18**), or SEQ ID NO: 32 (e.g., FIG. **19**).

(31) In certain embodiments, the fourth nucleic acid (e.g., engineered nucleic acid) encoding the inducing agent further comprises an SV-40-derived terminator sequence, including a sequence that is at least 70% identical to SEQ ID NO: 8.

(32) In certain embodiments, the inducing agent is capable of inducing expression from the inducible promoter in the presence of tetracycline (e.g., doxycycline). In certain embodiments, the inducing agent is rtTA (e.g., rtTA3, including rtTA3 with a sequence that is at least 70% identical to SEQ ID NO: 11, and rtTA4, including rtTA4 with a sequence that is at least 70% identical to SEQ ID NO: 13). In certain embodiments, the method further comprises administering tetracycline (e.g., doxycycline) to the cell, tissue, or subject. In certain embodiments, the method comprises removing tetracycline (e.g., doxycycline) from the cell, tissue, or subject.

(33) In certain embodiments, the inducing agent is capable of inducing expression from the inducible promoter in the absence of tetracycline (e.g., doxycycline). In certain embodiments, the inducing agent is a tetracycline transactivator (tTA). Without being bound by a particular theory, tetracycline (e.g., doxycycline) may bind to the tTA and prevent tTA from binding its cognate promoter (e.g., a promoter comprising a tetracycline response element (TRE)) and driving expression of an operably linked nucleic acid. Without being bound by a particular theory, a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent may not be on the same vector as any of the nucleic acids (e.g., engineered nucleic acids) encoding OCT4, KLF4, and SOX2 to reduce the size of a viral vector and improve viral titer.

(34) In certain embodiments, one or more expression vectors (e.g., AAV comprising an expression vector) is administered to a cell, tissue, or a subject in need thereof. The subject may have an injury or condition, is suspected of having a condition or injury, or is at risk for a condition or injury. Without being bound by a particular theory, expression of the transcription factors OCT4, SOX2, and KLF4 induces cellular reprogramming. In some embodiments, when the nucleic acid (e.g., engineered nucleic acid) encoding OCT4, SOX2, KLF4, or a combination thereof is operably linked to an inducible promoter, administration of an inducing agent (e.g., chemical, a protein, a nucleic acid (e.g., engineered nucleic acid) (e.g., a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent) under the appropriate conditions (e.g., in the presence or absence of tetracycline). In certain embodiments, an inducing agent (e.g., rtTA) is capable of binding a promoter and driving expression of an operably linked nucleic acid (e.g., engineered nucleic acid) only when the inducing agent is bound to tetracycline. In certain embodiments, an inducing agent (e.g., tTA) cannot bind a promoter and drive expression of an operably linked nucleic acid (e.g., engineered nucleic acid) when the inducing agent is bound to tetracycline. The condition may be an ocular disease, (e.g., a retinal disease, a corneal disease, or any disease affecting the eye), cancer,

aging, an age-related disease, injury, or a neurodegenerative disease. In certain embodiments, the cell or tissue is from eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine.

(35) In certain embodiments, the tissue is damaged (e.g., due to an injury, an accident, or an iatrogenic injury) and/or is aged tissue. In certain embodiments, the tissue may be considered healthy but suboptimal for performance or survival in current or future conditions (e.g., in agriculture or adverse conditions including toxic therapies, sun exposure, or travel outside the earth's atmosphere).

(36) In certain embodiments, the method comprises further comprises regulation of a biological process. In some embodiments, the methods described herein comprise regulating any biological process, including, cellular reprogramming, tissue repair, tissue survival, tissue regeneration, tissue growth, tissue function, organ regeneration, organ survival, organ function, or any combination thereof. In some embodiments, the methods comprise inducing cellular reprogramming, reversing aging, improving tissue function, improving organ function, promoting tissue repair, promoting tissue survival, promoting tissue regeneration, promoting tissue growth, promoting angiogenesis, reducing scar formation, reducing the appearance of aging including alopecia, hair thining, hair greying, sagging skin, and skin wrinkles, promoting organ regeneration, promoting organ survival, altering the taste and quality of agricultural products derived from animals, treating a disease, or any combination thereof, in vivo or in vitro. For example, the method may induce cellular reprogramming, cell survival, organ regeneration, tissue regeneration, or a combination thereof. In certain embodiments, the method comprises inducing and then stopping cellular reprogramming, cell survival, tissue regeneration, organ regeneration, aging, or a combination thereof. In certain embodiments, the method reverses aging of a cell, tissue, organ, or subject. In some embodiments, the method does not induce teratoma formation. In some embodiments, the method does not induce unwanted cell proliferation. In some embodiments, the method does not induce malignant cell growth. In some embodiments, the method does not induce cancer. In some embodiments, the method does not induce tumor growth or tumor formation. In some embodiments, the method does not induce glaucoma.

(37) In some embodiments, a method described herein reverses the epigenetic clock of a cell, a tissue, an organ, a subject, or any combination thereof. In some embodiments, the epigenetic clock is determined using a DNA methylation-based (DNAm) age estimator. In some embodiments, the method alters the expression of one or more genes associated with ageing. In some embodiments, the method reduces expression of one or more genes associated with ageing. In some embodiments, the method alters the expression of one or more genes associated with ageing. In some embodiments, the one or more genes is one or more sensory genes.

(38) In some embodiments, the method reduces expression of one or more genes associated with ageing. In some embodiments, the method reduces expression of 0610040J01Rik, 1700080N15Rik, 2900064F13Rik, 4833417C18Rik, 4921522P10Rik, 4930447C04Rik, 4930488N15Rik, Ace, Ackr1, Acot10, Acvr1, Adamts17, Adra1b, AI504432, Best3, Boc, Cadm3, Cand2, Ccl9, Cd14, Cd36, Cfh, Chrm3, Chrna4, Cntn4, Cracr2b, Cryaa, CT573017.2, Cyp26a1, Cyp27a1, D330050G23Rik, D930007P13Rik, Ddo, Dgkg, Dlk2, Dnajal-ps, Drd2, Dsel, Dytn, Ecscr, Edn1, Ednrb, Efemp1, Elfn2, Epha10, Ephx1, Erbb4, Fam20a, Fbxw21, Ffar4, Flt4, Fmod, Foxp4, Fzd7, Gabrd, Galntl5, Galnt18, Gfra2, Ggt1, Gm10416, Gm14964, Gm17634, Gm2065, Gm32352, Gm33172, Gm34280, Gm35853, Gm36298, Gm36356, Gm36937, Gm3898, Gm42303, Gm42484, Gm42537, Gm42743, Gm43151, Gm43843, Gm44545, Gm44722, Gm45516, Gm45532, Gm47494, Gm47982, Gm47989, Gm48398, Gm48495, Gm48593, Gm48958, Gm49089, Gm49326, Gm49331, Gm49760, Gm5796, Gm6374, Gm7276, Gm8237, Gm9796, Gm9954, Gpr75, Gprc5c, Grid2ip, Gsg112, Hapln4, Hcn3, Hcn4, Hhat1, Hs6st2, Htr3a, Il1rap, Il1rap12,

Inka1, Kbtbd12, Kcnj11, Kcnk4, Kdelc2, Klhl33, Lamc3, Lilra5, Lman11, Lrfn2, Lrrc38, Lrrn4c1, Ltc4s, Mansc1, Mir344c, Msr1, Mycbpap, Myoc, Ngfr, Nipal2, Olfr1372-ps1, Otop3, P2rx5, P2ry12, P4ha2, Pcdha12, Pcdha2, Pcdhac2, Pcdhb18, Pcdhb5, Pcsk2os1, Pcsk6, Perp, Pkp1, Plxna4, Prickle2, Qsox1, Rapgef4os2, Rbp4, Rcn3, Sec1415, Sel113, Serpinh1, Sgpp2, Shisa6, Siah3, Siglech, Slc12a4, Slc24a2, Slc2a5, Slc4a4, Slitrk3, Smagp, Smoc2, Speer4b, Spon2, Sstr2, Sstr3, St3gal3, Stc1, Stc2, Syndig1, Syt10, Thsd7a, Tlr8, Tmem132a, Tmem132d, Tmem200a, Tmem44, Trpc4, Trpv4, Unc5b, Vgf, Vmn1r90, Vwc21, Wfikkn2, Wnt11, Wnt6, Zeb2os, Zfp608, Zfp976, or any combination thereof. In some embodiments, the method reduces expression of Ace, Kcnk4, Lamc3, Edn1, Syt10, Ngfr, Gprc5c, Cd36, Chrna4, Ednrb, Drd2, or a combination thereof.
(39) In some embodiments, the method increases expression of one or more genes associated with ageing. In some embodiments, the method increases expression of 1700031P21Rik, 1810053B23Rik, 2900045020Rik, 2900060B14Rik, 4921504E06Rik, 4930402F11Rik, 4930453C13Rik, 4930455B14Rik, 4930500H12Rik, 4930549P19Rik, 4930555B11Rik, 4930556J02Rik, 4932442E05Rik, 4933431K23Rik, 4933438K21Rik, 6720475M21Rik, 9830132P13Rik, A430010J10Rik, A530064D06Rik, A530065N20Rik, Abcb5, Abhd17c, AC116759.2, AC131705.1, AC166779.3, Acot12, Adig, Akr1c1, Ankrd1, Asb15, Atp2c2, AU018091, AW822073, Btn110, C130093G08Rik, C730027H18Rik, Ccdc162, Chil6, Col26a1, Corin, Crls1, Cybrd1, Cyp2d12, Cyp7a1, D830005E20Rik, Dlx3, Dnah14, Dsc3, Dthd1, Eid2, Eps811, EU599041, Fam90ala, Fancf, Fau-ps2, Fezf1, Gja5, Gm10248, Gm10513, Gm10635, Gm10638, Gm10718, Gm10722, Gm10800, Gm10801, Gm11228, Gm11251, Gm11264, Gm11337, Gm11368, Gm11485, Gm11693, Gm12793, Gm13050, Gm13066, Gm13323, Gm13339, Gm13346, Gm13857, Gm14387, Gm14770, Gm15638, Gm16072, Gm16161, Gm16181, Gm17200, Gm17791, Gm18025, Gm18757, Gm18795, Gm18848, Gm19719, Gm20121, Gm20356, Gm2093, Gm21738, Gm21940, Gm22933, Gm24000, Gm24119, Gm25394, Gm26555, Gm27047, Gm28262, Gm28530, Gm29295, Gm29825, Gm29844, Gm3081, Gm32051, Gm32122, Gm33056, Gm33680, Gm34354, Gm34643, Gm3551, Gm36660, Gm36948, Gm37052, Gm37142, Gm37262, Gm37535, Gm37569, Gm37589, Gm37647, Gm37648, Gm37762, Gm38058, Gm38069, Gm38137, Gm38218, Gm39139, Gm42535, Gm42680, Gm42895, Gm42994, Gm43027, Gm43158, Gm43288, Gm43366, Gm44044, Gm44081, Gm44187, Gm44280, Gm44535, Gm45338, Gm45644, Gm45740, Gm46555, Gm46565, Gm4742, Gm47485, Gm47853, Gm47992, Gm48225, Gm48314, Gm48383, Gm48673, Gm48804, Gm48832, Gm4994, Gm5487, Gm5724, Gm595, Gm6012, Gm6024, Gm7669, Gm7730, Gm8043, Gm8953, Gm9348, Gm9369, Gm9495, H2a12a, Ido2, Igfbp1, Kif7, Klhl31, Lrrc31, Mc5r, Mgam, Msh4, Mucl2, Mug1, Mybl2, Myh15, Nek10, Neurod6, Nr1h5, Olfr1042, Olfr1043, Olfr1082, Olfr1090, Olfr1124, Olfr1167, Olfr1205, Olfr1206, Olfr1223, Olfr1263, Olfr1264, Olfr1269, Olfr127, Olfr1291-ps1, Olfr1406, Olfr1469, Olfr215, Olfr273, Olfr328, Olfr355, Olfr372, Olfr390, Olfr427, Olfr456, Olfr466, Olfr481, Olfr522, Olfr6, Olfr601, Olfr603, Olfr706, Olfr727, Olfr728, Olfr741, Olfr801, Olfr812, Olfr816, Olfr822, Olfr860, Olfr890, Olfr923, Olfr943, Otog1, Pi15, Pkhd1, Pkhd111, Platr6, Pou3f4, Prr9, Pvalb, Rhag, Sav1, Serpinb9b, Skint1, Skint3, Skint5, Slc10a5, Slc6a4, Smok2a, Tcaf3, Tomm20l, Trcg1, Trdn, Ugt1a6a, Usp17la, Vmn1r178, Vmn1r179, Vmn1r33, Vmn1r74, Vmn1r87, Vmn2r102, Vmn2r113, Vmn2r17, Vmn2r52, Vmn2r66, Vmn2r68, Vmn2r76, Vmn2r78, Wnt16, or any combination thereof. In some embodiments, the method increases expression of Olfr816, Olfr812, Olfr1264, Olfr727, Olfr923, Olfr1090, Olfr328, Olfr1124, Olfr522, Olfr1082, Olfr1206, Olfr1167, Olfr706, Olfr6, Pou3f4, Olfr603, Olfr127, Olfr1263, Olfr1269, Olfr1205, Olfr390, Olfr601, Olfr860, Olfr215, Olfr741, Olfr1469, Olfr355, Olfr481, Olfr456, Olfr1042, Olfr728, Olfr372, Olfr801, Olfr1223, Olfr822, Otog1, Olfr943, Olfr1406, Olfr273, Olfr466, Olfr1043, Olfr427, Olfr890, Rbp4, or any combination thereof.
(40) Further aspects of the disclosure relate to methods of reprogramming comprising rejuvenating the epigenetic clock of a cell, tissue, organ, subject, or any combination thereof.
(41) Further aspects of the disclosure relate to methods of reprogramming comprising altering the

expression of one or more genes associated with ageing.
(42) Further aspects of the disclosure relate to methods comprising resetting the transcriptional profile of an old cell, an old organ, an old tissue, and/or any combination thereof in vitro.
(43) Further aspects of the disclosure relate to methods comprising resetting the transcriptional profile of an old cell, an old organ, an old tissue, an old subject and/or any combination thereof in vivo.
(44) Further aspects of the disclosure relate to methods of transdifferentiating cells.
(45) Another aspect of the present disclosure provides engineered cells generated by any of the methods described herein. The methods described herein may be useful in the production of any engineered cell, including induced pluripotent stem cells. The engineered cells of the present disclosure may be produced ex vivo and the methods may further comprise generating an engineered tissue or engineered organ. In some embodiments, the methods of the present disclosure comprise administering an engineered cell, engineered tissue, and/or engineered organ of the present disclosure to a subject in need thereof. In some embodiments, the method further comprises treating a disease.
(46) Aspects of the present disclosure also provide compositions comprising any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, inducing agent, and/or SOX2 expression (e.g., expression vector), any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) OCT4, KLF4, an inducing agent, and/or SOX2, any of the antibodies activating (e.g., inducing expression of) OCT4, KLF4, an inducing agent, and/or SOX2, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination. In some embodiments, the pharmaceutical compositions of the present disclosure further comprise a pharmaceutically acceptable carrier. In some embodiments, the pharmaceutical composition further comprises a second engineered nucleic acid (e.g., engineered nucleic acid) (e.g., expression vector including viral vector) encoding an inducing agent (e.g., rtTA or tTA).
(47) Aspects of the present disclosure also provide compositions comprising any of the nucleic acids (e.g., engineered nucleic acid) acids capable of inducing OCT4, KLF4, and/or SOX2 (e.g., expression vector, including an inducible expression vector), any of the engineered proteins described herein, any of the chemical agents capable of activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, any of the antibodies capable of activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone or in combination. In some embodiments, the composition further comprises a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent, an engineered protein encoding an inducing agent, a chemical agent capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or a recombinant virus encoding an inducing agent. In some embodiments, the pharmaceutical compositions of the present disclosure further comprise a pharmaceutically acceptable carrier. In some embodiments, the pharmaceutical composition further comprises a second engineered nucleic acid (e.g., engineered nucleic acid) (e.g., expression vector including viral vector) encoding an inducing agent (e.g., rtTA or tTA).
(48) Aspects of the present disclosure also provide compositions comprising any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of one or more transcription factors selected from OCT4; SOX2; KLF4; and any combination thereof, any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) one or more transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof, any of the antibodies activating (e.g., inducing expression of) one or more transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described

herein, alone or in combination. In certain embodiments, a composition comprises any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of two or more transcription factors selected from OCT4; SOX2; KLF4; and any combination thereof, any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) two or more transcription selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof, any of the antibodies activating (e.g., inducing expression of) two or more transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone or in combination. The two or more transcription factors may comprise OCT4 and SOX2, OCT4 and KLF4, OCT4, KLF4, and SOX2, or SOX2 and KLF4. In certain embodiments, a composition comprises any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of three or more transcription factors selected from OCT4, SOX2, KLF4, and combinations thereof expression, any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) three or more transcription selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof, any of the antibodies activating (e.g., inducing expression of) three or more transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone or in combination. In certain embodiments, the three or more transcription factors may comprise OCT4, SOX2, and KLF4. In some embodiments, a pharmaceutical composition further comprises a nucleic acid (e.g., engineered nucleic acid) (e.g., expression vector) encoding an inducing agent (e.g., rtTA or tTA), any of the engineered proteins encoding an inducing agent, any of the chemical agents capable of activating (e.g., inducing expression of) an inducing agent, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) encoding an inducing agent. In some embodiments, the pharmaceutical compositions of the present disclosure further comprise a pharmaceutically acceptable carrier.

(49) In yet another aspect, the present disclosure provides kits comprising any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, inducing agent, and/or SOX2 expression (e.g., expression vector), any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) OCT4, KLF4, an inducing agent, and/or SOX2, any of the antibodies activating (e.g., inducing expression of) OCT4, KLF4, an inducing agent, and/or SOX2, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein.

(50) In yet another aspect, the present disclosure provides kits comprising any of the nucleic acids (e.g., engineered nucleic acid) acids capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, any of the antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein. In some embodiments, the kit further comprises a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent, an engineered protein encoding an inducing agent, a chemical agent capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or a recombinant virus encoding an inducing agent.

(51) In yet another aspect, the present disclosure provides kits comprising any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of one or more transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof, any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) one or more transcription factors selected from the group

consisting of OCT4; SOX2; KLF4; and any combinations thereof, any of the antibodies activating (e.g., inducing expression of) OCT4; SOX2; KLF4; and any combinations thereof, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein. In certain embodiments, a kit further comprises a nucleic acid (e.g., engineered nucleic acid) (e.g., expression vector) encoding an inducing agent (e.g., rtTA or tTA), any of the engineered proteins encoding an inducing agent, any of the chemical agents capable of activating (e.g., inducing expression of) an inducing agent, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) encoding an inducing agent.
(52) The details of one or more embodiments of the invention are set forth herein. Other features, objects, and advantages of the invention will be apparent from the Detailed Description, Examples, Figures, and Claims.
(53) References cited in this application are incorporated herein by reference.
Definitions
(54) Definitions of specific terms are described in more detail below. The disclosure is not intended to be limited in any manner by the exemplary listing of substituents described herein.
(55) "AAV" or "adeno-associated virus" is a nonenveloped virus that is capable of carrying and delivering nucleic acids (e.g., engineered nucleic acids) (e.g., nucleic acids (e.g., engineered nucleic acids) encoding OCT4; KLF4; SOX2; or any combination thereof) and belongs to the genus Dependoparvovirus. In some instances, an AAV is capable of delivering a nucleic acid encoding an inducing agent. In general, AAV does not integrate into the genome. The tissue-specific targeting capabilities of AAV is often determined by the AAV capsid serotype (see, e.g., Table 1 below for examples of AAV serotypes and their utility in tissue-specific delivery). Non-limiting serotypes of AAV include AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, and variants thereof. In certain embodiments, the AAV serotype is a variant of AAV9 (e.g., AAV PHP.b).
(56) A "recombinant virus" is a virus (e.g., lentivirus, adenovirus, retrovirus, herpes virus, alphavirus, vaccinia virus or adeno-associated virus (AAV))) that has been isolated from its natural environment (e.g., from a host cell, tissue, or a subject) or is artificially produced.
(57) The term "AAV vector" as used herein is a nucleic acid (e.g., engineered nucleic acid) that comprises AAV inverted terminal repeats (ITRs) flanking an expression cassette (e.g., an expression cassette comprising a nucleic acid (e.g., engineered nucleic acid) encoding OCT4, KLF4, and SOX2, each alone or in combination, or an expression cassette encoding rtTA or tTA). An AAV vector may further comprise a promoter sequence.
(58) The terms "administer," "administering," or "administration," as used herein refers to introduction of any of the compositions described herein, any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of one or more transcription factors selected from the group consisting of OCT4; KLF4; SOX2; and any combinations thereof, any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, any of the chemical agents activating (e.g., inducing expression of) one or more transcription factors selected from OCT4; KLF4; SOX2; and any combinations thereof, any of the antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, any of the antibodies activating (e.g., inducing expression of) one or more transcription factors selected from OCT4; KLF4; SOX2; and any combinations thereof, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination to any cell, tissue, organ, and/or subject. In some embodiments, a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent, an engineered protein encoding an inducing agent, a chemical agent capable of modulating (e.g., activating or inhibiting) the activity of an

inducing agent, and/or a recombinant virus encoding an inducing agent is also administered to the cell, tissue, organ and/or subject. Any of the compositions described herein, comprising any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), comprising any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of one or more transcription factors selected from OCT4; KLF4; SOX2; and any combinations thereof, any of the engineered proteins described herein, any of the chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, any of the engineered proteins encoding OCT4, SOX2, KLF4, or any combinations thereof, any of the chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, any of the antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, any of the antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination may be administered intravenously, intradermally, intraarterially, intralesionally, intratumorally, intracranially, intraarticularly, intraprostaticaly, intrapleurally, intranasally, intravitreally, intravaginally, intrarectally, topically, intratumorally, intramuscularly, intraperitoneally, subcutaneously, subconjunctival, intravesicularlly, mucosally, intrapericardially, intraumbilically, intraocularally, orally, topically, locally, systemically, injection, infusion, continuous infusion, localized perfusion bathing target cells directly, via a catheter, in creams, in lipid compositions (e.g., liposomes), or by other method or any combination of the forgoing as would be known to one of ordinary skill in the art (see, for example, *Remington's Pharmaceutical Sciences* (1990), incorporated herein by reference). In some embodiments, a composition comprising a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent, an engineered protein encoding an inducing agent, a chemical agent capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or a recombinant virus encoding an inducing agent is also administered to the cell, tissue, organ and/or subject is administered using any suitable method.

(59) The term “epigenome” or “epigenetics” refers to the modification and structural changes within a cell that control the expression of nucleic acids (e.g., engineered nucleic acids) or genomic information in a cell. Changes to the epigenome occur during, and drive the processes of embryonic development, disease progression, and aging.

(60) The term “epigenetic clock” may refer to an age estimator or an innate biological process. In some embodiments, rejuvenating or reversing the epigenetic clock refers to reducing the estimated age of a cell, tissue, organ, or a subject. The epigenetic clock may be partially or fully reversed or rejuvenated by any of the methods described herein. In some embodiments, an age estimator is an epigenetic age estimator. For example, an epigenetic age estimator may be sets of CpG dinucleotides that when used in combination with a mathematical algorithm may be used to estimate age of a DNA source, including cells, organs, or tissues. In some embodiments, an age estimator is a DNA methylation-based (DNAm) age estimator. In some embodiments, a DNAm age estimator is calculated as an age correlation using Pearson correlation coefficient r, between DNA methylation-based (DNam) age (also known as estimated age) and chronological age. In some embodiments, the DNA methylation-based (DNAm) age estimator is a single-tissue DNA methylation-based age estimator. In some embodiments, the DNA methylation-based age estimator is a multi-tissue DNA methylation-based age estimator. In some embodiments, the DNAm age estimator is DNAm PhenoAge. See, e.g., Horvath and Raj, Nat Rev Genet. 2018 June; 19(6):371-384; Levine et al., Aging (Albany NY). 2018 Apr. 18; 10(4):573-591; and the Examples below.

(61) “Epigenetic information” as used herein includes covalent modifications to DNA, such as 5-methylcytosine(5mC), hydroxymethylcytosine (5hmeC), 5-formylcytosine (fC), and 5-carboxylcytosine (caC), and to certain proteins, such as lysine acetylation, lysine and arginine methylation, serine and threonine phosphorylation, and lysine ubiquitination and sumoylation of

histone proteins, and the 3D architecture of cells, including TADs (topologically associated domains) and compartments. Epigenetic information is sometimes referred to as the “analog” information of the cell.

(62) “Restoring the expression” of at least one gene in Table 5 to youthful levels is meant to include increasing the expression of a downregulated gene or decreasing the expression of an upregulated gene that changes during aging.

(63) As used herein, the term “cell” is meant not only to include an individual cell but refers also to the particular tissue or organ from which it originates.

(64) The term “cellular senescence” refers to a cell that has exited the cell cycle, displays epigenetic markers consistent with senescence, or expressing senescence cell markers (e.g. senescence-associated beta-galactosidase, or inflammatory cytokines). Cellular senescence may be partial or complete.

(65) The term “gene expression” refers to the degree to which certain genes or all genes in a cell or tissue are transcribed into RNA. In some instances, the RNA is translated by the cell into a protein. The epigenome dictates gene expression patterns.

(66) The term “cellular reprogramming” refers to the process of altering the epigenome of a cell using reprogramming factors (e.g. reversing or preventing epigenetic changes in cells that are causes of dysfunction, deterioration, cell death, senescence or aging). Cellular reprogramming may be complete reprogramming, such that a differentiated cell (e.g., somatic cell) is reprogrammed to a pluripotent stem cell. Cellular reprogramming may be incomplete, such that a differentiated cell (e.g., somatic cell) retains its cellular identity (e.g., lineage-specific stem cell). Cellular reprogramming may be incomplete, e.g., a stem cell is not created, such that a cell is rejuvenated, or takes on more youthful attributes (e.g. increased survival, reduced inflammation, or ability to divide). Cellular reprogramming may provide additional cellular functions, or prevent cellular aging (e.g., transdifferentiation, or transition into cellular senescence). Cellular reprogramming may induce temporary or permanent gene expression changes. In some embodiments, incomplete cellular reprogramming is shown by the lack of Nanog expression. In some embodiments, cellular reprogramming prevents senescence from occurring.

(67) The term “rejuvenating a cell” as used herein is meant to include preventing or reversing the cellular causes of aging without inducing a pluripotent state. A rejuvenated cell as used herein includes for example a retinal ganglion cell that expresses RBPMS and or Brn3a.

(68) A “pluripotent state” as used herein is meant to include a state in which the cell expresses at least one stem cell marker such as but not limited to Esrrb, Nanog, Lin28, TRA-1-60/TRA-1-81/TRA-2-54, SSEA1, or SSEA4. Methods of measuring the expression of stem cell markers on the cell are known in the art and include the methods described herein.

(69) The term “transdifferentiation” refers to a process in which one cell type is changed into another cell type without entering a pluripotent state. Transdifferentiation may also be referred to as lineage reprogramming or lineage conversion. See, e.g., Cieślar-Pobuda et al., Biochim Biophys Acta Mol Cell Res. 2017 July; 1864(7):1359-1369, which is herein incorporated by reference in its entirety.

(70) The terms “condition,” “disease,” and “disorder” are used interchangeably. Non-limiting examples of conditions, diseases, and disorders include acute injuries, neurodegenerative diseases, chronic diseases, proliferative diseases, cardiovascular diseases, genetic diseases, inflammatory diseases, autoimmunue diseases, neurological diseases, hematological diseases, painful conditions, psychiatric disorders, metabolic disorders, chronic diseases, cancers, aging, age-related diseases, and diseases affecting any tissue in a subject. For example, age-related conditions include, heart failure, stroke, heart disease, atherosclerosis, neurodegenerative diseases (e.g., Parkinson's disease and Alzheimer's disease), cognitive decline, memory loss, diabetes, osteoporosis, arthritis, muscle loss, hearing loss (partial or total), eye-related conditions (e.g., poor eye sight or retinal disease), glaucoma, a progeroid syndrome (e.g., Hutchinson-Gilford progeria syndrome), and cancer. In

certain embodiments, the disease is a retinal disease (e.g., macular degeneration). In some embodiments, an age-related condition is senescence. As a non-limiting example, senescence of glial cells may be a cause of Alzheimer's disease. See e.g., Bussian, et al., Nature. 2018 Sep. 19. In some instances, the condition is nerve damage. In some instances, the condition is damage in the central nervous system (CNS). In some instances, the nerve damage is peripheral nerve damage. In some instances, the nerve damage is neurapraxia, axonotmesis, or neurotmesis.

(71) In some instances, a condition increases the DNA methylation-based age of a cell, a tissue, an organ, and/or a subject relative to a control. In some instances, a condition increases the DNA methylation-based age of a cell, a tissue, an organ, and/or a subject by at least 1%, at least 5%, at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 100%, at least 200%, at least 300%, at least 400%, at least 500%, at least 600%, at least 700%, at least 800%, at least 900%, or at least 1,000% relative to a control. In some instances, the control is a cell, a tissue, an organ, and/or a subject that does not have the condition. In some instances, the control is the same cell, tissue, organ, and/or subject prior to having the condition. Without being bound by a particular theory, any of the methods described herein may be useful in decreasing the DNA methylation-based age of a diseased cell, a diseased tissue, a diseased organ, and/or a subject who has, is at risk for, or is suspected of having a disease. In some instances, the disease increases the DNA-methylation-based age of the cell, tissue, organ, and/or subject. In some instances, the disease is an injury.

(72) In some instances, the condition is ageing. In some instances, aging is driven by epigenetic noise. See, e.g., Oberdoerffer and Sinclair. Nat Rev Mol Cell Biol 8, 692-702, doi:10.1038/nrm2238 (2007); Oberdoerffer et al. Cell 135, 907-918, doi:10.1016/j.cell.2008.10.025 (2008). Without being bound by a particular theory, mammalian cells may retain a faithful copy of epigenetic information from earlier in life, analogous to Shannon's “observer” system in Information Theory, essentially a back-up copy of the original signal to allow for its reconstitution at the receiving end if information is lost or noise is introduced during transmission. See, e.g., Shannon, The Bell System Technical Journal 27, 379-423 (1948) for a description of the observer system.

(73) As used herein, an “ocular disease” or “eye disease” is a disease or condition of the eye. Non-limiting examples of conditions that affect the eye include Ectropion, Lagophthalmos, Blepharochalasis, Ptosis, Stye, Xanthelasma, Dermatitis, Demodex, leishmaniasis, loiasis, onchocerciasis, phthiriasis, (herpes simplex), leprosy, molluscum contagiosum, tuberculosis, yaws, zoster, impetigo, Dacryoadenitis, Epiphora, exophthalmos, Conjunctivitis, Scleritis, Keratitis, Corneal ulcer/Corneal abrasion, Snow blindness/Arc eye, Thygeson's superficial punctate keratopathy, Corneal neovascularization, Fuchs' dystrophy, Keratoconus, Keratoconjunctivitis sicca, Iritis, iris, Uveitis, Sympathetic ophthalmia, Cataract, lens, Chorioretinal inflammation, Focal chorioretinal inflammation, chorioretinitis, choroiditis, retinitis, retinochoroiditis, Disseminated chorioretinal inflammation, exudative retinopathy, Posterior cyclitis, Pars planitis, chorioretinal inflammations, Harada's disease, Chorioretinal inflammation, choroid, Chorioretinal scars, Macula scars, posterior pole (postinflammatory) (post-traumatic), Solar retinopathy, Choroidal degeneration, Atrophy, Sclerosis, angioid streaks, choroidal dystrophy, Choroideremia, choroidal, areolar, (peripapillary), Gyrate atrophy, choroid, ornithinaemia, Choroidal haemorrhage, Choroidal detachment, Chorioretinal, Chorioretinal inflammation, infectious and parasitic diseases, Chorioretinitis, syphilitic, toxoplasma, tuberculosis, chorioretinal, Retinal detachment, retina, choroid, distorted vision, Retinoschisis, Hypertensive retinopathy, Diabetic retinopathy, Retinopathy, Retinopathy of prematurity, Age-related macular degeneration, macula, Macular degeneration, Bull's Eye Maculopathy, Epiretinal membrane, Peripheral retinal degeneration, Hereditary retinal dystrophy, Retinitis pigmentosa, Retinal haemorrhage, retinal layers, Central serous retinopathy, Retinal detachment, retinal disorders, Macular edema, macula, Retinal disorder, Diabetic retinopathy, Glaucoma, optic neuropathy, ocular hypertension, open-angle glaucoma,

angle-closure glaucoma, Normal Tension glaucoma, open-angle glaucoma, angle-closure glaucoma, Floaters, Leber's hereditary optic neuropathy, Optic disc drusen, Strabismus, Ophthalmoparesis, eye muscles, Progressive external ophthaloplegia, Esotropia, Exotropia, Disorders of refraction, accommodation, Hypermetropia, Myopia, Astigmatism, Anisometropia, Presbyopia, ophthalmoplegia, Amblyopia, Leber's congenital amaurosis, Scotoma, Anopsia, Color blindness, Achromatopsia/Maskun, cone cells, Nyctalopia, Blindness, River blindness, Micropthalmia/coloboma, optic nerve, brain, spinal cord, Red eye, Argyll Robertson pupil, pupils, Keratomycosis, Xerophthalmia, and Aniridia. In some embodiments, the ocular disease is acute or chronic eye injury.

(74) In some embodiments, the ocular disease is a scratched cornea.

(75) In some embodiments, the ocular disease is glaucoma.

(76) In some embodiments, an ocular disease is a corneal disease (e.g., a disease affecting the cornea or corneal cells). In some embodiments, an ocular disease is acanthamoeba keratitis, ectropion, lagoph amblyopia, anisocoria, astigmatism, Bell's Palsy, blepharitis, blurry vision, burning eyes, cataracts, macular degeneration, age-related macular degeneration, diabetic eye disease, glaucoma, dry eye, poor vision (e.g., low vision), astigmatism, blepharitis, cataract, chalazion, conjunctivitis, diabetic retinopathy, dry eye, glaucoma, keratitis, keratonconus, macular degeneration, ocular hypertension, pinquecula, pterygium, retinitis pigmentosa, or ocular cancer (e.g., retinoblastoma, melanoma of the eye, lymphoma of the eye, medulloepithelioma, squamous cell cancer of the conjunctiva). Examples of corneal diseases include, but are not limited to, corneal neovascularization (NV), corneal dystrophy, corneal inflammation, corneal abrasion, and corneal fibrosis. In some embodiments, the ocular disease is Keritaconus. In some embodiments, an ocular disease is macular degeneration. Additional non-limiting examples of eye diseases may be found in the International Statistical Classification of Diseases and Related Health Problems (e.g., VII Diseases of the eye and adnexa).

(77) An ocular disease may affect any part of the eye and/or adnexa. In some embodiments, the ocular disease is a disorder of the eyelid, lacrimal system and/or orbit. In some embodiments, the ocular disease is a disorders of conjunctiva. In some embodiments, the ocular disease is a disorder of sclera, cornea, iris, and/or ciliary body. In some embodiments, the ocular disease is a disorder of the lens. In some embodiments, the ocular disease is a disorder of choroid and/or retina. In some embodiments, the ocular disease is glaucoma. In some embodiments, the ocular disease is a disorder of vitreous body and/or globe. In some embodiments, the ocular disease is a disorder of optic nerve and/or visual pathways. In some embodiments, the ocular disease is a disorder of ocular muscles, binocular movement, accommodation, and/or refraction. In some embodiments, the ocular disease is a visual disturbance and/or blindness. In some embodiments, the ocular disease is associated with aging, for example, vision loss associated with aging, decline in visual acuity associated with aging, and/or decline in retinal function.

(78) Any suitable method may be used to measure ocular function. Non-limiting examples include visual acuity tests, pattern electroretinograms, and pathology.

(79) The term “genetic disease” refers to a disease caused by one or more abnormalities in the genome of a subject, such as a disease that is present from birth of the subject. Genetic diseases may be heritable and may be passed down from the parents' genes. A genetic disease may also be caused by mutations or changes of the DNAs and/or RNAs of the subject. In such cases, the genetic disease will be heritable if it occurs in the germline. Exemplary genetic diseases include, but are not limited to, Aarskog-Scott syndrome, Aase syndrome, achondroplasia, acrodysostosis, addiction, adreno-leukodystrophy, albinism, ablepharon-macrostomia syndrome, alagille syndrome, alkaptonuria, alpha-1 antitrypsin deficiency, Alport's syndrome, Alzheimer's disease, asthma, autoimmune polyglandular syndrome, androgen insensitivity syndrome, Angelman syndrome, ataxia, ataxia telangiectasia, atherosclerosis, attention deficit hyperactivity disorder (ADHD), autism, baldness, Batten disease, Beckwith-Wiedemann syndrome, Best disease, bipolar disorder,

brachydactyl), breast cancer, Burkitt lymphoma, chronic myeloid leukemia, Charcot-Marie-Tooth disease, Crohn's disease, cleft lip, Cockayne syndrome, Coffin Lowry syndrome, colon cancer, congenital adrenal hyperplasia, Cornelia de Lange syndrome, Costello syndrome, Cowden syndrome, craniofrontonasal dysplasia, Crigler-Najjar syndrome, Creutzfeldt-Jakob disease, cystic fibrosis, deafness, depression, diabetes, diastrophic dysplasia, DiGeorge syndrome, Down's syndrome, dyslexia, Duchenne muscular dystrophy, Dubowitz syndrome, ectodermal dysplasia Ellis-van Creveld syndrome, Ehlers-Danlos, epidermolysis bullosa, epilepsy, essential tremor, familial hypercholesterolemia, familial Mediterranean fever, fragile X syndrome, Friedreich's ataxia, Gaucher disease, glaucoma, glucose galactose malabsorption, glutaricaciduria, gyrate atrophy, Goldberg Shprintzen syndrome (velocardiofacial syndrome), Gorlin syndrome, Hailey-Hailey disease, hemihypertrophy, hemochromatosis, hemophilia, hereditary motor and sensory neuropathy (HMSN), hereditary non polyposis colorectal cancer (HNPCC), Huntington's disease, immunodeficiency with hyper-IgM, juvenile onset diabetes, Klinefelter's syndrome, Kabuki syndrome, Leigh's disease, long QT syndrome, lung cancer, malignant melanoma, manic depression, Marfan syndrome, Menkes syndrome, miscarriage, mucopolysaccharide disease, multiple endocrine neoplasia, multiple sclerosis, muscular dystrophy, myotrophic lateral sclerosis, myotonic dystrophy, neurofibromatosis, Niemann-Pick disease, Noonan syndrome, obesity, ovarian cancer, pancreatic cancer, Parkinson's disease, paroxysmal nocturnal hemoglobinuria, Pendred syndrome, peroneal muscular atrophy, phenylketonuria (PKU), polycystic kidney disease, Prader-Willi syndrome, primary biliary cirrhosis, prostate cancer, REAR syndrome, Refsum disease, retinitis pigmentosa, retinoblastoma, Rett syndrome, Sanfilippo syndrome, schizophrenia, severe combined immunodeficiency, sickle cell anemia, spina bifida, spinal muscular atrophy, spinocerebellar atrophy, sudden adult death syndrome, Tangier disease, Tay-Sachs disease, thrombocytopenia absent radius syndrome, Townes-Brocks syndrome, tuberous sclerosis, Turner syndrome, Usher syndrome, von Hippel-Lindau syndrome, Waardenburg syndrome, Weaver syndrome, Werner syndrome, Williams syndrome, Wilson's disease, xeroderma piginentosum, a progeroid syndrome (e.g., Hutchinson-Gilford progeria syndrome), and Zellweger syndrome.

(80) A “proliferative disease” refers to a disease that occurs due to abnormal growth or extension by the multiplication of cells (Walker, Cambridge Dictionary of Biology; Cambridge University Press: Cambridge, UK, 1990). A proliferative disease may be associated with: 1) the pathological proliferation of normally quiescent cells; 2) the pathological migration of cells from their normal location (e.g., metastasis of neoplastic cells); 3) the pathological expression of proteolytic enzymes such as the matrix metalloproteinases (e.g., collagenases, gelatinases, and elastases); or 4) the pathological angiogenesis as in proliferative retinopathy and tumor metastasis. Exemplary proliferative diseases include cancers (i.e., “malignant neoplasms”), benign neoplasms, angiogenesis, inflammatory diseases, and autoimmune diseases.

(81) The terms “neoplasm” and “tumor” are used herein interchangeably and refer to an abnormal mass of tissue wherein the growth of the mass surpasses and is not coordinated with the growth of a normal tissue. A neoplasm or tumor may be “benign” or “malignant,” depending on the following characteristics: degree of cellular differentiation (including morphology and functionality), rate of growth, local invasion, and metastasis. A “benign neoplasm” is generally well differentiated, has characteristically slower growth than a malignant neoplasm, and remains localized to the site of origin. In addition, a benign neoplasm does not have the capacity to infiltrate, invade, or metastasize to distant sites. Exemplary benign neoplasms include, but are not limited to, lipoma, chondroma, adenomas, acrochordon, senile angiomas, seborrheic keratoses, lentigos, and sebaceous hyperplasias. In some cases, certain “benign” tumors may later give rise to malignant neoplasms, which may result from additional genetic changes in a subpopulation of the tumor's neoplastic cells, and these tumors are referred to as “pre-malignant neoplasms.” An exemplary pre-malignant neoplasm is a teratoma. In contrast, a “malignant neoplasm” is generally poorly differentiated (anaplasia) and has characteristically rapid growth accompanied by progressive

infiltration, invasion, and destruction of the surrounding tissue. Furthermore, a malignant neoplasm generally has the capacity to metastasize to distant sites. The term “metastasis,” “metastatic,” or “metastasize” refers to the spread or migration of cancerous cells from a primary or original tumor to another organ or tissue and is typically identifiable by the presence of a “secondary tumor” or “secondary cell mass” of the tissue type of the primary or original tumor and not of that of the organ or tissue in which the secondary (metastatic) tumor is located. For example, a prostate cancer that has migrated to bone is said to be metastasized prostate cancer and includes cancerous prostate cancer cells growing in bone tissue.

(82) The term “cancer” refers to a class of diseases characterized by the development of abnormal cells that proliferate uncontrollably and have the ability to infiltrate and destroy normal body tissues. See, e.g., Stedman's Medical Dictionary, 25th ed.; Hensyl ed.; Williams & Wilkins: Philadelphia, 1990. Exemplary cancers include, but are not limited to, acoustic neuroma; adenocarcinoma; adrenal gland cancer; anal cancer; angiosarcoma (e.g., lymphangiosarcoma, lymphangioendotheliosarcoma, hemangiosarcoma); appendix cancer; benign monoclonal gammopathy; biliary cancer (e.g., cholangiocarcinoma); bladder cancer; breast cancer (e.g., adenocarcinoma of the breast, papillary carcinoma of the breast, mammary cancer, medullary carcinoma of the breast); brain cancer (e.g., meningioma, glioblastomas, glioma (e.g., astrocytoma, oligodendroglioma), medulloblastoma); bronchus cancer; carcinoid tumor; cervical cancer (e.g., cervical adenocarcinoma); choriocarcinoma; chordoma; craniopharyngioma; colorectal cancer (e.g., colon cancer, rectal cancer, colorectal adenocarcinoma); connective tissue cancer; epithelial carcinoma; ependymoma; endotheliosarcoma (e.g., Kaposi's sarcoma, multiple idiopathic hemorrhagic sarcoma); endometrial cancer (e.g., uterine cancer, uterine sarcoma); esophageal cancer (e.g., adenocarcinoma of the esophagus, Barrett's adenocarcinoma); Ewing's sarcoma; ocular cancer (e.g., intraocular melanoma, retinoblastoma); familiar hypereosinophilia; gall bladder cancer; gastric cancer (e.g., stomach adenocarcinoma); gastrointestinal stromal tumor (GIST); germ cell cancer; head and neck cancer (e.g., head and neck squamous cell carcinoma, oral cancer (e.g., oral squamous cell carcinoma), throat cancer (e.g., laryngeal cancer, pharyngeal cancer, nasopharyngeal cancer, oropharyngeal cancer)); hematopoietic cancers (e.g., leukemia such as acute lymphocytic leukemia (ALL) (e.g., B-cell ALL, T-cell ALL), acute myelocytic leukemia (AML) (e.g., B-cell AML, T-cell AML), chronic myelocytic leukemia (CML) (e.g., B-cell CML, T-cell CML), and chronic lymphocytic leukemia (CLL) (e.g., B-cell CLL, T-cell CLL)); lymphoma such as Hodgkin lymphoma (HL) (e.g., B-cell HL, T-cell HL) and non-Hodgkin lymphoma (NHL) (e.g., B-cell NHL such as diffuse large cell lymphoma (DLCL) (e.g., diffuse large B-cell lymphoma), follicular lymphoma, chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL), mantle cell lymphoma (MCL), marginal zone B-cell lymphomas (e.g., mucosa-associated lymphoid tissue (MALT) lymphomas, nodal marginal zone B-cell lymphoma, splenic marginal zone B-cell lymphoma), primary mediastinal B-cell lymphoma, Burkitt lymphoma, lymphoplasmacytic lymphoma (i.e., Waldenström's macroglobulinemia), hairy cell leukemia (HCL), immunoblastic large cell lymphoma, precursor B-lymphoblastic lymphoma and primary central nervous system (CNS) lymphoma; and T-cell NHL such as precursor T-lymphoblastic lymphoma/leukemia, peripheral T-cell lymphoma (PTCL) (e.g., cutaneous T-cell lymphoma (CTCL) (e.g., mycosis fungoides, Sezary syndrome), angioimmunoblastic T-cell lymphoma, extranodal natural killer T-cell lymphoma, enteropathy type T-cell lymphoma, subcutaneous panniculitis-like T-cell lymphoma, and anaplastic large cell lymphoma); a mixture of one or more leukemia/lymphoma as described above; and multiple myeloma (MM)), heavy chain disease (e.g., alpha chain disease, gamma chain disease, mu chain disease); hemangioblastoma; hypopharynx cancer; inflammatory myofibroblastic tumors; immunocytic amyloidosis; kidney cancer (e.g., nephroblastoma a.k.a. Wilms' tumor, renal cell carcinoma); liver cancer (e.g., hepatocellular cancer (HCC), malignant hepatoma); lung cancer (e.g., bronchogenic carcinoma, small cell lung cancer (SCLC), non-small cell lung cancer (NSCLC), adenocarcinoma of the lung); leiomyosarcoma

(LMS); mastocytosis (e.g., systemic mastocytosis); muscle cancer; myelodysplastic syndrome (MDS); mesothelioma; myeloproliferative disorder (MPD) (e.g., polycythemia vera (PV), essential thrombocytosis (ET), agnogenic myeloid metaplasia (AMM) a.k.a. myelofibrosis (MF), chronic idiopathic myelofibrosis, chronic myelocytic leukemia (CML), chronic neutrophilic leukemia (CNL), hypereosinophilic syndrome (HES)); neuroblastoma; neurofibroma (e.g., neurofibromatosis (NF) type 1 or type 2, schwannomatosis); neuroendocrine cancer (e.g., gastroenteropancreatic neuroendoctrine tumor (GEP-NET), carcinoid tumor); osteosarcoma (e.g., bone cancer); ovarian cancer (e.g., cystadenocarcinoma, ovarian embryonal carcinoma, ovarian adenocarcinoma); papillary adenocarcinoma; pancreatic cancer (e.g., pancreatic andenocarcinoma, intraductal papillary mucinous neoplasm (IPMN), Islet cell tumors); penile cancer (e.g., Paget's disease of the penis and scrotum); pinealoma; primitive neuroectodermal tumor (PNT); plasma cell neoplasia; paraneoplastic syndromes; intraepithelial neoplasms; prostate cancer (e.g., prostate adenocarcinoma); rectal cancer; rhabdomyosarcoma; salivary gland cancer; skin cancer (e.g., squamous cell carcinoma (SCC), keratoacanthoma (KA), melanoma, basal cell carcinoma (BCC)); small bowel cancer (e.g., appendix cancer); soft tissue sarcoma (e.g., malignant fibrous histiocytoma (MFH), liposarcoma, malignant peripheral nerve sheath tumor (MPNST), chondrosarcoma, fibrosarcoma, myxosarcoma); sebaceous gland carcinoma; small intestine cancer; sweat gland carcinoma; synovioma; testicular cancer (e.g., seminoma, testicular embryonal carcinoma); thyroid cancer (e.g., papillary carcinoma of the thyroid, papillary thyroid carcinoma (PTC), medullary thyroid cancer); urethral cancer; vaginal cancer; and vulvar cancer (e.g., Paget's disease of the vulva).

(83) The term “inflammatory disease” refers to a disease caused by, resulting from, or resulting in inflammation. The term “inflammatory disease” may also refer to a dysregulated inflammatory reaction that causes an exaggerated response by macrophages, granulocytes, and/or T-lymphocytes leading to abnormal tissue damage and/or cell death. An inflammatory disease can be either an acute or chronic inflammatory condition and can result from infections or non-infectious causes. Inflammatory diseases include, without limitation, atherosclerosis, arteriosclerosis, autoimmune disorders, multiple sclerosis, systemic lupus erythematosus, polymyalgia rheumatica (PMR), gouty arthritis, degenerative arthritis, tendonitis, bursitis, psoriasis, cystic fibrosis, arthrosteitis, rheumatoid arthritis, inflammatory arthritis, Sjogren's syndrome, giant cell arteritis, progressive systemic sclerosis (scleroderma), ankylosing spondylitis, polymyositis, dermatomyositis, pemphigus, pemphigoid, diabetes (e.g., Type I), myasthenia gravis, Hashimoto's thyroiditis, Graves' disease, Goodpasture's disease, mixed connective tissue disease, sclerosing cholangitis, inflammatory bowel disease, Crohn's disease, ulcerative colitis, pernicious anemia, inflammatory dermatoses, usual interstitial pneumonitis (UIP), asbestosis, silicosis, bronchiectasis, berylliosis, talcosis, pneumoconiosis, sarcoidosis, desquamative interstitial pneumonia, lymphoid interstitial pneumonia, giant cell interstitial pneumonia, cellular interstitial pneumonia, extrinsic allergic alveolitis, Wegener's granulomatosis and related forms of angiitis (temporal arteritis and polyarteritis nodosa), inflammatory dermatoses, hepatitis, delayed-type hypersensitivity reactions (e.g., poison ivy dermatitis), pneumonia, respiratory tract inflammation, Adult Respiratory Distress Syndrome (ARDS), encephalitis, immediate hypersensitivity reactions, asthma, hayfever, allergies, acute anaphylaxis, rheumatic fever, glomerulonephritis, pyelonephritis, cellulitis, cystitis, chronic cholecystitis, ischemia (ischemic injury), reperfusion injury, allograft rejection, host-versus-graft rejection, appendicitis, arteritis, blepharitis, bronchiolitis, bronchitis, cervicitis, cholangitis, chorioamnionitis, conjunctivitis, dacryoadenitis, dermatomyositis, endocarditis, endometritis, enteritis, enterocolitis, epicondylitis, epididymitis, fasciitis, fibrositis, gastritis, gastroenteritis, gingivitis, ileitis, iritis, laryngitis, myelitis, myocarditis, nephritis, omphalitis, oophoritis, orchitis, osteitis, otitis, pancreatitis, parotitis, pericarditis, pharyngitis, pleuritis, phlebitis, pneumonitis, proctitis, prostatitis, rhinitis, salpingitis, sinusitis, stomatitis, synovitis, testitis, tonsillitis, urethritis, urocystitis, uveitis, vaginitis, vasculitis, vulvitis, vulvovaginitis, angitis, chronic bronchitis,

osteomyelitis, optic neuritis, temporal arteritis, transverse myelitis, necrotizing fasciitis, and necrotizing enterocolitis. An ocular inflammatory disease includes, but is not limited to, post-surgical inflammation. In some embodiments, the inflammatory disease is inflammaging (e.g., inflammation that is a side effect of aging).

(84) An “autoimmune disease” refers to a disease arising from an inappropriate immune response of the body of a subject against substances and tissues normally present in the body. In other words, the immune system mistakes some part of the body as a pathogen and attacks its own cells. This may be restricted to certain organs (e.g., in autoimmune thyroiditis) or involve a particular tissue in different places (e.g., Goodpasture's disease which may affect the basement membrane in both the lung and kidney). The treatment of autoimmune diseases is typically with immunosuppression, e.g., medications which decrease the immune response. Exemplary autoimmune diseases include, but are not limited to, glomerulonephritis, Goodpasture's syndrome, necrotizing vasculitis, lymphadenitis, peri-arteritis nodosa, systemic lupus erythematosis, rheumatoid arthritis, psoriatic arthritis, systemic lupus erythematosis, psoriasis, ulcerative colitis, systemic sclerosis, dermatomyositis/polymyositis, anti-phospholipid antibody syndrome, scleroderma, pemphigus vulgaris, ANCA-associated vasculitis (e.g., Wegener's granulomatosis, microscopic polyangiitis), uveitis, Sjogren's syndrome, Crohn's disease, Reiter's syndrome, ankylosing spondylitis, Lyme disease, Guillain-Barré syndrome, Hashimoto's thyroiditis, and cardiomyopathy.

(85) The term “liver disease” or “hepatic disease” refers to damage to or a disease of the liver. Non-limiting examples of liver disease include intrahepatic cholestasis (e.g., alagille syndrome, biliary liver cirrhosis), fatty liver (e.g., alcoholic fatty liver, Reye's syndrome), hepatic vein thrombosis, hepatolenticular degeneration (i.e., Wilson's disease), hepatomegaly, liver abscess (e.g., amebic liver abscess), liver cirrhosis (e.g., alcoholic, biliary, and experimental liver cirrhosis), alcoholic liver diseases (e.g., fatty liver, hepatitis, cirrhosis), parasitic liver disease (e.g., hepatic echinococcosis, fascioliasis, amebic liver abscess), jaundice (e.g., hemolytic, hepatocellular, cholestatic jaundice), cholestasis, portal hypertension, liver enlargement, ascites, hepatitis (e.g., alcoholic hepatitis, animal hepatitis, chronic hepatitis (e.g., autoimmune, hepatitis B, hepatitis C, hepatitis D, drug induced chronic hepatitis), toxic hepatitis, viral human hepatitis (e.g., hepatitis A, hepatitis B, hepatitis C, hepatitis D, hepatitis E), granulomatous hepatitis, secondary biliary cirrhosis, hepatic encephalopathy, varices, primary biliary cirrhosis, primary sclerosing cholangitis, hepatocellular adenoma, hemangiomas, bile stones, liver failure (e.g., hepatic encephalopathy, acute liver failure), angiomyolipoma, calcified liver metastases, cystic liver metastases, fibrolamellar hepatocarcinoma, hepatic adenoma, hepatoma, hepatic cysts (e.g., Simple cysts, Polycystic liver disease, hepatobiliary cystadenoma, choledochal cyst), mesenchymal tumors (mesenchymal hamartoma, infantile hemangioendothelioma, hemangioma, peliosis hepatis, lipomas, inflammatory pseudotumor), epithelial tumors (e.g., bile duct hamartoma, bile duct adenoma), focal nodular hyperplasia, nodular regenerative hyperplasia, hepatoblastoma, hepatocellular carcinoma, cholangiocarcinoma, cystadenocarcinoma, tumors of blood vessels, angiosarcoma, Karposi's sarcoma, hemangioendothelioma, embryonal sarcoma, fibrosarcoma, leiomyosarcoma, rhabdomyosarcoma, carcinosarcoma, teratoma, carcinoid, squamous carcinoma, primary lymphoma, peliosis hepatis, erythrohepatic porphyria, hepatic porphyria (e.g., acute intermittent porphyria, porphyria cutanea tarda), and Zellweger syndrome.

(86) The term “spleen disease” refers to a disease of the spleen. Example of spleen diseases include, but are not limited to, splenomegaly, spleen cancer, asplenia, spleen trauma, idiopathic purpura, Felty's syndrome, Hodgkin's disease, and immune-mediated destruction of the spleen.

(87) The term “lung disease” or “pulmonary disease” refers to a disease of the lung. Examples of lung diseases include, but are not limited to, bronchiectasis, bronchitis, bronchopulmonary dysplasia, interstitial lung disease, occupational lung disease, emphysema, cystic fibrosis, acute respiratory distress syndrome (ARDS), severe acute respiratory syndrome (SARS), asthma (e.g., intermittent asthma, mild persistent asthma, moderate persistent asthma, severe persistent asthma),

chronic bronchitis, chronic obstructive pulmonary disease (COPD), emphysema, interstitial lung disease, sarcoidosis, asbestosis, aspergilloma, aspergillosis, pneumonia (e.g., lobar pneumonia, multilobar pneumonia, bronchial pneumonia, interstitial pneumonia), pulmonary fibrosis, pulmonary tuberculosis, rheumatoid lung disease, pulmonary embolism, and lung cancer (e.g., non-small-cell lung carcinoma (e.g., adenocarcinoma, squamous-cell lung carcinoma, large-cell lung carcinoma), small-cell lung carcinoma).

(88) A “hematological disease” includes a disease which affects a hematopoietic cell or tissue. Hematological diseases include diseases associated with aberrant hematological content and/or function. Examples of hematological diseases include diseases resulting from bone marrow irradiation or chemotherapy treatments for cancer, diseases such as pernicious anemia, hemorrhagic anemia, hemolytic anemia, aplastic anemia, sickle cell anemia, sideroblastic anemia, anemia associated with chronic infections such as malaria, trypanosomiasis, HTV, hepatitis virus or other viruses, myelophthisic anemias caused by marrow deficiencies, renal failure resulting from anemia, anemia, polycythemia, infectious mononucleosis (EVI), acute non-lymphocytic leukemia (ANLL), acute myeloid leukemia (AML), acute promyelocytic leukemia (APL), acute myelomonocytic leukemia (AMMoL), polycythemia vera, lymphoma, acute lymphocytic leukemia (ALL), chronic lymphocytic leukemia, Wilm's tumor, Ewing's sarcoma, retinoblastoma, hemophilia, disorders associated with an increased risk of thrombosis, herpes, thalassemia, antibody-mediated disorders such as transfusion reactions and erythroblastosis, mechanical trauma to red blood cells such as micro-angiopathic hemolytic anemias, thrombotic thrombocytopenic purpura and disseminated intravascular coagulation, infections by parasites such as *Plasmodium*, chemical injuries from, e.g., lead poisoning, and hypersplenism.

(89) The term “neurological disease” refers to any disease of the nervous system, including diseases and injuries that involve the central nervous system (brain, brainstem and cerebellum), the peripheral nervous system (including cranial nerves), and the autonomic nervous system (parts of which are located in both central and peripheral nervous system). Neurodegenerative diseases refer to a type of neurological disease marked by the loss of nerve cells, including, but not limited to, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, tauopathies (including frontotemporal dementia), and Huntington's disease. Examples of neurological diseases include, but are not limited to, vascular dementias, stroke, headache, stupor and coma, dementia, seizure, sleep disorders, trauma, infections, neoplasms, neuro-ophthalmology, movement disorders, demyelinating diseases, spinal cord disorders, and disorders of peripheral nerves, muscle and neuromuscular junctions. Addiction and mental illnesses include, but are not limited to, bipolar disorder and schizophrenia, are also included in the definition of neurological diseases. Further examples of neurological diseases include acquired epileptiform aphasia; acute disseminated encephalomyelitis; adrenoleukodystrophy; agenesis of the corpus callosum; agnosia; Aicardi syndrome; Alexander disease; Alpers' disease; alternating hemiplegia; Alzheimer's disease; amyotrophic lateral sclerosis; anencephaly; Angelman syndrome; angiomatosis; anoxia; aphasia; apraxia; arachnoid cysts; arachnoiditis; Arnold-Chiari malformation; arteriovenous malformation; Asperger syndrome; ataxia telangiectasia; attention deficit hyperactivity disorder; autism; autonomic dysfunction; back pain; Batten disease; Behcet's disease; Bell's palsy; benign essential blepharospasm; benign focal; amyotrophy; benign intracranial hypertension; Binswanger's disease; blepharospasm; Bloch Sulzberger syndrome; brachial plexus injury; brain abscess; brain injury; brain tumors (including glioblastoma multiforme); spinal tumor; Brown-Sequard syndrome; Canavan disease; carpal tunnel syndrome (CTS); causalgia; central pain syndrome; central pontine myelinolysis; cephalic disorder; cerebral aneurysm; cerebral arteriosclerosis; cerebral atrophy; cerebral gigantism; cerebral palsy; Charcot-Marie-Tooth disease; chemotherapy-induced neuropathy and neuropathic pain; Chiari malformation; chorea; chronic inflammatory demyelinating polyneuropathy (CIDP); chronic pain; chronic regional pain syndrome; Coffin Lowry syndrome; coma, including persistent vegetative state; congenital facial diplegia;

corticobasal degeneration; cranial arteritis; craniosynostosis; Creutzfeldt-Jakob disease; cumulative trauma disorders; Cushing's syndrome; cytomegalic inclusion body disease (CIBD); cytomegalovirus infection; dancing eyes-dancing feet syndrome; Dandy-Walker syndrome; Dawson disease; De Morsier's syndrome; Dejerine-Klumpke palsy; dementia; dermatomyositis; diabetic neuropathy; diffuse sclerosis; dysautonomia; dysgraphia; dyslexia; dystonias; early infantile epileptic encephalopathy; empty sella syndrome; encephalitis; encephaloceles; encephalotrigeminal angiomatosis; epilepsy; Erb's palsy; essential tremor; Fabry's disease; Fahr's syndrome; fainting; familial spastic paralysis; febrile seizures; Fisher syndrome; Friedreich's ataxia; frontotemporal dementia and other “tauopathies”; Gaucher's disease; Gerstmann's syndrome; giant cell arteritis; giant cell inclusion disease; globoid cell leukodystrophy; Guillain-Barre syndrome; HTLV-1 associated myelopathy; Hallervorden-Spatz disease; head injury; headache; hemifacial spasm; hereditary spastic paraplegia; heredopathia atactica polyneuritiformis; herpes zoster oticus; herpes zoster; Hirayama syndrome; HIV-associated dementia and neuropathy (see also neurological manifestations of AIDS); holoprosencephaly; Huntington's disease and other polyglutamine repeat diseases; hydranencephaly; hydrocephalus; hypercortisolism; hypoxia; immune-mediated encephalomyelitis; inclusion body myositis; incontinentia pigmenti; infantile; phytanic acid storage disease; Infantile Refsum disease; infantile spasms; inflammatory myopathy; intracranial cyst; intracranial hypertension; Joubert syndrome; Kearns-Sayre syndrome; Kennedy disease; Kinsbourne syndrome; Klippel Feil syndrome; Krabbe disease; Kugelberg-Welander disease; kuru; Lafora disease; Lambert-Eaton myasthenic syndrome; Landau-Kleffner syndrome; lateral medullary (Wallenberg) syndrome; learning disabilities; Leigh's disease; Lennox-Gastaut syndrome; Lesch-Nyhan syndrome; leukodystrophy; Lewy body dementia; lissencephaly; locked-in syndrome; Lou Gehrig's disease (aka motor neuron disease or amyotrophic lateral sclerosis); lumbar disc disease; lyme disease-neurological sequelae; Machado-Joseph disease; macrencephaly; megalencephaly; Melkersson-Rosenthal syndrome; Menieres disease; meningitis; Menkes disease; metachromatic leukodystrophy; microcephaly; migraine; Miller Fisher syndrome; mini-strokes; mitochondrial myopathies; Mobius syndrome; monomelic amyotrophy; motor neurone disease; moyamoya disease; mucopolysaccharidoses; multi-infarct dementia; multifocal motor neuropathy; multiple sclerosis and other demyelinating disorders; multiple system atrophy with postural hypotension; muscular dystrophy; myasthenia gravis; myelinoclastic diffuse sclerosis; myoclonic encephalopathy of infants; myoclonus; myopathy; myotonia congenital; narcolepsy; neurofibromatosis; neuroleptic malignant syndrome; neurological manifestations of AIDS; neurological sequelae of lupus; neuromyotonia; neuronal ceroid lipofuscinosis; neuronal migration disorders; Niemann-Pick disease; O'Sullivan-McLeod syndrome; occipital neuralgia; occult spinal dysraphism sequence; Ohtahara syndrome; olivopontocerebellar atrophy; opsoclonus myoclonus; optic neuritis; orthostatic hypotension; overuse syndrome; paresthesia; Parkinson's disease; paramyotonia congenita; paraneoplastic diseases; paroxysmal attacks; Parry Romberg syndrome; Pelizaeus-Merzbacher disease; periodic paralyses; peripheral neuropathy; painful neuropathy and neuropathic pain; persistent vegetative state; pervasive developmental disorders; photic sneeze reflex; phytanic acid storage disease; Pick's disease; pinched nerve; pituitary tumors; polymyositis; porencephaly; Post-Polio syndrome; postherpetic neuralgia (PHN); postinfectious encephalomyelitis; postural hypotension; Prader-Willi syndrome; primary lateral sclerosis; prion diseases; progressive; hemifacial atrophy; progressive multifocal leukoencephalopathy; progressive sclerosing poliodystrophy; progressive supranuclear palsy; pseudotumor cerebri; Ramsay-Hunt syndrome (Type I and Type II); Rasmussen's Encephalitis; reflex sympathetic dystrophy syndrome; Refsum disease; repetitive motion disorders; repetitive stress injuries; restless legs syndrome; retrovirus-associated myelopathy; Rett syndrome; Reye's syndrome; Saint Vitus Dance; Sandhoff disease; Schilder's disease; schizencephaly; septo-optic dysplasia; shaken baby syndrome; shingles; Shy-Drager syndrome; Sjogren's syndrome; sleep apnea; Soto's syndrome; spasticity; spina bifida; spinal cord injury; spinal cord tumors; spinal muscular atrophy; stiff-person syndrome; stroke;

Sturge-Weber syndrome; subacute sclerosing panencephalitis; subarachnoid hemorrhage; subcortical arteriosclerotic encephalopathy; sydenham chorea; syncope; syringomyelia; tardive dyskinesia; Tay-Sachs disease; temporal arteritis; tethered spinal cord syndrome; Thomsen disease; thoracic outlet syndrome; tic douloureux; Todd's paralysis; Tourette syndrome; transient ischemic attack; transmissible spongiform encephalopathies; transverse myelitis; traumatic brain injury; tremor; trigeminal neuralgia; tropical spastic paraparesis; tuberous sclerosis; vascular dementia (multi-infarct dementia); vasculitis including temporal arteritis; Von Hippel-Lindau Disease (VHL); Wallenberg's syndrome; Werdnig-Hoffman disease; West syndrome; whiplash; Williams syndrome; Wilson's disease; and Zellweger syndrome.

(90) The term “musculoskeletal disease” or “MSD” refers to an injury and/or pain in a subject's joints, ligaments, muscles, nerves, tendons, and structures that support limbs, neck, and back. In certain embodiments, an MSD is a degenerative disease. In certain embodiments, an MSD includes an inflammatory condition. Body parts of a subject that may be associated with MSDs include upper and lower back, neck, shoulders, and extremities (arms, legs, feet, and hands). In certain embodiments, an MSD is a bone disease, such as achondroplasia, acromegaly, bone callus, bone demineralization, bone fracture, bone marrow disease, bone marrow neoplasm, dyskeratosis congenita, leukemia (e.g., hairy cell leukemia, lymphocytic leukemia, myeloid leukemia, Philadelphia chromosome-positive leukemia, plasma cell leukemia, stem cell leukemia), systemic mastocytosis, myelodysplastic syndromes, paroxysmal nocturnal hemoglobinuria, myeloid sarcoma, myeloproliferative disorders, multiple myeloma, polycythemia vera, pearson marrow-pancreas syndrome, bone neoplasm, bone marrow neoplasm, Ewing sarcoma, osteochondroma, osteoclastoma, osteosarcoma, brachydactyly, Camurati-Engelmann syndrome, Craniosynostosis, Crouzon craniofacial dysostosis, dwarfism, achondroplasia, bloom syndrome, Cockayne syndrome, Ellis-van Creveld syndrome, Seckel syndrome, spondyloepiphyseal dysplasia, spondyloepiphyseal dysplasia congenita, Werner syndrome, hyperostosis, osteophyte, Klippel-Trenaunay-Weber syndrome, Marfan syndrome, McCune-Albright syndrome, osteitis, osteoarthritis, osteochondritis, osteochondrodysplasia, Kashin-Beck disease, Leri-Weill dyschondrosteosis, osteochondrosis, osteodystrophy, osteogenesis imperfecta, osteolysis, Gorham-Stout syndrome, osteomalacia, osteomyelitis, osteonecrosis, osteopenia, osteopetrosis, osteoporosis, osteosclerosis, otospondylomegaepiphyseal dysplasia, pachydermoperiostosis, Paget disease of bone, Polydactyly, Meckel syndrome, rickets, Rothmund-Thomson syndrome, Sotos syndrome, spondyloepiphyseal dysplasia, spondyloepiphyseal dysplasia congenita, syndactyly, Apert syndrome, syndactyly type II, or Werner syndrome. In certain embodiments, an MSD is a cartilage disease, such as cartilage neoplasm, osteochondritis, osteochondrodysplasia, Kashin-Beck disease, or Leri-Weill dyschondrosteosis. In certain embodiments, an MSD is hernia, such as intervertebral disk hernia. In certain embodiments, an MSD is a joint disease, such as arthralgia, arthritis (e.g., gout (e.g., Kelley-Seegmiller syndrome, Lesch-Nyhan syndrome), Lyme disease, osteoarthritis, psoriatic arthritis, reactive arthritis, rheumatic fever, rheumatoid arthritis, Felty syndrome, synovitis, Blau syndrome, nail-patella syndrome, spondyloarthropathy, reactive arthritis, Stickler syndrome, synovial membrane disease, synovitis, or Blau syndrome. In certain embodiments, an MSD is Langer-Giedion syndrome. In certain embodiments, an MSD is a muscle disease, such as Barth syndrome, mitochondrial encephalomyopathy, MELAS syndrome, MERRF syndrome, MNGIE syndrome, mitochondrial myopathy, Kearns-Sayre syndrome, myalgia, fibromyalgia, polymyalgia rheumatica, myoma, myositis, dermatomyositis, neuromuscular disease, Kearns-Sayre syndrome, muscular dystrophy, myasthenia, congenital myasthenic syndrome, Lambert-Eaton myasthenic syndrome, myasthenia gravis, myotonia, myotonia congenita, spinal muscular atrophy, tetany, ophthalmoplegia, or rhabdomyolysis. In certain embodiments, an MSD is Proteus syndrome. In certain embodiments, an MSD is a rheumatic diseases, such as arthritis (e.g., gout (e.g., Kelley-Seegmiller syndrome, Lesch-Nyhan lyme disease)), osteoarthritis, psoriatic arthritis, reactive arthritis, rheumatic fever, rheumatoid arthritis, Felty syndrome, synovitis, Blau syndrome, gout

(e.g., Kelley-Seegmiller syndrome, Lesch-Nyhan syndrome), polymyalgia rheumatica, rheumatic fever, rheumatic heart disease, or Sjogren syndrome. In certain embodiments, an MSD is Schwartz-Jampel syndrome. In certain embodiments, an MSD is a skeleton disease, such as Leri-Weill dyschondrosteosis, skeleton malformations, Melnick-Needles syndrome, pachydermoperiostosis, Rieger syndrome, spinal column disease, intervertebral disk hernia, scoliosis, spina bifida, spondylitis, ankylosing spondylitis, spondyloarthropathy, reactive arthritis, spondyloepiphyseal dysplasia, spondyloepiphyseal dysplasia congenita, or spondylosis. In some embodiments, the disease is a musculoskeletal disease.

(91) A “painful condition” includes, but is not limited to, neuropathic pain (e.g., peripheral neuropathic pain), central pain, deafferentiation pain, chronic pain (e.g., chronic nociceptive pain, and other forms of chronic pain such as post-operative pain, e.g., pain arising after hip, knee, or other replacement surgery), pre-operative pain, stimulus of nociceptive receptors (nociceptive pain), acute pain (e.g., phantom and transient acute pain), noninflammatory pain, inflammatory pain, pain associated with cancer, wound pain, burn pain, postoperative pain, pain associated with medical procedures, pain resulting from pruritus, painful bladder syndrome, pain associated with premenstrual dysphoric disorder and/or premenstrual syndrome, pain associated with chronic fatigue syndrome, pain associated with pre-term labor, pain associated with withdrawal symptoms from drug addiction, joint pain, arthritic pain (e.g., pain associated with crystalline arthritis, osteoarthritis, psoriatic arthritis, gouty arthritis, reactive arthritis, rheumatoid arthritis or Reiter's arthritis), lumbosacral pain, musculo-skeletal pain, headache, migraine, muscle ache, lower back pain, neck pain, toothache, dental/maxillofacial pain, visceral pain and the like. One or more of the painful conditions contemplated herein can comprise mixtures of various types of pain provided above and herein (e.g. nociceptive pain, inflammatory pain, neuropathic pain, etc.). In some embodiments, a particular pain can dominate. In other embodiments, the painful condition comprises two or more types of pains without one dominating. A skilled clinician can determine the dosage to achieve a therapeutically effective amount for a particular subject based on the painful condition.

(92) The term “psychiatric disorder” refers to a disease of the mind and includes diseases and disorders listed in the *Diagnostic and Statistical Manual of Mental Disorders—Fourth Edition* (DSM-IV), published by the American Psychiatric Association, Washington D. C. (1994). Psychiatric disorders include, but are not limited to, anxiety disorders (e.g., acute stress disorder agoraphobia, generalized anxiety disorder, obsessive-compulsive disorder, panic disorder, posttraumatic stress disorder, separation anxiety disorder, social phobia, and specific phobia), childhood disorders, (e.g., attention-deficit/hyperactivity disorder, conduct disorder, and oppositional defiant disorder), eating disorders (e.g., anorexia nervosa and bulimia nervosa), mood disorders (e.g., depression, bipolar disorder, cyclothymic disorder, dysthymic disorder, and major depressive disorder), personality disorders (e.g., antisocial personality disorder, avoidant personality disorder, borderline personality disorder, dependent personality disorder, histrionic personality disorder, narcissistic personality disorder, obsessive-compulsive personality disorder, paranoid personality disorder, schizoid personality disorder, and schizotypal personality disorder), psychotic disorders (e.g., brief psychotic disorder, delusional disorder, schizoaffective disorder, schizophreniform disorder, schizophrenia, and shared psychotic disorder), substance-related disorders (e.g., alcohol dependence, amphetamine dependence, cannabis dependence, cocaine dependence, hallucinogen dependence, inhalant dependence, nicotine dependence, opioid dependence, phencyclidine dependence, and sedative dependence), adjustment disorder, autism, delirium, dementia, multi-infarct dementia, learning and memory disorders (e.g., amnesia and age-related memory loss), and Tourette's disorder.

(93) The term “metabolic disorder” refers to any disorder that involves an alteration in the normal metabolism of carbohydrates, lipids, proteins, nucleic acids, or a combination thereof. A metabolic disorder is associated with either a deficiency or excess in a metabolic pathway resulting in an

imbalance in metabolism of nucleic acids, proteins, lipids, and/or carbohydrates. Factors affecting metabolism include, and are not limited to, the endocrine (hormonal) control system (e.g., the insulin pathway, the enteroendocrine hormones including GLP-1, PYY or the like), the neural control system (e.g., GLP-1 in the brain), or the like. Examples of metabolic disorders include, but are not limited to, diabetes (e.g., Type I diabetes, Type II diabetes, gestational diabetes), hyperglycemia, hyperinsulinemia, insulin resistance, and obesity.
(94) In some embodiments, a disease is characterized by cellular dysfunction. For example, a disease may be a mitochondrial disease. Non-limiting mitochondrial diseases include Freidrich's ataxia, alphers disease, barth syndrome, beta-oxidation defects, carnitine deficiency, CPT I deficiency, and mitochondrial DNA depletion. Cellular dysfunction may include mitochondria dysfunction, RNA replication dysfunction, DNA replication dysfunction, translation dysfunction, and/or protein folding dysfunction.
(95) In some embodiments, the disease or condition by a wood, bleeding out, injuries (e.g., broken bones, gunshot wound, cut, scarring during surgery (e.g., cesarean).
(96) In some embodiments, the disease is an infectious disease (e.g., a disease caused by a pathogen and/or virus). Non-limiting examples of infectious diseases include tuberculosis, HIV/AIDS, rabies, plague, cholera, dengue fever, measles, malaria, meningitis, whooping cough, Lyme disease, influenza, hepatitis C, typhoid fever, and poliomyelitis.
(97) “Cellular causes of aging” as used herein include loss or modification of epigenetic information.
(98) The terms “c-Myc” or “Myc” refer to a nuclear phosphoprotein that has been implicated in cell cycle progression. c-Myc is capable of forming a heterodimer with the transcription factor MAX and the heterodimer is capable of binding to an E box consequence sequence on nucleic acids (e.g., engineered nucleic acids) to regulate transcription of target genes. In certain embodiments, a nucleotide sequence encoding c-Myc comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence as described in the NCBI RefSeq database under accession number NM_001354870.1 or NM_002467.5. In certain embodiments, an amino acid sequence encoding c-Myc comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to NP_002458.2 or NP_001341799.1. In certain embodiments, the methods comprise inducing expression of OCT4; KLF4; SOX2; or any combination thereof in the absence of inducing c-Myc expression or in the absence of activating c-Myc. Absence of inducing c-Myc expression may refer to absence of substantial induction of c-Myc expression over endogenous levels of c-Myc expression in a cell, tissue, subject, or any combination thereof. Absence of substantial induction of c-Myc expression as compared to endogenous levels of c-Myc expression in a cell, tissue, subject, or any combination thereof, may refer to increasing c-Myc expression by less than 70%, less than 60%, less than 50%, less than 40%, less than 30%, less than 20%, less than 10%, or any values in between as compared to endogenous levels of c-Myc expression in the cell, tissue, subject, or any combination thereof. Absence of activating c-Myc expression may refer to absence of substantial activation of c-Myc (e.g., activity) over endogenous c-Myc activity in a cell, tissue, subject, or any combination thereof. Absence of substantial induction of c-Myc activity as compared to endogenous c-Myc activity in a cell, tissue, subject, or any combination thereof, may refer to increasing c-Myc activity by less than 70%, less than 60%, less than 50%, less than 40%, less than 30%, less than 20%, less than 10%, or any values in between as compared to endogenous c-Myc activity in the cell, tissue, subject, or any combination thereof.
(99) The terms “effective amount” and “therapeutically effective amount,” as used herein, refer to the amount or concentration of an inventive compound, that, when administered to a subject, is effective to at least partially treat a condition from which the subject is suffering.
(100) As used herein, a protein that is “functional” or “active” is one that retains its biological activity (e.g., capable of acting as a transcription factor or as an inducing agent). Conversely, a

protein that is not functional or is inactive is one that is not capable of performing one or more of its wild-type functions.

(101) The term “gene” refers to a nucleic acid (e.g., engineered nucleic acid) fragment that expresses a protein, including regulatory sequences preceding (5′ non-coding sequences) and following (3′ non-coding sequences) the coding sequence. “Native gene” refers to a gene as found in nature with its own regulatory sequences. “Chimeric gene” or “chimeric construct” refers to any gene or a construct, not a native gene, comprising regulatory and coding sequences that are not found together in nature. Accordingly, a chimeric gene or chimeric construct may comprise regulatory sequences and coding sequences that are derived from different sources, or regulatory sequences and coding sequences derived from the same source, but arranged in a manner different than that found in nature. “Endogenous gene” refers to a native gene in its natural location in the genome of an organism. A “foreign” gene refers to a gene not normally found in the host organism, but which is introduced into the host organism by gene transfer. Foreign genes can comprise native genes inserted into a non-native organism, or chimeric genes. A “transgene” is a gene that has been introduced into the genome by a transformation procedure.

(102) “Homolog” or “homologous” refers to sequences (e.g., nucleic acid (e.g., engineered nucleic acid) or amino acid sequences) that share a certain percent identity (e.g., at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% percent identity). Homologous sequences include but are not limited to paralogous or orthologous sequences. Paralogous sequences arise from duplication of a gene within a genome of a species, while orthologous sequences diverge after a speciation event. A functional homolog retains one or more biological activities of a wild-type protein. In certain embodiments, a functional homolog of OCT4, KLF4, or SOX2 retains at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or at least 100% of the biological activity (e.g., transcription factor activity) of a wild-type counterpart.

(103) “KLF4” may also be referred to as Kruppel-like factor 4, EZF, or GKLF and is a zinc-finger transcription factor. KLF4 has been implicated in regulation of differentiation and proliferation and is capable of interacting with co-activators, including members of the p300-CBP coactivator family. A KLF4 transcription factor, homolog (e.g., functional homolog), or variant thereof, as used herein, may be derived from any species, including humans. In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding human KLF4 comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a nucleic acid (e.g., engineered nucleic acid) described in the NCBI RefSeq database under accession number NM_004235.5 or NM_001314052.1. Non-limiting examples of KLF4 variants include Krueppel-like factor 4 transcript variant 1 and Krueppel-like factor 4 transcript variant 2. In certain embodiments, KLF4 comprises a nucleic acid (e.g., engineered nucleic acid) sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 5 or SEQ ID NO: 44. SEQ ID NO: 5 is a non-limiting example of a nucleotide sequence encoding KLF4 from *Mus musculus*. SEQ ID NO: 44 is a non-limiting example of a nucleotide sequence encoding human KLF4. In certain embodiments, KLF4 comprises an amino acid sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to NP_001300981.1 or NP_004226.3. In certain embodiments, KLF4 comprises an amino acid sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 6. In certain embodiments, KLF4 comprises an amino acid sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID

NO: 45. SEQ ID NO: 6 is a non-limiting example of an amino acid sequence encoding KLF4 from *Mus musculus*. SEQ ID NO: 45 is a non-limiting example of an amino acid sequence encoding human KLF4.

(104) "Inverted terminal repeats" or "ITRs" are nucleic acid (e.g., engineered nucleic acid) sequences that are reverse complements of one another. In general, in an AAV vector, ITRs are found on either side of a cassette (e.g., an expression cassette comprising a nucleic acid (e.g., engineered nucleic acid) encoding OCT4; KLF4; SOX2; or any combination thereof). In some instances, the cassette encodes an inducing agent. AAV ITRs include ITRs from AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, and AAV variants thereof.

(105) The terms "nucleic acid," "polynucleotide", "nucleotide sequence", "nucleic acid (e.g., engineered nucleic acid) molecule", "nucleic acid (e.g., engineered nucleic acid) sequence", and "oligonucleotide" refer to a series of nucleotide bases (also called "nucleotides") in DNA and RNA, and mean any chain of two or more nucleotides. The terms "nucleic acid" or "nucleic acid (e.g., engineered nucleic acid) sequence", "nucleic acid (e.g., engineered nucleic acid) molecule", "nucleic acid (e.g., engineered nucleic acid) fragment" or "polynucleotide" may be used interchangeably with "gene", "mRNA encoded by a gene" and "cDNA".

(106) The nucleic acids (e.g., engineered nucleic acids) can be chimeric mixtures or derivatives or modified versions thereof, single-stranded or double-stranded. The oligonucleotide can be modified at the base moiety, sugar moiety, or phosphate backbone, for example, to improve stability of the molecule, its hybridization parameters, etc. A nucleotide sequence typically carries genetic information, including the information used by cellular machinery to make proteins and enzymes. These terms include double- or single-stranded genomic and cDNA, RNA, any synthetic and genetically manipulated polynucleotide, and both sense and antisense polynucleotides. This includes single- and double-stranded molecules, i.e., DNA-DNA, DNA-RNA and RNA-RNA hybrids, as well as "protein nucleic acids (e.g., engineered nucleic acids)" (PNAs) formed by conjugating bases to an amino acid backbone. This also includes nucleic acids (e.g., engineered nucleic acids) containing carbohydrate or lipids. Exemplary DNAs include single-stranded DNA (ssDNA), double-stranded DNA (dsDNA), plasmid DNA (pDNA), genomic DNA (gDNA), complementary DNA (cDNA), antisense DNA, chloroplast DNA (ctDNA or cpDNA), microsatellite DNA, mitochondrial DNA (mtDNA or mDNA), kinetoplast DNA (kDNA), provirus, lysogen, repetitive DNA, satellite DNA, and viral DNA. Exemplary RNAs include single-stranded RNA (ssRNA), double-stranded RNA (dsRNA), small interfering RNA (siRNA), messenger RNA (mRNA), precursor messenger RNA (pre-mRNA), small hairpin RNA or short hairpin RNA (shRNA), microRNA (miRNA), guide RNA (gRNA), transfer RNA (tRNA), antisense RNA (asRNA), heterogeneous nuclear RNA (hnRNA), coding RNA, non-coding RNA (ncRNA), long non-coding RNA (long ncRNA or lncRNA), satellite RNA, viral satellite RNA, signal recognition particle RNA, small cytoplasmic RNA, small nuclear RNA (snRNA), ribosomal RNA (rRNA), Piwi-interacting RNA (piRNA), polyinosinic acid, ribozyme, flexizyme, small nucleolar RNA (snoRNA), spliced leader RNA, viral RNA, and viral satellite RNA.

(107) The nucleic acids (e.g., engineered nucleic acids) described herein may be synthesized by standard methods known in the art, e.g., by use of an automated DNA synthesizer (such as those that are commercially available from Biosearch, Applied Biosystems, etc.). As examples, phosphorothioate oligonucleotides may be synthesized by the method of Stein et al., *Nucl. Acids Res.*, 16, 3209, (1988), methylphosphonate oligonucleotides can be prepared by use of controlled pore glass polymer supports (Sarin et al., *Proc. Natl. Acad. Sci. U.S.A.* 85, 7448-7451, (1988)). A number of methods have been developed for delivering antisense DNA or RNA to cells, e.g., antisense molecules can be injected directly into the tissue site, or modified antisense molecules, designed to target the desired cells (antisense linked to peptides or antibodies that specifically bind receptors or antigens expressed on the target cell surface) can be administered systemically. Alternatively, RNA molecules may be generated by in vitro and in vivo transcription of DNA

sequences encoding the antisense RNA molecule. Such DNA sequences may be incorporated into a wide variety of vectors that incorporate suitable RNA polymerase promoters such as the T7 or SP6 polymerase promoters. Alternatively, antisense cDNA constructs that synthesize antisense RNA constitutively or inducibly, depending on the promoter used, can be introduced stably into cell lines. However, it is often difficult to achieve intracellular concentrations of the antisense sufficient to suppress translation of endogenous mRNAs. Therefore a preferred approach utilizes a recombinant DNA construct in which the antisense oligonucleotide is placed under the control of a strong promoter. The use of such a construct to transfect target cells in the patient will result in the transcription of sufficient amounts of single stranded RNAs that will form complementary base pairs with the endogenous target gene transcripts and thereby prevent translation of the target gene mRNA. For example, a vector can be introduced in vivo such that it is taken up by a cell and directs the transcription of an antisense RNA. Such a vector can remain episomal or become chromosomally integrated, as long as it can be transcribed to produce the desired antisense RNA. Such vectors can be constructed by recombinant DNA technology methods standard in the art. Vectors can be plasmid, viral, or others known in the art, used for replication and expression in mammalian cells. Expression of the sequence encoding the antisense RNA can be by any promoter known in the art to act in mammalian, preferably human, cells. Such promoters can be inducible or constitutive. Any type of plasmid, cosmid, yeast artificial chromosome, or viral vector can be used to prepare the recombinant DNA construct that can be introduced directly into the tissue site.
(108) The nucleic acids (e.g., engineered nucleic acids) may be flanked by natural regulatory (expression control) sequences or may be associated with heterologous sequences, including promoters, internal ribosome entry sites (IRES) and other ribosome binding site sequences, enhancers, response elements, suppressors, signal sequences, polyadenylation sequences, introns, 5′- and 3′-non-coding regions, and the like. The nucleic acids (e.g., engineered nucleic acids) may also be modified by many means known in the art. Non-limiting examples of such modifications include methylation, “caps”, substitution of one or more of the naturally occurring nucleotides with an analog, and internucleotide modifications, such as, for example, those with uncharged linkages (e.g., methyl phosphonates, phosphotriesters, phosphoroamidates, carbamates, etc.) and with charged linkages (e.g., phosphorothioates, phosphorodithioates, etc.). Polynucleotides may contain one or more additional covalently linked moieties, such as, for example, proteins (e.g., nucleases, toxins, antibodies, signal peptides, poly-L-lysine, etc.), intercalators (e.g., acridine, psoralen, etc.), chelators (e.g., metals, radioactive metals, iron, oxidative metals, etc.), and alkylators. The polynucleotides may be derivatized by formation of a methyl or ethyl phosphotriester or an alkyl phosphoramidate linkage. Furthermore, the polynucleotides herein may also be modified with a label capable of providing a detectable signal, either directly or indirectly. Exemplary labels include radioisotopes, fluorescent molecules, epitope tags, isotopes (e.g., radioactive isotopes), biotin, and the like.
(109) A “recombinant nucleic acid (e.g., engineered nucleic acid) molecule” or “engineered nucleic acid” is a nucleic acid (e.g., engineered nucleic acid) molecule that has undergone a molecular biological manipulation, i.e., non-naturally occurring nucleic acid (e.g., engineered nucleic acid) molecule or genetically engineered nucleic acid (e.g., engineered nucleic acid) molecule. Furthermore, the terms “recombinant DNA molecule” or “engineered nucleic acid” refer to a nucleic acid (e.g., engineered nucleic acid) sequence which is not naturally occurring, or can be made by the artificial combination of two otherwise separated segments of nucleic acid (e.g., engineered nucleic acid) sequence, i.e., by ligating together pieces of DNA that are not normally continuous. By “recombinantly produced” is meant artificial combination often accomplished by either chemical synthesis means, or by the artificial manipulation of isolated segments of nucleic acids (e.g., engineered nucleic acids), e.g., by genetic engineering techniques using restriction enzymes, ligases, and similar recombinant techniques as described by, for example, Sambrook et al., *Molecular Cloning*, second edition, Cold Spring Harbor Laboratory, Plainview, N.Y.; (1989), or

Ausubel et al., *Current Protocols in Molecular Biology*, Current Protocols (1989), and *DNA Cloning: A Practical Approach*, Volumes I and II (ed. D. N. Glover) IREL Press, Oxford, (1985); each of which is incorporated herein by reference.

(110) Such manipulation may be done to replace a codon with a redundant codon encoding the same or a conservative amino acid, while typically introducing or removing a sequence recognition site. Alternatively, it may be performed to join together nucleic acid (e.g., engineered nucleic acid) segments of desired functions to generate a single genetic entity comprising a desired combination of functions not found in nature. Restriction enzyme recognition sites are often the target of such artificial manipulations, but other site specific targets, e.g., promoters, DNA replication sites, regulation sequences, control sequences, open reading frames, or other useful features may be incorporated by design.

(111) “OCT4” may also be referred to as Octamer-binding transcription factor 4, OCT3, OCT3/4, POU5F1, or POU class 5 homeobox 1 and is a transcription factor that has been implicated in embryonic development and determination of cell fate. Similar to other OCT transcription factors, OCT4 is characterized by a bipartite DNA binding domain called a POU domain. An OCT4 transcription factor, homolog, or variant thereof, as used herein, may be derived from any species, including humans. In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding human OCT4 is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a nucleic acid (e.g., engineered nucleic acid) described in the NCBI RefSeq under accession number NM_002701, NM_203289, NM_001173531, NM_001285986, or NM_001285987. In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding an OCT4 comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a nucleic acid (e.g., engineered nucleic acid) sequence provided as SEQ ID NO: 1. SEQ ID NO: 1 is a non-limiting example of a nucleotide sequence encoding OCT4 from *Mus musculus*. In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding a human OCT4 comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a nucleic acid (e.g., engineered nucleic acid) sequence provided as SEQ ID NO: 40. SEQ ID NO: 40 is a non-limiting example of a nucleotide sequence encoding human OCT4. Non-limiting examples of OCT4 variants encompassed herein include POU5F1, transcript variant 1, POU5F1, transcript variant 2, POU5F1, transcript variant 3, POU5F1, transcript variant 4, and POU5F1 transcript variant 5. In certain embodiments, the amino acid sequence encoding human OCT4 is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a nucleic acid (e.g., engineered nucleic acid) described in the NCBI RefSeq under accession number NP_001167002.1, NP_001272915.1, NP_001272916.1, NP_002692.2, or NP_976034.4. In certain embodiments, an OCT4 comprises an amino acid sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 2. SEQ ID NO: 2 is a non-limiting example of an amino acid sequence encoding OCT4 from *Mus musculus*. In certain embodiments, an OCT4 comprises an amino acid sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 41. SEQ ID NO: 41 is a non-limiting example of an amino acid sequence encoding human OCT4. Other OCT4 transcription factors (e.g., from other species) are known and nucleic acids (e.g., engineered nucleic acids) encoding OCT4 transcription factors can be found in publically available databases, including GenBank.

(112) The term “promoter” refers to a control region of a nucleic acid (e.g., engineered nucleic acid) sequence at which initiation and rate of transcription of the remainder of a nucleic acid (e.g., engineered nucleic acid) sequence are controlled. A promoter may also contain sub-regions at which regulatory proteins and molecules may bind, such as RNA polymerase and other transcription factors. Promoters may be constitutive, inducible, activatable, repressible, tissue-specific, or any combination thereof. A promoter drives expression or drives transcription of the nucleic acid (e.g., engineered nucleic acid) sequence that it regulates. Herein, a promoter is

considered to be "operably linked" when it is in a correct functional location and orientation in relation to a nucleic acid (e.g., engineered nucleic acid) sequence it regulates to control ("drive") transcriptional initiation of that sequence, expression of that sequence, or a combination thereof.
(113) A promoter may promote ubiquitous expression or tissue-specific expression of an operably linked nucleic acid (e.g., engineered nucleic acid) sequence from any species, including humans. In some embodiments, the promoter is a eukaryotic promoter. Non-limiting examples of eukaryotic promoters include TDH3, PGK1, PKC1, TDH2, PYK1, TPI1, AT1, CMV, EF1 alpha, SV40, PGK1 (human or mouse), Ubc, human beta actin, CAG, TRE, UAS, Ac5, Polyhedrin, CaMKIIa, GAL1, GAL10, TEF1, GDS, ADH1, CaMV35S, Ubi, H1, and U6, as would be known to one of ordinary skill in the art (see, e.g., Addgene website: blog.addgene.org/plasmids-101-the-promoter-region).
(114) Non-limiting examples of ubiquitous promoters include tetracycline-responsive promoters (under the relevant conditions), CMV (e.g., SEQ ID NO: 48), EF1 alpha, a SV40 promoter, PGK1, Ubc, CAG, human beta actin gene promoter, a RSV promoter (e.g., SEQ ID NO: 47), an EFS promoter (e.g., SEQ ID NO: 49), and a promoter comprising an upstream activating sequence (UAS). In certain embodiments, the promoter is a mammalian promoter.
(115) Non-limiting examples of tissue-specific promoters include brain-specific, liver-specific, muscle-specific, nerve cell-specific, lung-specific, heart-specific, bone-specific, intestine-specific, skin-specific promoters, brain-specific promoters, and eye-specific promoters. As an example, a muscle-specific promoter is a desmin promoter (e.g., a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 29). Non-limiting examples of eye-specific promoters include human GRK1 (rhodopsin kinase) promoter (e.g., SEQ ID NO: 50), human CRX (cone rod homeobox transcription factor) promoter (e.g., SEQ ID NO: 51), and human NRL promoter (neural retina leucine zipper transcription factor enhancer upstream of the human TK terminal promoter).
(116) In some embodiments, a promoter is specific for senescent cells. For example, a promoter may specifically induce expression of an operably linked nucleic acid in a senescent cell and not in non-senescent cells. As a non-limiting example, the p16 promoter may be used to promote expression of a operably linked nucleic acid in senescent cells.
(117) In some embodiments, a promoter of the present disclosure is suitable for use in AAV vectors. See, e.g., U.S. Patent Application Publication No. 2018/0155789, which is hereby incorporated by reference in its entirety for this purpose.
(118) Non-limiting examples of constitutive promoters include CP1, CMV, EF1 alpha, SV40, PGK1, Ubc, human beta actin, beta tubulin, CAG, Ac5, Rosa26 promoter, COL1A1 promoter, polyhedrin, TEF1, GDS, CaM3 5S, Ubi, H1, U6, red opsin promoter (red promoter), rhodopsin promoter (rho promoter), cone arrestin promoter (car promoter), rhodopsin kinase promoter (rk promoter). An Ubc promoter may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 18. In some instances, the constitutive promoter is a Rosa26 promoter. In some instances, the constitutive promoter is a COL1A1 promoter. A red opsin promoter may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 101. A rho promoter may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 102. A cone arrestin promoter may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 103. A rhodopsin kinase promoter may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 104. A tissue-specific promoter may be used to drive expression of an engineered nucleic acid, including e.g., a nucleic acid encoding a rtTA, tTA, OCT4, KLF4, SOX2, or any combination thereof. In some embodiments, a tissue-specific promoter is used to drive expression of a rtTA or a rTA. Ins one embodiments, a tissue-specific promoter is used to drive expression of OCT4, KLF4, and SOX2. In some embodiments, the tissue-specific promoter is selected from the group

consisting of SEQ ID NOS: 101-104. In some embodiments, the hRK promoter is used to drive expression of OCT4, KLF4, and SOX2.
(119) An “inducible promoter” is one that is characterized by initiating or enhancing transcriptional activity when in the presence of, influenced by, or contacted by an inducing agent. An inducing agent may be endogenous or a normally exogenous condition, compound, agent, or protein that contacts an engineered nucleic acid (e.g., engineered nucleic acid) in such a way as to be active in inducing transcriptional activity from the inducible promoter. In certain embodiments, an inducing agent is a tetracycline-sensitive protein (e.g., tTA or rtTA, TetR family regulators).
(120) Inducible promoters for use in accordance with the present disclosure include any inducible promoter described herein or known to one of ordinary skill in the art. Examples of inducible promoters include, without limitation, chemically/biochemically-regulated and physically-regulated promoters such as alcohol-regulated promoters, tetracycline-regulated promoters (e.g., anhydrotetracycline (aTc)-responsive promoters and other tetracycline responsive promoter systems, which include a tetracycline repressor protein (TetR, e.g., SEQ ID NO: 26, or TetRKRAB, e.g., SEQ ID NO: 27), a tetracycline operator sequence (tetO) and a tetracycline transactivator fusion protein (tTA), and a tetracycline operator sequence (tetO) and a reverse tetracycline transactivator fusion protein (rtTA)), steroid-regulated promoters (e.g., promoters based on the rat glucocorticoid receptor, human estrogen receptor, moth ecdysone receptors, and promoters from the steroid/retinoid/thyroid 25 receptor superfamily), metal-regulated promoters (e.g., promoters derived from metallothionein (proteins that bind and sequester metal ions) genes from yeast, mouse and human), pathogenesis-regulated promoters (e.g., induced by salicylic acid, ethylene or benzothiadiazole (BTH)), temperature/heat-inducible promoters (e.g., heat shock promoters), pH-regulated promoters, and light-regulated promoters. A non-limiting example of an inducible system that uses a light-regulated promoter is provided in Wang et al., *Nat. Methods.* 2012 Feb. 12; 9(3):266-9.
(121) In certain embodiments, an inducible promoter comprises a tetracycline (Tet)-responsive element. For example, an inducible promoter may be a TRE3G promoter (e.g., a TRE3G promoter that comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 7). As an example, a TRE (e.g., TRE2) promoter may comprise a nucleic acid (e.g., engineered nucleic acid) sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 23. As an example, a TRE (e.g., P tight) promoter may comprise a nucleic acid (e.g., engineered nucleic acid) sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 24.
(122) Additional non-limiting examples of inducible promoters include mifepristone-responsive promoters (e.g., GAL4-E1b promoter) and coumermycin-responsive promoters. See, e.g., Zhao et al., *Hum Gene Ther.* 2003 Nov. 20; 14(17):1619-29.
(123) A “reverse tetracycline transactivator” (“rtTA”), as used herein, is an inducing agent that binds to a TRE promoter (e.g., a TRE3G, a TRE2 promoter, or a P tight promoter) in the presence of tetracycline (e.g., doxycycline) and is capable of driving expression of a transgene that is operably linked to the TRE promoter. rtTAs generally comprise a mutant tetracycline repressor DNA binding protein (TetR) and a transactivation domain (see, e.g., Gossen et al., *Science.* 1995 Jun. 23; 268(5218):1766-9 and any of the transactivation domains listed herein). The mutant TetR domain is capable of binding to a TRE promoter when bound to tetracycline. See, e.g., U.S. Provisional Application No. 62/738,894, entitled MUTANT REVERSE TETRACYCLINE TRANSACTIVATORS FOR EXPRESSION OF GENES, which was filed on Sep. 28, 2018, and is herein incorporated by reference in its entirety.
(124) “SRY-box 2” or “SOX2” is a member of the SRY-related HMG-box (SOX) family of transcription factors. SOX2 has been implicated in promoting embryonic development. Members of the SOX (SRY-related HMG-box) family of transcription factors are characterized by a high mobility group 5 (HMG)-box DNA sequence. This HMG box is a DNA binding domain that is

highly conserved throughout eukaryotic species. A SOX2 transcription factor, homolog or variant thereof, as used herein, may be derived from any species, including humans. In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding SOX2 comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a nucleic acid (e.g., engineered nucleic acid) described in the NCBI RefSeq under accession number NM_011443.4. In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding a human SOX2 comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a nucleic acid (e.g., engineered nucleic acid) described in the NCBI RefSeq under accession number NM_003106.4. In certain embodiments, SOX2 comprises a nucleic acid (e.g., engineered nucleic acid) sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 3 or SEQ ID NO: 42. SEQ ID NO: 3 is a non-limiting example of a nucleotide sequence encoding SOX2 from *Mus musculus*. SEQ ID NO: 42 is a non-limiting example of a nucleotide sequence encoding human SOX2. In certain embodiments, the nucleic acid (e.g., engineered nucleic acid) encoding human SOX2 comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to the amino acid sequence described in the NCBI RefSeq under accession number NP_003097.1. In some instances, SOX2 comprises an amino acid sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 4. In some instances, SOX2 comprises an amino acid sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 43. SEQ ID NO: 4 is a non-limiting example of an amino acid sequence encoding SOX2 from *Mus musculus*. SEQ ID NO: 43 is a non-limiting example of an amino acid sequence encoding human SOX2.

(125) A “multicistronic vector” is a vector that encodes more than one amino acid sequence (e.g., a vector encoding OCT4 and KLF4, OCT4 and SOX2, KLF4 and SOX2, or OCT4, SOX2, and KL4 (OSK)). A multicistronic vector allows for expression of multiple amino acid sequences from a nucleic acid (e.g., engineered nucleic acid) sequence. Nucleic acid (e.g., engineered nucleic acid) sequences encoding each transcription factor (e.g., OCT4, KLF4, or SOX2) may be connected or separated such that they produce unconnected proteins. For example, internal ribosome entry sites (IRES) or polypeptide cleavage signals may be placed between nucleic acid (e.g., engineered nucleic acid) sequences encoding each transcription factor in a vector. Exemplary polypeptide cleavage signals include 2A peptides (e.g., T2A, P2A, E2A, and F2A). A 2A peptide may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 9. In some embodiments, an expression vector of the present disclosure is a multicistronic expression vector.

(126) “Reversing aging” or “reversing ageing” as used herein refers to modifying the physical characteristics associated with aging. All animals typically go through a period of growth and maturation followed by a period of progressive and irreversible physiological decline ending in death. The length of time from birth to death is known as the life span of an organism, and each organism has a characteristic average life span. Aging is a physical manifestation of the changes underlying the passage of time as measured by percent of average life span.

(127) A “subject” to which administration is contemplated includes, but is not limited to, humans (i.e., a male or female of any age group, e.g., a pediatric subject (e.g., infant, child, adolescent) or adult subject (e.g., young adult, middle-aged adult, or senior adult)) and/or other non-human animals, for example, mammals (e.g., primates (e.g., cynomolgus monkeys, rhesus monkeys); commercially relevant mammals, such as cattle, pigs, horses, sheep, goats, cats, and/or dogs) and birds (e.g., commercially relevant birds, such as chickens, ducks, geese, and/or turkeys). In certain embodiments, the animal is a mammal. The animal may be a male or female and at any stage of development. A non-human animal may be a transgenic animal.

(128) One of ordinary skill in the art would recognize that the biological age of a pediatric subject

or adult subject may vary depending on the type of animal. As a non-limiting example, an adult mouse may be 1 year of age, while an adult human may be more than 21 years of age. In some embodiments, a pediatric subject is less than 21 years of age, less than 20 years of age, less than 15 years of age, less than 10 years of age, less than 9 years of age, less than 8 years of age, less than 7 years of age, less than 6 years of age, less than 5 years of age, less than 4 years of age, less than 3 years of age, less than 2 years of age, less than 1 year of age, less than 10 months of age, less than 9 months of age, less than 8 months of age, less than 7 months of age, less than 6 months of age, less than 5 months of age, less than 4 months of age, less than 2 months of age, or less than 1 month of age. In some embodiments, an adult subject is at least 3 weeks of age, 1 month of age, at least 2 months of age, at least 3 months of age, at least 4 months of age, at least 5 months of age, at least 6 months of age, at least 7 months of age, at least 8 months of age, at least 9 months of age, at least 10 months of age, at least 11 months of age, at least 1 year of age, at least 2 years of age, at least 3 years of age, at least 5 years of age, at least 10 years of age, at least 15 years of age, at least 20 years of age, at least 25 years of age, at least 30 years of age, at least 40 years of age, at least 50 years of age, at least 55 years of age, at least 60 years of age, at least 65 years of age, at least 70 years of age, at least 75 years of age, at least 80 years of age, at least 90 years of age, or at least 100 years of age. In some embodiments, a middle-aged adult subject is between 1 and 6 months of age, between 6 and 12 months of age, between 1 year and 5 years of age, between 5 years and 10 years of age, between 10 and 20 years of age, between 20 and 30 years of age, between 30 and 50 years of age, between 50 and 60 years of age, between 40 and 60 years of age, between 40 and 50 years of age, or between 45 and 65 years of age. In some embodiments, a senior adult subject is at least 1 month of age, at least 2 months of age, at least 3 months of age, at least 4 months of age, at least 5 months of age, at least 6 months of age, at least 7 months of age, at least 8 months of age, at least 9 months of age, at least 10 months of age, at least 11 months of age, at least 1 year of age, at least 2 years of age, at least 3 years of age, at least 5 years of age, at least 10 years of age, at least 15 years of age, at least 20 years of age, at least 25 years of age, at least 30 years of age, at least 40 years of age, at least 50 years of age, at least 55 years of age, at least 60 years of age, at least 65 years of age, at least 70 years of age, at least 75 years of age, at least 80 years of age, at least 90 years of age, or at least 100 years of age.

(129) A “terminator” or “terminator sequence,” as used herein, is a nucleic acid (e.g., engineered nucleic acid) sequence that causes transcription to stop. A terminator may be unidirectional or bidirectional. It is comprised of a DNA sequence involved in specific termination of an RNA transcript by an RNA polymerase. A terminator sequence prevents transcriptional activation of downstream nucleic acid (e.g., engineered nucleic acid) sequences by upstream promoters. Thus, in certain embodiments, a terminator that ends the production of an RNA transcript is contemplated.

(130) The most commonly used type of terminator is a forward terminator. When placed downstream of a nucleic acid (e.g., engineered nucleic acid) sequence that is usually transcribed, a forward transcriptional terminator will cause transcription to abort. In some embodiments, bidirectional transcriptional terminators may be used, which usually cause transcription to terminate on both the forward and reverse strand. In some embodiments, reverse transcriptional terminators may be used, which usually terminate transcription on the reverse strand only.

(131) Non-limiting examples of mammalian terminator sequences include bovine growth hormone terminator, and viral termination sequences such as, for example, the SV40 terminator, spy, yejM, secG-leuU, thrLABC, rrnB T1, hisLGDCBHAFI, metZWV, rrnC, xapR, aspA, and arcA terminator. In certain embodiments, the terminator sequence is SV40 and comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 8.

(132) A “Tet-Off” system, as used herein, is a type of inducible system that is capable of repressing expression of a particular transgene in the presence of tetracycline (e.g., doxycycline (DOX)). Conversely, a Tet-Off system is capable of inducing expression of a particular transgene in the

absence of tetracycline (e.g., doxycycline, DOX). In certain embodiments, a Tet-Off system comprises a tetracycline-responsive promoter operably linked to a transgene (e.g., encoding OCT4; KLF4; SOX2; or any combination thereof) and a tetracycline-controlled transactivator (tTA). The transgene with the tetracycline-responsive promoter (e.g., TRE3G, P tight, or TRE2) and the tetracycline-controlled transactivator may be encoded on the same vector or be encoded on separate vectors. See, e.g., U.S. Provisional Application No. 62/738,894, entitled MUTANT REVERSE TETRACYCLINE TRANSACTIVATORS FOR EXPRESSION OF GENES, which was filed on Sep. 28, 2018, and is herein incorporated by reference in its entirety.

(133) A “Tet-On” system, as used herein, is a type of inducible system that is capable of inducing expression of a particular transgene in the presence of tetracycline (e.g., doxycycline (DOX)). In certain embodiments, a Tet-On system comprises a tetracycline-responsive promoter operably linked to a transgene (e.g., encoding OCT4; KLF4; SOX2; or any combination thereof) and a reverse tetracycline-controlled transactivator (rtTA). For example, the rtTA may be rtTA3, rtTA4, or variants thereof. In certain embodiments, a nucleic acid (e.g., engineered nucleic acid) encoding rtTA3 comprises a sequence that is at least 70% (e.g., at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 100%) identical to SEQ ID NO: 10. In certain embodiments, an amino acid sequence encoding rtTA3 comprises a sequence that is at least 70% (e.g., at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or at least 100%) identical to (SEQ ID NO: 11). In certain embodiments, a nucleic acid (e.g., engineered nucleic acid) encoding rtTA4 comprises a sequence that is at least 70% (e.g., at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or at least 100%) identical to SEQ ID NO: 12. In certain embodiments, an amino acid sequence encoding rtTA4 comprises a sequence that is at least 70% (e.g., at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or at least 100%) identical to (SEQ ID NO: 13). The expression cassette encoding a tetracycline-responsive promoter (e.g., a promoter comprising a TRE, including TRE3G, P tight, and TRE2) and a reverse tetracycline-controlled transactivator may be encoded on the same vector or be encoded on separate vectors. See, e.g., U.S. Provisional Application No. 62/738,894, entitled MUTANT REVERSE TETRACYCLINE TRANSACTIVATORS FOR EXPRESSION OF GENES, which was filed on Sep. 28, 2018, and is herein incorporated by reference in its entirety.

(134) The term “tissue” refers to any biological tissue of a subject (including a group of cells, a body part, or an organ) or a part thereof, including blood and/or lymph vessels, which is the object to which a compound, particle, and/or composition of the invention is delivered. A tissue may be an abnormal, damaged, or unhealthy tissue, which may need to be treated. A tissue may also be a normal or healthy tissue that is under a higher than normal risk of becoming abnormal or unhealthy, which may need to be prevented. In certain embodiments, the tissue is considered healthy but suboptimal for performance or survival in current or future conditions. For example, in agricultural practice, environmental conditions including weather and growing conditions (e.g., nutrition) may benefit from any of the methods described herein. In certain embodiments, the tissue is the central nervous system. In certain embodiments, the tissue refers to tissue from the In certain embodiments, the cell or tissue is from eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine. In certain embodiments, the tissue is damaged (e.g., due to a congenital defect, an injury, an accident, or an iatrogenic injury) and/or is aged tissue. In certain embodiments, the tissue is a deep tissue that is reachable with a fiber optic probe.

(135) The term “tetracycline repressor” or “TetR” refers to a protein that is capable of binding to a Tet-O sequence (e.g., a Tet-O sequence in a TRE, e.g., a Tet-O sequence may comprise SEQ ID NO: 19) in the absence of tetracycline (e.g., doxycycline) and prevents binding of rtTA (e.g., rtTA3, rtTA4, or variants thereof) in the absence of tetracycline (e.g., doxycycline). TetRs prevent gene

expression from promoters comprising a TRE in the absence of tetracycline (e.g., doxycycline). In the presence of tetracycline, TetRs cannot bind promoters comprising a TRE, and TetR cannot prevent transcription. Non-limiting examples of TetRs include tetR (e.g., SEQ ID NO: 26), tetRKRAB (e.g., SEQ ID NO: 28). In some embodiments, a TetR is a TetR fusion (e.g., TRSID, which may be created by fusing TetR to a mSIN30interacting domain (SID) of Madl). See, e.g., Zhang et al., J Biol Chem. 2001 Nov. 30; 276(48):45168-74.

(136) As used herein, a "TRE promoter" is a promoter comprising a tetracycline-responsive element (TRE). As used herein, a TRE comprises at least one (e.g., at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20) Tet-O sequences. A non-limiting example of a Tet-O sequence is sequence that is at least 70% (e.g., 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 19. In some embodiments, a TRE promoter further comprises a minimal promoter located downstream of a tet-O sequence. A minimal promoter is a promoter that comprises the minimal elements of a promoter (e.g., TATA box and transcription initiation site), but is inactive in the absence of an upstream enhancer (e.g., sequences comprising Tet-O). As an example, a minimal promoter may be a minimal CMV promoter that comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 20. For example, a TRE promoter may be a TRE3G promoter (e.g., a TRE3G promoter that comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 7. In some embodiments, a TRE promoter is a TRE2 promoter comprising a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 23. In some embodiments, a TRE promoter is a P tight promoter comprising a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 24.

(137) The term "tissue repair" in the context of damaged tissue refers to restoration of tissue architecture, function following tissue damage, or a combination thereof. Tissue repair includes tissue regeneration, cell growth, tissue replacement, and/or rewiring of existing tissue (reprogramming).

(138) The term "tissue regeneration" refers to production of new tissue or cells within a tissue that are the same type as the tissue of interest (e.g., same type as the damaged tissue or cell). In some embodiments, the methods provided herein promote organ regeneration.

(139) The term "tissue replacement" refers to production of a different type of tissue compared to the tissue of interest (e.g., connective tissue to replace damaged tissue).

(140) As used herein, the terms "treatment," "treat," and "treating" refer to reversing, alleviating, delaying the onset of, or inhibiting the progress of a disease or disorder, or one or more symptoms thereof, as described herein. In certain embodiments, treatment may be administered after one or more symptoms have developed. In other embodiments, treatment may be administered in the absence of symptoms. For example, treatment may be administered to a susceptible individual prior to the onset of symptoms or may be treated with another damaging agent (e.g., in light of a history of symptoms, in light of genetic or other susceptibility factors, a disease therapy, or any combination thereof). Treatment may also be continued after symptoms have resolved, for example, to prevent or delay their recurrence.

(141) The term "variant" refers to a sequence that comprises a modification relative to a wild-type sequence. Non-limiting modifications in an amino acid sequence include insertions, deletions, and point mutations. Non-limiting modifications to nucleic acid (e.g., engineered nucleic acid) sequences include frameshift mutations, nucleotide insertions, and nucleotide deletions.

(142) The term "WPRE" refers to a Woodchuck Hepatitis Virus (WHP) Posttranscriptional Regulatory Element (WPRE). WPREs create tertiary structures in nucleic acids (e.g., expression vectors) and are capable of enhancing transgene expression (e.g., from a viral vector). In certain embodiments, a WPRE sequence is at least 70% (e.g., at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or at least 100%) identical to SEQ ID NO: 21.

(143) These and other exemplary substituents are described in more detail in the Detailed Description, Examples, and claims. The invention is not intended to be limited in any manner by the above exemplary listing of substituents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic with a linear representation of an expression vector encoding OCT4, SOX2, and KLF4. TRE3G is shown as an exemplary inducible promoter, and SV40 is shown as an exemplary terminator sequence.

(2) FIG. **2** is a vector map of TRE3G−OSK−SV40 pA, an AAV vector encoding OSK. Features including the location of sequences encoding OCT4, SOX2, and KLF4 and inverted terminal repeat sequences (ITRs) are indicated.

(3) FIG. **3** is a vector map showing the location of restriction enzyme digestion sites in TRE3G−OSK−SV40 pA.

(4) FIGS. **4**A-**4**AL include a series of schematics mapping the features shown in FIGS. **2** and **3** onto the nucleic acid (e.g., engineered nucleic acid) sequence of TRE3G−OSK−SV40 pA. FIGS. **4**A-**4**AL show SEQ ID NO: 16 from 5′ to 3′. FIGS. **4**G-**4**AC show the protein sequence, SEQ ID NO: 123.

(5) FIGS. **5**A-**5**D show the nucleotide positions and lengths of the nucleic acid (e.g., engineered nucleic acid) sequences of the features shown in FIGS. **4**A-**4**AL.

(6) FIGS. **6**A-**6**C include western blot data showing that different serotypes of AAVs encoding OSK (TRE3G−OSK−SV40 pA, SEQ ID NO: 16) were successfully used in a doxycycline (DOX)-inducible system to control OSK expression in 293T cells. OCT4, KLF4, and H3 expression were detected with antibodies. H3 refers to histone 3 and is a loading control. FIG. **6**A shows the effect of doxycycline on protein expression in cells infected with AAV9 virus harboring the TRE3G−OSK−SV40 pA vector and with AAV9 virus harboring a vector that encodes rtTA3 (tetracycline (Tet)-on system). FIG. **6**B shows the effect of DOX on protein expression in cells infected with AAV2 virus harboring the TRE3G−OSK−SV40 pA vector and with AAV2 virus harboring a vector that encodes tTA (Tet-Off system). FIG. **6**C shows the effect of DOX treatment and DOX removal on protein expression in cells infected with AAV.PHP.b virus harboring the TRE3G−OSK−SV40 pA vector and with AAV.PHP.b virus harboring a vector that encodes rtTA3 (Tet-On system). The length of DOX treatment (+DOX) or DOX removal (−DOX) in days is indicated in parenthesis.

(7) FIGS. **7**A-**7**F include data showing that AAV encoding OSK induced partial reprogramming and promoted regeneration of optic nerves after nerve crush in an inducible manner. FIG. **7**A includes a series of photos showing that injection of TRE−OSK−SV40 AAV virus and CAG-tTA AAV virus into mouse retina resulted in expression of KLF4 in mouse retina ganglion cells (RGCs). RBPMS (RGC marker) and KLF4 staining of an optical coherence tomography (OCT) section from mouse retina is shown. FIG. **7**B includes a series of photos showing that injection of TRE−OSK−SV40 AAV virus and CAG-tTA AAV virus resulted in inducible expression of KLF4 and OCT4 in mouse retina. OCT4 and KLF4 staining of a whole retina mount in the absence of doxycycline treatment (top two photos) and after four days of DOX treatment (bottom two photos) is shown. FIG. **7**C shows an experimental timeline to determine the effect of TRE−OSK−SV40 AAV virus alone or in combination with CAG-tTA AAV virus on optical nerve regeneration following optic nerve crush damage. CTB stands for cholera toxin P-subunit and allows for fluorescence imaging of axons. FIG. **7**D shows the co-localization staining of OCT4 and KLF4 from a whole mount retina with TRE−OSK−SV40 AAV virus injected in combination with CAG-tTA, RBPMS stains retina ganglion cells specifically. FIG. **7**E shows fluorescence imaging of

CTB-labeled axons in an optical nerve after crush damage in mouse retina injected with TRE−OSK−SV40 AAV virus alone (left) or TRE−OSK−SV40 AAV in combination with CAG-tTA AAV (right). Stars represent the site of the lesion. FIG. **7**F shows additional fluorescence images of optical nerves treated as in FIG. **7**E with viruses as indicated.

(8) FIGS. **8**A-**8**G shows administration of virus encoding OSK improved RGC axon regeneration after nerve crush injury. FIG. **8**A shows the effect of virus encoding tTA in combination with virus encoding TRE−OSK−SV40 in the absence of DOX (circles, n=9), with virus encoding TRE−OSK−SV40 in the presence of dox (triangles, n=5), or with virus encoding d2EGFP (squares, control, n=5) on RGC axon regeneration. The number of estimated axons per nerve is shown as a function of the distance from the site of injury (μm).

(9) FIG. **8**B is an experimental timeline to determine the effect of d2EGFP expression on RGC axon regeneration. FIG. **8**C is a series of images showing CTB-labeled axons from the experiment outlined in FIG. **8**B. FIG. **8**D is an experimental timeline to determine the effect of uninduced OSK expression on axon regeneration. FIG. **8**E is a series of images showing CTB-labeled axons from the experiment outlined in FIG. **4**D. FIG. **8**F is an experimental timeline to determine the effect of induced OSK expression on axon regeneration. FIG. **8**G is a series of images showing CTB-labeled axons from the experiment outlined in FIG. **8**F. Stars indicate the site of the lesion.

(10) FIGS. **9**A-**9**D show OSK-infected RGCs have a higher survival rate compared to cells not infected with OSK virus following nerve crush. FIG. **9**A shows staining for RBPMS and GFP in GFP AAV-infected uncrushed RGCs (upper left) and in crushed RGCs (upper right). staining for RBPMS and KLF4 in OSK AAV-infected uncrushed RGCs (lower left) and in crushed RGCs (lower right). FIG. **9**B shows the ratio of RBPMS (RNA binding protein with multiple splicing)-positive (+) cells for uncrushed and crushed RGCs infected with a destabilized form of GFP (d2EGFP) virus or OSK virus. GFP infected RGCs has the same survival rate as uninfected RGCs, therefore GFP+ RBPMs+% remains the same after crush injury. OSK infected RGCs had triple the survival rate compared to uninfected RGC, therefore, KLF4+ RBPMS+% increased after crush injury. FIG. **9**C shows survived RGCs under uncrushed (left) and crushed (right) condition, with OSK virus infection. FIG. **9**D shows the survival of RGCs (RBPMS+) under uncrushed and crushed condition, when they were infected with d2EGFP virus or OSK virus.

(11) FIGS. **10**A-**10**B show that OSK-mediated regeneration and protection is independent of mTOR activation. FIG. **10**A is a series of images showing RBPMS and pS6 staining of control and OSK-infected RGCs that were uncrushed or crushed. FIG. **10**B is a graph quantifying the percentage of pS6 positive cells from series of pictures like FIG. **10**A.

(12) FIGS. **11**A-**11**D show that an AAV Tet-On system comprising a CMV-rtTA vector (SEQ ID NO: 31) induces faster gene expression compared to an AAV Tet-Off system in retinal cells after nerve crush. FIG. **11**A shows an experimental timeline to test the effect of doxycycline removal on GFP expression in an AAV Tet-Off system. Lines indicate the length of DOX treatment. Treatments A-D as indicated correspond to photographs 1-4 of FIG. **7**B, respectively. FIG. **11**B is a series of photos showing results of the experiment outlined in FIG. **11**A. GFP-positive cells from mouse retina that was infected with virus encoding tTA and virus encoding TRE-d2EGFP at indicated days of DOX removal are shown. FIG. **11**C shows an experimental timeline to test the effect of doxycycline treatment on GFP expression in an AAV Tet-On system comprising a CMV-rtTA vector (SEQ ID NO: 31). Lines indicate the length of DOX treatment. Treatments A-C as indicated correspond to photographs 1-3 of FIG. **11**D, respectively. FIG. **11**D is a series of photos showing results of the experiment outlined in FIG. **11**C. GFP-positive cells from mouse retina that was infected with virus encoding rtTA and virus encoding TRE-d2EGFP at the indicated days of DOX treatment are shown.

(13) FIG. **12** is a vector map showing features in an adeno-associated virus (AAV) vector encoding reverse tetracycline-transactivator 4 (rtTA4). Ubc is a constitutive promoter that is operably linked to the nucleic acid (e.g., engineered nucleic acid) encoding rtTA4. SV40 pA is an SV40-derived

terminator sequence. The sequence of this vector is provided in SEQ ID NO: 17.
(14) FIGS. **13**A-**13**C include data showing that a Tet-On system comprising rtTA4 (SEQ ID NO: 13) has low leakiness in the liver of mice. FIG. **13**A is a series of immunofluorescence images showing expression of KLF4 in the livers of mice that have been administered AAVs harboring nucleic acids (e.g., engineered nucleic acids) shown in FIG. **13**B in the absence of doxycycline (no DOX) and in the presence of doxycycline (with DOX). DAPI is a nuclear stain that was used to visualize cells. FIG. **13**B is a schematic depicting the two nucleic acids (e.g., engineered nucleic acids) that were administered to mice in AAV9 viruses. FIG. **13**C is a western blot of liver samples from mice that received the constructs depicted in FIG. **13**B and were treated with no doxycycline or with doxycycline. OCT4, KLF4, and SOX2 levels were detected as indicated using antibodies. Actin is shown as a loading control.
(15) FIG. **14** is a graph comparing the body weights of mice under various treatments as indicated. WT indicates wild-type mice without exogenous OSK expression. All dead indicates that OSK transgenic mice treated with doxycycline were all dead.
(16) FIGS. **15**A-**15**B include data showing that induction of OCT4, KLF4, and SOX2 expression reversed aging of mice ear fibroblasts as indicated by expression of histone and Chaf (Chromatin assembly factor) genes but did not induce Nanog expression. The asterisk (*) indicates endogenous KLF4 expression from the 293T cell line.
(17) FIG. **16** is a western blot showing that an AAV vector comprising a nucleic acid (e.g., engineered nucleic acid) sequence that is greater than 4.7 kb between the two ITRs in the vector has low viral titer when incorporated into an AAV and produces non-functional AAV. The TRE2−OSK vector is provided as SEQ ID NO: 33. Expression of OCT4, KLF4 and H3 was detected using antibodies. H3 is shown as a loading control. Asterisk (*) indicates endogenous Klf4 from 293T cell line.
(18) FIG. **17** is a western blot showing that administration of modified mRNA encoding OCT4, SOX2, and KLF4 induced expression of KLF4 and OCT4 in mouse cells. Antibodies against KLF4, OCT4, GAPDH, and H3 were used to detect indicated proteins.
(19) FIG. **18** is a vector map of pAAV2_CMV_rtTA(VP16) (SEQ ID NO: 31). This vector is a non-limiting example of a vector encoding rtTA.
(20) FIG. **19** is a vector map of pAAV−MCS−tTA2 (or CAG-tTA) (SEQ ID NO: 32). This vector is a non-limiting example of a vector encoding tTA under a CAG promoter.
(21) FIG. **20** is a vector map of p-AAV−TetO−OSK−WPRE3-SV50LpA (TRE2−OSK, pAAV−TRE2−OSK−SV40LpA, or TRE2−OSK) (SEQ ID NO: 33). This vector is a non-limiting example of an AAV vector comprising a nucleic acid (e.g., engineered nucleic acid) sequence that is greater than 4.7 kb between the two ITRs in the vector.
(22) FIG. **21** is a series of images showing successful chemical reprogramming of mouse embryonic fibroblasts.
(23) FIG. **22** includes a schematic showing a non-limiting example of a Tet-Off system to express OCT4, SOX2, and KLF4 in the absence of tetracycline (top panel) and a schematic showing a non-limiting example of a Tet-ON system to express OCT4, SOX2, and KLF4 (OSK) in the presence of tetracycline (bottom panel).
(24) FIGS. **23**A-**23**C include data showing that administration of AAV2 virus encoding OCT4, SOX2, and KLF4 improved axon regeneration and RGC survival in adult and aged mice two weeks after optic nerve crush. FIG. **23**A is a series of images showing CTB-labeled axons from mice at indicated ages (in months) and comparing the effect of AAV2 virus encoding TRE−OSK−SV40 with the effect of AAV2 virus encoding GFP. Experiments were conducted in the absence of DOX using the Tet-Off system depicted in FIG. **22**, top panel. FIG. **23**B quantifies the number of estimated axons per nerve for mice with the indicated ages and treatments as a function of the distance from the site of injury (μm). FIG. **23**C is a chart showing that OSK increased the survival of RGCs after optic nerve injury in adult (3 month old) and aged (12 month old) mice compared to

control GFP. The survival of RGCs (RBPMS+) is shown for mice of the indicated ages receiving virus encoding d2EGFP or OSK.

(25) FIGS. **24**A-**24**B include data showing that increasing the time of reprogramming from two weeks to five weeks improved regeneration in aged mice. FIG. **24**A is a series of photos showing CTB-labeled axons from 12 month old mice five weeks after optic nerve crush injury. Mice were administered virus encoding GFP or encoding TRE-SV40-OSK and virus encoding tTA prior to nerve crush injury. FIG. **24**B is a graph quantifying the number of estimated axons per nerve as a function of the distance from the site of injury (μm) from FIG. **24**A.

(26) FIGS. **25**A-**25**C include data showing that induction of OSK expression using Tet-On and Tet-Off systems even after optic nerve crush injury improved regeneration and RGC cell survival in mice. FIG. **25**A includes schematics showing treatment timelines to determine the effect of OSK expression before or after optic nerve crush. In the Tet-On system, induction of OSK expression prior to optic nerve crush injury (pre-injury induction) and induction of OSK expression after optic nerve crush injury (post-injury induction) are shown (top panel). Doxycycline treatment was used to induce OSK expression. In the Tet-Off system, suppression of OSK induction with doxycycline treatment prior to optic nerve crush (pre-injury suppression) and suppression of OSK induction with doxycycline treatment after optic nerve crush (post-injury suppression) are shown (bottom panel). The shaded lines on the timeline indicate the length of doxycycline (DOX) treatment. Cholera toxin β-subunit (CTB) injection for imaging of axons is also shown. FIG. **25**B is a chart quantifying the number of estimated axons per nerve as a function of the distance from the site of injury (μm) for four-week old (young) mice with no OSK induction (n=4), OSK induction pre-injury only (n=5), OSK expression suppressed from injury (n=5), and OSK induction post injury (n=5). The protocols for pre-injury and post-injury induction used were as shown in FIG. **25**A. FIG. **25**C is a chart quantifying the number of RBPMS+ cells from four-week old (young) mice with no OSK induction, OSK induction pre-injury only, OSK suppressed from injury, and OSK induction post injury.

(27) FIGS. **26**A-**26**E include data showing that expression of OSK from a single transcript improved axon regeneration and retina ganglion cell (RGC) survival two weeks after optic nerve crush injury compared to expression of OCT4, SOX2, and KLF4 from separate transcripts. FIG. **26**A is a schematic showing the AAV combinations injected in each group two weeks before the crush injury and non-limiting exemplary expression cassettes in Tet-Off systems encoding OCT4, SOX2, and/or KLF4. FIG. **26**B is a chart showing that expression of OSK from a single transcript improved axon regeneration relative to expression of OCT4, SOX2, and KLF4 from separate transcripts. The number of estimated axons per nerve after optic nerve crush injury as a function of the distance from the site of injury (μm) was quantified for mice receiving tTA virus and one of the following (1) OCT4 virus, (2) SOX2 virus, (3) KLF4 virus, (4) virus with a vector encoding OCT4 and SOX2 under one promoter (OCT4-SOX2), (5) separate OCT4, SOX2, and KLF4 viruses (Oct4, Sox2, KLF4 or O,S,K), or (6) virus with a vector encoding OCT4, KLF4, and SOX2 under one promoter (Oct4-Sox2-KLF4 or OSK). The various vectors used are depicted in FIG. **26**A. FIG. **26**C is a chart showing that expression of OSK from a single transcript improved RGC survival relative to expression of OCT4, SOX2, and KLF4 from separate transcripts. FIG. **26**D includes whole mount staining of mouse retina showing that a heterogeneous population of cells with few RBPMS+ cells were detected when separate viral vectors encoding OCT4, SOX2, and KLF4 in separate viruses were delivered to the eye of mice. Arrows point to seven different types of cells expressing OCT4, SOX2, KLF4, and/or RBPMS in the upper left image under the schematic of the vectors. FIG. **26**E includes data showing that a more homogenous population of cells was detected when virus comprising a viral vector encoding OSK under one promoter was delivered to the eye of mice as compared to FIG. **26**D. More OSK-expressing cells that were also RBPMS+ were detected as compared to FIG. **26**D. In the upper left image under the schematic of the vector used, the long white arrow points to RBPMS+ cells expressing OCT4, SOX2, and KLF4 and the shorter

arrow indicates even some cells that did not express RBPMS expressed OSK.
(28) FIG. **27** is a chart showing that Tet1 and Tet2 DNA demethylases play a role in OSK-induced regeneration. The number of estimated axons per nerve after optic nerve crush was quantified in mice receiving (1) OSK virus and a short hairpin control, (2) OSK virus and a short hairpin against Tet1, or (3) OSK virus and a short hairpin against Tet2.
(29) FIG. **28** includes data showing that expression of OSK using a Tet-Off system reversed age-related visual acuity loss in aged mice one month post injection of AAV virus encoding TRE−OSK and AAV virus encoding tTA. FIG. **28** is a chart showing that intravitreal injection of mice with virus encoding tTA and virus encoding TRE−OSK in the absence of doxycycline (OSK induction condition) reversed the age-related decrease in the spatial frequency threshold (cycles/degree, visual acuity test) observed in aged mice (12 month old (12 m) and 18 month old (18 m) mice). A visual acuity test based on optomotor response (OMR) was used. As controls, age-matched mice received virus encoding virus encoding rtTA and virus TRE−OSK in the absence of doxycycline (uninduced control). Adult mice (3 month old (3 m)) were also used as a control.
(30) FIG. **29** includes data showing that expression of OSK reversed age-related decline in retina ganglion cell (RGC) function in aged mice. FIG. **29** is a chart showing the measurement of electrical waves generated from RGCs from adult (3 month old (3 m)) and aged (12 month old (12 m) and 18 month old (18 m)) mice. A pattern electroretinogram (pattern ERG) was used. Mice were injected with rtTA virus and TRE−OSK virus without doxycycline (uninduced control (ctl)) or with tTA virus and TRE−OSK virus (induced, OSK) without doxycycline. Results were obtained one month after virus injection.
(31) FIGS. **30**A-**30**E include data showing that expression of OSK improved glaucoma-induced declines in visual acuity and RGC function in one month-old mice. FIG. **30**A is a chart showing that polystyrene microbeads induced glaucoma as measured by an increase in intraocular pressure (IOP) compared to saline treatment in adult C57BL/6J mice. The IOP measurements are shown in the first four weeks after microbead injection. FIGS. **30**B-**30**C show that 4 weeks after microbeads injection into the anterior chamber of the eye, there was significant loss of axon density and RGC density. AAVs were intravitreally injected at 3 weeks post microbeads and it took another week before OSK expression was observed. FIG. **30**B includes a chart quantifying axon density (left panel) using p-phenylenediamine (PPD) staining (shown, for example, on the right). FIG. **30**C includes a chart quantifying RGC cell density (left panel) using Brn3a staining (shown, for example, on the right). FIG. **30**D is a chart showing visual acuity improvement by OSK AAV treatment in glaucoma-induced mice. Mice were intravitreally injected with microbeads to induce glaucoma or saline without microbeads (no glaucoma control; saline). Three weeks after microbeads injection, mice are then treated with (1) virus encoding rtTA and virus TRE−OSK in the absence of doxycycline (beads (OSK AAV OFF)); (2) virus encoding tTA and virus encoding TRE−OSK in the absence of doxycycline (beads (OSK AAV ON)). Results at 3 weeks after saline or microbead injection (pre-AAV injection) and 4 weeks after AAV injection (7 week post microbeads) are shown. FIG. **30**E is a chart showing RGC function results by pattern electroretinogram in mice treated as in FIG. **30**B.
(32) FIGS. **31**A-**31**C include data showing that expression of OSK promoted neuronal survival and axon regeneration of human SH-SY5Y neuronal cells following vincristine (VCS)-induced damage. FIG. **31**A includes a series of images showing the effect of inducing OSK expression (OSK On) compared to no induction of OSK expression (OSK Off) on the structure of neurons. Images were taken at day 3 and at day 9 after 24 hours of VCS treatment. The outlines of the neuronal cell area are shown at Day 9. FIG. **31**B is a chart quantifying neuron cell area (μm.sup.2) at indicated days after 24 hours of VCS treatment for cells in which OSK expression was induced (OSK On) and for cells in which OSK expression was not induced (OSK Off). FIG. **31**C is a chart quantifying neuron cell area (μm.sup.2) at indicated days after 48 hours of VCS treatment for cells in which OSK expression was induced (OSK On) and for cells in which OSK expression was not

induced (OSK Off).
(33) FIGS. **32**A-**32**G show that partial reprogramming with AAV-delivered polycistronic OSK is non-toxic and induces CNS axon regeneration. FIG. **32**A is a schematic of the Tet-On and Tet-Off AAV vectors used in the study to control OSK expression. FIG. **32**B shows body weight of WT mice, OSK transgenic mice, and AAV-mediated OSK-expressing mice (1.0×10.sup.12 gene copies) with or without doxycycline induction in the first 4 weeks (n=5, 3, 6, 4, 6, 3, respectively). FIG. **32**C is a schematic showing intravitreal AAV injection to target retina ganglion cells. Immunofluorescence of the whole-mounted display and cross section of retina, showing the infection rate and targeted retina layer. The scale bars represent 1 mm and 100 μm, respectively. FIG. **32**D shows an experimental outline of the optic nerve crush study using the Tet-Off system. FIG. **32**E shows quantification of the regenerating fibers by to d2EGFP, Oct4, Sox2, Klf4, OS, O+S+K, or OSK AAV at different distances distal to the lesion site. Error bars indicate s.e.m. (n=4-7). ****, P<0.0001, ANOVA with Bonferroni posttests. FIG. **32**F shows the survival of RBPMS-positive cells in the RGC layer transduced with different AAV vectors at day 14 post crush injury (n=4-8). ***, P<0.001, ****, P<0.0001, one-way ANOVA with Bonferroni post-tests, relative to d2EGFP.
(34) FIG. **32**G shows representative images of optic nerve sections showing CTB-labeled axons in wild-type mice with intravitreal injection of AAV2−tTA and TRE−OSK in the presence and absence of DOX at 2 weeks after optic nerve injury. The crush site is indicated by asterisks. The scale bars represent 200 μm.
(35) FIGS. **33**A-**33**K show that OSK expression promotes axon regeneration and neuronal survival through a Tet-dependent mechanism. FIG. **33**A shows experimental strategies for pre- and post-injury induction of OSK expression. FIG. **33**B shows RGC survival in retinas with pre- and post-injury OSK expression. FIG. **33**C shows the quantification of regenerating fibers from pre- and post-injury OSK expression models. FIG. **33**D shows representative images of optic nerves showing regenerating axons at 4 weeks after injury, with or without post injury OSK expression. The crush site is indicated with asterisks. The scale bars represent 200 μm. FIGS. **33**E-**33**F show the quantification of regenerating fibers and RGC survival in retinas co-transduced with AAV2 vectors encoding polycistronic OSK, tTA, and shRNA vectors with a scrambled sequence (Scr), Tet1, or Tet2 sequences to knockdown Tet DNA dioxygenases/demethylases. FIG. **33**G shows experimental outlines for examining axon regeneration in human neurons post vincristine (VCS) damage. FIG. **33**H shows that OSK rejuvenates human neurons according to the skin & blood clock. In the top panel of FIG. **33**H, P value is calculated by linear regression model to see if DNAmAge decrease with time. In the bottom panel of FIG. **33**H, DNA methylation age of human neurons with OSK expression pre (Day −) or after VCS damage (Day 1 and 9), estimated by skin and blood cell clocks is shown. FIG. **33**I shows the neurite area in each AAV treatment group. *p<0.05, **p<0.01, ****p<0.0001, one-way ANOVA with Tukey's multiple comparison test. FIG. **33**J shows representative images of human neurons and the neurite area after 9 days of recovery from VCS damage. FIG. **33**K shows rDNA methylation age of 1-month-old RGCs isolated from axon-intact retina infected with or without GFP, or from axon-injured retinas infected with GFP-AAV or OSK-AAV 4 days after nerve crush.
(36) FIGS. **34**A-**34**H show the reversal of glaucoma by OSK AAV treatment. FIG. **34**A is a schematic showing the experimental outline. FIG. **34**B shows Intraocular pressure measured weekly by rebound tonometry for the first 4 weeks post-microbead injection. FIG. **34**C shows Representative micrographs of PPD-stained optic nerve cross-sections at 4 wks post AAV2 or PBS injection. Scale bars, 50 μm. OSK Off (rtTA+TRE−OSK); OSK On (tTA+TRE−OSK). FIG. **34**D shows a quantification of healthy axons of the optic nerve at 4 weeks post PBS or AAV injection. FIG. **34**E is a Schematic of High-contrast visual stimulation assay to measure optomotor response. A reflexive head movement in response to the rotation of a moving stripe pattern that increases in spatial frequency was used to assess vision. FIG. **34**F shows Spatial frequency threshold response

of each mouse measured before treatment and 4 weeks after intravitreal injection of AAV vectors. FIG. **34**G shows Representative pERG waveforms recorded from the same eye at baseline before treatment and four weeks later after treatment with OSK-OFF AAV (top graph) or OSK-ON AAV (bottom graph). FIG. **34**H shows the Mean pERG amplitudes of recordings measured from each mouse at baseline before treatment and 4 weeks after intravitreal injection of AAVs. *$P<0.05$; **$P<0.01$; ***$P<0.001$, ****$P<0.0001$ Two-way ANOVA with Turkey posttests between groups was used for the overall effect of time and treatment. A paired t-test was used to compare before and after treatments.

(37) FIGS. **35**A-**35**I show that OSK AAV induces axon regeneration and restores visual function in aged mice. FIG. **35**A shows an Experimental outline for testing the effects of OSK AAV treatment in aged mice on axon regeneration following optic nerve crush and restoration of vision loss associated with physiological aging. FIG. **35**B shows Axon regeneration in 12-month-old mice with OSK AAV or control AAV (d2EGFP) treatment following 2 or 5 weeks post optic nerve crush. FIG. **35**C Representative confocal images of longitudinal sections through the optic nerve showing CTB-labeled axons after 5 weeks of OSK treatment. Scale bar represents 200 μm. FIG. **35**D The spatial frequency threshold in young mice (4 months) and old mice (12 months) treated with OSK-Off or OSK-On AAVs. FIGS. **35**E-**35**F show Spatial frequency threshold and pERG amplitudes in old mice (12 months) treated with: (i) OSK-Off, (ii) OSK-On, or (iii) OSK-On plus either: sh-Scr, sh-Tet1- or sh-Tet2-mediated knockdown of DNA demethylases. OSK-Off, (rtTA+TRE−OSK); OSK-On, (tTA+OSK). FIG. **35**G is hierarchical clustered heatmap showing RNA-Seq expression of 464 differentially expressed genes in cell sorted purified RGCs from intact young mice (5 months) or intact old mice (12 months), or old mice treated with either control AAV (TRE−OSK) or OSK-On AAV. FIG. **35**H is a scatter plot of OSK-induced changes in RNA levels versus age-associated changes in mRNA levels. Dots represent differentially expressed genes in RGCs. FIG. **35**I shows rDNA methylation age of 12-month-old RGCs FACS isolated from retinas infected for 4 weeks with −OSK or +OSK AAV together with short-hairpin DNAs with a scrambled sequence (sh-Scr) or targeted to Tet1 or Tet2 (sh-Tet1/sh-Tet2). Gene exclusion criteria for FIG. **35**G and FIG. **35**H: genes with low overall expression (log 2(CPM)<2), genes that did not significantly change with age (absolute log 2 fold-change <1) or genes altered by the virus (differentially expressed between intact old and old treatd with TRE−OSK AAV). *$P<0.05$; **$P<0.01$; ***$P<0.001$, ****$P<0.0001$. Two-way ANOVA in FIG. **35**D; One-way ANOVA in FIGS. **35**B, **35**E and **35**F.

(38) FIGS. **36**A-**36**H show an exploration of OSK (no Myc) effects on ageing and the safety of OSK AAV. FIG. **36**A is a schematic of an experimental outline of testing reprogramming effect in young and old transgenic mouse fibroblasts. FIG. **36**B shows OSKM expression rescues age-associated transcriptional changes without inducing pluripotency. For example, Nanog expression is not induced. FIG. **36**C shows OSK expression rescues age-associated transcriptional changes without inducing pluripotency. For example, Nanog expression is not induced. FIG. **36**D shows OSK AAV9 expression in the liver compared to transgenic mice. FIG. **36**E shows the body weight of WT mice and AAV-mediated OSK-expressing mice (1.0×10{circumflex over ( )}12 gene copies total) with or without doxycycline in the following 9 months after first 4 weeks (n=5, 3, 6, 4, respectively). FIG. **36**F shows AAV-UBC−rtTA and AAV−TRE−Luc vectors used for measuring tissue distribution. FIG. **36**G shows luciferase imaging of WT mice at 2 months after retro-orbital injections of AAV9-UBC−rtTA and AAV9−TRE−Luc (1.0×10{circumflex over ( )}12 gene copies total). Doxycycline was delivered in drinking water (1 mg/mL) for 7 days to the mouse shown on the right. FIG. **36**H shows luciferase imaging of eye (Ey), brain (Br), pituitary gland (Pi), heart (He), thymus (Th), lung (Lu), liver (Li), kidney (Ki), spleen (Sp), pancreas (Pa), testis (Te), adipose (Ad), muscle (Mu), spinal cord (SC), stomach (St), small intestine (In), and cecum (Ce) 2 months after retro-orbital injection of AAV9-UBC−rtTA and AAV9−TRE−Luc followed by treatment with doxycycline for 7 days. The luciferase signal is primarily in liver. Imaging the same

tissues with a longer exposure time (FIG. **36**H, lower panel) revealed lower levels of luciferase signal in pancreas (liver was removed).

(39) FIGS. **37**A-**37**D show the characterization of an inducible polycistronic AAV system. FIG. **37**A shows an Immunofluorescence analysis of the whole-mounted retina transduced with a polycistronic AAV vector expressing OCT4, SOX2, and KLF4 in the same cell. Arrows point at triple positive cells. FIG. **37**B shows an immunofluorescence analysis of the whole-mounted retina transduced with AAVs separately encoding OCT4, SOX2, and KLF4. Dotted arrows point to double-positive cells. Solid arrows point at single-positive cells, except for arrow in lower right corner of each image, which points at a triple positive cell. FIGS. **37**C-**37**D are images showing whole-mounted retina display of RBPMS and Klf4 immunofluorescence. FIG. **37**C shows that expression from AAV2 Tet-Off system can be turned off by Dox drinking water (2 mg/mL 3 days). FIG. **37**D shows that expression from AAV2 Tet-On system can be turned on by Dox drinking water (2 mg/mL 2 days). Scale bars represent 1 mm.

(40) FIGS. **38**A-**38**C show that OSK induces long-term axon regeneration post injury without RGC proliferation. FIG. **38**A shows retina whole-mount staining showing OSK infected RGCs have no proliferation marker Ki67 (left), while proliferating 293T cells have Ki67 signal (right). The scale bars represent 100 μm. FIG. **38**B shows whole nerve imaging of optic nerves showing regenerating axons from control (no AAV) or OSK AAV treatment at 3 months after injury. The scale bars represent 200 μm. FIG. **38**C shows whole nerve imaging showing CTB-labeled regenerative axons at 16 weeks post-injury (wpc) in wild-type mice with intravitreal injection of AAV2−tTA and TRE−OSK. Scale bars represent 200 m.

(41) FIGS. **39**A-**39**D show Tet-On system has better turn on rate and OSK transduced RGCs have higher survival rate. FIG. **39**A shows Representative images showing the d2EGFP expression in retina from Tet-Off AAV system with different Dox treatment. When pre-treated with DOX to suppress expression (on DOX), the GFP expression only showed up sparsely after DOX been withdrawal for 8 days, much weaker compared to peak expression (Never DOX). FIG. **39**B are representative images showing the d2EGFP in retina from Tet-On AAV system. No GFP expression was observed in the absence of DOX, and GFP expression reached peak in 2 days after Dox induction and didn't get stronger with 5 days of DOX induction. FIG. **39**C shows representative Immunofluorescence image of GFP-positive or KLF4-positive RGCs in intact and crushed samples. FIG. **39**D shows quantification of GFP- or KLF4-positive cells indicating higher survival rate of OSK expressing RGCs after crush. Scale bars represent 200 μm in FIG. **39**A, FIG. **39**B, and FIG. **39**C.

(42) FIGS. **40**A-**40**F show identification of epigenetic mechanism underlying OSK effect. FIG. **40**A Representative images of retinal whole mounts transduced with d2EGFP- or OSK-encoding AAV2 in the presence or absence of crush injury. The retinal whole mounts were immunostained for RGC marker RBPMS and mTOR activation marker pS6. FIG. **40**B shows the quantification of pS6 positive RGC % in intact and crushed samples. FIG. **40**C quantification of transduction rate of shRNA-YFP AAV in the OSK expressed RGCs. FIG. **40**D shows representative images of retinal whole mounts transduced with OSK-encoding AAV2 in the combination with sh-Scr, sh-Tet1 or sh-Tet2 YFP AAV. The retinal whole mounts were immunostained for Klf4. Scale bars represent 100 μm in FIG. **40**A and FIG. **40**D. FIG. **40**E shows Tet1 versus GAPDH mRNA level with sh-Scr or sh-Tet1 treatment in mouse RGCs in the presence of OSK expression. FIG. **40**F shows Tet2 versus GAPDH mRNA level with sh-Scr or sh-Tet2 AAV in mouse RGCs in the presence of OSK expression.

(43) FIGS. **41**A-**41**K show that OSK robustly induces human neuron axon regeneration independent of mTOR pathway. FIG. **41**A shows immunofluorescence of differentiated human neurons with transduction of AAV-DJ vectors encoding TRE−OSK and tTA (OSK On) or TRE−OSK alone (OSK Off). FIG. **41**B shows mRNA level of Oct4, Sox2 and Klf4 of human neurons transduced with AAV-DJ vectors as in FIG. **41**A. FIG. **41**C shows the FACS profile of G1,

S, and G2 phases in undifferentiated cells and differentiated cells with OSK On or Off. FIG. **41**D shows the quantification of cell population that are in proliferating S phase. FIG. **41**E shows representative images and the neurite area of human neurons post vincristine damage with or without OSK expression. FIG. **41**F shows the quantification of neurite area at different time points post vincristine damage. FIG. **41**G shows Tet2 mRNA level with sh-Scr and sh-Tet2 AAV treatment in human neurons. FIG. **41**H shows the phosphorylation level of S6 in human neurons with Rapamycin treatment (10 nM) for 5 days. FIG. **41**I shows the effect of mTOR inhibition on axon regeneration of differentiated neurons with OSK Off or OSK On. FIG. **41**J shows DNA methylation age of human neurons before vincristine (VCS) damage (Day −) or 1 and 9 days post-damage in the absence of OSK expression, estimated using a skin or a blood cell clock. FIG. **41**K shows mouse Oct4 mRNA level with sh-Scr or sh-Tet2 AAV in human neurons in the absence or presence of OSK expression.

(44) FIGS. **42**A-**42**C show the effect of OSK in a Microbead-induced mouse model. FIG. **42**A shows the quantification of RGCs and representative confocal microscopic images from retinal flat-mounts stained with anti-Brn3a, an RGC-specific marker, and DAPI, a nuclear stain, at 4 weeks post-microbead or post-saline injection. The scale bar represents 75 mm. FIG. **42**B shows the quantification of healthy axons of optic nerve and representative photomicrographs of PPD-stained optic nerve cross-sections, at 4 weeks post-microbead or post-saline injection. The scale bars represent 10 μm. FIG. **42**C shows the quantification of RGCs and representative confocal microscopic images at 4 weeks post AAV injection and 8 weeks post-microbead or post-saline injection.

(45) FIG. **43**A-**43**G show the effect of OSK in aged mice. FIG. **43**A shows the effect of OSK expression on RGC survival in young, adult, and aged mice after optic nerve crush. FIG. **43**B shows the axon regeneration promoted by OSK expression compared to the d2EGFP controls in young (1 month old), adult (3 months old), and aged (12 months old) mice at 2 weeks post injury. FIG. **43**C shows a comparison of pERG measurements in different ages at one month post OSK off or OSK On treatment. OSK Off, rtTA+TRE=OSK AAV; OSK On, tTA+OSK AAV. FIG. **43**D shows comparison of RGC cell density in 4 m- and 12 m-old mice at one month post OSK off or OSK On treatment. FIG. **43**E shows a comparison of axon density in 4 m- and 12 m-old-mice at one month post OSK off or OSK On treatment. FIG. **43**F shows comparison of pERG measurement in different ages at one-month after −OSK or +OSK treatment. −OSK: AAV−rtTA+AAV−TRE−OSK; +OSK: AAVtTA+AAV−TRE−OSK. FIG. **43**G shows spatial frequency threshold in 18-month-old mice treated with −OSK or +OSK AAV for 4 weeks.

(46) FIGS. **44**A-**44**D show RNA-seq analysis of genes that reset their expression by Reviver treatment. FIG. **44**A is a scatter plot of OSK-induced changes in RNA levels versus age-associated changes in mRNA levels. Dots represent differentially expressed genes in RGCs are shown. Gene exclusion criteria: genes with low overall expression (log 2(CPM)<2), genes that did not significantly change with age (absolute log 2 fold-change <1) or genes altered by the virus (differentially expressed between intact old and old treatd with TRE−OSK AAV). FIG. **44**B is a hierarchical clustered heatmap showing RNA-Seq expression of sensory genes in cell sorted purified RGCs from intact young mice (5 months) or intact old mice (12 months), or old mice treated with either control AAV (TRE−OSK) or OSK-On AAV. FIG. **44**C shows the top 10 biological process that are lower in old compared to young and restored by OSK. FIG. **44**D shows the top 10 biological process that are higher in old compared to young and reduced by OSK.

(47) FIGS. **45**A-**45**C show methylation clock analysis of mouse RGCs and human neurons. FIG. **45**A shows correlation between rDNA methylation age and chronological age of sorted mouse RGCs. FIG. **45**B shows average DNA methylation levels of RGCs from different ages and treatments. FIG. **45**C shows average DNA methylation levels of human neurons treated with OSK before treatment with vincristine (VCS) (−) or days post-VCS damage (1 and 9).

(48) FIGS. **46**A-**46**B show that OSK mediates axon regeneration in a Tet2-dependent manner. A

Tet2 conditional knockout mouse was used. Mouse eyes were injected with (1) AAV−CRE (Tet2 cKO); (2) AAV-tTA+AAV−TRE−OSK: OSK (Tet2 WT); or (3) AAV-tTA+AAV−TRE−OSK+AAV−CRE: OSK (Tet2 cKO). Axon regeneration was assayed after optic nerve crush. FIG. **46**A are representative optic nerve images. FIG. **46**B is a graph quantifying axon numbers.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS OF THE INVENTION

(49) The present disclosure is based, at least in part, on the unexpected results demonstrating that expression of OCT4, SOX2, and KLF4 in the absence of exogenous c-Myc expression can be used to promote partial reprogramming and tissue regeneration in vivo. Surprisingly, using the eye as a model tissue, as described herein, in some embodiments, it was determined that the combination of OCT4, SOX2, and KLF4 (OSK) could be used to reset the youthful gene expression patterns and epigenetic age of retinal ganglion cells to promote optic nerve regrowth and the restoration of vision in a rodent model of glaucoma and in old animals. In some embodiments, the DNA demethylases Tet1 and Tet2 are required for these restorative activities, which without being bound by a particular theory, suggests that the DNA methylation clock is not just a correlate of age but a regulator of it.

(50) Provided herein, in certain embodiments, are engineered nucleic acids (e.g., expression vectors, including viral vectors) encoding OCT4, SOX2, and KLF4, each alone or in combination, recombinant viruses (e.g., lentivirus, alphavirus, vaccinia virus, retrovirus, adenovirus, herpes virus, or AAV) comprising the same, pharmaceutical compositions comprising the engineered nucleic acids and/or recombinant viruses, kits comprising the engineered nucleic acids and/or recombinant viruses, and methods of regulating (e.g., inducing, inducing and then stopping, etc.) cellular reprogramming, reversing aging, tissue repair, organ regeneration, and tissue regeneration.

(51) In certain embodiments, the expression of one of more of the genes is transient (e.g., using an inducible promoter to regulate gene expression). Expression of one or more of the genes (e.g., OCT4, SOX2, KLF4, or a combination thereof) may be modulated by altering the activity of an inducing agent. As a non-limiting example, tetracycline transactivator (tTA) is capable of inducing expression from a tetracycline-responsive promoter in the absence of tetracycline. When tetracycline is added, tTA can no longer bind to the promoter and induce cannot expression. As another non-limiting example, reverse tetracycline transactivator (rtTA) is capable of inducing expression from a tetracycline-responsive promoter in the presence of tetracycline. When tetracycline is removed, rtTA can no longer bind to the promoter and cannot induce expression. As described herein, an inducible AAV vector encoding OCT4, SOX2, and KLF4 (OSK) promoted optic regeneration in vivo following damage. Therefore, the expression of these three genes may be useful in tissue and organ regeneration, tissue and organ repair, reversing aging, treating neurodegenerative diseases and conditions, cellular reprogramming, As described below, the vectors described herein may be packaged, in some instances, into viruses with a titer of more than 2×10.sup.12 particles per preparation and allow for precise control of OSK expression in mammalian cells in vitro and in vivo.

(52) Cellular reprograming allows for the production of numerous cell types from existing somatic cells. Although the Yamanaka factors (OCT4, SOX2, KLF4 and c-Myc, also known collectively as OSKM) have been shown to induce pluripotency in differentiated cells, administration of these factors may induce teratomas or other cancers in vivo (Takahashi et al., Cell. 2006 Aug. 25; 126(4):663-76); (Abad et al., Nature. 2013 Oct. 17; 502(7471):340-5). As a result of these safety concerns, use of the Yamanaka factors has largely been limited to in vitro applications. Furthermore, existing methods of gene therapy are plagued by inefficient and inconsistent gene transduction of target cells. The engineered nucleic acids, recombinant viruses comprising the same, pharmaceutical compositions thereof and kits provided herein overcome many of these limitations.

(53) Engineered Nucleic Acids

(54) The engineered nucleic acids of the present disclosure may encode OCT4, SOX2, KLF4, and homologs or variants (e.g., functional variants) thereof, each alone or in combination. In certain embodiments, an engineered nucleic acid (e.g., engineered nucleic acid) does not encode c-Myc. In certain embodiments, an engineered nucleic acid (e.g., engineered nucleic acid) does not encode a functional c-Myc because it lacks a c-Myc sequence. Assays to determine transcription factor (e.g., OCT4, SOX2, KLF4, or any combination thereof) activity are known in the art and include cell-based transcription assays and in vitro transcription assays. Transcription factor expression may also be determined using other methods including enzyme-linked immunosorbent assays (ELISAs), western blots, and quantification of RNA (e.g., using reverse transcription polymerase chain reaction).
(55) A transcription factor (e.g., OCT4, SOX2, KLF4, or homologs or variants thereof, including mammalian OCT4, mammalian SOX2, and mammalian KLF4) may be encoded by a single nucleic acid, or a single nucleic acid (e.g., engineered nucleic acid) may encode two or more transcription factors (e.g., each operably linked to a different promoter, or both operably linked to the same promoter). For example, in certain embodiments, a nucleic acid (e.g., engineered nucleic acid) may encode OCT4; SOX2; KLF4; OCT4 and SOX2; OCT4 and KLF4; SOX2 and KLF4; or OCT4, SOX2, and KLF4, in any order.
(56) In certain embodiments, an engineered nucleic acid (e.g., engineered nucleic acid) encodes an inducing agent (e.g., tTA or rtTA). In certain embodiments, a nucleic acid (e.g., engineered nucleic acid) may encode one or more transcription factors (e.g., one, two or three transcription factors) and an inducing agent. In certain embodiments, an inducing agent is encoded by a separate nucleic acid (e.g., engineered nucleic acid) that does not also encode a transcription factor (e.g., OCT4, SOX2, or KLF4). In certain embodiments, an inducing agent is encoded by a the nucleic acid (e.g., engineered nucleic acid) that also encodes a transcription factor (e.g., OCT4, SOX2, and/or KLF4). In certain embodiments, an inducing agent is encoded by a nucleic acid (e.g., engineered nucleic acid) that also encodes one or more transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof (e.g., OCT4; SOX2; KLF4; OCT4 and SOX2; OCT4 and KLF4; SOX2 and KLF4; or OCT4, SOX2, and KLF4).
(57) The transcription factors described herein (e.g., OCT4, SOX2, KLF4, or any combination thereof) or inducing agents may comprise one or more amino acid substitutions. Variants can be prepared according to methods for altering polypeptide sequences known to one of ordinary skill in the art such as those found in references which compile such methods, e.g. *Molecular Cloning: A Laboratory Manual*, J. Sambrook, et al., eds., Second Edition, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, New York, 1989, or *Current Protocols in Molecular Biology*, F. M. Ausubel et al., eds., John Wiley & Sons, Inc., New York. Conservative substitutions of amino acids include substitutions made amongst amino acids within the following groups: (a) M, I, L, V; (b) F, Y, W; (c) K, R, H; (d) A, G; (e) S, T; (f) Q, N; and (g) E, D.
(58) In certain embodiments, the engineered nucleic acids of the present disclosure comprise RNA (e.g., mRNA) and/or DNA. In some embodiments, the RNA and/or DNA is further modified. As a non-limiting example, an nucleic acid (e.g., engineered nucleic acid) of the present disclosure, may be modified RNA (e.g., mRNA) encoding OCT4, KLF4, SOX2, an inducing, or any combination thereof. See, e.g., Warren et al., Cell Stem Cell. 2010 Nov. 5; 7(5):618-30. As a non-limiting example, the engineered nucleic acids (e.g., RNA, including mRNA, or DNA) of the present disclosure may be formulated in a nanoparticle for delivery. See, e.g., Dong et al., Nano Lett. 2016 Feb. 10; 16(2):842-8. In some embodiments, the nanoparticle comprises acetylated galactose. See, e.g., Lozano-Torres et al., J Am Chem Soc. 2017 Jul. 5; 139(26):8808-8811. In some embodiments, the engineered nucleic acids (e.g., RNA, including mRNA, or DNA) is electroporated or transfected into a cell. In certain embodiments, the engineered nucleic acids are delivered as a naked nucleic acid (e.g., naked DNA or naked RNA).
(59) In some embodiments, an engineered nucleic acid that is formulated in a nanoparticle for

delivery is not an AAV vector. Suitable vector backbones for formulation in a nanoparticle include, but are not limited to, NANOPLASMID™ vectors and NTC '8' Series Mammalian Expression Vectors. Non-limiting examples of vector backbones for formulation in a nanoparticle include NTC9385R and NTC8685. Without being bound by a particular theory, NTC '8' Series Mammalian Expression Vectors may be useful as they are generally cleared by cells within weeks. The NTC '8' Series Mammalian Expression Vector comprises a CMV promoter, which can be operably linked to a sequence encoding OCT4, KLF4, SOX2, or a combination thereof. Without being bound by a particular theory, the NANOPLASMID™ vector may be less immunogenic than other vectors and express at a higher level and may express for a long time, which could be useful in long-term expression of an operably linked nucleic acid. In some embodiments, the NANOPLASMID™ vector may be useful in long term expression of OCT4, KLF4, SOX2, or a combination thereof.

(60) In some embodiments, engineered nucleic acids encoding OSK may be useful in making induced pluripotent stem cells). Without being bound by a particular theory, modified RNA (e.g., mRNA) may have an advantage of minimal activation of innate immune responses and limited cytotoxicity, thereby allowing robust and sustained protein expression. In some embodiments, the RNA (e.g., mRNA) comprises modifications including complete substitution of either 5-methylcytidine (5mC) for cytidine or pseudouridine (psi) for uridine.

(61) In some embodiments, OCT4, KLF4, and/or SOX2 expression may be activated using a CRISPR-activating system. In some embodiments, expression of one or more transcription factors selected from the group consisting of OCT4, KLF4, SOX2, and combinations thereof may be activated using a CRISPR-activating system. See, e.g., Liao et al., Cell. 2017 Dec. 14; 171(7):1495-1507.e15; Liu et al., 2018, Cell Stem Cell 22, 1-10 Feb. 1, 2018. In general, a CRISPR-activating system comprises an enzymatically dead Cas9 nuclease (or nuclease-deficient Cas9 (dCas9)) fused to a transcription activation complex (e.g., comprising VP64, P65, Rta, and/or MPH). Non-limiting examples of sequences encoding VP64, P65, Rta, and/or MPH are provided below. A VP64, P65, Rta, or MPH may be encoded by a sequence that comprises a sequence that is at least 70% (e.g., 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to any of the VP64, P65, Rta, and/or MPH sequences described herein. This Cas9 fusion protein may be referred to as a CRISPR activator. A guide RNA targeting the promoter and/or enhancer region of a gene of interest is used in a CRISPR-activating system to target the dCas9-transcription activation complex and drive expression of the endogenous gene.

(62) In some embodiments, expression of OCT4; KLF4; SOX2; or any combination thereof may be activated using a transcription activator-like effector nucleases (TALEN) or a Zinc-finger nuclease (ZFN) system.

(63) The engineered nucleic acids of the present disclosure may encode sgRNA to target and the promoter and/or enhancer region of the endogenous locus of OCT4, SOX2, and/or KLF4 in a cell. The engineered nucleic acids of the present disclosure may encode sgRNA to target and the promoter and/or enhancer region of the endogenous locus of one or more transcription factors selected from OCT4; SOX2; KLF4; and any combinations thereof in a cell. In some embodiments, the engineered nucleic acid (e.g., expression vector) further encodes a dCas9 (dead Cas9) and a transcriptional activation complex (e.g., VP64, P65, Rta, MPH). In some embodiments, the dCas9 (dead Cas9) and a transcriptional activation complex (e.g., VP64, P65, Rta, MPH) is administered to a cell on a engineered nucleic acid (e.g. expression vector). In some embodiments, the vector encoding the sgRNA and/or a dCas9 (dead Cas9) and a transcriptional activation complex (e.g., VP64, P65, Rta, MPH) is a viral vector (e.g., AAV vector). In some embodiments, dCas9 (dead Cas9) and a transcriptional activation complex (e.g., VP64, P65, Rta, MPH) is introduced into a cell as protein.

(64) In some embodiments, guide RNA targeting the enhancer and/or promoter region of OCT4, SOX2, and/or KLF4 is formulated in a nanoparticle and injected with dCas9-VP64 protein. In some embodiments, guide RNA targeting the enhancer and/or promoter region of OCT4, SOX2,

KLF4, or any combination thereof is formulated in a nanoparticle and injected with dCas9-VP64 protein. In some embodiments, the guide RNA and/or nucleic acid encoding dCas9 (dead Cas9) and a transcriptional activation complex (e.g., VP64, P65, Rta, MPH) is administered as naked nucleic acid (e.g., naked DNA formulated in a nanoparticle). In some embodiments, the guide RNA and/or nucleic acid encoding dCas9 (dead Cas9) and a transcriptional activation complex (e.g., VP64, P65, Rta, MPH) is delivered via a recombinant virus (e.g., lentivirus, adenovirus, retrovirus, herpes virus, alphavirus, vaccinia virus or adeno-associated virus (AAV)).

(65) Non-limiting example, sequences of guide RNAs targeting the endogenous OCT4 locus or SOX2 locus are provided in Liu et al., Cell Stem Cell. 2018 Feb. 1; 22(2):252-261.e4. Non-limiting examples of guide RNAs targeting OCT4, SOX2, and/or KLF4 are also provided in Weltner et al., Nat Commun. 2018 Jul. 6; 9(1):2643.

(66) Without being bound by a particular theory, use of a CRISPR-CAS9 system to activation endogenous expression of OCT4, KLF4, and/or SOX2 in the absence of c-Myc expression may obviate potential toxicity associated with exogenous gene expression and/or superphysiological gene expression.

(67) Nucleic acids (e.g., engineered nucleic acids) encoding a transcription factor (OCT4, SOX2, KLF4, or any combination thereof) or encoding an inducing agent) may be introduced into an expression vector using conventional cloning techniques. Suitable expression vectors include vectors with a promoter (e.g., a constitutive or inducible promoter, including a TRE promoter) operably-linked to a nucleic acid (e.g., engineered nucleic acid) encoding OCT4, SOX2, KLF4, or any combination thereof, and a terminator sequence (e.g., a SV40 sequence as described herein). In some embodiments, a nucleic acid (e.g., engineered nucleic acid) encodes a promoter operably linked to a nucleic acid encoding an inducing agent. In some embodiments, a vector comprises a WPRE sequence. Expression vectors containing the necessary elements for expression are commercially available and known to one of ordinary skill in the art (see, e.g., Sambrook et al., Molecular Cloning: A Laboratory Manual, Fourth Edition, Cold Spring Harbor Laboratory Press, 2012).

(68) Vectors of the invention may further comprise a marker sequence for use in the identification of cells that have or have not been transformed or transfected with the vector, or have been reprogrammed. Markers include, for example, genes encoding proteins that increase or decrease either resistance or sensitivity to antibiotics (e.g., ampicillin resistance genes, kanamycin resistance genes, neomycin resistance genes, tetracycline resistance genes and chloramphenicol resistance genes) or other compounds, genes encoding enzymes with activities detectable by standard assays known in the art (e.g., β-galactosidase, senescence-associated beta-galactosidase, luciferase or alkaline phosphatase), and genes that visibly affect the phenotype of transformed or transfected cells, hosts, colonies or plaques (e.g., green fluorescent protein). In some embodiments, the vectors used herein are capable of autonomous replication and expression of the structural gene products present in the DNA segments to which they are operably linked.

(69) In certain embodiments, the expression vector comprises an inducible promoter (e.g., a tetracycline-responsive promoter) operably linked to a sequence encoding a transcription factor (e.g., OCT4, SOX2, KLF4, or any combination thereof). In certain embodiments, the promoter operably linked to a sequence encoding a transcription factor (e.g., OCT4, SOX2, KLF4, or any combination thereof) is a tissue-specific or cell type-specific promoter (e.g., brain-specific, liver-specific, muscle-specific, nerve cell-specific, glial cell-specific, endothelial cell-specific, lung-specific, heart-specific, bone-specific, intestine-specific, skin-specific promoters, or eye-specific promoter). As an example, the muscle-specific promoter may be a desmin promoter (e.g., a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 29). In some embodiments, an eye-specific promoter may be a promoter that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence selected from SEQ ID NOs: 101-104.

(70) In certain embodiments, the promoter operably linked to a sequence encoding a transcription factor (e.g., OCT4, SOX2, KLF4, or any combination thereof) is age- or senescence-specific (e.g. the age- or senescence-specific promoter may be a p16 promoter or a Cas9-directed transcription factor that binds to methylated DNA, which is known to accumulate with age).
(71) In certain embodiments, an expression vector comprises a constitutive promoter operably linked to a nucleic acid (e.g., engineered nucleic acid) encoding OCT4, SOX2, KLF4, or any combination thereof. In some embodiments, such a vector may be inactivated using a Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)/guide RNA system. For example, a guide RNA may be complementary to the vector and is capable of targeting a Cas9 nuclease to the vector. In some embodiments, the guide RNA is complementary to a transgene (e.g. transgene encoding OCT4, KLF4, SOX2, or a combination thereof) in any of the expression vectors described herein. Cas9 may then generate double-stranded breaks in the vector and/or mutate the vector, rendering the vector inactive.
(72) In certain embodiments, the promoter operably linked to a sequence encoding an inducing agent is a constitutive promoter (e.g., CMV, EF1 alpha, a SV40 promoter, PGK1, UBC, CAG, human beta actin gene promoter, or UAS). In certain embodiments, the promoter operably linked to a sequence encoding an inducing agent is a tissue-specific promoter (e.g., brain-specific, liver-specific, muscle-specific, nerve cell-specific, lung-specific, heart-specific, bone-specific, intestine-specific, skin-specific promoters, or eye-specific promoter). As an example, the muscle-specific promoter may be a desmin promoter (e.g., a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 29).
(73) A nucleic acid (e.g., engineered nucleic acid) (e.g., an expression vector) may further comprise a separator sequence (e.g., an IRES or a polypeptide cleavage signal). Exemplary polypeptide cleavage signals include 2A peptides (e.g., T2A, P2A, E2A, and F2A). A 2A peptide may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 9. For nucleic acids (e.g., engineered nucleic acids) (e.g., expression vectors) encoding more than one transcription factor (e.g., OCT4, SOX2, KLF4, or any combination thereof), each transcription factor may be operably linked to a different promoter or to the same promoter. The transcription factors may be separated (e.g., by peptide separator sequence) on the nucleic acid. Expression of the nucleic acid (e.g., engineered nucleic acid) results in separate amino acid sequences encoding each transcription factor.
(74) In certain embodiments, an expression vector (e.g., an expression vector encoding OCT4, KLF4, SOX2, or a combination thereof) of the present disclosure may further comprise a selection agent (e.g., an antibiotic, including blasticidin, geneticin, hygromycin B, mycophenolic acid, puromycin, zeocin, actinomycin D, ampicillin, carbenicillin, kanamycin, and neomycin) and/or detectable marker (e.g., GFP, RFP, luciferase, CFP, mCherry, DsRed2FP, mKate, biotin, FLAG-tag, HA-tag, His-tag, Myc-tag, V5-tag, etc.).
(75) In certain embodiments, an expression vector encoding an inducing agent of the present disclosure may further comprise a selection agent (e.g., an antibiotic, including blasticidin, geneticin, hygromycin B, mycophenolic acid, puromycin, zeocin, actinomycin D, ampicillin, carbenicillin, kanamycin, and neomycin) and/or detectable marker (e.g., GFP, RFP, luciferase, CFP, mCherry, DsRed2FP, mKate, biotin, FLAG-tag, HA-tag, His-tag, Myc-tag, V5-tag, etc.).
(76) In certain embodiments, an expression vector (e.g., encoding OCT4, SOX2, KLF4, or any combination thereof) is present on a viral vector (e.g., AAV vector). In certain embodiments, an expression vector encoding an inducing agent is present on a viral vector (e.g., AAV vector). An AAV vector, as used herein, generally comprises ITRs flanking an expression cassette (e.g., a nucleic acid (e.g., engineered nucleic acid) comprising a promoter sequence operably linked to a sequence encoding OCT4, SOX2, KLF4, or any combination thereof and a terminator sequence, a nucleic acid (e.g., engineered nucleic acid) comprising a promoter sequence operably linked to a sequence encoding an inducing agent, or a combination thereof).

(77) In certain embodiments, the number of base pairs between two ITRs in an AAV vector of the present disclosure is less than 5 kilobases (kb) (e.g., less than 4.9 kb, less than 4.8 kb, less than 4.7 kb, less than 4.6 kb, less than 4.5 kb, less than 4.4 kb, less than 4.3 kb, less than 4.2 kb, less than 4.1 kb, less than 4 kb, less than 3.5 kb, less than 3 kb, less than 2.5 kb, less than 2 kb, less than 1.5 kb, less than 1 kb, or less than 0.5 kb). In certain embodiments, an AAV vector with a distance of less than 4.7 kb between two ITRs is capable of being packaged into virus at a titer of at least 0.5×10{circumflex over ( )}10 particle forming units per ml (pfu/ml), at least 1×10{circumflex over ( )}10 pfu/ml, at least 5×10{circumflex over ( )}10 pfu/ml, at least 1×10{circumflex over ( )}11 pfu/ml, at least 5×10{circumflex over ( )}11 pfu/ml, at least 1×10{circumflex over ( )}12 pfu/ml, at least 2×10{circumflex over ( )}12 pfu/ml, at least 3×10{circumflex over ( )}12 pfu/ml, at least 4×10{circumflex over ( )}12 pfu/ml, at least 5×10{circumflex over ( )}12 pfu/ml, at least 6×10{circumflex over ( )}12 pfu/ml, at least 7×10{circumflex over ( )}12 pfu/ml, at least 8×10{circumflex over ( )}12 pfu/ml, at least 9×10{circumflex over ( )}12 pfu/ml, or at least 1×10{circumflex over ( )}13 pfu/ml.
(78) In certain embodiments, an expression vector of the present disclosure is at least 1 kilobase (kb) (e.g., at least 1 kb, 2 kb, 3 kb, 4 kb, 5 kb, 6 kb, 7 kb, 8 kb, 9 kb, 10 kb, 50 kb, or 100 kb). In certain embodiments, an expression vector of the present disclosure is less than 10 kb (e.g., less than 9 kb, less 8 kb, less than 7 kb, less than 6 kb, less than 5 kb, less than 4 kb, less than 3 kb, less than 2 kb, or less than 1 kb).
(79) Without being bound by a particular theory, an expression vector (e.g., an AAV vector) that encodes OCT4, SOX2, and KLF4 under one promoter results in more efficient transduction of all three transcription factors in vivo compared to separate nucleic acids (e.g., engineered nucleic acids) encoding one or two of the transcription factors. In certain embodiments, the infection efficiency of a recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, retrovirus, adenovirus, herpes virus, or AAV) harboring a vector of the present disclosure in cells (e.g., animal cells, including mammalian cells) is at least 20% (e.g., at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%, or 100%).
(80) Recombinant Viruses
(81) Aspects of the present disclosure provide recombinant viruses (e.g., lentiviruses, alphaviruses, vaccinia viruses, adenoviruses, herpes viruses, retroviruses, or AAVs). The recombinant viruses (e.g., lentiviruses, alphaviruses, vaccinia viruses, adenoviruses, herpes viruses, retroviruses, or AAVs) may harbor a nucleic acid (e.g., engineered nucleic acid) (e.g., expression vector) encoding a transcription factor (e.g., OCT4, SOX2, KLF4, or any combination thereof), or a combination thereof. In some embodiments, a recombinant virus harbors a nucleic acid encoding at least two transcription factors selected from OCT4, SOX2, and KLF4 (e.g., OCT4 and SOX2; KLF4 and SOX2; OCT4, KLF4, and SOX2; or OCT4 and KLF4). In some embodiments, a recombinant virus harbors a nucleic acid encoding at least three transcription factors selected from OCT4, SOX2, and KLF4 (e.g., OCT4, SOX2, and KLF4). In some instances, a recombinant virus of the present disclosure comprises a nucleic acid encoding an inducing agent.
(82) In certain embodiments, recombinant virus is a recombinant AAV. In some embodiments, a recombinant AAV has tissue-specific targeting capabilities, such that a transgene of the AAV will be delivered specifically to one or more predetermined tissue(s). Generally, the AAV capsid is a relevant factor in determining the tissue-specific targeting capabilities of an AAV. An AAV capsid may comprise an amino acid sequence derived from AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, AAV11, and variants thereof. Non-limiting examples of the tissue-specificity of AAV serotypes are provided in Table 1. An “x” indicates that the indicated AAV serotype is capable of delivering a transgene to a specific tissue.
(83) TABLE-US-00001 TABLE 1 Non-limiting examples of AAV serotypes and their utility in specific tissues. Relevant Tissue Immune Central System Nervous (T-cells, Muscle Central System B-cells (e.g., Nervous (Blood- and AAV Skeletal System brain Dendritic serotype Liver Heart

Muscle) Eye (CNS) barrier) Pancreas Lung Cells) AAV1 x x x AAV2 x x x x AAV3 x x x x AAV4 x x x AAV5 x x x x AAV6 (e.g., x x x x AAV6.2) AAV7 x x AAV8 x x x x AAV9 x x x x x x x x AAV10 (e.g., x x x x x x x x AAVrh10) AAVDJ x x x AAVPHP.B x x

(84) Recombinant AAVs comprising a particular capsid protein may be produced using any suitable method. See, e.g., U.S. Patent Application Publication, US 2003/0138772, which is incorporated herein by reference. AAV capsid protein sequences also known in the art. See, e.g., Published PCT Application, WO 2010/138263, which is incorporated herein by reference. Generally, recombinant AAV is produced in a host cell with the following components: (1) a nucleic acid (e.g., engineered nucleic acid) sequence encoding an AAV capsid protein or a fragment thereof, (2) a nucleic acid (e.g., engineered nucleic acid) encoding a functional rep gene, (3) a recombinant AAV vector comprising AAV inverted terminal repeats flanking a transgene (e.g., nucleic acids (e.g., engineered nucleic acids) encoding OCT4, KLF4, SOX2, or a combination thereof), and (4) helper functions that allow for packaging of the recombinant AAV vector into AAV capsid proteins. In some instances, a recombinant AAV vector comprises a nucleic acid encoding an inducing agent. In certain embodiments, the helper functions are introduced via a helper vector that is known in the art.

(85) In some instances, a suitable host cell line (e.g., HEK293T cells) may be used for producing a recombinant AAV disclosed herein following routine practice. One or more expression vectors encoding one or more of the components described above may be introduced into a host cell by exogenous nucleic acids (e.g., engineered nucleic acids), which can be cultured under suitable conditions allowing for production of AAV particles. When needed, a helper vector can be used to facilitate replication, to facilitate assembly of the AAV particles, or any combination thereof. In certain embodiments, the recombinant AAV vector is present on a separate nucleic acid (e.g., engineered nucleic acid) from the other components (e.g., a nucleic acid (e.g., engineered nucleic acid) sequence encoding an AAV capsid protein or a fragment thereof, a nucleic acid (e.g., engineered nucleic acid) encoding a functional rep gene, and helper functions that allow for packaging of the recombinant AAV vector into AAV capsid proteins. In certain embodiments, a host cell may stably express one or more components needed to produce AAV virus. In that case, the remaining components may be introduced into the host cell. The supernatant of the cell culture may be collected, and the viral particles contained therein can be collected via routine methodology.

(86) Methods of Activating OCT4, SOX2, and KLF4, Each Alone or in Combination, and Replacements Thereof

(87) Aspects of the present disclosure, in some embodiments, relate to activating OCT4, SOX2, and KLF4, each alone or in combination, in a cell, tissue and/or organ. In some embodiments, OCT4, SOX2, and KLF4, each alone or in combination, is activated in the absence of c-Myc activation. The cell, tissue, and/or organ may be in vivo (e.g., in a subject) or be ex vivo. As used herein, activation includes any nucleic acid (e.g., nucleic acid comprising RNA, comprising DNA, or any combination thereof), protein, antibody, chemical agent, or any combination thereof that is capable of increasing the biological activity of a protein of interest (e.g., OCT4, SOX2, and/or KLF4). Biological activity (e.g., gene expression, reprogramming ability, transcription factor activity, etc.) may be measured using any routine method known in the art. In some embodiments, any nucleic acid (e.g., nucleic acid comprising RNA, comprising DNA, or any combination thereof), protein, antibody, chemical agent, or any combination thereof described herein replaces OCT4, SOX2 and/or KLF4. In some embodiments, any nucleic acid (e.g., nucleic acid comprising RNA, comprising DNA, or any combination thereof), protein, antibody, chemical agent, or any combination thereof described herein replaces OCT4, SOX2, KLF4, or any combination thereof. In some embodiments, any of the nucleic acids (e.g., engineered nucleic acid) encoding an inducing agent, engineered proteins encoding an inducing agent, chemical agents capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or recombinant viruses

encoding an inducing agent described herein is used to activate an inducing agent.
(88) Activation of OCT4, SOX2, and KLF4, each alone or in combination includes increasing expression (e.g., RNA and/or protein expression) of OCT4, SOX2, and KLF4, each alone or in combination. In some embodiments, the expression of OCT4, SOX2, and KLF4, each alone or in combination is increased by at least 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, or 1000% after administration of a nucleic acid (e.g., nucleic acid comprising RNA, comprising DNA, or any combination thereof) encoding OCT4, SOX2, and/or KLF4, protein encoding OCT4, SOX2, and/or KLF4, antibody capable of activating encoding OCT4, SOX2, and/or KLF4, chemical agent capable of activating encoding OCT4, SOX2, and/or KLF4, or any combination thereof to a cell, tissue, organ, and/or subject compared to before administration. In some embodiments, the expression of OCT4, SOX2, and KLF4, each alone or in combination is increased by at least 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, or 1000% after administration of a nucleic acid (e.g., nucleic acid comprising RNA, comprising DNA, or any combination thereof) encoding OCT4, SOX2, KLF4, or any combination thereof, protein encoding OCT4, SOX2, KLF4, or any combination thereof, antibody capable of activating encoding OCT4, SOX2, KLF4, or any combination thereof, chemical agent capable of activating encoding OCT4, SOX2, KLF4, or any combination thereof, or any combination thereof to a cell, tissue, organ, and/or subject compared to before administration.
(89) Activation of a inducing agent includes increasing expression (e.g., RNA and/or protein expression) of an inducing agent. In some embodiments, the expression of an inducing agent, is increased by at least 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, or 1000% after administration of a nucleic acid (e.g., nucleic acid comprising RNA, comprising DNA, or any combination thereof) encoding the inducing agent, protein encoding the inducing agent, chemical agent capable of modulating the activity of the inducing agent, or any combination thereof to a cell, tissue, organ, and/or subject compared to before administration.
(90) Expression may be measured by any routine method known in the art, including quantification of the level of a protein of interest (e.g., using an ELISA, and/or western blot analysis with antibodies that recognize a protein of interest) or quantification of RNA (e.g., mRNA) levels for a gene of interest (e.g., using reverse transcription polymerase chain reaction).
(91) In addition to the engineered nucleic acids discussed herein, OCT4, SOX2, KLF4, alone or in combination may be activated in a cell, tissue, organ, and/or subject through the use of engineered proteins. For example, protein encoding OCT4, SOX2, and/or KLF4 may be generated (e.g., recombinantly or synthetically) and administered to a cell, tissue, organ, and/or subject through any suitable route. For example, protein encoding one or more transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof may be generated (e.g., recombinantly or synthetically) and administered to a cell, tissue, organ, and/or subject through any suitable route.
(92) In some embodiments, activating expression of OCT4; SOX2; KLF4; a replacement thereof; or any combination thereof from a tetracycline-inducible expression vector comprises administering a tetracycline (e.g., doxycycline) to a cell, organ, tissue, or a subject. As one of ordinary skill in the art would appreciate, the route of tetracycline administration may be dependent on the type of cell, organ, tissue, and/or characteristics of a subject. In some embodiments, tetracycline is administered directly to a cell, organ, and/or tissue. As a non-limiting example, tetracycline may be administered to the eye of a subject through any suitable method, including eye drops comprising tetracycline, sustained release devices (e.g., micropumps, particles, and/or drug depots), and medicated contact lenses comprising a tetracycline). In some embodiments, tetracycline is administered systemically (e.g., through drinking water or intravenous injection) to a subject. Tetracycline may be administered topically (e.g., in a cream) or through a subcutaneous

pump (e.g., to deliver tetracycline to a particular tissue). Tetracycline may be administered intravenously, intradermally, intraarterially, intralesionally, intratumorally, intracranially, intraarticularly, intraprostaticaly, intrapleurally, intranasally, intravitreally, intravaginally, intrarectally, topically, intratumorally, intramuscularly, intraperitoneally, subcutaneously, subconjunctival, intravesicularlly, mucosally, intrapericardially, intraumbilically, intraocularally, orally, topically, locally, systemically, injection, infusion, continuous infusion, localized perfusion bathing target cells directly, via a catheter, in creams, in particles (e.g., nanoparticles, microparticles), in lipid compositions (e.g., liposomes), or by other method or any combination of the forgoing as would be known to one of ordinary skill in the art (see, for example, *Remington's Pharmaceutical Sciences* (1990), incorporated herein by reference).

(93) As a non-limiting example, an engineered protein may be further modified or formulated for delivery to a cell, tissue, organ, and/or subject. For example, protein transduction domains (i.e., PTD or cell-penetrating peptides) may be attached to an engineered protein (e.g., OCT4, SOX2, and/or KLF4). As a non-limiting example, a protein transduction domain (i.e., PTD or cell-penetrating peptide) may be attached to an engineered protein encoding an inducing agent. Without being bound by a particular theory, a protein transduction domain facilitate delivery of a cargo (e.g., a protein, nucleic acids, nanoparticles, viral particles, etc.) across cellular membranes. Protein transduction domains include cationic peptides, hydrophobic peptides, and/or a cell specific peptides. See, e.g., Zhou et al., Cell Stem Cell. 2009 May 8; 4(5):381-4; Zahid et al., Curr Gene Ther. 2012 October; 12(5):374-80.

(94) In some embodiments, a protein encoding OCT4, SOX2, and/or KLF4, and/or an inducing agent is formulated in a nanoparticle (e.g., for nuclear delivery). In some embodiments, a protein encoding OCT4, SOX2, KLF4, or any combination thereof (e.g., OCT4 and SOX2; KLF4 and SOX2; OCT4 and KLF4; or KLF4, SOX2, and OCT4) is formulated in a nanoparticle (e.g., for nuclear delivery). In certain embodiments, a nanoparticle further comprises a protein encoding an inducing agent. For example, chitosan [poly(N-acetyl glucosamine)] is a biodegradable polysaccharide and may be used to formulate nanoparticles by several methods. In some embodiments, a chitosan polymeric nanoparticle is loaded with protein encoding OCT4, SOX2, and/or KLF4, and/or an inducing agent and is delivered to the nucleus of a cell. See, e.g., Tammam et al., Oncotarget. 2016 Jun. 21; 7(25):37728-37739.

(95) In some embodiments, a chemical agent, antibody and/or protein replaces OCT4, SOX2, and/or KLF4. In some embodiments, a chemical agent, antibody, a protein, or any combination thereof replaces OCT4, SOX2, KLF4, or any combination thereof (e.g., OCT4 and SOX2; OCT4 and KLF4; KLF4 and SOX2; or KLF4, SOX2, and OCT4). For example, a chemical agent, antibody and/or protein may promote expression of OCT4, SOX2, and/or KLF4. In certain instances, a chemical agent, antibody and/or protein may promote expression of one or more transcription factors selected from OCT4; SOX2; KLF4; and any combinations thereof. In some embodiments, a chemical agent, antibody and/or protein may activate target genes downstream of OCT4, SOX2, and/or KLF4. In some embodiments, a chemical agent, antibody, a protein, or any combination thereof may activate target genes downstream of one or more transcription factors selected from the group consisting of OCT4; SOX2; KLF4; and any combinations thereof. In some embodiments, a chemical agent, antibody and/or protein is said to replace OCT4, SOX2, and/or KLF4 if the chemical agent, antibody and/or protein may be used together with the other two transcription factors and promote cellular reprogramming. In some embodiments, a chemical agent, antibody, protein, or any combination thereof is said to replace OCT4, SOX2, KLF4, or any combination thereof if the chemical agent, antibody, protein or any combination thereof may be used together with the other two transcription factors and promote cellular reprogramming. For example, cellular reprogramming may be determined by measuring gene expression (e.g., expression of embryonic markers and/or pluripotency markers). In some embodiments, pluripotency markers include AP, SSEA1, and/or Nanog.

(96) In some embodiments, an antibody is used to activate OCT4, SOX2, and/or KLF4. In some embodiments, an antibody is used to activate one or more transcription factors selected from OCT4, SOX2, KLF4, or any combination thereof. In some embodiments, the antibody does not target OCT4, SOX2, and/or KLF4. In some embodiments, the antibody does not target OCT4, SOX2, KLF4, or any combination thereof. In some embodiments, the antibody increases expression of OCT4, SOX2, and/or KLF4. In some embodiments, the antibody increases expression of OCT4, SOX2, KLF4, or any combination thereof. In some embodiments, the antibody does not increase expression of OCT4, SOX2, and/or KLF4. In some embodiments, an antibody replaces OCT4, SOX2, and/or KLF4. In some embodiments, the antibody does not increase expression of OCT4, SOX2, KLF4, or any combination thereof. In some embodiments, an antibody replaces OCT4, SOX2, KLF4, or any combination thereof. Any suitable method of identifying antibodies that can replace a transcription factor (e.g., OCT4, SOX2, and/or KLF4) may be used. Any suitable method of identifying antibodies that can replace a transcription factor (e.g., OCT4, SOX2, KLF4, or any combination thereof) may be used. See, e.g., Blanchard et al., Nat Biotechnol. 2017 October; 35(10):960-968.

(97) In some embodiments, another protein (e.g., a nucleic acid encoding the protein or a polypeptide encoding the protein) may be used to replace OCT4, SOX2, and/or KLF4. In some embodiments, another protein (e.g., a nucleic acid encoding the protein or a polypeptide encoding the protein) may be used to replace OCT4, SOX2, KLF4, or a combination thereof. For example, OCT4 may be replaced by Tet1, NR5A-2, Sa114, E-cadherin, NKX3-1, or any combination thereof. In some embodiments, OCT4, SOX2, and/or KLF4 may be replaced by NANOG and/or TET2. In some embodiments, OCT4, SOX2, KLF4, or any combination thereof may be replaced by NANOG and/or TET2. See, e.g., Nat Cell Biol. 2018 August; 20(8):900-908; Gao et al., Cell Stem Cell. 2013 Apr. 4; 12(4):453-69. Nanog and Lin28 can replace Klf4. See, e.g., Yu et al, Science. 318, 1917-1920, 2007). In some embodiments, OCT4, SOX2, and/or KLF4 is replaced by Tet3 (tet methylcytosine dioxygenase 3). In some embodiments, OCT4, SOX2, KLF4, or any combination thereof is replaced by Tet3 (tet methylcytosine dioxygenase 3). In some embodiments, a nucleic acid encoding a Tet1 DNA demethylase comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to NM_030625.3 or NM_001253857.2. In some embodiments, an amino acid encoding a Tet1 DNA demethylase comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to NP_085128.2 or NP_001240786.1. In some embodiments, a nucleic acid encoding a Tet2 DNA demethylase comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to NM_001127208.2, NM_001040400.2, NM_001346736.1, or NM_017628.4. In some embodiments, an amino acid encoding a Tet2 DNA demethylase comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to NP_060098.3, NP_001035490.2, NP_001333665.1, or NP_001120680.1. In some embodiments, a nucleic acid encoding a Tet3 DNA demethylase comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to NM_001287491.2, NM_001347313.1, NM_183138.2, or NM_001366022.1. In some embodiments, an amino acid encoding a Tet3 DNA demethylase comprises a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to NP_001274420.1, NP_001334242.1, NP_898961.2, or NP_001352951.1. Tet1, Tet2, and/or Tet3 may be derived from any species. In some embodiments, Tet1, Tet2, and/or Tet3 is a truncated form of a wild-type counterpart. As a non-limiting example, Tet1, Tet2, and/or Tet3 is N-terminally truncated compared to a wild-type Tet1, Tet2, and/or Tet3 counterpart and is catalytically active. In some embodiments, Tet1, Tet2, and/or Tet3 only comprises the catalytic domain of Tet1, Tet2, and/or Tet3. In some embodiments, Tet1, Tet2, and/or Tet3 comprises the catalytic domain of Tet1, Tet2, Tet3, or any combination thereof. Non-limiting examples of functional truncated Tet1 may be found in Hrit et al., Elife. 2018 Oct. 16; 7. pii: e34870.

(98) Additional methods of replacing OCT4, SOX2, and/or KLF4 to promote cellular reprogramming are known in the art. See, e.g., Heng et al., Cell Stem Cell 6, 167-174 (2010); Eguchi et al., Proc. Natl Acad. Sci. USA 113, E8257-E8266 (2016); Gao et al., Cell Stem Cell 12, 453-469 (2013); Long et al., Cell Res. 25, 1171-1174 (2015); Hou et al., Science 341, 651-654 (2013); Redmer et al., EMBO Rep. 12, 720-726 (2011); Tan et al., J. Biol. Chem. 290, 4500-4511 (2014); Anokye-Danso et al., Cell Stem Cell 8, 376-388 (2011); Miyoshi et al., Cell Stem Cell 8, 633-638 (2011); Shu et al., Cell 153, 963-975 (2013); Yu, J. et al., Science 318, 1917-1920 (2007).
(99) In some embodiments, a chemical agent replaces OCT4, SOX2, and/or KLF4 (e.g., can be used in place of OCT4, SOX2, and/or KLF4 along with the other two transcription factors to promote cellular reprogramming). In some embodiments, a chemical agent replaces OCT4, SOX2, KLF4, or any combination thereof (e.g., can be used in place of OCT4, SOX2, KLF4, or any combination thereof, along with the other two transcription factors to promote cellular reprogramming). For example, SOX2 may be replaced by CHIR, FSK, or 616452. OCT4 may be replaced by DZNep. Since Sa114 may be used to replace OCT4 as mentioned above, any compound that replaces Sa114 may also be used to replace OCT4. For example, CHIR, FSK, and 616452 may be used to replace Sa114. Nanog may be replaced with 2i medium. See, e.g., Hou et al., Science. 2013 Aug. 9; 341(6146):651-4. See, also, e.g., Zhao et al., Cell. 2015 Dec. 17; 163(7):1678-91.
(100) In some embodiments, chemical reprogramming comprises using chemicals that reduce the toxicity of chemical agents that induce reprogramming. Non-limiting examples of chemicals that reduce the toxicity of chemical reprogramming include ROCK inhibitors (e.g., Y27632 and Fasudil) and P38 MAPK inhibitors (e.g., SB203580 and BIRB796). See, e.g., Li et al., Cell Stem Cell. 2015 Aug. 6; 17(2):195-203.
(101) OCT4, KLF4, SOX2, replacements, or any combination thereof may be activated (e.g., expression may be induced) in combination with activating an enhancer of reprogramming and/or inhibiting a barrier of reprogramming. An enhancer of reprogramming may be activated using any suitable method known in the art, including overexpression of the enhancer, increasing expression of an endogenous gene encoding the enhancer (e.g., using CRISPR technology), use of a chemical agent and/or antibody to increase the biological activity of the enhancer, and use a chemical agent and/or antibody to promote expression of the enhancer. A barrier of reprogramming may be inhibited using any suitable method known in the art, including knocking down expression of the inhibitor (e.g., with siRNAs, miRNAs, shRNAs), knocking out an endogenous copy of the inhibitor (e.g., using CRISPR technology, TALENs, zinc finger nucleases, etc.), using a chemical agent and/or antibody to decrease the biological activity of the inhibitor, and using a chemical agent and/or antibody to decrease expression of the inhibitor.
(102) Non-limiting examples of enhancers and barriers of reprogramming are provided in Table 2. See also, e.g., Ebrahimi, Cell Regen (Lond). 2015 Nov. 11; 4:10, which is incorporated by reference in its entirety for this purpose.
(103) TABLE-US-00002 TABLE 2 Non-limiting examples of strategies to enhance reprogramming. Reprogramming Enhancing Strategy Enhancers Activation of C/EBPα; UTF1; Mef2c; Tdgf1; Enhancers FOXH1; GLIS1; mutated reprogramming factors, MDM2; Bcl-2; CCL2; Kdm3a, Kdm3b, Kdm4c, and Kdm4b/2b; Jhdm1a/1b; MOF; Mbd1-4 (or their small molecule activators); Wnt/β-catenin signaling; small molecule Pitstops 1 and 2; vitamin C, palbiociclib; cytokines, e.g. IL-6; CDK4, CDK8, CDK19; lincU Inhibition of Barriers Barriers p53, p57, p38, p16.sup.Ink4a/p19.sup.Arf, p21.sup.Cip1, Rb TGF-β, MAP kinase, Aurora A kinase, MEK/ERK, Gsk3, Wnt/β-catenin signaling pathways, LATS2, PKC, IP3K, CDK8, CDK19. Native/somatic gene or transcriptional regulatory network (GRN/TRN). Specific members of ADAM family (e.g., ADAM7, ADAM21, ADAM29), endocytosis: (e.g., DRAM1, SLC17A5, ARSD), phosphatase: (e.g., PTPRJ, PTPRK, PTPN11). Chromatin regulators: (e.g., ATF7IP, MacroH2A, Mbd1-4, Setdb1a. Transcription factors: (e.g., TTF1, TTF2, TMF1, T), Bright. Fbxw7 (a member of

ubiquitin-proteasome system (UPS)) Lzts1, Ssbp3, Arx, Tfdp1, Nfe2, Ankrd22, Msx3, Dbx1, Lasp1, and Hspa8. Cytokines e.g., TNFα Cells (e.g., senescent cells and NK cells) (e.g., navitoclax, BAY117082) NuRD, Mbd1-4, Gatad2a, Chd4 (see, e.g., Mor et al., Cell Stem Cell. 2018 Sep. 6; 23(3):412-425.e10) KDM1a Kaiso (see, e.g., Kaplun et al., Biochemistry (Mosc). 2019 March; 84(3):283-290)

(104) Additional reprogramming enhancers that may be activated in combination with activation of OCT4, KLF4, SOX2, replacements thereof, or any combination thereof, include histone lysine demethylases (e.g., KDM2, KDM3, and KDM4). Histone lysine demethylases may be activated by being overexpressed in a cell, tissue, organ, and/or a subject. Chemical activators of histone lysine demethylases are also encompassed by the present disclosure. For example, vitamin C may be used to activate KDM3 and/or KDM4.

(105) In some embodiments, OCT4, SOX2, KLF4, replacements thereof, or any combination thereof, is activated along with activation of C/EBPα and Tfcp2l1. Without being bound by a particular theory, C/EBPα, and Tfcp2l1 together with Klf4 may drive Tet2-mediated enhancer demethylation and activation during reprogramming.

(106) In some embodiments, OCT4, SOX2, KLF4, replacements thereof, or any combination thereof are activated in a cell, tissue, organ and/or a subject in combination with a cytokine that facilitates reprogramming. IL6 is a non-limiting example of a cytokine. See, e.g., Mosteiro et al, Science. 2016 Nov. 25; 354(6315), which is hereby incorporated by reference in its entirety for this purpose.

(107) In some embodiments, OCT4, SOX2, KLF4, replacements thereof, or any combination thereof are activated in a cell, tissue, organ and/or a subject in combination with activation of a miRNA (e.g., administration of a miRNA and/or expression of a miRNA). For example, a miRNA that promotes cell cycle progression may be introduced to a cell, tissue, organ, and/or subject. Non-limiting examples of miRNAs that promote cell cycle progression include miR 302-367, miR 371-373, miR-200b, miR-200c, miR-205, miR 290-295, miR-93, miR-106, and miR 135b.

(108) As a non-limiting example, nerve regeneration may be enhanced by combining activation of OCT4, SOX2, KLF4, replacements thereof, or any combination thereof with activation of an enhancer. Non-limiting activation of enhancers include overexpression of a member of the KLF family (e.g., KLF7), overexpression of c-Myc, STAT3 activation, SOX11 overexpression, overexpression of Lin28, overexpression of or delivery of soluble protein encoding insulin-like growth factor 1 (IGF1) and osteopontin (OPN), and activation of B-RAF (e.g., introduction of a gain of function mutation). See also, e.g., Blackmore et al., Proc Natl Acad Sci USA. 2012 May 8; 109(19):7517-22; Belin et al., Neuron. 2015 May 20; 86(4):1000-1014; Bareyre et al., Proc Natl Acad Sci USA. 2011 Apr. 12; 108(15):6282-7; Norsworthy et al., Neuron. 2017 Jun. 21; 94(6):1112-1120.e4; Wang et al., Cell Rep. 2018 Sep. 4; 24(10):2540-2552.e6; Liu et al., Neuron. 2017 Aug. 16; 95(4):817-833; O'Donovan et al., J Exp Med, 2014, 211(5): p. 801-14, which is each hereby incorporated by reference in its entirety for this purpose.

(109) In some embodiments, OCT4, SOX2, KLF4, replacements thereof, or any combination thereof, are activated in a cell, tissue, organ, and/or a subject in combination with suppression or knockdown of reprogramming barriers. Non-limiting examples of reprogramming barriers include Chaf1a, Chaf1b, Ube2i, sumo2, and/or Nudt21. See, e.g., Brumbaugh et al., Cell. 2018 Jan. 11; 172(1-2):106-120.e21; Cheloufi et al., Nature. 2015 Dec. 10; 528(7581):218-24; and Borkent et al., Stem Cell Reports, 2016. 6(5): p. 704-716, which is each hereby incorporated by reference in its entirety for this purpose.

(110) As a non-limiting example, a reprogramming barrier may be a DNA methyltransferase (DNMT) may be and a DNMT may be inhibited to promote reprogramming of a tissue, cell, and/or organ. Most DNA methyltransferases use S-adenosyl-L-methionine as a methyl donor. DNMT may be from any species. There are at least three different types of methyltransferases. m6A methyltransferases are capable of methylating the amino group at the c-6 position of adenines in

DNA (e.g., Enzyme Commission (EC) No. 2.1.1.72). m4C methyltransferases are capable of generating N4-methylcytosine (e.g., Enzyme Commission (EC) No. 2.1.1.113). M5C methyltransferases are capable of generating C5-methylcytosine (e.g., Enzyme Commission (EC) No. 2.1.1.37).

(111) Non-limiting examples of mammalian DNA methyltransferases (DNMTs) include DNMT1 and its isoforms DNMT1b and DNMT1o (oocytes-specific), DNMT3a, DNMT3b, DNMT3L. GenBank Accession Nos. NM_001130823.3 (isoform a), NM_001318730.1 (isoform c), NM_001318731.1 (isoform d), and NM_001379.3 (isoform b) are non-limiting examples of nucleotide sequences encoding human DNMT1. A nucleic acid encoding a DNMT1 may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence set forth in GenBank Accession Nos. NM_001130823.3 (isoform a), NM_001318730.1 (isoform c), NM_001318731.1 (isoform d), and/or NM_001379.3 (isoform b). GenBank Accession Nos. NP_001124295.1 (isoform a), NP_001305659.1 (isoform c), NP_001305660.1 (isoform d), and NP_001370.1 (isoform b) are non-limiting examples of amino acid sequences encoding human DNMT1. An amino acid sequence encoding a DNMT1 may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence set forth in GenBank Accession Nos. NP_001124295.1 (isoform a), NP_001305659.1 (isoform c), NP_001305660.1 (isoform d), and/or NP_001370.1 (isoform b). A nucleic acid encoding human DNMT3A includes GenBank Accession No. NM_001320892.1, NM_001320893.1, NM_022552.4, NM_153759.3, NM_175629.2, and NM_175630.1. A nucleic acid encoding a DNMT3A may be at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence set forth in GenBank Accession Nos. NM_001320892.1, NM_001320893.1, NM_022552.4, NM_153759.3, NM_175629.2, and/or NM_175630.1. An amino acid sequence encoding human DNMT3A includes GenBank Accession Nos. NP_001307821.1, NP_001307822.1, NP_072046.2, NP_715640.2, NP_783328.1, and NP_783329.1. An amino acid sequence encoding a DNMT3A may be at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence set forth in GenBank Accession Nos. NP_001307821.1, NP_001307822.1, NP_072046.2, NP_715640.2, NP_783328.1, and/or NP_783329.1. A nucleic acid encoding human DNMT3B includes GenBank Accession No. NM_001207055.1, NM_001207056.1, NM_006892.3, NM_175848.1, NM_175849.1, and NM_175850.2. A nucleic acid encoding a DNMT3B may be at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence set forth in GenBank Accession Nos. NM_001207055.1, NM_001207056.1, NM_006892.3, NM_175848.1, NM_175849.1, and/or NM_175850.2. An amino acid sequence encoding human DNMT3B includes GenBank Accession Nos. NP_001193984.1, NP_001193985.1, NP_008823.1, NP_787044.1, NP_787045.1, and NP_787046.1. An amino acid sequence encoding a DNMT3B may be at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence set forth in GenBank Accession Nos. NP_001193984.1, NP_001193985.1, NP_008823.1, NP_787044.1, NP_787045.1, and/or NP_787046.1. A nucleic acid encoding human DNMT3L includes GenBank Accession No. NM_013369.3 and NM_175867.2. A nucleic acid encoding a DNMT3L may be at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence set forth in GenBank Accession Nos. NM_013369.3 and/or NM_175867.2. An amino acid sequence encoding human DNMT3L includes GenBank Accession Nos. NP_037501.2 and NP_787063.1. An amino acid sequence encoding a DNMT3L may be at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to a sequence set forth in GenBank Accession Nos. NP 037501.2 and/or NP 787063.1.

(112) A DNMT may be inhibited using any suitable method known in the art. Suitable methods include knockdown of a DNMT mRNA, genetically knocking out a DNMT, and use of a DNMT inhibitor (e.g., chemical inhibitors). DNMT inhibitors are being investigated in clinical trials (e.g., phase III clinical trials) in the United States of America and beyond. Non-limiting examples of

DNMT inhibitors include VIDAZA™ (azacitidine) (e.g., for the treatment of Myelodysplastic Syndromes and treatment of acute myeloid leukemia (AML)), DACOGEN™ (decitabine) (e.g., for treatment of AML and treatment of Chronic myeloid leukemia (CML)), and Guadecitabine (SGI-110) (e.g., for treatment of AML). In 2012, the European Union approved DACOGEN™ (decitabine) for use in patients with AML.

(113) A DNMT may be inhibited by inhibiting a DNMT stabilizer. Suitable methods of inhibiting a DNMT stabilizer include knockdown of the mRNA encoding the stabilizer, genetically knocking out the gene that encodes the stabilizer and use of an inhibitor (e.g., chemical inhibitors). As a non-limiting example, KDM1a, which is also referred to as Lsd1 or Aof2, is a stabilizer of DNMT1. See, e.g., Wang et al., Nat Genet. 2009 January; 41(1):125-9. In some embodiments, KDM1a expression is knocked down using a shRNA disclosed herein or known in the art. In some embodiments, KDM1a is inhibited to prevent injury induced by hypermethylation from DNMTs, which could be useful in promoting reprogramming.

(114) In some embodiments, a histone methyltransferase is a reprogramming barrier and is inhibited to facilitate reprogramming of a cell, tissue and/or organ. Histone methyltransferases may be inhibited by any suitable method, including use of chemical inhibitors. For example, 3-deazaneplanocin A (Dznep), epz004777, and BIX-01294 are examples of histone methyltransferase inhibitors.

(115) In some embodiments, a reprogramming barrier is a histone deacetylase (HDAC) and a HDAC is inhibited to facilitate reprogramming of a cell, tissue, and/or organ. Non-limiting examples of HDAC inhibitors include valproic acid (VPA), trichostatin A (TSA), suberoylanilide hydroxamic Acid (SAHA), sodium butyrate (SB), Belinostat (PXD101), Panobinostat (LBH589), Quisinostat (JNJ-26481585), Abexinostat (PCI-24781), Givinostat (ITF2357), Resminostat (4SC-201), Phenylbutyrate (PBA), Depsipeptide (romidepsin), Entinostat (MS-275), Mocetinostat (MGCD0103), and Tubastatin A (TBA).

(116) In some embodiments, a reprogramming barrier is a NF-κB, and it is inhibited to facilitate reprogramming of a cell, tissue, and/or organ. Non-limiting examples of NF-κB inhibitor includes BAY 11-7082, TPCA 1, and p65 siRNA. See, e.g., the NF-κB small molecule guide compiled by Abcam, which is available on the Abcam website (www.abcam.com/reagents/nf-kb-small-molecule-guide).

(117) In some embodiments, a repogramming barrier is a cytokine secreted from senescent cells in which a cytokine is inhibited to facilitate reprogramming of a cell, tissue, and/or organ. None limiting examples of cytokines inhibitors include Anti-TNFα (Mahmoudi et al, Biorxiv, 2018) and drugs, including Navitoclax, that kill senescence cells.

(118) In some embodiments, a reprogramming barrier is a microRNA (miRNA) and a microRNA is inhibited to facilitate reprogramming of a cell, tissue, and/or organ. Non-limiting examples of microRNAs that are reprogramming barriers include miR Let-7 and miR-34. Without being bound by a particular theory, inhibition of miR Let-7 may increase the efficiency of reprogramming because miR Let-7 inhibits the cell cycle and inhibition of miR-34 may facilitate reprogramming because miR-34 inhibits the translation of p53.

(119) In some embodiments, OCT4, SOX2, KLF4, replacements thereof, or any combination thereof is activated in a cell, tissue, organ and/or a subject in combination with inhibition of PTEN, SOCS3, RhoA, and/or ROCK to enhance nerve regeneration. In some embodiments, PTEN is deleted, SOCS3 is deleted, RhoA is knocked down, and/or ROCK is knocked down in a cell, tissue, organ and/or subject. See, e.g., Park et al., Science. 2008 Nov. 7; 322(5903):963-6; Smith et al., Neuron. 2009 Dec. 10; 64(5):617-23; Koch et al., Front Cell Neurosci. 2014 Sep. 5; 8:273; Koch et al., Cell Death Dis. 2014 May 15; 5:e1225 for descriptions of inhibition of PTEN, SOCS3, RhoA, and/or ROCK. Each reference is hereby incorporated by reference in its entirety for this purpose.

(120) In some embodiments, OCT4, SOX2, KLF4, replacements thereof, or any combination thereof is activated in a cell, tissue, organ and/or a subject in combination with neuronal electrical

stimulation (e.g., high-contrast visual stimulation) to promote nerve regeneration. See, e.g., Lim et al., Nat Neurosci. 2016 August; 19(8):1073-84 for a description of high-contrast visual stimulation. This reference is hereby incorporated by reference in its entirety for this purpose.

(121) In some embodiments, OCT4, SOX2, KLF4, replacements thereof, or any combination thereof is activated in a cell, tissue, organ and/or a subject in combination with gamma band light stimulation to promote nerve regeneration. See, e.g., McDermott et al., J Alzheimers Dis. 2018; 65(2): 363-392 for a description of gamma band light stimulation. This reference is hereby incorporated by reference in its entirety for this purpose.

(122) Engineered Cells

(123) Engineered cells and method of producing engineered cells are also encompassed by the present disclosure. The engineered cells, for example, may be useful in cell-based therapies (e.g., stem cell therapies). Although stem cell therapy is currently in clinical trials (see, e.g., David Cyranoski, Nature 557, 619-620 (2018), toxicity (e.g., off-target toxicity) is a concern, Without being bound by a particular theory, the engineered cells of the present disclosure (e.g., cells engineered using AAV vectors encoding OCT4, KLF4, and/or SOX2, and/or an inducing agent) may have a lower toxicity because AAV is does not integrate into the genome of host cells and use of the inducible systems described herein to control expression of OCT4, KLF4, and/or SOX2 may allow for precise control (e.g., amount and timing) of gene expression.

(124) Any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered proteins described herein, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination may be introduced into a host cell, host tissue, or organ to produce an engineered cell, an engineered tissue, or an engineered organ. Any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, engineered proteins described herein, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination may be introduced into a host cell, host tissue, or organ to produce an engineered cell, an engineered tissue, or an engineered organ. In some embodiments, a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent, an engineered protein encoding an inducing agent, a chemical agent capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or a recombinant virus encoding an inducing agent is also introduced into a host cell, host tissue, or organ to produce an engineered cell, an engineered tissue, or an engineered organ.

(125) In some embodiments, the engineered cell is an induced pluripotent stem cell (iPSC).

(126) In some embodiments, a viral vector (e.g., an AAV vector, including a vector with a TRE promoter operably linked to a nucleic acid encoding OCT4, KLF4, and SOX2) is packaged into a virus with an AAV-DJ capsid. In some embodiments, the AAV-DJ capsid increases the transduction efficiency into cultured cells compared to cells without the AAV-DJ capsid. In some embodiments, the AAV virus encoding OSK is administered to a cell. In some embodiments, an AAV virus (e.g., AAV-DJ virus) encoding the inducing agent or a protein encoding the inducing agent is administered to the same cells. In some embodiments, this system produces an engineered cell (e.g., an induced pluripotent stem cell). In some embodiments, the engineered cell is further differentiated into (e.g., differentiated into an eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine cell). In some embodiments, the differentiated cell is used for

transplantation purposes. In some embodiments, the engineered cell is cultured to create an engineered tissue. In some embodiments, the engineered cell is cultured to create an engineered organ. In some embodiments, the engineered cells are retina pigment epithelium cells, neuron cells, pancreatic beta-cells, or cardiac cells.

(127) Compositions

(128) The compositions of the disclosure may comprise at least one of any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered proteins, engineered cells, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein alone, or in combination. In certain embodiments, the compositions of the disclosure comprise at least one of any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, engineered proteins, engineered cells, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein alone, or in combination. In some embodiments, a composition comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more different nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vectors encoding OCT4, KLF4, and/or SOX2). In some embodiments, a composition comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more different nucleic acids (e.g., engineered nucleic acids) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof (e.g., expression vectors encoding OCT4; KLF4; SOX2; or any combination thereof). In some embodiments, a composition comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more different viruses (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) each having one or more different transgenes. In some embodiments, a composition comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more different chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2. In some embodiments, a composition comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more different chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof. In some embodiments, a composition further comprises one or more nucleic acids (e.g., engineered nucleic acids) encoding an inducing agent, one or more engineered proteins encoding an inducing agent, one or more chemical agents capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or one or more recombinant viruses encoding an inducing agent. In some embodiments, a composition comprises engineered cells (e.g., induced pluripotent stem cells and/or differentiated cells). In some embodiments, a composition comprises an engineered protein encoding OCT4, SOX2, and/or KLF4. In some embodiments, a composition comprises an engineered protein encoding OCT4, SOX2, KLF4, or any combination thereof. In some embodiments, a composition further comprises an engineered protein encoding an inducing agent.

(129) In some embodiments, a composition further comprises a pharmaceutically acceptable carrier. Suitable carriers may be readily selected by one of skill in the art in view of the indication for which the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, engineered proteins, engineered cells, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) e.g. is directed. Suitable carriers may be readily selected by one of skill in the art in view of the indication for which the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vectors) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating

(e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, engineered proteins, engineered cells, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) e.g. is directed. Suitable carriers may also be readily selected by one of skill in the art in view of the indication for which the nucleic acids (e.g., engineered nucleic acids) encoding an inducing agent, engineered proteins encoding an inducing agent, chemical agents capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) comprising an inducing agent e.g. is directed. For example, one suitable carrier includes saline, which may be formulated with a variety of buffering solutions (e.g., phosphate buffered saline). Other exemplary carriers include sterile saline, lactose, sucrose, calcium phosphate, gelatin, dextran, agar, pectin, peanut oil, sesame oil, and water. The selection of the carrier is not a limitation of the present disclosure.

(130) Optionally, the compositions of the disclosure may comprise, in addition to the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered cells comprising OCT4, KLF4, and/or SOX2, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) e.g. and carrier(s), other pharmaceutical ingredients, such as preservatives, or chemical stabilizers. Optionally, the compositions of the disclosure may comprise, in addition to the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, engineered cells comprising OCT4; KLF4; SOX2; or any combination thereof, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) e.g. and carrier(s), other pharmaceutical ingredients, such as preservatives, or chemical stabilizers. Suitable exemplary preservatives include chlorobutanol, potassium sorbate, sorbic acid, sulfur dioxide, propyl gallate, the parabens, ethyl vanillin, glycerin, phenol, and parachlorophenol. Suitable chemical stabilizers include gelatin and albumin. The compositions of the present disclosure may further comprise a nucleic acid (e.g., engineered nucleic acids) encoding an inducing agent, an engineered protein encoding an inducing agent, chemical agents capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or recombinant viruses encoding an inducing agent.

(131) The nucleic acid (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, engineered cells, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, engineered proteins encoding OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) encoding the same described herein are administered in sufficient amounts to transfect the cells of a desired tissue (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine tissue) and to provide sufficient levels of gene transfer and expression without undue adverse effects. Any of the nucleic acids (e.g., engineered nucleic acids) encoding an inducing agent, an engineered protein encoding an inducing agent, chemical agents capable of modulating (e.g., activating or inhibiting) the activity of an inducing agent, and/or recombinant viruses encoding an inducing agent are administered in sufficient amounts to transfect the cells of a desired tissue (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin

including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine tissue) and to provide sufficient levels of gene transfer and expression without undue adverse effects. Examples of pharmaceutically acceptable routes of administration include, but are not limited to, direct delivery to the selected organ (e.g., direct delivery to eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine). Any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered cells, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, engineered proteins, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein may be delivered intravenously, intradermally, intraarterially, intralesionally, intratumorally, intracranially, intraarticularly, intraprostaticaly, intrapleurally, intranasally, intravitreally, intravaginally, intrarectally, topically, intratumorally, intramuscularly, intraperitoneally, subcutaneously, subconjunctival, intravesicularlly, mucosally, intrapericardially, intraumbilically, intraocularally, orally, topically, locally, systemically, injection, infusion, continuous infusion, localized perfusion bathing target cells directly, via a catheter, in creams, in lipid compositions (e.g., liposomes), or by other method or any combination of the forgoing as would be known to one of ordinary skill in the art. Any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vectors) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, engineered cells, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, engineered proteins, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein may be delivered intravenously, intradermally, intraarterially, intralesionally, intratumorally, intracranially, intraarticularly, intraprostaticaly, intrapleurally, intranasally, intravitreally, intravaginally, intrarectally, topically, intratumorally, intramuscularly, intraperitoneally, subcutaneously, subconjunctival, intravesicularlly, mucosally, intrapericardially, intraumbilically, intraocularally, orally, topically, locally, systemically, injection, infusion, continuous infusion, localized perfusion bathing target cells directly, via a catheter, in creams, in lipid compositions (e.g., liposomes), or by other method or any combination of the forgoing as would be known to one of ordinary skill in the art. Any of the nucleic acids encoding an inducing agent, chemical agents capable of modulating the activity of an inducing agent, engineered proteins encoding an inducing agent, and/or recombinant viruses encoding an inducing agent may be may be delivered intravenously, intradermally, intraarterially, intralesionally, intratumorally, intracranially, intraarticularly, intraprostaticaly, intrapleurally, intranasally, intravitreally, intravaginally, intrarectally, topically, intratumorally, intramuscularly, intraperitoneally, subcutaneously, subconjunctival, intravesicularlly, mucosally, intrapericardially, intraumbilically, intraocularally, orally, topically, locally, systemically, injection, infusion, continuous infusion, localized perfusion bathing target cells directly, via a catheter, in creams, in lipid compositions (e.g., liposomes), or by other method or any combination of the forgoing as would be known to one of ordinary skill in the art. Routes of administration may be combined, if desired.

(132) In some embodiments, a nucleic acid is delivered non-virally (e.g., not on a viral vector and/or not in a virus). In some embodiments, a nucleic acid (e.g., RNA or DNA) encoding OCT4, SOX2, and/or KLF4 and/or an inducing agent is administered in a liposome. In some embodiments, a nucleic acid (e.g., RNA or DNA) encoding OCT4, SOX2, KLF4, or any combination thereof, and/or an inducing agent is administered in a liposome. In some embodiments, a nucleic acid (e.g., RNA or DNA) encoding OCT4, SOX2, and/or KLF4 and/or an inducing agent is administered in a particle. In some embodiments, a nucleic acid (e.g., RNA or DNA) encoding OCT4, SOX2, KLF4,

or any combination thereof, and/or an inducing agent is administered in a particle. In some embodiments, the nucleic acid is RNA (e.g., mRNA).

(133) In some embodiments, a pharmaceutical composition comprising an expression vector encoding OCT4, KLF4, and/or SOX2 or a pharmaceutical composition comprising a virus harboring the expression vector is administered to a cell, tissue, organ or a subject. In some embodiments, a pharmaceutical composition comprising an expression vector encoding an inducing agent or a pharmaceutical composition comprising a virus harboring the expression vector is administered to a cell, tissue, organ or a subject. In some embodiments, the virus and/or expression vector encoding OCT4, KLF4, and/or SOX2 is administered systemically. In some embodiments, the virus and/or expression vector encoding an inducing agent is administered systemically. In some embodiments, the virus and/or expression vector encoding OCT4, KLF4, and/or SOX2 is administered locally (e.g., directly to a tissue or organ of interest, including eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine). In some embodiments, a virus and/or expression vector encoding an inducing agent is administered locally (e.g., directly to a tissue or organ of interest, including eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine). In some embodiments, the inducing agent (e.g., a nucleic acid encoding the inducing agent, a protein encoding the inducing agent, or a virus encoding the inducing agent) and/or chemical agent capable of modulating (e.g., activating or inhibiting) the activity of the inducing agent is administered using the same route of administration as the OCT4, KLF4, and/or SOX2 (e.g., nucleic acid encoding OCT4, KLF4, and/or SOX2). In some embodiments, the inducing agent (e.g., a nucleic acid encoding the inducing agent, a protein encoding the inducing agent, or a virus encoding the inducing agent) and/or chemical agent capable of modulating (e.g., activating or inhibiting) the activity of the inducing agent is administered via a different route of administration as the OCT4, KLF4, and/or SOX2 (e.g., nucleic acid encoding OCT4, KLF4, and/or SOX2).

(134) In some embodiments, a pharmaceutical composition comprising an expression vector encoding OCT4; KLF4; SOX2; or any combination thereof, or a pharmaceutical composition comprising a virus harboring the expression vector is administered to a cell, tissue, organ, or subject. In some embodiments, a pharmaceutical composition comprising an expression vector encoding an inducing agent or a pharmaceutical composition comprising a virus harboring the expression vector is administered to a cell, tissue, organ, or subject. In some embodiments, the virus and/or expression vector encoding OCT4; KLF4; SOX2; or any combination thereof is administered systemically. In some embodiments, the virus and/or expression vector encoding an inducing agent is administered systemically. In some embodiments, the virus and/or expression vector encoding OCT4; KLF4; SOX2; or any combination thereof is administered locally (e.g., directly to a tissue or organ of interest, including eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine). In some embodiments, a virus and/or expression vector encoding an inducing agent is administered locally (e.g., directly to a tissue or organ of interest, including eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine). In some embodiments, the inducing agent (e.g., a nucleic acid encoding the inducing agent, a protein encoding the inducing agent, or a virus encoding the inducing agent) and/or chemical agent capable of modulating (e.g., activating or inhibiting) the activity of the inducing agent is administered using

the same route of administration as the OCT4; KLF4; SOX2; or any combination thereof (e.g., nucleic acid encoding OCT4; KLF4; SOX2; OCT4 and SOX2; OCT4 and KLF4; KLF4 and SOX2; or KLF4, OCT4, and SOX2). In some embodiments, the inducing agent (e.g., a nucleic acid encoding the inducing agent, a protein encoding the inducing agent, or a virus encoding the inducing agent) and/or chemical agent capable of modulating (e.g., activating or inhibiting) the activity of the inducing agent is administered via a different route of administration as the OCT4; KLF4; SOX2; or any combination thereof (e.g., nucleic acid encoding nucleic acid encoding OCT4; KLF4; SOX2; OCT4 and SOX2; OCT4 and KLF4; KLF4 and SOX2; or KLF4, OCT4, and SOX2).

(135) In some embodiments, the expression vector is an inducible vector in which a nucleic acid encoding OCT4, KLF4, and/or SOX2 and/or inducing agent, is operably linked to an inducible TRE promoter (e.g., TRE3G, TRE2, or P tight). In some embodiments, the expression vector is an inducible vector in which a nucleic acid encoding OCT4; KLF4; SOX2; or any combination thereof, and/or inducing agent, is operably linked to an inducible TRE promoter (e.g., TRE3G, TRE2, or P tight). In some embodiments, the virus and/or inducible vector is administered with tetracycline (e.g., doxycycline). In some embodiments, the virus and/or expression vector comprising a TRE promoter is administered separately from tetracycline (e.g., doxycycline). For example, any of the viruses and/or expression vectors comprising a TRE promoter described herein may be administered systemically and the tetracycline may be administered locally (e.g., to an organ or tissue of interest). In some embodiments, any of the viruses and/or expression vectors comprising a TRE promoter described herein may be administered locally (e.g., to directly to a tissue or organ of interest, including eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine) and the tetracycline may be administered systemically. As a non-limiting example, a virus and/or expression vector comprising a TRE promoter is administered directly (e.g., injected) into the eye of a subject and the tetracycline (e.g., doxycycline) is administered systemically (e.g., orally as a pill).

(136) In some embodiments, tetracycline is administered intravenously, intradermally, intraarterially, intralesionally, intratumorally, intracranially, intraarticularly, intraprostaticaly, intrapleurally, intranasally, intravitreally, intravaginally, intrarectally, topically, intratumorally, intramuscularly, intraperitoneally, subcutaneously, subconjunctival, intravesicularlly, mucosally, intrapericardially, intraumbilically, intraocularally, orally, topically, locally, systemically, injection, infusion, continuous infusion, localized perfusion bathing target cells directly, via a catheter, in creams, or in lipid compositions. In some embodiments, tetracycline is administered directly to a cell, organ, and/or tissue. As a non-limiting example, tetracycline may be administered to the eye of a subject through any suitable method, including eye drops comprising tetracycline, sustained release devices (e.g., micropumps, particles, and/or drug depots), and medicated contact lenses comprising tetracycline. In some embodiments, tetracycline is administered systemically (e.g., through drinking water or intravenous injection) to a subject. Tetracycline may be administered topically (e.g., in a cream) or through a subcutaneous pump (e.g., to deliver tetracycline to a particular tissue).

(137) As an example, the dose of recombinant virus (e.g., lentivirus, alphaviruses, vaccinia viruses, adenovirus, retrovirus, herpes virus, or AAV) virions required to achieve a particular therapeutic effect, e.g., the units of dose in genome copies/per kilogram of body weight (GC/kg), will vary based on several factors including, but not limited to: the route of recombinant virus (e.g., lentivirus, alphaviruses, vaccinia viruses, adenovirus, retrovirus, herpes virus, or AAV) virion administration, the level of gene or RNA expression required to achieve a therapeutic effect, the specific disease or disorder being treated, and the stability of the gene or RNA product. One of skill in the art can readily determine a recombinant virus (e.g., lentivirus, alphaviruses, vaccinia viruses,

adenovirus, retrovirus, herpes virus, or AAV virion) dose range to treat a patient having a particular disease or disorder based on the aforementioned factors, as well as other factors.
(138) An effective amount of a recombinant virus (e.g., lentivirus, alphaviruses, vaccinia viruses, adenovirus, retrovirus, herpes virus, or AAV) is an amount sufficient to target infect an animal, target a desired tissue. In some embodiments, an effective amount of an recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) is an amount sufficient to produce a stable somatic transgenic animal model. The effective amount will depend primarily on factors such as the species, age, weight, health of the subject, and the tissue to be targeted, and may thus vary among animal and tissue. For example, an effective amount of the recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) is generally in the range of from about 1 ml to about 100 ml of solution containing from about 10.sup.9 to 10.sup.16 genome copies. In some cases, a dosage between about 10.sup.11 to 10.sup.13 recombinant virus (e.g., lentivirus, adenovirus, retrovirus, alphavirus, vaccinia virus, herpes virus, or AAV) genome copies is appropriate. In certain embodiments, 10.sup.10 or 10.sup.11 recombinant virus (e.g., lentivirus, adenovirus, retrovirus, alphavirus, vaccinia virus, herpes virus, or AAV) genome copies is effective to target ocular tissue (e.g., retinal tissue). In some cases, stable transgenic animals are produced by multiple doses of a recombinant virus (e.g., lentivirus, adenovirus, retrovirus, herpes virus, alphavirus, vaccinia virus, or AAV).
(139) In some embodiments, a dose of recombinant virus (e.g., lentivirus, adenovirus, retrovirus, herpes virus, alphavirus, vaccinia virus, or AAV) is administered to a subject no more than once per calendar day (e.g., a 24-hour period). In some embodiments, a dose of recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) is administered to a subject no more than once per 2, 3, 4, 5, 6, or 7 calendar days. In some embodiments, a dose of recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) is administered to a subject no more than once per calendar week (e.g., 7 calendar days). In some embodiments, a dose of recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) is administered to a subject no more than bi-weekly (e.g., once in a two calendar week period). In some embodiments, a dose of recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) is administered to a subject no more than once per calendar month (e.g., once in 30 calendar days). In some embodiments, a dose of recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) is administered to a subject no more than once per six calendar months. In some embodiments, a dose of recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) is administered to a subject no more than once per calendar year (e.g., 365 days or 366 days in a leap year).
(140) In some embodiments, recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) compositions are formulated to reduce aggregation of AAV particles in the composition, particularly where high recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) concentrations are present (e.g., ˜10.sup.13 GC/ml or more). Appropriate methods for reducing aggregation of may be used, including, for example, addition of surfactants, pH adjustment, salt concentration adjustment, etc. (See, e.g., Wright F R, et al., Molecular Therapy (2005) 12, 171-178, the contents of which are incorporated herein by reference.)
(141) As a non-limiting example, delivery of transgenes via AAV have been shown to be feasible and non-toxic in humans. For example, AAV may be delivered to the eye. See, e.g., Smalley Nat Biotechnol. 2017 Nov. 9; 35(11):998-999.
(142) Formulation of pharmaceutically-acceptable excipients and carrier solutions is well-known to those of skill in the art, as is the development of suitable dosing and treatment regimens for using the particular compositions described herein in a variety of treatment regimens. Typically, these formulations may contain at least about 0.1% of the active compound or more, although the

percentage of the active ingredient(s) may, of course, be varied and may conveniently be between about 1 or 2% and about 70% or 80% or more of the weight or volume of the total formulation. Naturally, the amount of active compound in each therapeutically-useful composition may be prepared is such a way that a suitable dosage will be obtained in any given unit dose of the compound. Factors such as solubility, bioavailability, biological half-life, route of administration, product shelf life, as well as other pharmacological considerations will be contemplated by one skilled in the art of preparing such pharmaceutical formulations, and as such, a variety of dosages and treatment regimens may be desirable.

(143) In some embodiments, the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered cells comprising OCT4, KLF4, and/or SOX2, engineered proteins encoding Oct4, KLF4, and/or SOX2, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) e.g. in suitably formulated pharmaceutical compositions disclosed herein are delivered directly to target tissue, e.g., direct to a tissue of interest (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine).

(144) In some embodiments, the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, engineered cells comprising OCT4; KLF4; SOX2; or any combination thereof, engineered proteins encoding Oct4, KLF4, SOX2, or a combination thereof, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) e.g. in suitably formulated pharmaceutical compositions disclosed herein are delivered directly to target tissue, e.g., direct to a tissue of interest (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine).

(145) In some embodiments, the nucleic acids (e.g., engineered nucleic acid) encoding an inducing agent (e.g., an expression vector), engineered cells comprising an inducing agent, engineered proteins encoding a inducing agent, chemical agents capable of modulating the activity of an inducing agent, and/or recombinant viruses (e.g., lentiviruses, adenoviruses, alphaviruses, vaccinia viruses, retroviruses, herpes viruses, or AAVs) encoding an inducing agent e.g. in suitably formulated pharmaceutical compositions disclosed herein are delivered directly to target tissue, e.g., direct to a tissue of interest (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine).

(146) However, in certain circumstances it may be desirable to separately or in addition deliver any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector) and/or nucleic acid encoding an inducing agent, nucleic acids (e.g., engineered nucleic acid) capable of inducing expression of a combination of transcription factors selected from OCT4, KLF4, and/or nucleic acid encoding an inducing agent, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, chemical agents activating (e.g., inducing expression of) a combination of transcription factors selected from OCT4, KLF4, and SOX2, chemical agents capable of modulating (e.g., inhibiting or activating) the activity of an inducing agent, antibodies activating

(e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) via another route, e.g., subcutaneously, intraopancreatically, intranasally, parenterally, intravenously, intramuscularly, intrathecally, or orally, intraperitoneally, or by inhalation. In some embodiments, the administration modalities as described in U.S. Pat. Nos. 5,543,158; 5,641,515 and 5,399,363 (each specifically incorporated herein by reference in its entirety) may be used to deliver recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAVs). In some embodiments, a preferred mode of administration is by intrastromal injection.

(147) In some embodiments, a nucleic acid (e.g., mRNA) encoding OCT4, SOX2, KLF4, or any combination thereof is nanoformulated into a polyplex, which may be useful, for example, for noninvasive aerosol inhalation and delivery of the nucleic acid to the lung (e.g., lung epithelium). See, e.g., Patel et al., Adv Mater. 2019 Jan. 4:e1805116. doi: 10.1002/adma.201805116 for description of nanoformulated mRNA polyplexes, which is hereby incorporated by reference in its entirety for this purpose.

(148) The pharmaceutical forms suitable for injectable use include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. Dispersions may also be prepared in glycerol, liquid polyethylene glycols, and mixtures thereof and in oils. Under ordinary conditions of storage and use, these preparations contain a preservative to prevent the growth of microorganisms. In many cases the form is sterile and fluid to the extent that easy syringability exists. It must be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms, such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (e.g., glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and/or vegetable oils. Proper fluidity may be maintained, for example, by the use of a coating, such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, sorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars or sodium chloride. Prolonged absorption of the injectable compositions can be brought about by the use in the compositions of agents delaying absorption, for example, aluminum monostearate and gelatin.

(149) For administration of an injectable aqueous solution, for example, the solution may be suitably buffered, if necessary, and the liquid diluent first rendered isotonic with sufficient saline or glucose. These particular aqueous solutions are especially suitable for intravenous, intramuscular, subcutaneous and intraperitoneal administration. In this connection, a suitable sterile aqueous medium may be employed. For example, one dosage may be dissolved in 1 ml of isotonic NaCl solution and either added to 1000 ml of hypodermoclysis fluid or injected at the proposed site of infusion, (see for example, “Remington's Pharmaceutical Sciences” 15th Edition, pages 1035-1038 and 1570-1580). Some variation in dosage will necessarily occur depending on the condition of the host. The person responsible for administration will, in any event, determine the appropriate dose for the individual host.

(150) Sterile injectable solutions are prepared by incorporating the nucleic acid (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, an and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or active recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) e.g. in the required amount in the appropriate solvent with various of the other ingredients enumerated herein, as required, followed by filtered sterilization. Sterile injectable solutions are prepared by incorporating the nucleic acid (e.g., engineered nucleic

acid) (e.g., expression vector) capable of inducing expression of OCT4, KLF4, SOX2, or any combination thereof, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or active recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) e.g. in the required amount in the appropriate solvent with various of the other ingredients enumerated herein, as required, followed by filtered sterilization. In certain embodiments, the sterile injectable solutions are prepared by incorporating a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent, engineered protein encoding an inducing agent, chemical agents capable of modulating the activity of an inducing agent and/or active recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) encoding an inducing agent e.g. in the required amount in the appropriate solvent with various of the other ingredients enumerated herein, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the various sterilized active ingredients into a sterile vehicle which contains the basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum-drying and freeze-drying techniques which yield a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof.
(151) The compositions comprising nucleic acids (e.g., engineered nucleic acids) encoding OCT4, KLF4, and/or SOX2 (e.g., expression vector), engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) disclosed herein may also be formulated in a neutral or salt form. The compositions comprising nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) encoding OCT4; KLF4; SOX2; or any combination thereof, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) disclosed herein may also be formulated in a neutral or salt form. The compositions may comprise an inducing agent (e.g., a nucleic acid encoding an inducing agent or a protein encoding an inducing agent and/or a recombinant virus encoding an inducing agent) and/or a chemical agent capable of modulating the activity of an inducing agent. Pharmaceutically-acceptable salts, include the acid addition salts (formed with the free amino groups of the protein) and which are formed with inorganic acids such as, for example, hydrochloric or phosphoric acids, or such organic acids as acetic, oxalic, tartaric, mandelic, and the like. Salts formed with the free carboxyl groups can also be derived from inorganic bases such as, for example, sodium, potassium, ammonium, calcium, or ferric hydroxides, and such organic bases as isopropylamine, trimethylamine, histidine, procaine and the like. Upon formulation, solutions will be administered in a manner compatible with the dosage formulation and in such amount as is therapeutically effective. The formulations are easily administered in a variety of dosage forms such as injectable solutions, drug-release capsules, and the like.
(152) A carrier includes any and all solvents, dispersion media, vehicles, coatings, diluents, antibacterial and antifungal agents, isotonic and absorption delaying agents, buffers, carrier solutions, suspensions, colloids, and the like. The use of such media and agents for pharmaceutical active substances is well known in the art. Supplementary active ingredients can also be incorporated into the compositions.
(153) Delivery vehicles such as liposomes, nanocapsules, microparticles, microspheres, lipid particles, vesicles, and the like, may be used for the introduction of the compositions of the present

disclosure into suitable host cells. In particular, any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), any of the engineered proteins, any of the chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, any of the antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, engineered cells, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) may be encapsulated in a lipid particle, a liposome, a vesicle, a nanosphere, or a nanoparticle or the like. In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, any of the engineered proteins, any of the chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, any of the antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, engineered cells, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) may be encapsulated in a lipid particle, a liposome, a vesicle, a nanosphere, or a nanoparticle or the like. An inducing agent (e.g., a nucleic acid encoding an inducing agent or a protein encoding an inducing agent and/or a recombinant virus encoding an inducing agent) and/or a chemical agent capable of modulating the activity of an inducing agent may be encapsulated in a lipid particle, a liposome, a vesicle, a nanosphere, or a nanoparticle or the like.

(154) In some embodiments, the delivery vehicle targets the cargo. For example, any of the nucleic acids, engineered proteins, chemical agents, antibodies, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein may be delivered via a nanoparticle that delivers the cargo to a certain tissue or cell type. Nanoparticles coated in galactose polymers, for example, are known to release their cargo within senescent cells as a result of their endogenous beta-galactosidase activity. See e.g., Lozano-Torres et al., J Am Chem Soc. 2017 Jul. 5; 139(26):8808-8811.

(155) In some embodiments, any of the nucleic acids, engineered proteins, chemical agents, antibodies, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) is formulated in a poly(glycoamidoamine) brush nanoparticles. See, e.g., Dong et al., Nano Lett. 2016 Feb. 10; 16(2):842-8.

(156) In some embodiments, any of the nucleic acids, engineered proteins, chemical agents, antibodies, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) is formulated in a lipid nanoparticle. See, e.g., Cullis and Hope Mol Ther. 2017 Jul. 5; 25(7):1467-1475. In some embodiments, the lipid nanoparticle comprises one or more membrane fusion proteins, which deliver plasmids directly into the cytoplasm or the factors OCT4; KLF4; SOX2; or any combination thereof may be fused directly to the targeting protein with or without nanoparticle encapsulation. In some embodiments, the lipid nanoparticle is a Fusogenix lipid nanoparticle. In some embodiments, the lipid nanoparticle is a “Wrapped Liposomes” (WL). See, e.g., Yamauchi et al., Biochim Biophys Acta. 2006 January; 1758(1):90-7. In some embodiments, the lipid nanoparticle is a PEGylated liposome (e.g., DOXIL™) (e.g., Allen & Hansen, Biochim Biophys Acta. 1991 Jul. 1; 1066(1):29-36.), 1, 2-dioleoyl-sn-glycerol-3 phosphatidylethanolamine (DOPE), a neutral helper lipid phosphatidylethanolamine (PE), or combinations thereof (e.g., Farhood et al., Biochim Biophys Acta. 1995 May 4; 1235(2):289-95; Zhou & Huang, Biochim Biophys Acta. 1994 Jan. 19; 1189(2):195-203.). In some embodiments, the lipid nanoparticle or fusion protein comprises employs a molecule or protein to mimic methods employed by viruses for intracellular delivery of macromolecules (e.g., Kobayashi et al., Bioconjug Chem. 2009 May 20; 20(5):953-9), e.g., using a variety of pH sensitive peptides such as vesicular stomatitis virus proteins (VSV G), phage coat proteins and/or shGALA, and/or Fusion associated small transmembrane (FAST) proteins, e.g., avian reovirus (ARV), nelson bay reovirus (NBV), and baboon reovirus (BBV), aquareovirus reovirus (AQV) and reptilian reovirus (RRV), and/or Bombesin targeting peptide. See, e.g., Peisajovich et al., Eur J Biochem. 2002 September;

269(17):4342-50.; Sakurai et al., 2011. See also Nesbitt, Targeted Intracellular Therapeutic Delivery Using Liposomes Formulated with Multifunctional FAST proteins, Western University Thesis, 2012. https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=14&ved=2ahUKEwi X-YW5puzfAhXGTd8KHUmCATOQFjANegQIAhAB&url=http %3A %2F %2Fir.lib.uwo.ca %2Fcgi %2Fviewcontent.cgi %3Farticle %3D1571%26context %3Detd&usg=AOvVaw3A20aOefHfJIJSZRR_-kPD

(157) In some embodiments, a nucleic acid (e.g., RNA or DNA, including a plasmid) encoding OCT4, KLF4, SOX2, or a combination thereof is encapsulated in a Fusogenix lipid nanoparticle. In some embodiments, a nucleic acid encoding an inducing agent (e.g., rtTA or tTA) is encapsulated in a Fusogenix lipid nanoparticle. In some embodiments, a lipid nanoparticle comprises a viral membrane protein. Without being bound by a particular theory, a lipid nanoparticle may be non-toxic because it comprises a membrane fusion protein that is not a viral membrane fusion protein. Non-limiting examples of membrane fusion proteins include membrane fusion proteins disclosed in U.S. Pat. Nos. 7,851,595, 8,252,901, International Application Publication No. WO 2012/040825, and International Application Publication No. WO 2002/044206.

(158) In some embodiments, a composition of the present disclosure (e.g., comprising a nucleic acid encoding OCT4, KLF4, SOX2, or a combination thereof) is delivered non-virally. Methods of non-viral delivery of nucleic acids include lipofection, nucleofection, microinjection, biolistics, virosomes, liposomes, immunoliposomes, polycation or lipid:nucleic acid conjugates, naked nucleic acid (e.g., RNA or DNA), artificial virions, and agent-enhanced uptake of a nucleic acid (e.g., RNA or DNA).

(159) In some embodiments, a cationic lipid is used to deliver a nucleic acid. A cationic lipid is a lipid which has a cationic, or positive, charge at physiologic pH. Cationic lipids can take a variety of forms including, but not limited to, liposomes or micelles. Cationic lipids useful for certain aspects of the present disclosure are known in the art, and, generally comprise both polar and non-polar domains, bind to polyanions, such as nucleic acid molecules or negatively supercharged proteins, and are typically known to facilitate the delivery of nucleic acids into cells. Examples of useful cationic lipids include polyethylenimine, polyamidoamine (PAMAM) starburst dendrimers, Lipofectin (a combination of DOTMA and DOPE, see, e.g., U.S. Pat. Nos. 5,049,386, 4,946,787; and 4,897,355), Lipofectase, LIPOFECTAMINE® (e.g., LIPOFECTAMINE® 2000, LIPOFECTAMINE® 3000, LIPOFECTAMINE® RNAiMAX, LIPOFECTAMINE® LTX), SAINT-RED (Synvolux Therapeutics, Groningen Netherlands), DOPE, Cytofectin (Gilead Sciences, Foster City, Calif.), and Eufectins (JBL, San Luis Obispo, Calif.). Exemplary cationic liposomes can be made from N-[1-(2,3-dioleoloxy)-propyl]-N,N,N-trimethylammonium chloride (DOTMA), N-[1-(2,3-dioleoloxy)-propyl]-N,N,N-trimethylammonium methylsulfate (DOTAP), 30-[N—(N′,N′-dimethylaminoethane)carbamoyl]cholesterol (DC-Chol), 2,3,-dioleyloxy-N-[2(sperminecarboxamido)ethyl]-N,N-dimethyl-1-propanaminium trifluoroacetate (DOSPA), 1,2-dimyristyloxypropyl-3-dimethyl-hydroxyethyl ammonium bromide; and dimethyldioctadecylammonium bromide (DDAB). Cationic lipids have been used in the art to deliver nucleic acid molecules to cells (see, e.g., U.S. Pat. Nos. 5,855,910; 5,851,548; 5,830,430; 5,780,053; 5,767,099; 8,569,256; 8,691,750; 8,748,667; 8,758,810; 8,759,104; 8,771,728; Lewis et al. 1996. Proc. Natl. Acad. Sci. USA 93:3176; Hope et al. 1998. Molecular Membrane Biology 15:1).

(160) In addition, other lipid compositions are also known in the art and include, e.g., those taught in U.S. Pat. Nos. 4,235,871; 4,501,728; 4,837,028; 4,737,323. Cationic and neutral lipids that are suitable for efficient receptor-recognition lipofection of polynucleotides include those of Feigner, WO 91/17424; WO 91/16024. Delivery can be to cells (e.g. in vitro or ex vivo administration) or target tissues (e.g. in vivo administration).

(161) The preparation of lipid:nucleic acid complexes, including targeted liposomes such as

immunolipid complexes, is well known to one of skill in the art (see, e.g., Crystal, Science 270:404-410 (1995); Blaese et al., Cancer Gene Ther. 2:291-297 (1995); Behr et al., Bioconjugate Chem. 5:382-389 (1994); Remy et al., Bioconjugate Chem. 5:647-654 (1994); Gao et al., Gene Therapy 2:710-722 (1995); Ahmad et al., Cancer Res. 52:4817-4820 (1992); U.S. Pat. Nos. 4,186,183, 4,217,344, 4,235,871, 4,261,975, 4,485,054, 4,501,728, 4,774,085, 4,837,028, and 4,946,787).

(162) Polymer-based delivery systems may also be used to deliver a nucleic acid. Polymers including polyethylenimine (PEI), chitosan, Poly (DL-Lactide) (PLA) and Poly (DL-Lactide-co-glycoside) (PLGA), dedrimers, and Polymethacrylate may be used. See, e.g., Yang et al., Macromol Biosci. 2012 December; 12(12):1600-14; Ramamoorth et al., J Clin Diagn Res. 2015 January; 9(1): GE01-GE06. As a non-limiting example, a cationic polymer may be used. A cationic polymer is a polymer having a net positive charge. Cationic polymers are well known in the art, and include those described in Samal et al., Cationic polymers and their therapeutic potential. Chem Soc Rev. 2012 Nov. 7; 41(21):7147-94; in published U.S. patent applications U.S. 2014/0141487 A1, U.S. 2014/0141094 A1, U.S. 2014/0044793 A1, U.S. 2014/0018404 A1, U.S. 2014/0005269 A1, and U.S. 2013/0344117 A1; and in U.S. Pat. Nos. 8,709,466; 8,728,526; 8,759,103; and 8,790,664; the entire contents of each are incorporated herein by reference. Exemplary cationic polymers include, but are not limited to, polyallylamine (PAH); polyethyleneimine (PEI); poly(L-lysine) (PLL); poly(L-arginine) (PLA); polyvinylamine homo- or copolymer; a poly(vinylbenzyl-tri-C1-C4-alkylammonium salt); a polymer of an aliphatic or araliphatic dihalide and an aliphatic N,N,N′,N′-tetra-C1-C4-alkyl-alkylenediamine; a poly(vinylpyridin) or poly(vinylpyridinium salt); a poly(N,N-diallyl-N,N-di-C1-C4-alkyl-ammoniumhalide); a homo- or copolymer of a quaternized di-C1-C4-alkyl-aminoethyl acrylate or methacrylate; POLYQUAD™; a polyaminoamide; and the like.

(163) Such formulations may be preferred for the introduction of pharmaceutically acceptable formulations of any of the nucleic acids, engineered proteins, chemical agents, antibodies, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) disclosed herein. The formation and use of liposomes is generally known to those of skill in the art. Recently, liposomes were developed with improved serum stability and circulation half-times (U.S. Pat. No. 5,741,516). Further, various methods of liposome and liposome like preparations as potential drug carriers have been described (U.S. Pat. Nos. 5,567,434; 5,552,157; 5,565,213; 5,738,868; and 5,795,587).

(164) Liposomes have been used successfully with a number of cell types that are normally resistant to transfection by other procedures. In addition, liposomes are free of the DNA length constraints that are typical of viral-based delivery systems. Liposomes have been used effectively to introduce genes, drugs, radiotherapeutic agents, viruses, transcription factors and allosteric effectors into a variety of cultured cell lines and animals. In addition, several successful clinical trials examining the effectiveness of liposome-mediated drug delivery have been completed.

(165) Liposomes are formed from phospholipids that are dispersed in an aqueous medium and spontaneously form multilamellar concentric bilayer vesicles (also termed multilamellar vesicles (MLVs). MLVs generally have diameters of from 25 nm to 4 μm. Sonication of MLVs results in the formation of small unilamellar vesicles (SUVs) with diameters in the range of 200 to 500. ANG., containing an aqueous solution in the core.

(166) Alternatively, nanocapsule formulations of the recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) may be used. Nanocapsules can generally entrap substances in a stable and reproducible way. To avoid side effects due to intracellular polymeric overloading, such ultrafine particles (sized around 0.1 μm) should be designed using polymers able to be degraded in vivo. Biodegradable polyalkyl-cyanoacrylate nanoparticles that meet these requirements are contemplated for use.

(167) Kits and Related Compositions

(168) Any of the nucleic acids, engineered proteins, chemical agents, antibodies, and/or recombinant viruses described herein may, in some embodiments, be assembled into pharmaceutical or diagnostic or research kits to facilitate their use in therapeutic, diagnostic or research applications. A kit may include one or more containers housing the components of the disclosure and instructions for use. Specifically, such kits may include one or more agents described herein, along with instructions describing the intended application and the proper use of these agents. In certain embodiments agents in a kit may be in a pharmaceutical formulation and dosage suitable for a particular application and for a method of administration of the agents. Kits for research purposes may contain the components in appropriate concentrations or quantities for running various experiments.
(169) In some embodiments, the instant disclosure relates to a kit for producing a recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) and/or engineered cells, the kit comprising a container housing an engineered nucleic acid (e.g., engineered nucleic acid) encoding OCT4, KLF4, SOX2, or a combination thereof and/or host cells. In some embodiments, the kit further comprises instructions for producing the recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, retrovirus, herpes virus, or AAV) and/or instructions for producing engineered cells. In some embodiments, the kit further comprises at least one container housing a recombinant AAV vector, wherein the recombinant AAV vector comprises a transgene (e.g., a gene associated with ocular disease, such as corneal disease).
(170) In some embodiments, the instant disclosure relates to a kit comprising a container housing any of the engineered nucleic acids (e.g., expression vectors), chemical agents, antibodies, engineered cells, or recombinant viruses described herein. For example, an expression vector or recombinant virus encoding KLF4, SOX2, OCT4, or a combination thereof may comprise a sequence that is at least 70% (e.g., at least 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 100%) identical to SEQ ID NO: 16, SEQ ID NO: 105, or SEQ ID NO: 121. In some embodiments, an expression vector or recombinant virus encoding KLF4, SOX2, OCT4, or a combination thereof comprises SEQ ID NO: 16, SEQ ID NO: 105, or SEQ ID NO: 121. In some embodiments, the expression vector encoding these three transcription factors consists of SEQ ID NO: 16, SEQ ID NO: 105, or SEQ ID NO: 121. The kit may further comprise an expression vector or recombinant virus encoding an inducing agent. In some embodiments, an expression vector encoding an inducing agent comprises SEQ ID NO: 17, SEQ ID NO: 31, or SEQ ID NO: 32. In some embodiments, the expression vector encoding an inducing agent consists of SEQ ID NO: 17, SEQ ID NO: 31, or SEQ ID NO: 32. See, e.g., U.S. Provisional Application No. 62/738,894, entitled MUTANT REVERSE TETRACYCLINE TRANSACTIVATORS FOR EXPRESSION OF GENES, which was filed on Sep. 28, 2018, and is herein incorporated by reference in its entirety.
(171) The kit may be designed to facilitate use of the methods described herein by researchers and can take many forms. Each of the compositions of the kit, where applicable, may be provided in liquid form (e.g., in solution), or in solid form, (e.g., a dry powder). In certain cases, some of the compositions may be constitutable or otherwise processable (e.g., to an active form), for example, by the addition of a suitable solvent or other species (for example, water or a cell culture medium), which may or may not be provided with the kit. As used herein, “instructions” can define a component of instruction and/or promotion, and typically involve written instructions on or associated with packaging of the disclosure. Instructions also can include any oral or electronic instructions provided in any manner such that a user will clearly recognize that the instructions are to be associated with the kit, for example, audiovisual (e.g., videotape, DVD, etc.), Internet, and/or web-based communications, etc. The written instructions may be in a form prescribed by a governmental agency regulating the manufacture, use or sale of pharmaceuticals or biological products, which instructions can also reflect approval by the agency of manufacture, use or sale for animal administration.
(172) The kit may contain any one or more of the components described herein in one or more

containers. As an example, in one embodiment, the kit may include instructions for mixing one or more components of the kit and/or isolating and mixing a sample and applying to a subject. The kit may include a container housing agents described herein. The agents may be in the form of a liquid, gel or solid (powder). The agents may be prepared sterilely, packaged in syringe and shipped refrigerated. Alternatively it may be housed in a vial or other container for storage. A second container may have other agents prepared sterilely. Alternatively the kit may include the active agents premixed and shipped in a syringe, vial, tube, or other container. The kit may have one or more or all of the components required to administer the agents to an animal, such as a syringe, topical application devices, or iv needle tubing and bag, particularly in the case of the kits for producing specific somatic animal models.
(173) The kit may have a variety of forms, such as a blister pouch, a shrink-wrapped pouch, a vacuum sealable pouch, a sealable thermoformed tray, or a similar pouch or tray form, with the accessories loosely packed within the pouch, one or more tubes, containers, a box or a bag. The kit may be sterilized after the accessories are added, thereby allowing the individual accessories in the container to be otherwise unwrapped. The kits can be sterilized using any appropriate sterilization techniques, such as radiation sterilization, heat sterilization, or other sterilization methods known in the art. The kit may also include other components, depending on the specific application, for example, containers, cell media, salts, buffers, reagents, syringes, needles, a fabric, such as gauze, for applying or removing a disinfecting agent, disposable gloves, a support for the agents prior to administration etc.
(174) The instructions included within the kit may involve methods for detecting a latent AAV in a cell. In addition, kits of the disclosure may include, instructions, a negative and/or positive control, containers, diluents and buffers for the sample, sample preparation tubes and a printed or electronic table of reference AAV sequence for sequence comparisons.
(175) Therapeutic Applications
(176) Any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein may be used for regulating (e.g., inducing or inducing and stopping) cellular reprogramming, tissue repair, tissue regeneration, organ regeneration, reversing aging, treating a disease, or any combination thereof. Any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing expression of a combination of transcription factors selected from OCT4, KLF4, and SOX2 (e.g., OCT4 and KLF4, OCT4 and SOX2, SOX2 and KLF4, or KLF4, OCT4, and SOX2), engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) a combination of transcription factors selected from OCT4, KLF4, and SOX2 (e.g., OCT4 and KLF4, OCT4 and SOX2, SOX2 and KLF4, or KLF4, OCT4, and SOX2), antibodies activating (e.g., inducing expression of) combination of transcription factors selected from OCT4, KLF4, and SOX2 (e.g., OCT4 and KLF4, OCT4 and SOX2, SOX2 and KLF4, or KLF4, OCT4, and SOX2), and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein may be used for regulating (e.g., inducing or inducing and stopping) cellular reprogramming, tissue repair, tissue regeneration, organ regeneration, reversing aging, treating a disease, or any combination thereof. In some embodiments, any of the nucleic acid (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), any of the engineered cells, any of the engineered proteins, any of the chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, any of the antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) may be useful in regulating cellular reprogramming, tissue repair, tissue survival, tissue

regeneration, tissue growth, tissue function, organ regeneration, organ survival, organ function, or any combination thereof, optionally wherein regulating comprises inducing cellular reprogramming, reversing aging, improving tissue function, improving organ function, tissue repair, tissue survival, tissue regeneration, tissue growth, angiogenesis, scar formation, the appearance of aging, organ regeneration, organ survival, altering the taste and quality of agricultural products derived from animals, treating a disease, or any combination thereof, in vivo or in vitro may be administered to a cell, tissue, or organ that is in vivo (e.g., part of a subject), or may be administered to a cell, tissue, or organ ex vivo. In some embodiments, any of the nucleic acid (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, any of the engineered cells, any of the engineered proteins, any of the chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, any of the antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or any of the recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) may be useful in regulating cellular reprogramming, tissue repair, tissue survival, tissue regeneration, tissue growth, tissue function, organ regeneration, organ survival, organ function, or any combination thereof, optionally wherein regulating comprises inducing cellular reprogramming, reversing aging, improving tissue function, improving organ function, tissue repair, tissue survival, tissue regeneration, tissue growth, angiogenesis, scar formation, the appearance of aging, organ regeneration, organ survival, altering the taste and quality of agricultural products derived from animals, treating a disease, or any combination thereof, in vivo or in vitro may be administered to a cell, tissue, or organ that is in vivo (e.g., part of a subject), or may be administered to a cell, tissue, or organ ex vivo. As used herein, regulating may refer to any type of modulation, including inducing or promoting, inhibiting, and/or stopping. Angiogenesis refers to growth of new blood vessels, including capillaries.

(177) In some instances, a viral vector (e.g., lentivirus vector, alphavirus vector, vaccinia virus vector, adenovirus vector, herpes virus vector, retrovirus vector, or AAV vector) is administered in a recombinant virus (e.g., lentivirus, alphavirus, vaccinia virus, adenovirus, herpes virus, retrovirus, or AAV). Without being bound by a particular theory, transient expression of OCT4, SOX2, and KLF4 may result in partial reprogramming of a cell. For example, partial reprogramming may induce a fully differentiated cell to rejuvenate and gain pluripotency. In some embodiments, transient expression of OCT4, SOX2, and/or KLF4 does not induce expression of stem cell markers (e.g., Nanog).

(178) In some embodiments, transient expression of OCT4, SOX2, KLF4, or a combination thereof does not induce expression of stem cell markers (e.g., Nanog). Without being bound by any particular theory, Nanog activation may induce teratomas and cause death of the host. In some embodiments, the method does not induce teratoma formation. In some embodiments, the method does not induce unwanted cell proliferation. In some embodiments, the method does not induce malignant cell growth. In some embodiments, the method does not induce cancer. In some embodiments, the method does not induce glaucoma. In some embodiments, transient expression is at most 1 hour, 5 hours, 24 hours, 2 days, 3 days, 4 days, 5, days, or 1 week. In some instances, prolonged expression (e.g., continued expression for at least 5 days, at least 1 week, or at least 1 month) of OCT4, SOX2, and KLF4, results in full reprogramming of a cell. For example, a cell may be fully reprogrammed into a pluripotent cell (e.g., induced pluripotent cell). In some instances, prolonged expression (e.g., continued expression for at least 5 days, at least 1 week, or at least 1 month) of OCT4, SOX2, KLF4, or a combination thereof, results in full reprogramming of a cell. For example, a cell may be fully reprogrammed into a pluripotent cell (e.g., induced pluripotent cell).

(179) Without being bound by a particular theory, expression of OCT4, SOX2, and KLF4 may promote cellular reprogramming, promote tissue regeneration, promote organ regeneration, reverse

aging, treat a disease, or any combination thereof because OCT4, SOX2, and KLF4 induce partial reprogramming. As used herein, partial or incomplete reprogramming of a cell refers to a cell that are not stem cells, but have youthful characteristics. In some embodiments, a youthful characteristic is an epigenome that is similar to a young cell. In some embodiments, a stem cell shows higher levels of Nanog expression compared to a cell that is not a stem cell. In some embodiments, youthful characteristics refers to rejuvenation of a cell without changing cell identity. See, e.g., shown in FIG. **16**, in which the expression of histone and Chaf (Chromatin assembly factor) genes decline during aging in ear fibroblasts from aged mice (12 months or 15 months) compared to those from young mice, short term of OSKM (3 days) or OSK expression (5 days) induction can reset their gene expression level to young state, without making the cells into a stem cell (e.g., Nanog expression is not induced in these cells).

(180) To practice this embodiment, an effective amount of any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) are administered to a cell, a tissue, organ, and/or subject. In some embodiments, an effective amount of any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) are administered to a cell, a tissue, organ, and/or subject. Engineered cells may be administered to any tissue, organ, and/or subject. When the expression vector comprises an inducible promoter (e.g., a TRE promoter, including a TRE3G, TRE2, or P tight), the inducing agent may also be introduced into the cell (e.g., simultaneously or sequentially with one or more nucleic acids (e.g., engineered nucleic acids) encoding OCT4, SOX2, KLF4, or any combination thereof). In one embodiment, OCT4, SOX2, and KLF4 are encoded by one expression vector that is separate from an expression vector encoding the inducing agent. In some instances, the inducing agent is encoded by the same expression vector that encodes OCT4, SOX2, KLF4, or any combination thereof.

(181) In some instances, an inducing agent (e.g., a nucleic acid encoding an inducing agent, an engineered protein encoding an inducing agent, or a virus encoding an inducing agent) and/or a chemical agent (e.g., tetracycline) that is capable of modulating (e.g., activating or inhibiting) activity of the inducing agent is also introduced into a cell, tissue, organ, and/or subject. In certain embodiments, a cell, tissue, subject, and/or organ is further cultured in the presence or absence of a chemical agent that is capable of modulating the activity of an inducing agent (e.g., tetracycline, which includes doxycycline). For a Tet-On system, the inducing agent may be rtTA (e.g., rtTA3 or rtTA4), and the inducing agent promotes expression of OCT4, SOX2, KLF4, or any combination thereof in the presence of tetracycline. For a Tet-Off system, the inducing agent may be tTA, and the inducing agent promotes expression of OCT4, SOX2, KLF4, or any combination thereof in the absence of tetracycline.

(182) Administration of an expression vector encoding a transcription factor described herein and in some cases the inducing agent (e.g., a nucleic acid (e.g., engineered nucleic acid) encoding an inducing agent or the inducing agent as protein) and/or chemical agent that is capable of modulating the activity of the inducing agent under suitable conditions for expression may increase expression of the transgene by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 100%, at least 200%, at least 300%, at least 400%, at least 500%, at least 500%, at least 600%, at least 700%, at least 800%, at least 900%, or at least 1,000% in a cell. Gene expression may be determined by routine methods

including enzyme-linked immunosorbent assays (ELISAs), western blots, and quantification of RNA (e.g., reverse transcription polymerase chain reaction).

(183) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered proteins described herein, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination may be introduced to a tissue, cell, or organ ex vivo (e.g., not in a subject) and the tissue, cell, and/or organ may be further cultured ex vivo. In some embodiments, any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, engineered proteins described herein, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination may be introduced to a tissue, cell, or organ ex vivo (e.g., not in a subject) and the tissue, cell, and/or organ may be further cultured ex vivo. In some instances, an inducing agent and/or a chemical agent capable of modulating the activity of the inducing agent is introduced to a tissue, cell, and/or organ ex vivo and the tissue, cell, and/or organ may be further cultured ex vivo. In some embodiments, engineered cells are cultured to produce an engineered tissue. In some embodiments, engineered cells are cultured to produce an engineered organ. In some embodiments, an engineered tissue is cultured to produce an engineered organ. These methods may be useful in producing an engineered (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine) cell, engineered (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine) tissue or organ (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine) for transplantation into a subject. In some embodiments, the engineered cell, tissue, and/or organ is transplanted into a subject.

(184) In some embodiments, cells, tissues, organs, or any combination thereof to be engineered are autologous to the subject, e.g., obtained from a subject in need thereof. Administration of autologous cells, autologous tissues, autologous organs, or any combination thereof may result in reduced rejection of the cells, tissues, organs, or any combination thereof compared to administration of non-autologous cells, non-autologous tissue and/or non-autologous organs. Alternatively, the cells, tissues, or organs to be engineered may be allogenic cells, allogenic tissues, or allogenic organs. For example, allogenic cells, allogenic tissue, allogenic organs, or any combination thereof may be derived from a donor (e.g., from a particular species) and administered to a recipient (e.g., from the same species) who is different from the donor. In some embodiments, allogenic cells, allogenic tissue, allogenic organs, or any combination thereof may be derived from a donor subject from a particular species and administered to a recipient subject from a different species from the donor.

(185) In some embodiments, the engineered cell is a stem cell (iPSC) including naïve iPSC that can different into three germ layers. In some embodiments, the iPSC is further differentiated into another cell type (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or

intestine). The iPSC may be further differentiated using methods known in the art (e.g., ex vivo)
(186) In some embodiments, engineered cells comprise more than one cell type (e.g., eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine).
(187) As a non-limiting example, any of the engineered nucleic acids (e.g., naked nucleic acids or nucleic acids formulated in a delivery vehicle, including a viral vector and/or nanoparticle) encoding OCT4, KLF4, and SOX2, may be delivered to a cell (e.g., a differentiated cell) to produce an induced pluripotent stem cell. In some embodiments, the induced pluripotent stem cell is further differentiated (e.g., differentiated into an eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine cell). In some embodiments, cells are engineered ex vivo and administered to a subject in need thereof. In some embodiments, an organ or a tissue may be regenerated in vitro using iPSCs and the organ or tissue is transplanted into an individual.
(188) As a non-limiting example, the methods described herein may be used to produce engineered skin, an engineered liver, an engineered eye, an engineered liver, any engineered cell, any engineered organ, or any engineered tissue ex vivo. The engineered organ, engineered tissue, engineered organ, or any combination thereof may be administered to a subject. In some embodiments, administration of an engineered cell, engineered tissue, engineered organ, or a combination thereof improves survival of a subject (e.g., increases the lifespan of a subject relative to not receiving the engineered cell, tissue, or organ).
(189) A pharmaceutical composition described herein may be administered to a subject in need thereof. Non-limiting examples of subjects include any animal (e.g., mammals, including humans). A subject may be suspected of having, be at risk for or have a condition. For example, the condition may be an injury or a disease and the condition may affect any tissue (e.g., ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine). Non-limiting examples of conditions, diseases, and disorders include acute injuries, neurodegenerative disease, chronic diseases, proliferative diseases, cardiovascular diseases, genetic diseases, inflammatory diseases, autoimmunue diseases, neurological diseases, hematological diseases, painful conditions, psychiatric disorders, metabolic disorders, cancers, aging, age-related diseases, and diseases affecting any tissue in a subject. In some embodiments, the disease is an ocular disease.
(190) In certain embodiments, any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered proteins described herein, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination may be introduced to a subject prior to the onset of a disease (e.g., to prevent a disease or to prevent damage to a cell, tissue, or organ). In certain embodiments, any of the nucleic acids (e.g., engineered nucleic acid) (e.g., expression vector) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, engineered proteins described herein, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination may be introduced to a subject prior to the onset of a disease (e.g., to prevent a disease or to prevent damage to a cell, tissue, or organ). In certain embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent may be introduced to

a subject prior to the onset of a disease. In some embodiments, the subject may be a healthy subject. In certain embodiments, any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered proteins described herein, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone, or in combination may be introduced to a subject following the onset of disease (e.g., to alleviate the damage or symptoms associated with a disease). In certain embodiments, any of the nucleic acids (e.g., engineered nucleic acid) capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof expression, engineered proteins described herein, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein, alone or in combination, may be introduced to a subject following the onset of disease (e.g., to alleviate the damage or symptoms associated with a disease). In some embodiments, OCT4, KLF4, and/or SOX2 expression is induced prior to the onset of a disease. In some embodiments, expression of OCT4; KLF4; SOX2; or any combination thereof is induced prior to the onset of a disease. In some embodiments, OCT4, KLF4, and/or SOX2 expression is induced after the onset of a disease. In some embodiments, expression of OCT4; KLF4; SOX2; or any combination thereof is induced after the onset of a disease. In some embodiments, OCT4, KLF4, and/or SOX2 expression is induced in a young subject, young cell, young tissue, and/or young organ. In some embodiments, OCT4, KLF4, and/or SOX2 expression is induced in an aged subject, aged cell, aged tissue, and/or aged organ. In some embodiments, expression of OCT4; KLF4; SOX2; or any combination thereof is induced in a young subject, young cell, young tissue, and/or young organ. In some embodiments expression of, OCT4; KLF4; SOX2; or any combination thereof is induced in an aged subject, aged cell, aged tissue, and/or aged organ. In certain embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent may be introduced to a subject following the onset of a disease.

(191) In certain embodiments, the tissue may be considered healthy but suboptimal for performance or survival in current or future conditions (e.g., in agriculture, or adverse conditions including disease treatments, toxic therapies, sun exposure, or space travel outside the earth's atmosphere).

(192) In certain embodiments, the condition is aging. All animals typically go through a period of growth and maturation followed by a period of progressive and irreversible physiological decline ending in death. The length of time from birth to death is known as the life span of an organism, and each organism has a characteristic average life span. Aging is a physical manifestation of the changes underlying the passage of time as measured by percent of average life span.

(193) In some cases, characteristics of aging can be quite obvious. For example, characteristics of older humans include skin wrinkling, graying of the hair, baldness, and cataracts, as well as hypermelanosis, osteoporosis, cerebral cortical atrophy, lymphoid depletion, thymic atrophy, increased incidence of diabetes type II, atherosclerosis, cancer, and heart disease. Nehlin et al. (2000), *Annals NY Acad Sci* 980:176-79. Other aspects of mammalian aging include weight loss, lordokyphosis (hunchback spine), absence of vigor, lymphoid atrophy, decreased bone density, dermal thickening and subcutaneous adipose tissue, decreased ability to tolerate stress (including heat or cold, wounding, anesthesia, and hematopoietic precursor cell ablation), liver pathology, atrophy of intestinal villi, skin ulceration, amyloid deposits, and joint diseases. Tyner et al. (2002), *Nature* 415:45-53.

(194) Those skilled in the art will recognize that the aging process is also manifested at the cellular level, as well as in mitochondria. Cellular aging is manifested in loss of doubling capacity,

increased levels of apoptosis, changes in differentiated phenotype, and changes in metabolism, e.g., decreased levels of protein synthesis and turnover.

(195) Given the programmed nature of cellular and organismal aging, it is possible to evaluate the "biological age" of a cell or organism by means of phenotypic characteristics that are correlated with aging. For example, biological age can be deduced from patterns of gene expression, resistance to stress (e.g., oxidative or genotoxic stress), rate of cellular proliferation, and the metabolic characteristics of cells (e.g., rates of protein synthesis and turnover, mitochondrial function, ubiquinone biosynthesis, cholesterol biosynthesis, ATP levels within the cell, levels of a Krebs cycle intermediate in the cell, glucose metabolism, nucleic acid (e.g., engineered nucleic acid) metabolism, ribosomal translation rates, etc.). As used herein, "biological age" is a measure of the age of a cell or organism based upon the molecular characteristics of the cell or organism. Biological age is distinct from "temporal age," which refers to the age of a cell or organism as measured by days, months, and years.

(196) The rate of aging of an organism, e.g., an invertebrate (e.g., a worm or a fly) or a vertebrate (e.g., a rodent, e.g., a mouse) can be determined by a variety of methods, e.g., by one or more of: a) assessing the life span of the cell or the organism; (b) assessing the presence or abundance of a gene transcript or gene product in the cell or organism that has a biological age-dependent expression pattern; (c) evaluating resistance of the cell or organism to stress, e.g., genotoxic stress (e.g., etoposide, UV irradiation, exposure to a mutagen, and so forth) or oxidative stress; (d) evaluating one or more metabolic parameters of the cell or organism; (e) evaluating the proliferative capacity of the cell or a set of cells present in the organism; and (f) evaluating physical appearance or behavior of the cell or organism. In one example, evaluating the rate of aging includes directly measuring the average life span of a group of animals (e.g., a group of genetically matched animals) and comparing the resulting average to the average life span of a control group of animals (e.g., a group of animals that did not receive the test compound but are genetically matched to the group of animals that did receive the test compound). Alternatively, the rate of aging of an organism can be determined by measuring an age-related parameter. Examples of age-related parameters include: appearance, e.g., visible signs of age; the expression of one or more genes or proteins (e.g., genes or proteins that have an age-related expression pattern); resistance to oxidative stress; metabolic parameters (e.g., protein synthesis or degradation, ubiquinone biosynthesis, cholesterol biosynthesis, ATP levels, glucose metabolism, nucleic acid (e.g., engineered nucleic acid) metabolism, ribosomal translation rates, etc.); and cellular proliferation (e.g., of retinal cells, bone cells, white blood cells, etc.).

(197) Aging can also be determined by the rate of change of biomarkers (e.g., epigenetic marks including DNA methylation level of CpG island in the genome (known as the "Horvath Clock") beta-galactosidase-positive cells in cells, gene expression changes, or certain changes to the abundance of molecules in the bloodstream). An example is an algorithm from Segterra Inc. that determines "InnerAge" based on blood biomarkers (see InsideTracker.com).

(198) As shown in the Examples herein, recombinant viruses (e.g., AAVs) encoding OCT4, KLF4, and SOX2 promoted regeneration of axons, which may be used to prevent or alleviate neurodegeneration that is often associated with aging. The methods may be used to prevent or alleviate neurodegeneration and peripheral neuropathies associated. Neurodegenerative diseases include Parkinson's disease, Alzheimer's disease, multiple sclerosis, amniotropic lateral sclerosis (ALS), Huntington's disease, and muscular dystrophy. Neurodegeneration may be quantified using any method known in the art. For example, the executive function of an individual may be determined (Moreira et al., Front Aging Neurosci. 2017 Nov. 9; 9:369).

(199) In some embodiments, expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof as described herein increases the number of axons per nerve in a tissue, organ, or a subject relative to a control. In some embodiments, a method described herein increases the number of axons per nerve by at least 1.5 fold, by at least 2 fold, by at least 3 fold, by at least 5

fold, by at least 6 fold, by at least 7 fold, by at least 8 fold, by at least 9 fold, by at least 10 fold, by at least 20 fold, by at least 30 fold, by at least 40 fold, by at least 50 fold, by at least 60 fold, by at least 70 fold, by at least 80 fold, by at least 90 fold, or by at least 100 fold relative to a control. In some embodiments, the control is the number of axons per nerve in the tissue, organ, or subject prior to expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof.

(200) Additional age-related conditions which may be treated include heart failure, stroke, diabetes, liver diseases, fibrotic diseases, osteoporosis, arthritis, hearing loss (partial or total), eye-related conditions (e.g., poor eye sight, retinal disease, any ocular disease (e.g., any condition affecting the eye)), glaucoma, muscle diseases (e.g., sarcopenia and muscular dystrophies), frailty, a progeroid syndrome (e.g., Hutchinson-Gilford progeria syndrome), and cancer. In certain embodiments, the disease is a retinal disease (e.g., macular degeneration).

(201) In some embodiments, expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof in a neuron increases neurite area of the neuron by at least 1.5 fold, by at least 2 fold, by at least 3 fold, by at least 5 fold, by at least 6 fold, by at least 7 fold, by at least 8 fold, by at least 9 fold, by at least 10 fold, by at least 20 fold, by at least 30 fold, by at least 40 fold, by at least 50 fold, by at least 60 fold, by at least 70 fold, by at least 80 fold, by at least 90 fold, or by at least 100 fold relative to the neuron without expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof.

(202) In some embodiments, expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof as described herein increases the axon density in a tissue, organ, or a subject relative to a control. In some embodiments, a method described herein increases axon density at least 1.5 fold, by at least 2 fold, by at least 3 fold, by at least 5 fold, by at least 6 fold, by at least 7 fold, by at least 8 fold, by at least 9 fold, by at least 10 fold, by at least 20 fold, by at least 30 fold, by at least 40 fold, by at least 50 fold, by at least 60 fold, by at least 70 fold, by at least 80 fold, by at least 90 fold, or by at least 100 fold relative to a control. In some embodiments, the control is the axon density in the tissue, organ, or subject prior to expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof.

(203) In some embodiments, expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof in a subject increases the visual acuity of the subject relative to a control. In some embodiments, a method described herein increases the visual acuity of a subject by at least 1.5 fold, by at least 2 fold, by at least 3 fold, by at least 5 fold, by at least 6 fold, by at least 7 fold, by at least 8 fold, by at least 9 fold, by at least 10 fold, by at least 20 fold, by at least 30 fold, by at least 40 fold, by at least 50 fold, by at least 60 fold, by at least 70 fold, by at least 80 fold, by at least 90 fold, or by at least 100 fold relative to a control. In some embodiments, the control is the visual acuity of the subject prior to expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof. In some embodiments, visual acuity is measured by optomotor acuity. In some embodiments, visual acuity is measured using a pattern electroretinogram response. In some embodiments, visual acuity is measured using a distance visual acuity test, which may include the use of a Snellen chart or E chart. See, e.g., Marsden et al., Community Eye Health. 2014; 27(85): 16 and the Examples below.

(204) In some embodiments, expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof in a subject decreases the intraocular pressure of the subject relative to a control. In some embodiments, a method described herein decreases the intraocular pressure of a subject by at least 1.5 fold, by at least 2 fold, by at least 3 fold, by at least 5 fold, by at least 6 fold, by at least 7 fold, by at least 8 fold, by at least 9 fold, by at least 10 fold, by at least 20 fold, by at least 30 fold, by at least 40 fold, by at least 50 fold, by at least 60 fold, by at least 70 fold, by at least 80 fold, by at least 90 fold, or by at least 100 fold relative to a control. In some embodiments, the control is the intraocular pressure of the subject prior to expression, induction, or activation of OCT4, SOX2, KLF4, or a combination thereof. See, e.g., the Examples below.

(205) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) capable of

inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein may be used to treat and/or prevent any of the diseases described herein. In some embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent is also used.

(206) As a non-limiting example, an engineered cell of the present disclosure may be used to replace a dysfunctional cell in a subject in need thereof. As another non-limiting example, any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) capable of inducing expression of a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, chemical agents activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) may be used to (e.g., incompletely or fully) reprogram a cell in vivo or in vitro. In some embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent is also used. For example, any of the any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) may be used to produce an engineered cell (e.g., an induced pluripotent stem cell). For example, any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) may be used to produce an engineered cell (e.g., an induced pluripotent stem cell). The engineered cell (e.g., induced pluripotent stem cell) may then be administered to a subject in need thereof. In some embodiments, the engineered cell is cultured in the presence of an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent. In some embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent is also administered to the subject.

(207) Non-limiting uses of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) capable of inducing expression of a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, chemical agents activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, engineered cells, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) include wound healing, bleed out, injuries, broken bones, gunshot wounds, cuts, scarring during surgery (e.g., cesarean). In some

embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent is also used.

(208) In some embodiments, any of the of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) capable of inducing expression of a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, an KLF4, and/or SOX2, chemical agents activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) are used to treat disease that affects a non-human subject (e.g., a disease affecting livestock, domesticated pets, and/or other non-human animals). In some embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent is also used. For example, the disease may be a cattle disease, a primate (e.g., cynomolgus monkeys, rhesus monkeys) disease, a disease affecting a commercially relevant animal, such as cattle, pigs, horses, sheep, goats, cats, and/or dogs) and/or a disease affecting birds (e.g., commercially relevant birds, such as chickens, ducks, geese, and/or turkeys).

(209) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) capable of inducing expression of a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, chemical agents activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein are used to promote wound healing (e.g., for a cut), treat an injury (e.g., broken bones, bleeding out, gun shot injury, and/or reduce scarring during surgery). In some embodiments, surgery includes cesarean. In some embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent is also used.

(210) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vector), nucleic acids (e.g., engineered nucleic acids) (e.g., expression vector) capable of inducing expression of a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, chemical agents activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) a combination of at least two (e.g., at least three) transcription factors selected from OCT4, KLF4, and SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein are useful in healing an injury and/or inflammation. In some embodiments, an inducing agent and/or a chemical agent capable of modulating activity of the inducing agent is also used. In some embodiments, the inflammation is hyperinflammation, which may be a side effect of aging. In some embodiments, the hyperinflammation is inflammaging.

(211) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) capable of inducing OCT4, KLF4, and/or SOX2 expression (e.g., expression vectors), engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, antibodies activating (e.g., inducing expression of) OCT4, KLF4, and/or SOX2, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein provide a healing capacity.
(212) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vectors) capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein provide a healing capacity.
(213) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vectors) capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein are useful in enhancing or rejuvenating optimal or sub-optimal organs. As a non-limiting example, any of the compositions described herein (e.g., recombinant viruses including recombinant AAV viruses) encoding OCT4, KLF4, SOX2, or a combination thereof may be useful in enhancing or rejuvenating suboptimal organs (e.g., from older individuals) that are used for transplantation or to promote organ survival during transport or to promote organ survival after reimplantation of the organ into a subject.
(214) Any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vectors) capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein may be used to rejuvenate or increase the survival and longevity of cells (e.g., hematopoietic stem cells, T-cells, etc.) that are used for transplantation. In some embodiments, recombinant viruses (e.g., AAV viruses) encoding OCT4, KLF4, SOX2, or a combination thereof are useful in rejuvenating or increasing the survival and longevity of cells (e.g., hematopoietic stem cells, T-cells, etc.) that are used for transplantation.
(215) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vectors) capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein is used to prevent or relieve the side effects of a toxin and/or a drug (e.g., a chemotherapy) in a subject. Non-limiting examples of side effects include hair loss and peripheral neuropathy. Chemotherapies include vincristine (VCS). See, e.g., example 15. In certain embodiments, a composition comprising a recombinant virus (e.g., AAV virus) encoding SOX2, KLF4, OCT4, or a combination thereof, is administered to treat (e.g., recover from) or prevent the side effects induced by a toxin and/or damaging drug therapy (e.g., a chemotherapy drug including VCS).
(216) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vectors) capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression

of) OCT4, KLF4, SOX2, or a combination thereof, antibodies activating (e.g., inducing expression of) OCT4, KLF4, SOX2, or a combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein is administered to a subject to prevent or relieve the side effects of a toxin and/or a drug (e.g., a chemotherapy).

(217) In some embodiments, any of the nucleic acids (e.g., engineered nucleic acids) (e.g., expression vectors) capable of inducing expression of OCT4, KLF4, SOX2, or a combination thereof, engineered cells, engineered proteins, chemical agents activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, antibodies activating (e.g., inducing expression of) OCT4; KLF4; SOX2; or any combination thereof, and/or recombinant viruses (e.g., lentivirus, adenovirus, alphavirus, vaccinia virus, retrovirus, herpes virus, or AAV) described herein is administered to a subject to protect a tissue, organ, and/or entire body of the subject from radiation (e.g., prevent the damaging effects of radiation). In certain embodiments, AAV encoding OCT4, SOX2, KLF4, or any combination thereof, is administered to a subject to protect a tissue, organ, and/or entire body of the subject from radiation protect (e.g., prevent the damaging effects of radiation).

(218) Methods for identifying subjects suspected of having a condition may include physical examination, subject's family medical history, subject's medical history, biopsy, genetic testing, DNA sequencing of pathogens or the microbiome, proteomics, or a number of imaging technologies such as ultrasonography, computed tomography, magnetic resonance imaging, magnetic resonance spectroscopy, or positron emission tomography.

(219) Effective amounts of the engineered nucleic acids (e.g., expression vectors, including viral vectors), viruses (e.g., lentiviruses, retroviruses, adenoviruses, retroviruses, alphaviruses, vaccinia viruses, or AAVs) or compositions thereof vary, as recognized by those skilled in the art, depending on route of administration, excipient usage, and co-usage with other active agents. The quantity to be administered depends on the subject to be treated, including, for example, the age of the subject, the gravity of the condition, the weight of the subject, the genetics of the subject, the cells, tissue, or organ to be targeted, or any combination thereof.

(220) Expression of one or more transcription factors of the present disclosure (e.g., OCT4; KLF4; SOX2; or any combination thereof) may result in reprogramming of a cell, tissue repair, tissue regeneration, increase blood flow, organ regeneration, improved immunity, reversal of aging, counter senescence, or any combination thereof. Cellular reprogramming may be determined by determining the extent of differentiation of a cell (e.g., by determining the expression of one or more lineage markers or pluripotency markers, including OCT4, KLF4, SOX2, NANOG, ESRRB, NR4A2, and C/EBPα). The differentiation potential of a cell may also be determined using routine differentiation assays or gene expression patterns. Tissue repair may be determined by tissue replacement and tissue regeneration assays. For example, tissue replacement assays include wound healing assays in cell culture or in mice. Tissue regeneration may be determined by quantifying a particular cell type following expression of one or more transcription factors compared to before expression of OCT4, KLF4, and SOX2 (see, e.g., the Examples provided below). Tissue regeneration may be determined by quantifying a particular cell type following expression of one or more transcription factors compared to before expression of OCT4; KLF4; SOX2; or any combination thereof. In some instances, the methods described herein promote organ regeneration (e.g. liver regeneration or reversal of liver fibrosis and regrowth). In some instances, the methods described herein promote tissue and cell survival. Cell survival in the face of adversity and damage may be determined using assays for cell viability that are standard in the art (e.g., testing neuronal survival with the nano-glo live cell assay from Promega corp.). In some instances, the methods described herein may prevent axonal or Wallerian degeneration, which may be determined by quantifying the rate of axonal degeneration after nerve crush in vitro using nerve cell cultures or in rat and mouse nerve crush models known to those skilled in the art.

(221) In some embodiments, the methods described herein do not induce teratoma formation. In some embodiments, expression of OCT4, SOX2, KLF4, or a combination thereof or activation of OCT4, SOX2, KLF4, or a combination thereof in a subject, tissue, or organ, results in at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or at least 100% reduction in teratoma formation as compared to expression of OCT4, SOX2, KLF4, or a combination thereof and c-MYC or activation of OCT4, SOX2, KLF4, or a combination thereof and c-MYC in the subject, tissue, or organ. In some embodiments, expression of OCT4, SOX2, and KLF4 or activation of OCT4, SOX2, and KLF4 in a subject, tissue, or organ, results in at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or at least 100% reduction in teratoma formation as compared to expression of OCT4, SOX2, and KLF4, and c-MYC or activation of OCT4, SOX2, KLF4, and c-MYC in the subject, tissue, or organ. In some embodiments, the number of teratomas or the size of a teratoma in a subject, tissue, or organ is the same or is reduced following expression of OCT4, SOX2, KLF4, or a combination thereof or activation of OCT4, SOX2, KLF4, or a combination thereof in a subject, tissue, or organ as compared to the number of teratomas or the size of a teratoma in the subject, tissue, or organ prior to activation or expression of OCT4, SOX2, KLF4, or a combination thereof.
(222) In some embodiments, the methods described herein do not induce unwanted cell proliferation. In some embodiments, the unwanted cell proliferation is aberrant cell proliferation, which may be benign or cancerous. In some embodiments, expression of OCT4, SOX2, KLF4, or a combination thereof or activation of OCT4, SOX2, KLF4, or a combination thereof in a subject, tissue, or organ reduces unwanted cell proliferation in a subject, tissue, or organ, by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or at least 100% as compared to the same method with c-Myc expression or activation. In some embodiments, unwanted cell proliferation in a subject, tissue, or organ is the same or is reduced following expression of OCT4, SOX2, KLF4, or a combination thereof or activation of OCT4, SOX2, KLF4, or a combination thereof in a subject, tissue, or organ as compared to the amount of unwanted cell proliferation in the subject, tissue, or organ prior to activation or expression of OCT4, SOX2, KLF4, or a combination thereof.
(223) In some embodiments, the methods described herein do not induce tumor formation or tumor growth. In some embodiments, expression of OCT4, SOX2, KLF4, or a combination thereof or activation of OCT4, SOX2, KLF4, or a combination thereof in a subject, tissue, or organ reduces the number of tumors or the size of a tumor in a subject, tissue, or organ, by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or at least 100% as compared to the same method with c-Myc expression or activation. In some embodiments, the number of tumors or the size of a tumor in a subject, tissue, or organ is the same or is reduced following expression of OCT4, SOX2, KLF4, or a combination thereof or activation of OCT4, SOX2, KLF4, or a combination thereof in a subject, tissue, or organ as compared to the number of tumors or the size of a tumor in the subject, tissue, or organ prior to activation or expression of OCT4, SOX2, KLF4, or a combination thereof. In some embodiments, a method described herein does not induce cancer. In some embodiments, a method described herein does not induce glaucoma.
(224) Methods of reprogramming are also provided herein. In some embodiments, a method of reprogramming described herein comprises reversing or rejuvenating the epigenetic clock of a cell, tissue, organ, or a subject. In some embodiments, the epigenetic clock may be partially or fully reversed. In some embodiments, the epigenetic clock of a cell, tissue, organ, or a subject is measured using DNA methylation-based age (DNAmAGE or DNAm age). In some embodiments, a method described herein reduces the DNAmAge age of a cell by 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 100%.
(225) In some embodiments, a method of reprogramming described herein comprises altering the

expression of one or more genes associated with ageing. In some embodiments, expression of a gene is increased by at least 1%, at least 5%, at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or at least 100%. In some embodiments, expression of a gene is reduced by at least 1%, at least 5%, at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or at least 100%. In some embodiments, expression of one or more genes following performance of a method is determined relative to expression of the one or more genes prior to performance of the method. In some embodiments, expression of one or more genes is determined relative to expression of the one or more genes in a young cell, a young subject, a young tissue, a young organ, or any combination thereof. In some embodiments, expression of one or more genes is determined relative to expression of the one or more genes in an old cell, an old subject, an old tissue, an old organ, or any combination thereof.

(226) A gene associated with ageing may be a gene whose expression is altered in an old, an old tissue, an old organ, an old subject, or any combination thereof as compared to a young counterpart. In some embodiments, the gene associated with ageing is 1700031P21Rik, 1810053B23Rik, 2900045020Rik, 2900060B14Rik, 4921504E06Rik, 4930402F11Rik, 4930453C13Rik, 4930455B14Rik, 4930500H12Rik, 4930549P19Rik, 4930555B11Rik, 4930556J02Rik, 4932442E05Rik, 4933431K23Rik, 4933438K21Rik, 6720475M21Rik, 9830132P13Rik, A430010J10Rik, A530064D06Rik, A530065N20Rik, Abcb5, Abhd17c, AC116759.2, AC131705.1, AC166779.3, Acot12, Adig, Akr1cl, Ankrd1, Asb15, Atp2c2, AU018091, AW822073, Btn110, C130093G08Rik, C730027H18Rik, Ccdc162, Chil6, Col26a1, Corin, Crls1, Cybrd1, Cyp2d12, Cyp7a1, D830005E20Rik, Dlx3, Dnah14, Dsc3, Dthd1, Eid2, Eps811, EU599041, Fam90ala, Fancf, Fau-ps2, Fezf1, Gja5, Gm10248, Gm10513, Gm10635, Gm10638, Gm10718, Gm10722, Gm10800, Gm10801, Gm11228, Gm11251, Gm11264, Gm11337, Gm11368, Gm11485, Gm11693, Gm12793, Gm13050, Gm13066, Gm13323, Gm13339, Gm13346, Gm13857, Gm14387, Gm14770, Gm15638, Gm16072, Gm16161, Gm16181, Gm17200, Gm17791, Gm18025, Gm18757, Gm18795, Gm18848, Gm19719, Gm20121, Gm20356, Gm2093, Gm21738, Gm21940, Gm22933, Gm24000, Gm24119, Gm25394, Gm26555, Gm27047, Gm28262, Gm28530, Gm29295, Gm29825, Gm29844, Gm3081, Gm32051, Gm32122, Gm33056, Gm33680, Gm34354, Gm34643, Gm3551, Gm36660, Gm36948, Gm37052, Gm37142, Gm37262, Gm37535, Gm37569, Gm37589, Gm37647, Gm37648, Gm37762, Gm38058, Gm38069, Gm38137, Gm38218, Gm39139, Gm42535, Gm42680, Gm42895, Gm42994, Gm43027, Gm43158, Gm43288, Gm43366, Gm44044, Gm44081, Gm44187, Gm44280, Gm44535, Gm45338, Gm45644, Gm45740, Gm46555, Gm46565, Gm4742, Gm47485, Gm47853, Gm47992, Gm48225, Gm48314, Gm48383, Gm48673, Gm48804, Gm48832, Gm4994, Gm5487, Gm5724, Gm595, Gm6012, Gm6024, Gm7669, Gm7730, Gm8043, Gm8953, Gm9348, Gm9369, Gm9495, H2a12a, Ido2, Igfbp1, Kif7, Klhl31, Lrrc31, Mc5r, Mgam, Msh4, Muc12, Mug1, Mybl2, Myh15, Nek10, Neurod6, Nrlh5, Olfr1042, Olfr1043, Olfr1082, Olfr1090, Olfr1124, Olfr1167, Olfr1205, Olfr1206, Olfr1223, Olfr1263, Olfr1264, Olfr1269, Olfr127, Olfr1291-ps1, Olfr1406, Olfr1469, Olfr215, Olfr273, Olfr328, Olfr355, Olfr372, Olfr390, Olfr427, Olfr456, Olfr466, Olfr481, Olfr522, Olfr6, Olfr601, Olfr603, Olfr706, Olfr727, Olfr728, Olfr741, Olfr801, Olfr812, Olfr816, Olfr822, Olfr860, Olfr890, Olfr923, Olfr943, Otog1, Pi15, Pkhd1, Pkhd111, Platr6, Pou3f4, Prr9, Pvalb, Rhag, Sav1, Serpinb9b, Skint1, Skint3, Skint5, Slc10a5, Slc6a4, Smok2a, Tcaf3, Tomm201, Trcg1, Trdn, Ugt1a6a, Usp17la, Vmn1r178, Vmn1r179, Vmn1r33, Vmn1r74, Vmn1r87, Vmn2r102, Vmn2r113, Vmn2r17, Vmn2r52, Vmn2r66, Vmn2r68, Vmn2r76, Vmn2r78, Wnt16, 0610040J01Rik, 1700080N15Rik, 2900064F13Rik, 4833417C18Rik, 4921522P10Rik, 4930447C04Rik, 4930488N15Rik, Ace, Ackr1, Acot10, Acvr1, Adamts17, Adra1b, A1504432, Best3, Boc, Cadm3, Cand2, Cc19, Cd14, Cd36, Cfh, Chrm3, Chrna4, Cntn4, Cracr2b, Cryaa, CT573017.2, Cyp26a1, Cyp27a1, D330050G23Rik, D930007P13Rik, Ddo, Dgkg, Dlk2, Dnajal-ps, Drd2, Dsel, Dytn, Ecscr, Edn1, Ednrb, Efemp1, Elfn2, Epha10, Ephx1, Erbb4,

Fam20a, Fbxw21, Ffar4, Flt4, Fmod, Foxp4, Fzd7, Gabrd, Galnt15, Galntl8, Gfra2, Ggt1, Gm10416, Gm14964, Gm17634, Gm2065, Gm32352, Gm33172, Gm34280, Gm35853, Gm36298, Gm36356, Gm36937, Gm3898, Gm42303, Gm42484, Gm42537, Gm42743, Gm43151, Gm43843, Gm44545, Gm44722, Gm45516, Gm45532, Gm47494, Gm47982, Gm47989, Gm48398, Gm48495, Gm48593, Gm48958, Gm49089, Gm49326, Gm49331, Gm49760, Gm5796, Gm6374, Gm7276, Gm8237, Gm9796, Gm9954, Gpr75, Gprc5c, Grid2ip, Gsg112, Hapln4, Hcn3, Hcn4, Hhat1, Hs6st2, Htr3a, Il1rap, Il1rap12, Inka1, Kbtbd12, Kcnj11, Kcnk4, Kdelc2, Klhl33, Lamc3, Lilra5, Lman11, Lrfn2, Lrrc38, Lrrn4c1, Ltc4s, Mansc1, Mir344c, Msr1, Mycbpap, Myoc, Ngfr, Nipal2, Olfr1372-ps1, Otop3, P2rx5, P2ry12, P4ha2, Pcdha12, Pcdha2, Pcdhac2, Pcdhb18, Pcdhb5, Pcsk2os1, Pcsk6, Perp, Pkp1, Plxna4, Prickle2, Qsox1, Rapgef4os2, Rbp4, Rcn3, Sec1415, Sel113, Serpinh1, Sgpp2, Shisa6, Siah3, Siglech, Slc12a4, Slc24a2, Slc2a5, Slc4a4, Slitrk3, Smagp, Smoc2, Speer4b, Spon2, Sstr2, Sstr3, St3gal3, Stc1, Stc2, Syndig1, Syt10, Thsd7a, Tlr8, Tmem132a, Tmem132d, Tmem200a, Tmem44, Trpc4, Trpv4, Unc5b, Vgf, Vmn1r90, Vwc21, Wfikkn2, Wnt11, Wnt6, Zeb2os, Zfp608, Zfp976, or any combination thereof. In some embodiments, the gene is a sensory gene.

(227) In some embodiments, a method described herein reduces expression of 0610040J01Rik, 1700080N15Rik, 2900064F13Rik, 4833417C18Rik, 4921522P10Rik, 4930447C04Rik, 4930488N15Rik, Ace, Ackr1, Acot10, Acvr1, Adamts17, Adra1b, A1504432, Best3, Boc, Cadm3, Cand2, Cc19, Cd14, Cd36, Cfh, Chrm3, Chrna4, Cntn4, Cracr2b, Cryaa, CT573017.2, Cyp26a1, Cyp27a1, D330050G23Rik, D930007P13Rik, Ddo, Dgkg, Dlk2, Dnajal-ps, Drd2, Dsel, Dytn, Ecscr, Edn1, Ednrb, Efemp1, Elfn2, Epha10, Ephx1, Erbb4, Fam20a, Fbxw21, Ffar4, Flt4, Fmod, Foxp4, Fzd7, Gabrd, Galntl5, Galntl8, Gfra2, Ggt1, Gm10416, Gm14964, Gm17634, Gm2065, Gm32352, Gm33172, Gm34280, Gm35853, Gm36298, Gm36356, Gm36937, Gm3898, Gm42303, Gm42484, Gm42537, Gm42743, Gm43151, Gm43843, Gm44545, Gm44722, Gm45516, Gm45532, Gm47494, Gm47982, Gm47989, Gm48398, Gm48495, Gm48593, Gm48958, Gm49089, Gm49326, Gm49331, Gm49760, Gm5796, Gm6374, Gm7276, Gm8237, Gm9796, Gm9954, Gpr75, Gprc5c, Grid2ip, Gsg112, Hapln4, Hcn3, Hcn4, Hhat1, Hs6st2, Htr3a, Il1rap, Il1rap12, Inka1, Kbtbd12, Kcnj11, Kcnk4, Kdelc2, Klhl33, Lamc3, Lilra5, Lman11, Lrfn2, Lrrc38, Lrrn4c1, Ltc4s, Mansc1, Mir344c, Msr1, Mycbpap, Myoc, Ngfr, Nipal2, Olfr1372-ps1, Otop3, P2rx5, P2ry12, P4ha2, Pcdha12, Pcdha2, Pcdhac2, Pcdhb18, Pcdhb5, Pcsk2os1, Pcsk6, Perp, Pkp1, Plxna4, Prickle2, Qsox1, Rapgef4os2, Rbp4, Rcn3, Sec1415, Sel113, Serpinh1, Sgpp2, Shisa6, Siah3, Siglech, Slc12a4, Slc24a2, Slc2a5, Slc4a4, Slitrk3, Smagp, Smoc2, Speer4b, Spon2, Sstr2, Sstr3, St3gal3, Stc1, Stc2, Syndig1, Syt10, Thsd7a, Tlr8, Tmem132a, Tmem132d, Tmem200a, Tmem44, Trpc4, Trpv4, Unc5b, Vgf, Vmn1r90, Vwc21, Wfikkn2, Wnt11, Wnt6, Zeb2os, Zfp608, Zfp976, or any combination thereof. See, e.g., Table 5 for genes associated with ageing.

(228) In some embodiments, a method described herein increases expression of 1700031P21Rik, 1810053B23Rik, 2900045020Rik, 2900060B14Rik, 4921504E06Rik, 4930402F11Rik, 4930453C13Rik, 4930455B14Rik, 4930500H12Rik, 4930549P19Rik, 4930555B11Rik, 4930556J02Rik, 4932442E05Rik, 4933431K23Rik, 4933438K21Rik, 6720475M21Rik, 9830132P13Rik, A430010J10Rik, A530064D06Rik, A530065N20Rik, Abcb5, Abhd17c, AC116759.2, AC131705.1, AC166779.3, Acot12, Adig, Akr1c1, Ankrd1, Asb15, Atp2c2, AU018091, AW822073, Btn110, C130093G08Rik, C730027H18Rik, Ccdc162, Chil6, Col26a1, Corin, Crls1, Cybrd1, Cyp2d12, Cyp7a1, D830005E20Rik, Dlx3, Dnah14, Dsc3, Dthd1, Eid2, Eps811, EU599041, Fam90ala, Fancf, Fau-ps2, Fezf1, Gja5, Gm10248, Gm10513, Gm10635, Gm10638, Gm10718, Gm10722, Gm10800, Gm10801, Gm11228, Gm11251, Gm11264, Gm11337, Gm11368, Gm11485, Gm11693, Gm12793, Gm13050, Gm13066, Gm13323, Gm13339, Gm13346, Gm13857, Gm14387, Gm14770, Gm15638, Gm16072, Gm16161, Gm16181, Gm17200, Gm17791, Gm18025, Gm18757, Gm18795, Gm18848, Gm19719, Gm20121, Gm20356, Gm2093, Gm21738, Gm21940, Gm22933, Gm24000, Gm24119, Gm25394, Gm26555, Gm27047, Gm28262, Gm28530, Gm29295, Gm29825, Gm29844, Gm3081, Gm32051,

Gm32122, Gm33056, Gm33680, Gm34354, Gm34643, Gm3551, Gm36660, Gm36948, Gm37052, Gm37142, Gm37262, Gm37535, Gm37569, Gm37589, Gm37647, Gm37648, Gm37762, Gm38058, Gm38069, Gm38137, Gm38218, Gm39139, Gm42535, Gm42680, Gm42895, Gm42994, Gm43027, Gm43158, Gm43288, Gm43366, Gm44044, Gm44081, Gm44187, Gm44280, Gm44535, Gm45338, Gm45644, Gm45740, Gm46555, Gm46565, Gm4742, Gm47485, Gm47853, Gm47992, Gm48225, Gm48314, Gm48383, Gm48673, Gm48804, Gm48832, Gm4994, Gm5487, Gm5724, Gm595, Gm6012, Gm6024, Gm7669, Gm7730, Gm8043, Gm8953, Gm9348, Gm9369, Gm9495, H2a12a, Ido2, Igfbp1, Kif7, Klhl31, Lrrc31, Mc5r, Mgam, Msh4, Muc12, Mug1, Mybl2, Myh15, Nek10, Neurod6, Nrlh5, Olfr1042, Olfr1043, Olfr1082, Olfr1090, Olfr1124, Olfr1167, Olfr1205, Olfr1206, Olfr1223, Olfr1263, Olfr1264, Olfr1269, Olfr127, Olfr1291-ps1, Olfr1406, Olfr1469, Olfr215, Olfr273, Olfr328, Olfr355, Olfr372, Olfr390, Olfr427, Olfr456, Olfr466, Olfr481, Olfr522, Olfr6, Olfr601, Olfr603, Olfr706, Olfr727, Olfr728, Olfr741, Olfr801, Olfr812, Olfr816, Olfr822, Olfr860, Olfr890, Olfr923, Olfr943, Otog1, Pi15, Pkhd1, Pkhd111, Platr6, Pou3f4, Prr9, Pvalb, Rhag, Sav1, Serpinb9b, Skint1, Skint3, Skint5, Slc10a5, Slc6a4, Smok2a, Tcaf3, Tomm20l, Trcg1, Trdn, Ugt1a6a, Usp17la, Vmn1r178, Vmn1r179, Vmn1r33, Vmn1r74, Vmn1r87, Vmn2r102, Vmn2r113, Vmn2r17, Vmn2r52, Vmn2r66, Vmn2r68, Vmn2r76, Vmn2r78, Wnt16, or any combination thereof.

(229) Aspects of the present disclosure relate to methods comprising resetting the transcriptional profile of an old cell, an old organ, an old tissue, and/or any combination thereof in vitro. Aspects of the present disclosure relate to methods comprising resetting the transcriptional profile of an old cell, an old organ, an old tissue, an old subject and/or any combination thereof in vivo. In some embodiments, resetting the transcriptional profile an old cell, an old organ, an old tissue, an old subject and/or any combination thereof comprises altering the gene expression of one or more genes associated with ageing. In some embodiments, resetting the transcriptional profile an old cell, an old organ, an old tissue, an old subject and/or any combination thereof comprises reversing the epigenetic clock. In some embodiments, the transcription profile of an old cell is reset. In some embodiments, the transcriptional profile of an old cell, an old organ, an old tissue, an old subject, or any combination thereof is reset to that of a young cell, a young tissue, a young organ, a young subject, or any combination thereof. In some embodiments, a method described herein reverses one or more changes in gene expression that are detected between an old cell, an old organ, an old tissue, an old subject, or any combination thereof and a control. In some embodiments, the control is a young cell, a young organ, a young tissue, a young subject, or any combination thereof. In some embodiments, the transcriptional profile of an old cell, an old organ, an old tissue, an old subject, or any combination thereof is changed from a young counterpart. In some embodiments, a method described herein resets at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or 100% of the gene expression change in an old cell, an old organ, an old tissue, an old subject, or any combination thereof to a young level. In some embodiments, a sensory gene is a sensory receptor gene. Without being bound by a particular theory, resetting of a sensor receptor gene expression level in an aged cell to a young level may be indicative of an improvement of retina ganglion cell function.

(230) In some aspects, the cellular reprogramming methods described herein may be used to promote the transdifferentiation of cells, which may be useful in treatment of disease. In some embodiments, the methods described herein may improve the efficiency of existing methods of transdifferentiation. For example, OCT4, SOX2, KLF4, or a combination thereof may be activated (e.g., expressed) in one cell type along with one or more perturbations of genes that affect cell fate to promote lineage reprogramming or conversion to another cell type. In some embodiments, the perturbation is reducing expression of a lineage determining factor. In some embodiments, the perturbation is expression of a lineage determining factor. In some embodiments, the lineage determining factor is a lineage transcription factor.

(231) As a non-limiting example, night blindness is caused by rod death and daytime blindness is

caused by cone death. Cell types including cones, rods, and muller cells could be reprogrammed into another cell type needed to restore vision. For example, loss of Nrl promotes transdifferentiation of adult rods into cone cells. See, e.g., Montana et al., Proc Natl Acad Sci USA. 2013 Jan. 29; 110(5):1732-7. In some embodiments, transcription factors that promote rod cell fate include Otx2, Crx and Nrl. As a non-limiting example, Müller glia (MG) can be reprogrammed into rod cells by expressing β-catenin, Otx2, Crx, and Nrl. See, e.g., Yao et al., Nature. 2018 August; 560(7719):484-488.

(232) As another non-limiting example, pancreatic alpha may be reprogrammed into beta cells for treating autoimmune diseases and diabetes. Transcription factors including Pdx1 and MafA can be used to reprogram mouse alpha cells into beta cells. See, e.g., Xiao et al., Cell Stem Cell. 2018 Jan. 4; 22(1):78-90.e4.

(233) Additional non-limiting examples of transdifferentiation inducing factors for production of various cell types may be found in Cieślar-Pobuda et al., Biochim Biophys Acta Mol Cell Res. 2017 July; 1864(7):1359-1369, which is herein incorporated by reference in its entirety. See e.g., Table 4 of Cieślar-Pobuda et al., Biochim Biophys Acta Mol Cell Res. 2017 July; 1864(7):1359-1369.

(234) Induction of OCT4, SOX2, KLF4, or a combination thereof may increase the efficiency of trandifferentiation of cells by at least 1%, at least 5%, at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 100%, at least 200%, at least 300%, at least 400%, at least 500%, at least 600%, at least 700%, at least 800%, at least 900%, or at least 1000%, including all values in between, as compared to a control. The efficiency of transdifferentiation may be measured by any suitable method including comparing the percentage of cells that were transdifferentiated when OCT4, SOX2, KLF4, or a combination thereof was activated as compared to control cells in which OCT4, SOX2, KLF4, or a combination thereof was not activated.

(235) These and other aspects of the present invention will be further appreciated upon consideration of the following Examples, which are intended to illustrate certain particular embodiments of the invention but are not intended to limit its scope, as defined by the claims.

EXAMPLES

(236) In order that the present disclosure may be more fully understood, the following examples are set forth. The synthetic and biological examples described in this application are offered to illustrate the compounds, pharmaceutical compositions, and methods provided herein and are not to be construed in any way as limiting their scope.

Example 1: Development of an Adenovirus-Associated Virus (AAV) Vector for Inducible Expression of OCT4, SOX2, and KLF4 (OSK) in Mammalian Cells

(237) An AAV vector that is capable of expressing OCT4, SOX2, and KLF4 in mammalian cells was developed as described herein. As shown in FIG. **1**, such a vector comprises a TRE3G promoter (SEQ ID NO: 7), nucleic acids (e.g., engineered nucleic acids) encoding OCT4, SOX2, KLF4, and an SV40 polyA (SV40 pA) terminator sequence (SEQ ID NO: 8). This vector will be referred to as TRE3G−OSK−SV40 pA. Nucleic acid (e.g., engineered nucleic acid) sequences encoding self-cleaving peptides (T2A, a 2A peptide, SEQ ID NO: 9) were used to separate the nucleic acids (e.g., engineered nucleic acids) encoding OCT4, SOX2, and KLF4. As shown in FIG. **2**, the entire vector is 7,408 base pairs in length and two inverted terminal repeats (ITRs) flank the OSK sequences. The restriction enzyme digestion sites in the vector are depicted in FIG. **3**. A schematic mapping the features shown in the vector maps of FIGS. **4**A-**4**AL onto the nucleic acid (e.g., engineered nucleic acid) sequence of the vector is shown in FIGS. **2-3**. The restriction enzyme cut sites are shown in Table 3 below. As shown in FIGS. **5**A-**5**D, the open reading frame (ORF frame 3) encoding OSK and intervening 2A peptides (T2A peptides) is 3,610 base pairs.

(238) TABLE-US-00003 TABLE 3 Restriction Enzyme Cut Sites in the TRE3G-OSK-SV40pA vector. Enzymes Sites Location AatII 1 4033 Acc65I 1 6074 AfeI 1 5333 AflII 1 2847 AleI 1 5656

BbvCI 1 5098 BclI 1* 4246* BmtI 1 3349 Bpu10I 1 5098 BsaBI 1* 6098* BspQI 1 1436 BsrDI 1 371 BstAPI 1 4016 BstXI 1 4667 BstZ17I 1 5078 EcoRI 1 1893 KpnI 1 6078 NheI 1 3345 NotI 1 2276 PaeR7I 1 4449 PflFI 1 3546 RsrII 1 3542 SacII 1 3765 SapI 1 1436 ScaI 1 7358 SexAI 1* 2330* SpeI 1 5907 Tth111I 1 3546 XhoI 1 4449 ZraI 1 4031

(239) The vector shown in FIGS. **3** and **4**A-**4**AL was cloned using routine methods. Briefly, a TRE3G promoter sequence (SEQ ID NO: 7) from Clonetech was synthesized using flanking restriction sites, primers were designed to clone OSK out of a TetO−FUW−OSKM plasmid, and a stop codon was added. To make the vector shorter, a short SV40 sequence was synthesized with flanking restriction cut sites. Whereas conventional AAV vectors encoding OSK is over the packaging limit of AAV, could only be packaged into AAV9 capsid with low titer (less than 2×10{circumflex over ( )}12 particles per viral prep), and the low titer virus is not functional due to possible truncation as shown in FIG. **17**. The vector depicted in FIGS. **3** and **4**A-**4**AL produced virus with more than 2×10{circumflex over ( )}12 viral partial per prep or 1×10{circumflex over ( )}13 per mL (data not shown).

(240) To determine whether the OSK vector could be used for inducible OSK expression in mammalian cells, the OSK vector and was packaged into different serotypes of AAV virus (AAV9 (FIG. **6**A), AAV2 (FIG. **6**B), and AAV.PhP.b (FIG. **6**C)) using routine methods. Additional batches of AAV9 and AAV.PHP.b virus with a vector encoding rtTA3 (Tet-On system) and AAV2 virus with a vector encoding tTA (Tet-Off system) were produced. Then, mammalian 293T cells were co-infected with the OSK virus along with the same serotype of rtTA3 or tTA virus. Cells were subsequently treated with or without doxycycline (DOX) and expression of OCT4, KLF4 and the loading control H3 was determined by western blot with antibodies against OCT4, KLF4, and H3.

(241) As shown with the Tet-On system in FIG. **6**A, doxycycline treatment increased OCT4 and KLF4 expression in 293T cells infected with AAV9 viruses encoding OSK and rtTA3. The OSK expression could also be controlled with a Tet-Off system. DOX treatment decreased OCT4 and KLF4 expression in 293T cells infected with OSK AAV2 and AAV2 with a vector driving tTA expression under a constitutive CAG promoter (FIG. **6**B). Furthermore, OSK expression could be tightly controlled even after stimulation of transgene expression. As shown in the fourth lane of FIG. **6**C, one day of DOX treatment is sufficient to increase OCT4 and KLF4 expression in 293T cells infected with TRE3G−OSK−SV40 pA AAV.PHP.b virus and with Ubc-rTtA3-p2a-mkate AAV.PHP.b virus. Removal of DOX for three days after one day of DOX treatment, however, returns OCT4 and KLF4 expression back to uninduced levels (last lane of FIG. **6**C). The Ubc-rtTA3-p2a-mkate vector comprises a constitutive Ubc promoter that drives expression of rtTA3, a self-cleaving 2A peptide, and a far-red fluorescent protein (mKate).

(242) Therefore, an AAV vector that allows for controlled expression of OSK in mammalian cells (e.g., in vivo) was developed. Furthermore, the AAV vector was packaged into different AAV serotypes that successfully delivered a functional vector into 293T cells.

Example 2: AAV Encoding OSK Promoted Optic Nerve Regeneration and Survival of Retina Ganglion Cells (RGCs) Nerves after Nerve Crush in an Inducible Manner

(243) To determine whether OSK could be delivered by AAV and inducibly expressed in vivo, AAV virus with the TRE−OSK−SV40 vector and AAV virus encoding tTA under the CAG constitutive promoter were produced through routine methods and injected into the retina of mice. Next, an optical coherence tomography (OCT) section was stained with antibody against RBPMS to identify retina ganglion cells (RGCs) and with an antibody against KLF4 to detect KLF4 expression. As shown in FIG. **7**A, KLF4 was expressed in RGCs (RBPMS-positive cells), which suggested that the vectors were functional.

(244) The inducibility of the system was also tested in vivo. In the absence of DOX treatment, OCT4 and KLF4 were expressed in the mouse retina as determined by whole retina mount staining (FIG. **7**B, top). After four days of DOX treatment, however, OCT4 and KLF4 staining was significantly reduced, indicating that expression from the TRE−OSK−SV40 vector was turned off

(FIG. **7**B, bottom). Therefore, OSK vector expression could be tightly controlled.

(245) To determine whether inducible OSK expression could induce partial reprogramming and promote regeneration following nerve damage, AAV virus with the TRE−OSK−SV40 vector and AAV virus encoding tTA under the CAG constitutive promoter were injected into the retina of 4-week old mice (n=6) as shown in the experimental timeline provided in FIG. **7**C. As a control, a separate cohort of mice (n=2) were only injected with the OSK virus. Mechanical damage was induced through optic nerve crush in both cohorts two weeks after virus injection. To trace axon regeneration by fluorescent microscopy of the optic nerve, fluorescently labeled cholera toxin β-subunit (CTB) was intraocularly injected into mice and perfusion was performed two days after CTB injection. Axon regeneration and axon survival analysis was subsequently conducted.

(246) Axon regeneration was determined by estimating the number of axons per nerve. As shown in FIG. **7**D, co-administration of OSK and tTA virus significantly promoted optic nerve regeneration away from the site of the optic nerve crush compared to administration of OSK virus alone. This effect was also visually apparent when comparing the fluorescence intensity of optic nerves from mice receiving both OSK and tTA virus compared to mice receiving OSK virus alone. The fluorescence intensity of optic nerves from mice receiving both viruses was higher than that of mice receiving OSK virus alone, indicating that nerve regeneration was higher with combination treatment (FIGS. **7**E-**7**F).

(247) To show that the observed axon regeneration after crush injury was specifically mediated by OSK, an axon regeneration experiment was used to compare the effects of tTA virus in combination with (1) TRE−OSK virus with no DOX treatment, (2) TRE-d2EGFP virus with no DOX treatment, and (3) TRE−OSK virus with DOX treatment. The experimental timeline of treatments (1)-(3) are indicated in FIGS. **8**B, **8**D, and **8**F, respectively. Fluorescently-labeled CTB was used to visualize axons. As shown in FIGS. **8**A and **8**G, the extent of optic nerve regeneration in mice in which OSK expression was induced (mice receiving OSK and tTA viruses in the absence of DOX) was very significant at 200 μm and 500 μm from crush site. In contrast, even when d2EGFP expression was induced (mice receiving d2EGFP and tTA viruses in the absence of DOX), minimal regeneration was observed (FIG. **8**A and FIG. **8**C). Notably, axon regeneration was dependent on induction of OSK expression. When mice were treated with DOX to inhibit OSK expression as outlined in FIG. **8**D, administration of tTA and OSK viruses did not induce axon regeneration (FIG. **8**A). The intensity of CTB-labeled axons in these DOX-treated mice were similar to mice receiving control d2EGFP virus (compare FIG. **8**E with FIG. **8**C). Therefore, administration of an AAV-based inducible OSK expression system could be used to promote regeneration following optic nerve damage.

(248) The effect of OSK on the survival rate of retina ganglion cells (RGCs) was also assessed. As shown in FIGS. **9**A-**9**D, OSK significantly increases RGC survival rate. RGCs (RBPMS positive cells) that were infected with OSK and tTA virus (green) or uninfected with both (red) shown following optic nerve crush, OSK-infected RGC had 3 times higher survival rate (54% vs 18%) after crush compared to cells without OSK infection, quantification from a series of pictures like shown (FIG. **9**A). Therefore, the percentage of RBPMS-positive cells expressing KLF4 (OSK-infected cells) was lower than 40% before crush, but significantly increased to around 70% following optic nerve crush due to its higher survival rate. While the percentage of d2EGFP-infected cells maintained at 35-40% after crush. This indicates a strong cell protection effect from OSK expression (FIG. **9**B). As shown in FIG. **9**C, in d2EGFP or OSK plus CAG-tTA (SEQ ID NO: 32) AAV infected retina, there is no significant difference in RGC number (RBPMs positive) without uncrushed, but after crush there is clearly more RGCs survived when they infected with OSK and CAG-tTA compared to those infected with d2EGFP and CAG-tTA. FIG. **9**D shows the quantification of survived RGC numbers from each group. Though lower than 40% cells infected with both OSK and CAG-tTA AAV, it increases the total survival RGC number compared to d2EGFP(542 compared to 323).

(249) mTOR activation has reported as a pathway for optic nerve regeneration (Parker et al., *Science*, 322(5903), 963-966 November 2008). To determine whether OSK expression activated the mTOR pathway, control and OSK virus-infected cells were imaged using antibodies against RBPMS and phosphorylated S6 (pS6) in the absence of damage (uncrushed) and after damage (crushed). Representative images of the staining is shown in FIG. **10**A, and as quantified in FIG. **10**B, the percentage of pS6-positive cells was not significantly different between control cells and OSK-infected cells following optic nerve crush.

Example 3. An AAV Tet-on System Induces Faster Gene Expression Compared to an AAV Tet-Off System in Retinal Cells after Nerve Crush

(250) To compare the rate of gene expression between AAV-based Tet-On and Tet-Off systems, TRE-d2EGFP virus and (1) virus encoding tTA (Tet-Off) or (2) virus encoding rtTA (Tet-On) were administered into the retina of 4-week old mice. In the Tet-Off system, mice were given DOX starting from virus injection and DOX was removed for 3 days, 5 days or 8 days (FIG. **11**A). As a control, a cohort of mice in the Tet-Off system were given no DOX. Approximately 8 days of DOX removal was needed to induce the same level of GFP expression as no DOX treatment in the Tet-Off system (FIG. **11**B). In the Tet-On system, mice were treated as indicated in FIG. **11**C. GFP expression was observed after only 2 days of DOX treatment in the Tet-On system (FIG. **11**D). Therefore, a shorter period of time was needed to induce transgene expression in mice retina infected with an AAV-based Tet-On system compared to infection with an AAV-based Tet-Off system.

Example 4: An AAV Vector Encoding Mutant Reverse Tetracycline Transactivator (rtTA) Showed Low Leakiness in the Liver of Mice and Low Toxicity

(251) As shown in FIGS. **13**A-**13**C, OCT4, SOX2, and KLF4 through AAV9 delivery (TRE−OSK with UBC−rtTA4) can be successfully induced in liver of the mice with DOX treatment, shown with both western blot and immune staining. While mice with transgene of OCT4, SOX2, KLF4 died after 2 days-induction from doxycycline water (FIG. **14**) due to generalized cytological and architectural dysplasia in the intestinal epithelium, expression from the OCT4, SOX2, and KLF4 AAV described herein did not cause toxicity or teratoma in vivo even with continuous expression through doxycycline administration in their drinking water. No teratoma or body weight loss were detected for three months when AAV9 encoding these three transcription factors were delivered to the entire body of mice (FIG. **14**).

Example 5: Expression of OCT4, SOX2, and KLF4 Induced Partial Reprogramming in Mice

(252) FIGS. **15**A-**15**B show that the expression of histone and Chaf (Chromatin assembly factor) genes declined during aging in ear fibroblasts from aged mice (12 months or 15 months) compared to those from young mice, short term of OSKM (3 days) or OSK expression (5 days) induction reset their gene expression level to young state, without making them into stem cell (e.g., Nanog was not been turned on).

(253) Conventional AAV vectors encoding OSK is over the packaging limit of AAV (e.g., 4.7 Kb), could only be packaged into AAV9 capsid with low titer (less than 2×10{circumflex over ( )}12 particles per viral prep), and the low titer virus is not functional (e.g., no overexpression of OCT4 or KLF4 was detected) due to possible truncation as shown in FIG. **16**.

Example 6: An AAV Vector Encoding Mutant Reverse Tetracycline Transactivator (rtTA) Showed Low Leakiness in the Liver of Mice

(254) A Tet-On system comprising rtTA4 (SEQ ID NO: 13) was also tested in vivo using recombinant AAV9 viruses. Two AAV vectors comprising components shown in FIG. **13**B were used. AAV virus encoding rtTA4 operably linked to a UBC promoter (pAAV-UBC−rtTA4-WPRE3-SV40 pA vector is provided as SEQ ID NO: 17 and an exemplary vector map of SEQ ID NO: 17 is provided in FIG. **12**) and AAV virus encoding an AAV TRE3G−OSK−SV40 pA vector (SEQ ID NO: 16) with a vector map depicted in FIG. **3** were administered to mice. Mice were treated without doxycycline or with doxycycline and liver samples were collected. As shown in the

immunofluorescence images of FIG. **13**A, in the absence of doxycycline, KLF4 expression was not detectable in the liver. When mice were treated with doxycycline through their drinking water, KLF4 expression was detected in the liver (FIG. **13**A). These results were also evident by western blot using antibodies against OCT4, KLF4, and SOX2 to determine expression of these protein (FIG. **13**C). Actin was used as a loading control (FIG. **13**C). OCT4, KLF4, and SOX2 were only detected in the liver when mice were treated with doxycycline (FIG. **13**C).

Example 7. Modified mRNAs Encoding OCT4, SOX2, and KLF4(OSK) Induced Expression of OSK in Mouse Fibroblasts

(255) Mouse fibroblasts were successfully transfected with modified mRNA encoding OCT4, SOX2, KLF4, and c-MYC (OSKM). Lipofectamine™ MessengerMAX™ Transfection Reagent from Invitrogen was used to transfect the modified mRNAs. The modifications were complete substitution of either 5-methylcytidine (5mC) for cytidine or pseudouridine (psi) for uridine. See, e.g., Warren et al., Cell Stem Cell. 2010 Nov. 5; 7(5):618-30; Mandal et al., Nat Protoc. 2013 March; 8(3):568-82. The dose of each RNA that was used is provided in Table 4 below. The numbers 1-5 in the first column of Table 4 correspond to the numbers 1-5 in FIG. **17**.

(256) TABLE-US-00004 TABLE 4 Doses of mRNA administered. mRNA (μg) NDG O S K M Total 1 0 0 0 0 0 0 0 2 1X 0.2 0.6 0.2 0.2 0.2 1.4 3 2X 0.4 1.2 0.4 0.4 0.4 2.8 4 4X 0.8 2.4 0.8 0.8 0.8 5.6 5 6X 1.2 3.6 1.2 1.2 1.2 8.4

(257) A western blot was used to confirm that administration of the modified mRNA induced expression of protein in the mouse fibroblasts. As shown in FIG. **17**, transfection of OSK modified mRNA into mouse fibroblasts cells to induce expression of OCT4, KLF4, and SOX2 protein (NDG and zsGreen are modified mRNA that express green fluorescent protein to indicate the efficiency of transfection).

(258) This example shows that delivery of RNA (e.g., mRNA, modified RNA, modified mRNA, etc.) encoding OCT4, KLF4, and SOX2 to mouse cells is feasible. These findings may be extended to in vivo delivery of mRNA encoding OCT4, KLF4, and SOX2. As an example, for in vivo muscle delivery, electroporation, is used. As an example, for liver and other internal organ delivery, nanoparticles comprising RNA encoding OCT4, KLF4, and SOX2, nanoparticles are used. See, e.g., Dong et al., Nano Lett. 2016 Feb. 10; 16(2):842-8.

Example 8. Chemical Reprogramming of Cells

(259) A non-limiting of a protocol to chemically reprogram a mouse embryonic fibroblast to an induced pluripotent stem cell is provided below. A similar protocol may be found at Zhao et al., Cell. 2015 Dec. 17; 163(7):1678-91. FIG. **21** shows the results after using the protocol provided below.

(260) Stage 1

(261) 100 ng/ml bFGF 0.5 mM VPA, 20 μM CHIR99021, 10 μM 616452, 5 μM tranylcypromine, 50 μM forskolin, 0.05 μM AM580 5 μM EPZ004777 On day 12, the cells were trypsinized, harvested and then re-plated at 50,000-200,000 cells per well of a 6-well plate (1:10-15) During days 12-16, concentrations of bFGF, CHIR, and forskolin were reduced to 25 ng/ml, 10 μM, and 10 μM, respectively. On day 16, XEN-like epithelial colonies were formed and the culture was changed into stage 2 medium

Stage 2 25 ng/ml bFGF, 0.5 mM VPA, 10 μM CHIR99021, 10 μM 616452, 5 μM tranylcypromine, 10 μM forskolin, 0.05 μM AM580, 0.05 μM DZNep, 0.5 μM 5-aza-dC, 5 μM SGC0946

(262) On day 28, the culture was transferred into stage 3 medium.

(263) Stage 3

(264) N2B27-2iL medium 3 μM CHIR99021, 1 μM PD0325901, 1,000 U/ml LIF

(265) After another 8-12 days, 2i-competent, ESC-like, and GFP-positive (if using pOct4-GFP reporter) CiPSC colonies emerged and were then picked up for expansion and characterization.

Example 9. Expression of OCT4, SOX2, and KLF4 Improved Axon Regeneration in Adult and Aged Mice after Optic Nerve Crush Injury

(266) The Tet-Off system depicted in FIG. **22**, top panel was used to determine whether a vector encoding TRE−OSK−SV40 (SEQ ID NO: 16) could be used to promote optic nerve axon regeneration in adult (3 month old) and aged (12 month old) mice.

(267) AAV2 virus with the TRE−OSK−SV40 vector and AAV2 virus encoding tTA under the CAG constitutive promoter were injected into the retina of 1 month old, 3 month old, or 12 month old mice (n=5-9), similar to the experimental timeline provided in FIG. **7**C. As a control, a separate cohort of 1 month old mice (n=5-6) were injected with AAV2 virus with a AAV2 vector TRE-d2EGFP-SV40 and the AAV2 virus encoding tTA. Mechanical damage was induced through optic nerve crush in both cohorts two weeks after virus injection. To trace axon regeneration by fluorescent microscopy of the optic nerve, fluorescently labeled cholera toxin β-subunit (CTB) was intraocularly injected into mice two weeks after optic nerve crush injury and perfusion was performed two days after CTB injection. Axon regeneration analysis was subsequently conducted.

(268) As shown in FIGS. **23**A-**23**B, administration of AAV2 virus encoding OSK increased the number of estimated axons per nerve in 1 month old (young), 3 month old (adult), and 12 month old (aged) mice relative to administration of control virus encoding d2EGFP. Furthermore, TRE−OSK virus also increased the survival of RGCs after optic nerve injury in adult (3 month old) and aged (12 month old) mice compared to control GFP (FIG. **23**C). Therefore, OSK expression surprisingly promoted axon regeneration and RGC survival after nerve crush injury in young, adult, and aged mice.

(269) Next, the impact of the length of time of OSK expression on axon regeneration in aged mice was determined. Mice were administered tTA virus and either TRE−OSK virus or TRE-GFP virus 2 weeks prior to optic nerve crush. Then, fluorescently labeled cholera toxin β-subunit (CTB) was intraocularly injected into mice that were five weeks instead of two weeks after optic nerve crush injury. As shown in FIGS. **24**A-**24**B, increasing the length of time of post-injury OSK expression to five weeks increased the number of estimated axons per nerve in the 12 month old mice compared to two weeks post-injury of OSK expression in FIG. **23**B. In contrast, increasing the length of time of post injury GFP expression had no effect on axon regeneration (compare results with GFP in FIGS. **24**A-**24**B with those shown in FIGS. **23**A-**23**B). Therefore, the data suggests that a longer time of OSK expression may be beneficial in promoting axon regeneration and RGC survival after nerve crush injury in aged mice.

Example 10. Induction of OSK Expression Following Optic Nerve Crush Injury Increased Axon Regeneration and RGC Survival in Mice

(270) It was also determined whether induction of OSK expression after optic nerve crush injury would promote axon regeneration and RGC survival. Both the Tet-On and Tet-Off systems depicted in the panel of FIG. **22** were used. In the Tet-On system, AAV virus with the TRE−OSK−SV40 vector and AAV virus encoding rtTA under the CMV constitutive promoter were produced through routine methods and injected into the retina of mice. As depicted in FIG. **25**A, in the Tet-On system (top panel), OSK expression was induced by giving mice doxycycline either prior to optic nerve crush injury or after optic nerve crush injury. A cohort of mice were not treated with doxycycline as a control (no induction). In the Tet-Off system, AAV virus with the TRE−OSK−SV40 vector and AAV virus encoding tTA under the CAG constitutive promoter were produced through routine methods and injected into the retina of mice. As depicted in FIG. **25**A, in the Tet-Off system (bottom panel), OSK expression was suppressed after optic nerve crush injury. Fluorescently labeled cholera toxin β-subunit (CTB) injection was used to visualize axons.

(271) As shown in FIG. **25**B, induction of OSK expression post injury through Tet-On system significantly increased the number of estimated axons per nerve compared to no induction of OSK or induction of OSK prior to injury (pre-injury) only through either Tet-On or Tet-Off system. Furthermore, induction of OSK expression post injury significantly increased the survival of RBPMS+ cells compared to no induction of OSK expression or compared to OSK induction pre-injury only through either Tet-On or Tet-Off system (FIG. **25**C). Therefore, the Tet-On system

depicted in FIG. **25**A, top panel, allowed for temporal control of OSK expression and induction of OSK after optic nerve crush injury promoted axon regeneration and RGC survival. Without being bound by a particular theory, induction of OCT4, KLF4, and SOX2 expression using a Tet-Off system following an injury may promote regeneration when recovery from an injury does not require immediate expression of OCT4, KLF4, and/or SOX2.

Example 11. Superior Effect of OCT4, SOX2, and KLF4 (OSK) Expression from a Single Transcript Compared to Individual Transcripts in Promoting Axon Regeneration

(272) This example explored the effect of expressing OCT4, SOX2, and KLF4 under one promoter as compared to expression of OCT4, SOX2, KLF4 alone or in combination under separate promoters. AAV virus encoding tTA under the CAG constitutive promoter and AAV virus or viruses encoding (1) OCT4 under the TRE promoter, (2) SOX2 under a TRE promoter, (3) KLF4 under a TRE promoter, (4) OCT4 and SOX2 under one TRE promoter, (5) OCT4, SOX2, and KLF4 each under separate promoters, or (6) OCT4, SOX2, and KLF4 under the same promoter were injected into the retina of mice. A schematic showing the various vectors used in this study is shown in FIG. **26**A. Optic nerve crush injury was induced 2 weeks after virus administration. Fluorescently labeled cholera toxin (3-subunit (CTB) injection 2 weeks after optic nerve crush was used to image axons.

(273) As shown in FIG. **26**B, when all three transcription factors (OSK) were expressed under one promoter, the number of estimated axons per nerve was at least four times higher than when OCT4, SOX2, and KLF4 were each expressed under a separate promoter (e.g., compare OCT4, SOX2, KLF4 (5), and OCT4-SOX2-KLF4 (6) results). Similarly, the number of estimated axons per nerve was also at least four times higher when OSK was expressed on a single transcript than when OCT4, SOX2, and KLF4 expression alone (FIG. **26**B) (e.g., compare OCT4 (1), SOX2 (2), and KLF4 (3) with OCT4-SOX2-KLF4 (6) results). The increase in axon regeneration was likely attributed to expression of all three transcription factors (OSK) under one promoter, as expression of OCT4 and SOX2 under one promoter did not significantly increase the number of estimated axons per nerve relative to expression of each transcription factor alone (FIG. **26**B) (e.g., compare OCT4-SOX2 (4) with OCT4-SOX2-KLF4 (6) results).

(274) Analysis of retina ganglion cell (RGC) survival was also conducted by quantifying RBPMS+ cells. As shown in FIG. **26**C, expression of OSK from one promoter increased the survival of RBPMS+ cells relative to expression of OCT4, SOX2, or KLF4 alone and relative to expression of OCT4 and SOX2 under one promoter. Expression of OSK from one promoter also increased the survival of RBPMS+ cells relative to expression of OCT4, SOX2, or KLF4 from separate vectors in separate viruses.

(275) As shown by the fluorescence staining depicted in FIG. **26**D, expression of OCT4, SOX2, and KLF4 in separate vectors in separate viruses resulted in a heterogeneous population of RGCs. Some cells only expressed OCT4, SOX2, or KLF4. Some cells expressed a combination of only two out of the three transcription factors and only a few three-factor positive RGCs were detected (white color cell in the bottom right corner of the top left panel in FIG. **26**D). In contrast, as shown in FIG. **26**E, expression of OCT4, SOX2, and KLF4 from a single vector resulted in a more homogenous population. All of the cells expressed all three of the OSK transcription factors (white color cells in the top left panel). Even in cells that were not pure white, expression of all three transcription factors were detected as shown in FIG. **26**E, suggesting that the results were due to differences in staining intensity for the three transcription factors.

(276) Therefore, this example shows that expression of OCT4, SOX2, and KLF4 using one promoter had greater therapeutic effect (e.g., increased axon regeneration and a greater survival of retina ganglion cells) compared to expression of each transcription factor alone, expression of all three transcription factors under separate promoters, or expression of only two of the transcription factors (e.g., OCT4 and SOX2) under one promoter.

Example 12. Knockdown of Tet1 or Tet2 Abrogated OSK-Induced Axon Regeneration Following

Optic Nerve Crush Injury

(277) This example determined the effect of knocking down DNA demethylases Tet1 and Tet2 on OSK-induced axon regeneration. A Tet-Off system was used. AAV2 of CAG-tTA+TRE−OSK−SV40 were injected into mice through intravitreal injection two weeks before crush together with AAV2 of U6-shRNA. Mice were one month old with four mice in each group.

(278) Addgene AAV plasmids encoding shRNA sequences were used. Control shRNA comprised the sequence 5′-GTTCAGATGTGCGGCGAGT-3′ (plasmid #85741 from Addgene). mTETl (Tet1 shRNA) comprised the sequence 5′-GCTCATGGAGACTAGGTTTGG-3′ (plasmid #85742 from Addgene). mTet2 (Tet2 shRNA) comprised the sequence 5′-GGATGTAAGTTTGCCAGAAGC-3′ (Plasmid #85743 from Addgene).

(279) As shown in FIG. **27**, knockdown of either Tet1 or Tet2 significantly reduced the number of estimated axons per nerve in animals also treated with OSK virus and subjected to optic nerve crush injury compared to the control hairpin (sh-cntl).

(280) These results suggest that Tet DNA methylases may be involved in OSK-induced axon regeneration and overexpression of Tet (e.g., Tet1 or Tet2) alone or in combination with OSK expression may promote regeneration.

(281) As a non-limiting example, mTet3 comprising the sequence 5′-GCTCCAACGAGAAGCTATTTG-3′ (Plasmid #85740 from Addgene) may be used to knockdown Tet3.

Example 13. Expression of OSK Reversed Age-Related Decline in Visual Acuity and Reversed Age-Related Decline in Retina Ganglion Cell (RGC) Function

(282) To determine whether age-related visual acuity loss may be reversed with OSK expression, an optomotor response (OMR) assay was conducted on adult mice (3 month old mice) and aged mice (12 month old and 18 month old mice). OMR is a reflexive head movement used to assess visual acuity. To induce OMR, individual mice are placed on a platform in the middle of an arena surrounded by computer monitors displaying stripes. The rotation of the striped pattern elicits mouse head tracking in the same direction by reflexive neck movements. Tracking is monitored by two independent masked observers. Visual acuity is quantified by increasing the spatial frequency of the stripes until an OMR cannot be elicited.

(283) Mice were retinally injected with AAV virus encoding tTA and AAV virus encoding TRE−OSK in the absence of doxycycline (OSK induction condition). In this Tet-Off system, OSK is expressed from a single promoter in the absence of doxycycline. As controls, age-matched mice were administered virus encoding virus encoding rtTA and virus TRE−OSK in the absence of doxycycline (uninduced control, ctl). In the control Tet-On system, OSK expression requires doxycycline treatment. Adult mice (3 month old (3 m)) were also used as a control. An OMR study was conducted to measure the spatial frequency threshold one month after virus injection.

(284) As shown in FIG. **28**, in the absence of OSK expression (control (ctl) condition) the aged mice (12 month old and 18 month old mice) had vision loss compared to the adult mice (3 month old mice). The decrease in the spatial frequency threshold for the aged mice relative to the 3 month old mice indicated vision loss in the absence of OSK expression. When OSK was expressed, however, the spatial frequency threshold on average increased for the 12 month old and 18 month old mice relative to no OSK expression. Furthermore, the spatial frequency thresholds for the 12 month old and 18 month old mice with OSK expression were similar to that of the 3 month old control mice in the presence and absence of OSK expression. These results demonstrate that induction of OSK expression reversed age-related vision loss in mice.

(285) To determine whether age-related decline in retina ganglion cell (RGC) function could also be reversed by OSK treatment, electrical waves from RGCs were measured using pattern electroretinograms (pattern ERGs or pERGs). In pERG assays, a checkerboard light and dark pattern stimulus is projected via electrodes placed on the cornea of mice of various ages (3 month old, 12 month old, or 18 month old mice). A contrast reversing pattern is displayed with no overall

change in luminance. Electrical waves generated from the RGCs are then measured.
(286) Mice were retinally injected with AAV virus encoding tTA and AAV virus encoding TRE−OSK in the absence of doxycycline (OSK induction condition). In this Tet-Off system, OSK is expressed from a single promoter in the absence of doxycycline. As controls, age-matched mice were administered virus encoding virus encoding rtTA and virus TRE−OSK in the absence of doxycycline (uninduced control, ctl). In the control Tet-On system, OSK expression requires doxycycline treatment. Adult mice (3 month old (3 m)) were also used as a control. A pERG study was conducted to measure the amplitude of the electrical waves in the RGCs following the pattern stimulus one month after virus injection.
(287) As shown in FIG. **29**, electrical waves generated from RGCs declined in aged mice (3 month old mice compared to 12 month old and 18 month old mice) in the absence of OSK expression (ctl condition). In contrast, administration of AAV virus encoding tTA and AAV virus encoding TRE−OSK in the absence of doxycycline (OSK induction condition) restored RGC electrical waves in 12 month old mice. For 18 month old mice, however, RGC function was likely not restored because corneal opacity blocked the pattern stimulus. These results suggest that expression of OSK improved RGC function in aged (12 month old) mice.
(288) Therefore, this example demonstrates that induction of OSK expression can improve vision acuity and RGC function that is caused by aging.
Example 14. Expression of OSK Reversed Glaucoma-Induced Decline in Visual Acuity and Reversed Glaucoma-Induced Decline in Retina Ganglion Cell (RGC) Function
(289) To determine whether OSK expression could be used to reverse glaucoma-induced declines in visual acuity and RGC function, a mouse model of glaucoma was used. Chronic elevation of intraocular pressure (IOP) was induced unilaterally in adult C57BL/6J mice by injecting polystyrene microbeads to the anterior chamber. IOP was measured in the first four weeks. As shown in FIG. **30**A, microbead injection increased IOP 4-21 days after microbead injection. Axon density was quantified using p-phenylenediamine (PPD) staining (FIG. **30**B). FIG. **30**C includes a chart quantifying RGC cell density (left panel) using Brn3a staining (shown, for example, on the right). FIGS. **30**B-**30**C show that 4 weeks after microbeads injection into the anterior chamber of the eye, there was significant loss of axon density and RGC density in wild-type (WT) mice that were not treated with AAV virus encoding TRE−OSK.
(290) In these experiments, glaucoma was induced with microbead injection and then three weeks later, OMR and pERG assays were conducted (pre AAV injection measurements in FIGS. **30**D-**30**E). Then, mice were divided into two treatment groups. One group of mice were retinally injected with AAV virus encoding rtTA and AAV virus encoding TRE−OSK in the absence of tetracycline (OSK AAV OFF) or with AAV virus encoding tTA and AAV virus encoding TRE−OSK (OSK AAV ON). Four weeks post AAV virus injection, OMR and pERG assays were conducted again (4w post AAV) measurements in FIGS. **30**D-**30**E). As a control, experiments were also conducted with injection of saline instead of microbeads (no glaucoma control).
(291) As shown in FIG. **30**D, induction of OSK expression (OSK AAV ON) increased the spatial frequency threshold compared to no induction of OSK expression (OSK AAV OFF) for mice with glaucoma (mice injected with microbeads). These results suggest that induction of OSK expression can improve glaucoma-related vision loss.
(292) As shown in FIG. **30**E, induction of OSK expression restored the electrical wave amplitude in mice with microbead-induced glaucoma. These results suggest that induction of OSK expression can also reverse glaucoma-related decline in RGC function.
(293) Therefore, induction of OSK expression can improve the symptoms induced by glaucoma.
Example 15. Expression of Human OSK Promoted Survival of Human Neurons and Axon Regrowth Following Vincristine-Induced Neuronal Damage
(294) To determine whether expression of human OCT4, human KLF4, and human SOX2 (human OSK) could protect human neuronal cells and regenerate axons in vitro, a neurite regeneration

assay was used as described below. SH-SY5Y cells, which are human neuroblastoma cells, were differentiated into neurons and were transduced with a AAV.DJ vector encoding human OCT4, human KLF4, and human SOX2 under a Tet-inducible promoter (using a Tet-Off system). In the OSK Off condition, OSK expression was not induced in cells. In the OSK On condition, OSK expression was induced in cells. Five days after transduction, vincristine (VCS) was used to induce neurite degeneration. Cells were treated with VCS for 24 hours or 48 hours. A schematic of a treatment timeline (with 24 hour VCS treatment) is provided in the left portion of FIG. **31**A. VCS is a chemotherapy drug that disrupts microtubules. It is often used in vitro to determine whether treatments maintain and/or promote cellular function (e.g., neuronal function) after damage. As described herein, VCS was used determine the effect of OSK treatment on neuronal survival and axon regrowth. After VCS treatment, cells were grown in differentiation medium and neurite outgrowth was assayed.

(295) In FIG. **31**A, cells were assayed for neuronal outgrowth nine days after cells were treated with VCS for 24 hours. Cells in which OSK expression was induced (OSK On condition) showed increased neuronal survival and axon outgrowth relative to cells in which OSK expression was not induced (OSK Off condition) (FIG. **31**A). Quantification of neuronal cell area similarly showed that OSK expression increased the cell area of neurons by at least 8 times compared to no OSK expression (FIG. **31**B). Similar results were also observed with 48 hours of VCS treatment (FIG. **31**C).

(296) These results show that expression of human OSK protected human neuron cells against VCS-induced neuron degeneration.

(297) Methods

(298) Cell Culture and Differentiation Protocol

(299) SH-SY5Y neuroblastoma cells were obtained from the American Tissue Culture Collection (ATCC, CRL-2266) and maintained according to ATCC recommendations. The cells were cultured in a 1:1 mixture of Eagle's Minimum Essential Medium (EMEM, ATCC, 30-2003) and F12 medium (ThermoFisher Scientific, 11765054), supplemented with 10% fetal bovine serum (FBS, Sigma, F0926) and 1×penicillin/streptomycin (ThermoFisher Scientific, 15140122). Cells were cultured at 37° C. with 5% CO.sub.2 and 3% O.sub.2. Cells were passaged at ˜80% confluency.

(300) SH-SY5Y cells were differentiated into neurons as previously described (Encinas et al., J Neurochem. 2000 September; 75(3):991-1003; Shipley et al., J Vis Exp. 2016 Feb. 17; (108):53193), with some modifications. Briefly, 1 day after plating, cells started to be differentiated in EMEM/F12 medium (1:1) containing 2.5% FBS, 1× penicillin/streptomycin, and 10 μM all-trans retinoic acid (ATRA, Stemcell Technologies, 72264) (Differentiation Medium 1) for 3 days, followed by treating the cells in EMEM/F12 (1:1) containing 1% FBS, 1×penicillin/streptomycin, and 10 μM ATRA (Differentiation Medium 2) for 3 days. Cells were then split into 35 mm cell culture plates coated with poly-D-lysine (ThermoFisher Scientific, A3890401). One day after splitting, neurons were matured in serum-free neurobasal/B27 μlus culture medium (ThermoFisher Scientific, A3653401) containing 1×Glutamax (ThermoFisher Scientific, 35050061), 1×penicillin/streptomycin, and 50 ng/ml BDNF (Alomone labs) (Differentiation Medium 3) for at least 5 days.

(301) Neurite Regeneration Assay

(302) The differentiated neurons from SH-SY5Y cells were transduced with AAV.DJ vectors at 10.sup.6 genome copy per cell. Five days after transduction, 100 nM vincristine (Sigma, V8879) was added to the cells for 24 hours or 48 hours to induce neurite degeneration. After vincristine treatment, neurons were washed in PBS twice and fresh differentiation medium was added back to the plates. Neurons were followed for neurite outgrowth for up to 2 weeks.

Example 16. Recovery from Injury and Restoration of Vision by Tet-Dependent Resetting of the Epigenetic Clock

(303) To determine whether mammalian cells might retain a faithful copy of epigenetic information

from earlier in life, it was tested whether the three gene combination of OSK was sufficient to reset age. The three-gene OSK combination into fibroblasts from old mice and measured its effect on RNA levels of genes known to be altered with age, such as H2A, H2B, LaminB1, and Chaf1b. OSK treatment of fibroblasts from old mice restored youthful gene expression patterns, similar to what OSKM does, with no apparent loss of cellular identity or the induction of Nanog, an early embryonic transcription factor that can induce teratomas (FIG. **36**A-**36**C).

(304) To deliver and control OSK expression in vivo, a tightly regulated Tet-ON and Tet-OFF adeno-associated viral (AAV) vector system was developed to accommodate all three reprogramming genes in one viral particle (Smalley et al., First AAV gene therapy poised for landmark approval. Nat Biotechnol, 2017. 35(11): p. 998-999; Senis et al., AAV vector-mediated in vivo reprogramming into pluripotency. Nat Commun, 2018. 9(1): p. 2651) (FIG. **32**A). First, to test if induction of OSK AAVs caused toxicity in vivo, 5-month-old C57BL/6J mice were infected with rtTA and TRE−OSK AAV9s and induced expression to levels comparable to those of transgenic mice (FIG. **36**D). Surprisingly, continuous induction of OSK for over a year had no discernable negative effect on the mice for over a year (FIG. **32**B and FIG. **36**E). Without being bound by a particular theory, there was ostensibly no discernable negative effect on the mice because high-level expression in the intestine was avoided (FIGS. **36**F-**36**H), thus avoiding the dysplasia and weight loss seen in other studies, including Abad et al., Nature 502, 340-345, doi:10.1038/naturel2586 (2013).

(305) Almost all species experience a decline in regenerative potential during ageing. In mammals, one of the first systems to lose this potential is the central nervous system (CNS). A canonical CNS cell type, the retinal ganglion cell, projects an axon away from the retina towards the brain, forming the optic nerve. During embryogenesis and in neonates, RGCs can regenerate if damaged, but this capacity is soon lost (Goldberg et al., Science, 2002. 296(5574): p. 1860-4). Over time, as organisms age, the overall function and resilience of the CNS continues to decline (Geoffroy et al., Cell Rep, 2016. 15(2): p. 238-46). To explore whether it is possible to restore an early epigenetic profile in adult RGCs, OSK expression was induced in a nerve crush injury model in adult mice of various ages. The Tet-Off system (Tet-Off tTA−AAV2) carrying OSK, either in separate AAVs or in the same AAV, was injected into the vitreous body, resulting in efficient, selective, and doxycycline-responsive gene expression in RGCs. As a negative control, a group of mice were also treated with doxycycline to repress the AAVs (FIG. **32**C and FIG. **37**C). Two weeks post-injection, optic nerve crush was performed, and, two weeks after that, axon length and optic nerve density were calculated (FIG. **32**D).

(306) Induction of the polycistronic OSK-AAV2 caused a significant increase in RGC survival and long-distance axonal regeneration (FIG. **32**E and FIG. **37**D) without any sign of RGC proliferation (FIG. **38**A). In contrast, when introduced on separate AAVs, OCT4, SOX2, KLF4 had no effect on regenerative capacity (FIG. **32**E), ostensibly due to the lower frequency of co-infection (FIG. **37**A and FIG. **37**B). Because Klf4 can repress axonal growth (Moore et al., Science, 2009. 326(5950): p. 298-301; Qin et al., Nat Commun, 2013. 4: p. 2633), OCT4, SOX2, and KLF4 were also individually and a dual-cistron of Oct4 and Sox2 was tested. No regenerative effect, however, was observed in the absence of Klf4. Remarkably, if poly-cistronic OSK was induced for 3-months, RGC axon fibers extended all the way to the chiasm, a distance of over 3 mm (FIG. **38**B). Indeed, when polycistronic OSK was induced for 12-16 weeks, regenerating RGC axon fibers further extended into the chiasm (5 mm away from crush site), where optic nerve connects to brain (FIGS. **38**B-**38**C).

(307) Next, the requisite timing of OSK expression was tested to promote neuronal survival and regeneration. For these experiments, the Tet-On AAV system was utilized due to its rapid on-rate (FIG. **37**D and FIGS. **39**A-**39**B). Significant improvement in axon regeneration only occurred when OSK expression was induced after injury and the longer OSK was induced, the greater distance the neurons extended, with no increase in the total number of RGCs (FIGS. **33**B, **33**C, and

33D). By co-staining for OSK and performing neuronal counts, survival rate was estimated to be 2.5-3 times of uninfected or GFP-infected RGCs (52 vs. 17%-20%) (FIGS. **39**C and **39**D), suggesting OSK effect is cell-intrinsic. The Pten-mTOR-S6K pathway, previously shown to improve neuronal survival in vivo, was not activated in OSK-infected cells post-injury (FIG. **40**A and FIG. **40**B), indicating a new pathway might be involved.

(308) It was determined whether neuronal injury advanced epigenomic age and whether OSK's benefits were due to the preservation of a younger epigenome. Genomic DNA from RGCs was FACS-isolated before injury or 4-days after injury in the presence or absence of OSK induction, and subjected reduced-representation bisulfite sequencing (RRBS-Seq). Without being bound by a particular theory, rDNAme clock (Wang et al., *Genome Res* 29, 325-333, doi:10.1101/gr.241745.118 (2019)) provided the best site coverage (70/72 CpG sites) relative to other available mouse clocks (Meer et al., Elife 7, doi:10.7554/eLife.40675 (2018); Thompson et al., Aging (Albany NY) 10, 2832-2854, doi:10.18632/aging.101590 (2018)) and its age estimate remained highly correlated with chronological age of RGCs (FIG. **45**A and Methods). In the absence of global methylation changes, injured RGCs experienced an acceleration of the epigenetic clockand OSK expression counteracted this effect (FIG. **33**K and FIG. **45**B).

(309) It was determined whether that the effect of OSK on neuronal survival and regeneration occurred by restoring a younger epigenome. If so, these effects should be dependent on the reversal of the epigenetic clock, which would require the removal of methyl groups from DNA via the activity of Ten-Eleven-Translocation (TET) dioxygenases. Previously characterized AAVs expressing short-hairpin RNAs against Tet1 and Tet2 (sh-Tet1 and sh-Tet2) (Guo et al., *Cell* 145, 423-434, doi:10.1016/j.cell.2011.03.022 (2011); Yu et al., *Nat Neurosci* 18, 836-843, doi:10.1038/nn.4008 (2015); Weng et al., *Neuron* 94, 337-346.e336, doi:10.1016/j.neuron.2017.03.034 (2017)) were utilized, and the transduction rate and knockdown efficiency in vivo was validated (FIGS. **40**C-**40**F). Knockdown of either Tet1 or Tet2 (sh-Tet1 and sh-Tet2 AAV2, at 1/5 titer of OSK AAV), which transduced around 70% of OSK positive cells (FIGS. **40**C and **40**D), efficiently blocked OSK from regenerating axons and improved RGC survival (FIGS. **33**E and **33**F).

(310) To test whether neuronal rejuvenation by OSK is specific for mouse RGCs, axon regeneration assays were performed in human neurons in vitro (FIG. **33**G). Human neuroblastoma SH-SY5Y cells were differentiated into neurons and transduced them with AAV-DJ vectors to express OSK (FIG. **33**G, FIG. **41**A, and FIG. **41**B). Similar to mouse RGCs in vivo (FIG. **38**A), OSK did not induce cell proliferation (FIGS. **41**C-**41**D). Axon degeneration was then induced by a 24 hour treatment with vincristine (VCS), a chemotherapeutic agent, and cells were then allowed to recover for 9 days. The epigenetic clock of these neurons were measured using the skin and blood cell clock (Horvath and Raj, Nat Rev Genet. 2018 June; 19(6):371-384). Similarly, DNA methylation age is significantly increased after VCS damage in human neurons (FIG. **41**J), and OSK expression not only prevented this increase of DNA methylation age, but also restored a younger DNA methylation age without a global reduction of DNA methylation (FIG. **33**H, bottom panel and FIG. **45**C). DNAmAge is significantly decreased with experiment day 9 post VCS damage in OSK treated cells, but not in cells not treated by OSK (FIG. **33**H). At Day 9 post damage, the neurite area was 15-fold greater in the rejuvenated OSK-transduced cells than controls (FIG. **41**E and FIG. **41**F) and the recovery from damage was dependent on the Tet2 demethylase (FIG. **33**I, FIG. **33**J, and FIG. **41**G), even in presence of high OSK expression (FIG. **41**K) but not the mTOR-S6K pathway, paralleling mouse retinal ganglia cells (FIG. **41**H and FIG. **41**I). Thus, the ability of OSK to reprogram neurons and promote axon growth is cell intrinsic, conserved in mammals, and requires epigenetic rejuvenation through DNA demethylation. This process is referred to herein as the recovery of information via epigenetic reprogramming, or “REVIVER” for short.

(311) Glaucoma, a progressive loss of RGCs and their axons, most often due to increased

intraocular pressure, is a leading cause of age-related blindness worldwide. Although some treatments can slow down disease progression, it is currently not possible to restore vision once it has been lost. Given the ability of OSK to regenerate axons after acute nerve damage, we decided to test whether REVIVER treatment could restore the function of RGCs in achronic setting like glaucoma (FIG. **34**A). Elevated intraocular pressure (IOP) was induced unilaterally for 4-21 days by injection of microbeads into the anterior chamber. OSK AAVs or PBS were then injected intravitreally, and express at a time point when glaucomatous damage was established, with a significant decrease in RGCs and axonal density (FIG. **34**B, FIG. **42**A, and FIG. **42**B) (Krishnan et al., J Immunol, 2016. 197(12): p. 4626-4638). At four weeks post-AAV injection, OSK-On treated mice presented with a significant increase in axon density when compared to PBS and OSK-Off treated mice. The increased axon density observed was equivalent to the axon density in the saline-only, non-glaucomatous mice (FIGS. **34**C and **34**D), and was not associated with proliferation of RGCs (FIG. **42**C).
(312) To determine whether the increased axon density observed in OSK treated mice coincided with increased vision, a behavior assay, optomotor response (OMR), was used (FIG. **34**E) to track the visual acuity of each mouse. Compared to mice that received either PBS or the OSK-Off AAV, OSK treatment significantly increased visual acuity relative to the pre-treatment baseline measurement, restoring more than half of the vision loss (FIG. **34**F). A readout of electrical waves generated by RGCs in response to a reversing contrast checkerboard pattern, known as Pattern electroretinogram response (pERG) analysis, showed that OSK treatment significantly improved RGC function relative to the pre-treatment baseline measurements, as well as, compared with either PBS or OSK-Off AAV treated mice (FIGS. **34**G and H). Without being bound by a particular theory, treatment with OSK AAV, as shown herein, may be the first treatment to reverse vision loss in any glaucoma model. Notably, OSK reversed vision loss in a glaucoma model.
(313) Given the ability of OSK to induce axon regeneration following optic nerve crush and to restore vision after glaucomatous damage in young mice, it was determined whether OSK could also restore vision loss associated with physiological aging and regenerate axons following optic nerve injury in aged mice. This is particularly important since a recently reported retinal rod photoreceptor regenerative approach that was successful when treating young mice was significantly diminished when treating older mice (Yao, K., et al., Restoration of vision after de novo genesis of rod photoreceptors in mammalian retinas. Nature, 2018. 560(7719): p. 484-488.).
(314) To determine whether OSK AAV treatment could induce axon regeneration in aging mice, the optic nerve crush injury model was performed on 12-month-old mice using the same protocol as in FIG. **32**D with the experimental design shown (FIG. **35**A). In aged mice, OSK AAV treatment for two weeks post-injury showed doubled RGC survival, similar to that observed in young mice (FIG. **43**A). Though the axon regeneration is slightly less than young mice at two weeks post injury (FIG. **43**B), OSK AAV treatment in aged mice for five weeks post-injury showed a significant increase in axon regeneration (FIGS. **35**B and **35**C), similar to that observed in young mice. These data indicate that aging does not diminish the effectiveness of OSK AAV treatment in inducing axon regeneration following an optic nerve crush injury.
(315) To test whether OSK treatment could reverse vision loss associated with physiological aging, 4- and 12-month-old mice received intravitreal injections of OSK-Off or OSK-On AAV. As expected, at one year of age, mice showed a significant reduction in visual acuity and RGC function as measured by OMR and pERG, which was restored by OSK AAV treatment (FIG. **35**D and FIG. **43**C). Such restoration was not observed in 18 month-old mice (FIG. **43**F-**43**G) likely due to spontaneous corneal opacity developed at this age (McClellan et al., *Am J Pathol* 184, 631-643, doi:10.1016/j.ajpath.2013.11.019 (2014)), suggesting the restoration effect is specifically contributed by AAV-infected RGC layer.
(316) Next, it was determined whether restoration of youthful transcriptome by OSK indicates a youthful epigenome and thus would requires Tet enzymes. Remarkably, Tet1 or Tet2 knockdown

completely blocked the rejuvenation effect of OSK-On AAV treatment as measured by both OMR and pERG analyses (FIG. **35**E and FIG. **35**F), consistent with DNA methylation as the key process for vision restoration. Notably, there is no obvious RGC and axon density increase by OSK in aged mice (FIG. **43**D and FIG. **43**E), suggesting functional improvement of existing RGCs. The rDNA methylation age of FACS-sorted RGCs from 12 month-old mice was measured. OSK AAV expression for 4 weeks significantly decreased the DNA methylation age and Tet1 and Tet2 knockdown blocked such rejuvenation (FIG. **35**I). Together, these results demonstrate that Tet-dependent in vivo reprogramming can restore youthful gene expression patterns, reverse the DNA methylation clock, and restore the function and regenerative capacity of a complex tissue.

(317) To further determine whether Tet2 knockout can block the effect of OSK on axon regeneration, mouse OSK and Tet2 conditional knockout mice (B6; 129S-Tet2tm1.1Iaai/J) were used. Mouse eyes were injected with (1) AAV−CRE (Tet2 cKO); (2) AAV-tTA+AAV−TRE−OSK: OSK (Tet2 WT); or (3) AAV-tTA+AAV−TRE−OSK+AAV−CRE: OSK (Tet2 cKO). After two weeks, optic nerve crush was conducted. CTB was administered two weeks after optic nerve crush and mice were sacrificed two days after CTB administration to determine the extent of axon regeneration following injury. As shown in FIGS. **46**A-**46**B, the number of axons per nerve up to at least 500 μm from the injury site was significantly higher in Tet2 wild-type mice that were administered OSK as compared to Tet2 knockout mice that were administered OSK. These results suggest that OSK-mediated axon regeneration is Tet2-dependent.

(318) In order to determine the effect of reprogramming on the transcriptome in the retina, FACS-purified RGCs from intact old mice (12 month) and those that were either treated with empty control AAV (TRE−OSK) or OSK-On (tTA+TRE−OSK) were analyzed by genome-wide RNA-seq. Compared to RGCs from intact young mice (5 month), 464 genes were identified that were differentially-expressed during ageing (FIG. **35**G, FIG. **35**I, FIG. **44**A, and Table 5) and not induced by empty AAV alone. Of these, 268 genes were down-regulated during aging which were enriched in sensory perception genes (FIG. **35**I), suggesting a decline of signaling receptors/sensory function during aging (FIGS. **44**B and **44**C). Interestingly, 116 of these genes appear uncharacterized, lacking an official gene name. The other 196 genes that are slightly up-regulated during aging are enriched of ion transporter genes (FIG. **44**D).

(319) Remarkably, consistent with OSK resetting the epigenomic landscape, the vast majority (90%, 418) of the 464 genes that change in expression during aging were restored towards youthful levels after treatment (FIGS. **35**G and **35**H). Together, these results demonstrate that Tet-dependent in vivo reprogramming can restore youthful gene expression patterns, reverse the epigenetic clock, and restore the function of a tissue as complex as the retina.

(320) Post-mitotic neurons in the central nervous system are some of the first cells in the body to lose their ability or regenerate. In this study, it was shown that in vivo reprogramming of aged neurons can reverse epigenetic age and allow them to regenerate and function as though they were young again. The requirement of the DNA demethylases Tet1 and Tet2 for this process indicates that DNA methylation at clock sites are not merely an indicator of ageing, but an active participant in it. It was concluded that mammalian cells retain a set of original epigenetic information, in the same way Shannon's observer stores information to ensure the recovery of lost information at a later time (SHANNON, C. E., *A Mathematical Theory of Communication*. The Bell System Technical Journal, 1948. 27: p. 379-423). How cells are able to find and remove the appropriate DNA methylation moieties and restore youthful gene expression patterns is still an open question, but even in the absence of this knowledge, our data indicate that the reversal of epigenetic age could be an effective translational strategy, not just to restore vision, but to give other tissues the ability to recover from injury and resist age-related decline.

(321) TABLE-US-00005 TABLE 5 Genes that were differentially expressed during ageing in mice RGCs. Downregulated genes Upregulated Genes 1700031P21Rik 0610040J01Rik 1810053B23Rik 1700080N15Rik 2900045O20Rik 2900064F13Rik 2900060B14Rik 4833417C18Rik

4921504E06Rik 4921522P10Rik 4930402F11Rik 4930447C04Rik 4930453C13Rik 4930488N15Rik 4930455B14Rik Ace 4930500H12Rik Ackr1 4930549P19Rik Acot10 4930555B11Rik Acvr1 4930556J02Rik Adamts17 4932442E05Rik Adra1b 4933431K23Rik AI504432 4933438K21Rik Best3 6720475M21Rik Boc 9830132P13Rik Cadm3 A430010J10Rik Cand2 A530064D06Rik Ccl9 A530065N20Rik Cd14 Abcb5 Cd36 Abhd17c Cfh AC116759.2 Chrm3 AC131705.1 Chrna4 AC166779.3 Cntn4 Acot12 Cracr2b Adig Cryaa Akr1c1 CT573017.2 Ankrd1 Cyp26a1 Asb15 Cyp27a1 Atp2c2 D330050G23Rik AU018091 D930007P13Rik AW822073 Ddo Btn110 Dgkg C130093G08Rik Dlk2 C730027H18Rik Dnaja1-ps Ccdc162 Drd2 Chi16 Dsel Col26a1 Dytn Corin Ecscr Crls1 Edn1 Cybrd1 Ednrb Cyp2d12 Efemp1 Cyp7a1 Elfn2 D830005E20Rik Epha10 Dlx3 Ephx1 Dnah14 Erbb4 Dsc3 Fam20a Dthd1 Fbxw21 Eid2 Ffar4 Eps811 Flt4 EU599041 Fmod Fam90a1a Foxp4 Fancf Fzd7 Fau-ps2 Gabrd Fezf1 Galnt15 Gja5 Galnt18 Gm10248 Gfra2 Gm10513 Ggt1 Gm10635 Gm10416 Gm10638 Gm14964 Gm10718 Gm17634 Gm10722 Gm2065 Gm10800 Gm32352 Gm10801 Gm33172 Gm11228 Gm34280 Gm11251 Gm35853 Gm11264 Gm36298 Gm11337 Gm36356 Gm11368 Gm36937 Gm11485 Gm3898 Gm11693 Gm42303 Gm12793 Gm42484 Gm13050 Gm42537 Gm13066 Gm42743 Gm13323 Gm43151 Gm13339 Gm43843 Gm13346 Gm44545 Gm13857 Gm44722 Gm14387 Gm45516 Gm14770 Gm45532 Gm15638 Gm47494 Gm16072 Gm47982 Gm16161 Gm47989 Gm16181 Gm48398 Gm17200 Gm48495 Gm17791 Gm48593 Gm18025 Gm48958 Gm18757 Gm49089 Gm18795 Gm49326 Gm18848 Gm49331 Gm19719 Gm49760 Gm20121 Gm5796 Gm20356 Gm6374 Gm2093 Gm7276 Gm21738 Gm8237 Gm21940 Gm9796 Gm22933 Gm9954 Gm24000 Gpr75 Gm24119 Gprc5c Gm25394 Grid2ip Gm26555 Gsg112 Gm27047 Hapln4 Gm28262 Hcn3 Gm28530 Hon4 Gm29295 Hhat1 Gm29825 Hs6st2 Gm29844 Htr3a Gm3081 Il1rap Gm32051 Il1rapl2 Gm32122 Inka1 Gm33056 Kbtbd12 Gm33680 Konj11 Gm34354 Kcnk4 Gm34643 Kdelc2 Gm3551 Klhl33 Gm36660 Lamc3 Gm36948 Lilra5 Gm37052 Lman11 Gm37142 Lrfn2 Gm37262 Lrrc38 Gm37535 Lrrn4cl Gm37569 Ltc4s Gm37589 Mansc1 Gm37647 Mir344c Gm37648 Msr1 Gm37762 Mycbpap Gm38058 Myoc Gm38069 Ngfr Gm38137 Nipal2 Gm38218 Olfr1372-ps1 Gm39139 Otop3 Gm42535 P2rx5 Gm42680 P2ry12 Gm42895 P4ha2 Gm42994 Pcdha12 Gm43027 Pcdha2 Gm43158 Pcdhac2 Gm43288 Pcdhb18 Gm43366 Pcdhb5 Gm44044 Pcsk2os1 Gm44081 Pcsk6 Gm44187 Perp Gm44280 Pkp1 Gm44535 Plxna4 Gm45338 Prickle2 Gm45644 Qsox1 Gm45740 Rapgef4os2 Gm46555 Rbp4 Gm46565 Rcn3 Gm4742 Sec1415 Gm47485 Sell13 Gm47853 Serpinh1 Gm47992 Sgpp2 Gm48225 Shisa6 Gm48314 Siah3 Gm48383 Siglech Gm48673 Slc12a4 Gm48804 Slc24a2 Gm48832 Slc2a5 Gm4994 Slc4a4 Gm5487 Slitrk3 Gm5724 Smagp Gm595 Smoc2 Gm6012 Speer4b Gm6024 Spon2 Gm7669 Sstr2 Gm7730 Sstr3 Gm8043 St3gal3 Gm8953 Stc1 Gm9348 Stc2 Gm9369 Syndig1 Gm9495 Syt10 H2al2a Thsd7a Ido2 Tlr8 Igfbp1 Tmem132a Kif7 Tmem132d Klhl31 Tmem200a Lrrc31 Tmem44 Mc5r Trpc4 Mgam Trpv4 Msh4 Unc5b Muc12 Vgf Mug1 Vmn1r90 Myb12 Vwc21 Myh15 Wfikkn2 Nek10 Wnt11 Neurod6 Wnt6 Nr1h5 Zeb2os Olfr1042 Zfp608 Olfr1043 Zfp976 Olfr1082 Olfr1090 Olfr1124 Olfr1167 Olfr1205 Olfr1206 Olfr1223 Olfr1263 Olfr1264 Olfr1269 Olfr127 Olfr1291-ps1 Olfr1406 Olfr1469 Olfr215 Olfr273 Olfr328 Olfr355 Olfr372 Olfr390 Olfr427 Olfr456 Olfr466 Olfr481 Olfr522 Olfr6 Olfr601 Olfr603 Olfr706 Olfr727 Olfr728 Olfr741 Olfr801 Olfr812 Olfr816 Olfr822 Olfr860 Olfr890 Olfr923 Olfr943 Otogl Pi15 Pkhd1 Pkhd111 Platr6 Pou3f4 Prr9 Pvalb Rhag Sav1 Serpinb9b Skint1 Skint3 Skint5 Slc10a5 Slc6a4 Smok2a Tcaf3 Tomm201 Trcg1 Trdn Ugt1a6a Usp17la Vmn1r178 Vmn1r179 Vmn1r33 Vmn1r74 Vmn1r87 Vmn2r102 Vmn2r113 Vmn2r17 Vmn2r52 Vmn2r66 Vmn2r68 Vmn2r76 Vmn2r78 Wnt16

Methods

Mouse Lines

(322) C57BL6/J wild type mice are purchased from Jackson Laboratory(000664) for optic nerve crush and glaucoma model experiment. For ageing experiment, females from NIA Aged Rodent Colonies (https://www.nia.nih.gov/research/dab/aged-rodent-colonies-handbook) are used. Col1a1-tetOP-OKS-mCherry/Rosa26-M2rtTA alleles are described in Bar-Nur et al., Nat Methods. 2014.

11(11): p. 1170-6. All animal work was approved by Harvard Medical School, Boston Children's Hospital, Mass Eye and Ear Institutional animal care and use committees.

(323) Production of AAVs

(324) Vectors of AAV−TRE−OSK were made by cloning mouse Oct4, Sox2 and Klf4 cDNA into an AAV plasmid consisting of the a Tet Response Element (TRE3G promoter) and SV40 element. The other vectors were directly chemically synthesized. All pAAVs, as listed in Table 6, were then packaged into AAVs of serotype 2/2 or 2/9 (titers: >5×10.sup.12 genome copies per milliliter). Adeno associated viruses were produced by Boston Children's Hospital Viral Core.

(325) Systemical Delivery of AAV9 to Internal Organs

(326) Expression in internal organs was achieved through retro-orbital injection of AAV9 (3×10.sup.11 TRE−OSK plus 7×10.sup.11 UBC−rtTA4). 1 mg/mL doxycycline was treated 3 weeks post injection continuously to induce OSK expression.

(327) Cell Culture and Differentiation

(328) Ear fibroblasts (EFs) were isolated from Reprogramming 4F (Jackson Laboratory 011011) or 3F (Hochedlinger lab) mice and cultured at 37° C. in DMEM (Invitrogen) containing Gluta-MAX, non-essential amino acids, and 10% fetal bovine serum (FBS). EFs of WT 4F and WT 3F mice were passaged to P3 and treated with doxycycline (2 mg/ml) for the indicated time periods in the culture medium.

(329) SH-SY5Y neuroblastoma cells were obtained from the American Tissue Culture Collection (ATCC, CRL-2266) and maintained according to ATCC recommendations. Basically, the cells were cultured in a 1:1 mixture of Eagle's Minimum Essential Medium (EMEM, ATCC, 30-2003) and F12 medium (ThermoFisher Scientific, 11765054), supplemented with 10% fetal bovine serum (FBS, Sigma, F0926) and 1× penicillin/streptomycin (ThermoFisher Scientific, 15140122). Cells were cultured at 37° C. with 5% CO.sub.2 and 3% O.sub.2. Cells were passaged when reaching ˜80% confluency.

(330) SH-SY5Y cells were differentiated into neurons as previously describedl,2, with some modifications. Briefly, 1 day after plating, cells started to be differentiated in EMEM/F12 medium (1:1) containing 2.5% FBS, lx penicillin/streptomycin, and 10 μM all-trans retinoic acid (ATRA, Stemcell Technologies, 72264) (Differentiation Medium 1) for 3 days, followed by treating the cells in EMEM/F12 (1:1) containing 1% FBS, 1×penicillin/streptomycin, and 10 μM ATRA (Differentiation Medium 2) for 3 days. Cells were then splitted into 35 mm cell culture plates coated with poly-D-lysine (ThermoFisher Scientific, A3890401). 1 day after splitting, neurons were matured in serum-free neurobasal/B27 μlus culture medium (ThermoFisher Scientific, A3653401) containing 1×Glutamax (ThermoFisher Scientific, 35050061), 1×penicillin/streptomycin, and 50 ng/ml BDNF (Alomone labs) (Differentiation Medium 3) for at least 5 days.

(331) Neurite Regeneration Assay

(332) The differentiated neurons from SH-SY5Y cells were transduced with AAV.DJ vectors at 10.sup.6 genome copy per cell. 5 days after transduction, 100 nM vincristine (Sigma, V8879) was added to the cells for 24 hours to induce neurite degeneration. After vincristine treatment, neurons were washed in PBS twice and fresh Differentiation medium 3 was added back to the plates. Neurons were followed for neurite outgrowth for 2-3 weeks. Phase-contrast images were taken at 100× magnification every three to four days. Neurite area was quantified using Image J.

(333) Cell Cycle Analysis

(334) Cells were harvested and fixed with 70% cold ethanol for 16 hours at 4° C. After fixation, cells were washed twice with PBS, followed by incubation with PBS containing 50 g/mL propidium iodide (Biotium, 40017) and 100 μg/mL RNase A (Omega) for 1 hour at room temperature. PI stained samples were analyzed on BD LSR II analyzer, and only single cells were gated for analysis. Cell cycle profiles were analyzed using FCS Express 6 (De Novo Software).

(335) Human Neuron Methylation Studies and Epigenetic Clock

(336) DNA was extracted from cells using the Zymo Quick DNA mini-prep plus kit (D4069) according to the manufacturer's instructions and DNA methylation levels were measured on Illumina 850 EPIC arrays according to the manufacturer's instructions. The Illumina BeadChip (EPIC) measures bisulfite-conversion-based, single-CpG resolution DNAm levels at different CpG sites in the human genome. These data were generated by following the standard protocol of Illumina methylation assays, which quantifies methylation levels by the β value using the ratio of intensities between methylated and un-methylated alleles. Specifically, the β value is calculated from the intensity of the methylated (M corresponding to signal A) and un-methylated (U corresponding to signal B) alleles, as the ratio of fluorescent signals $\beta = Max(M,0)/[Max(M,0)+Max(U,0)+100]$. Thus, β values range from 0 (completely un-methylated) to 1 (com-pletely methylated). We used the "noob" normalization method, which is implemented in the "minfi" R package (Triche et al., NAR 2013, Fortin et al., Bioinformatics 2017). The mathematical algorithm and available software underlying the skin & blood clock (based on 391 CpGs) is presented in Horvath et al., Aging 2018.
(337) AAV2 Virus Intravitreal Injection
(338) For intravitreal injection, adult animals were anesthetized with ketamine/xylazine (100/10 mg/kg) and then AAV (1-3 μl) was injected intravitreally, just posterior to the limbus-parallel conjunctival vessels, with a fine glass pipette attached to the Hamilton syringe using plastic tubing. In elevated IOP model, mice received a 1 μl intravitreal injection between 3-4 weeks post microbead injection.
(339) Optic Nerve Crush
(340) For optic nerve crush in anesthetized animals, the optic nerve was accessed intraorbitally and crushed using a pair of Dumont #5 forceps (FST), two weeks after AAV injection. Alexa-conjugated cholera toxin beta subunit (CTB-555, 1 mg/ml; 1-2 μl) injection was performed 2-3 days before euthanasia to trace regenerating RGC axons. More detailed surgical methods were described by Park et al., Science, 2008. 322(5903): p. 963-6.
(341) In Vivo Doxycycline Induction or Suppression
(342) Induction of Tet-On system or suppression of Tet-Off system in the retina were performed by administration of doxycycline hyclate(2 mg/ml) (Sigma) in the drinking water. Induction of Tet-On system in the whole body were performed by administration of doxycycline (1 mg/ml) (USP grade, MP Biomedicals 0219895505) in the drinking water.
(343) Axon Regeneration Quantification
(344) Number of regenerating axons in the optic nerve was estimated by counting the number of CTB-labeled axons at different distances from the crush site as described previously (Park, K. K., et al., *Promoting axon regeneration in the adult CNS by modulation of the PTEN/mTOR pathway*. Science, 2008. 322(5903): p. 963-6).
(345) Whole-Mount Optic Nerve Preparation
(346) Optic nerves and connecting chiasm were dehydrated in methanol for 5 min, then incubated overnight with Visikol® HISTO-1™. Next day nerves were transferred to Visikol® HISTO-2™ and then incubated for 3 hr. Finally, optic nerves and connecting chiasm were mounted with Visikol® HISTO-2™.
(347) Immunofluorescence
(348) Whole-mount retinas were blocked with horse serum 4° C. overnight then incubated at 4° C. for 3 days with primary antibodies: Mouse anti-Oct4 (1:100, BD bioscience, 611203), Rabbit anti-Sox2 (1:100, Cell signaling, 14962), Goat anti-Klf4 (1:100, R&D system, AF3158), Rabbit anti-Brn3a (1:200, EMD Millipore, MAB1585), and Guinea pig anti-RBPMS (1:400, Raygene custom order A008712 to peptide GGKAEKENTPSEANLQEEEVRC) diluted in PBS, BSA (3%) Triton X-100 (0.5%). Then, tissues were incubated at 4° C. overnight with appropriate Alexa Fluor conjugate secondary antibodies (Alexa 405, 488, 567, 674; Invitrogen) diluted with the same blocking solution as the primary antibodies, generally used at 1:400 final dilution. For section

staining, primary overnight at 4° C. and then secondary at room temperature for 2 h. Sections or whole-mount retinas were mounted with VECTASHIELD Antifade Mounting Medium.
(349) Western Blot
(350) SDS-PAGE and western blot analysis was performed according to standard procedures and detected with the ECL detection kit. Antibody used: Rabbit anti-Sox2 (1:100, EMD Millipore, AB5603), Mouse anti-Klf4 (1:1000, ReproCell, 09-0021), Rabbit anti-p-S6 (S240/244) (1:1000, CST, 2215), Mouse anti-S6 (1:1000, CST, 2317), Mouse anti-β-Tubulin (1:1000, Sigma-Aldrich, 05-661), Mouse anti-β-Actin-Peroxidase antibody (1:20,000, Sigma-Aldrich, A3854).
(351) RGCs Survival and Phospho-S6 Signal
(352) RBPMS-positive cells in the ganglion layer were counted using a fluorescent microscope after immunostaining whole-mount retinas with anti-RBPMs antibodies. A total of four random fields per retina were enumerated. The average number per field was determined, and the percentages of viable RGCs were obtained by comparing the values determined from the uninjured contralateral retinas. In the same condition, after phospho-S6 staining, the densities of phopsho-S6-positive RGCs were obtained by comparing the value from the uninjured contralateral retinas.
(353) RGC Enrichment
(354) Retinas were fresh dissected and dissociated in AMES media using papain, then triturated carefully and stained with Thy1.2-PE-Cy7 anti-body (Invitrogen 25-0902-81) and Calcine Blue live-dead cell stain, then flow sorted on a BD FACS Aria using an 130 μm nozzle to collect over 10,000 Thy1.2+ and Clacine blue+ cells (1-2% of total events). Freezed cells were sent to Genewiz for RNA extraction and ultra low input RNA-seq sequencing, or to Zymo research for DNA extraction and ultra low input RRBS sequencing.
(355) Classic RRBS Library Preparation
(356) DNA was extracted using Quick-DNA Plus Kit Microprep Kit. 2-10 ng of starting input genomic DNA was digested with 30 units of MspI (NEB). Fragments were ligated to pre-annealed adapters containing 5′-methyl-cytosine instead of cytosine according to Illumina's specified guidelines. Adaptor-ligated fragments ≥50 bp in size were recovered using the DNA Clean & Concentrator™-5 (Cat #: D4003). The fragments were then bisulfite-treated using the EZ DNA Methylation-Lightning™ Kit (Cat #: D5030). Preparative-scale PCR was performed and the resulting products were purified with DNA Clean & Concentrator™-5 (Cat #: D4003) for sequencing on an Illumina HiSeq using 2×125 bp PE.
(357) DNA Methylation Age Analysis of Mouse RGC
(358) Reads were filtered using trim galore v0.4.1 and mapped to the mouse genome GRCm38 using Bismark v0.15.0. Methylation counts on both positions of each CpG site were combined. Only CpG sites covered in all samples were considered for analysis. This resulted in total of 708156 sites. For the rDNA methylation clock reads were mapped to BK000964 and the coordinates were adjusted accordingly (Wang et al., Genome Res 29, 325-333, doi:10.1101/gr.241745.118 (2019)). 70/72 sites were covered for rDNA clock, compared to 102/435 sites of whole lifespan multi-tissue clock (Meer et al., Elife 7, doi:10.7554/eLife.40675 (2018)), or 248/582 and 77,342/193,651 sites (ridge) of two entire lifespan multi-tissue clocks (Thompson et al., Aging (Albany NY) 10, 2832-2854, doi:10.18632/aging.101590 (2018)).
(359) Microbead-Induced Mouse Model of Elevated IOP
(360) Mice were anesthetized by intraperitoneal injection of a mixture of ketamine (100 mg/kg; Ketaset; Fort Dodge Animal Health, Fort Dodge, IA) and xylazine (9 mg/kg; TranquiVed; Vedco, Inc., St. Joseph, MO) supplemented by topical application of proparacaine (0.5%; Bausch & Lomb, Tampa, FL). Elevation of IOP was induced unilaterally by injection of polystyrene microbeads (FluoSpheres; Invitrogen, Carlsbad, CA; 15-μm diameter) to the anterior chamber of the right eye of each animal under a surgical microscope, as previously reported (Krishnan et al., J Immunol, 2016. 197(12): p. 4626-4638). Briefly, microbeads were prepared at a concentration of 5.0×10.sup.6 beads/mL in sterile physiologic saline. The right cornea was gently punctured near the

center using a sharp glass micropipette (World Precision Instruments Inc., Sarasota, FL). A 2 μL volume of microbeads was injected through the preformed hole into the anterior chamber followed by injection of an air bubble via the micropipette connected with a Hamilton syringe. Any mice that developed signs of inflammation (clouding of the cornea, edematous cornea etc) were excluded from the study.

(361) IOP (Intraocular Pressure) Measurements

(362) IOPs were measured with a rebound TonoLab tonometer (Colonial Medical Supply, Espoo, Finland), as previously described (Krishnan et al., J Immunol, 2016. 197(12): p. 4626-4638; Mukai et al., PLoS One, 2019. 14(1): p. e0208713). Mice were anesthetized by 3% isoflurane in 100% oxygen (induction) followed by 1.5% isoflurane in 100% oxygen (maintenance) delivered with a precision vaporizer. IOP measurement was initiated within 2 to 3 min after the loss of a toe pinch reflex or tail pinch response. Anesthetized mice were placed on a platform and the tip of the pressure sensor was placed approximately ⅛ inch from the central cornea. Average IOP was displayed automatically after 6 measurements after elimination of the highest and lowest values. The machine-generated mean was considered as one reading, and six readings were obtained for each eye. All IOPs were taken at the same time of day (between 10:00 and 12:00 hours) due to the variation of IOP throughout the day.

(363) Optomotor Response

(364) Visual acuity of mice was measured using an optomotor re-flex-based spatial frequency threshold test (Gao et al., Am J Pathol, 2016. 186(4): p. 985-1005; Sun et al., Glia, 2013. 61(8): p. 1218-1235). Mice would be able to freely move and were placed on a pedestal located in the center of an area formed by four computer monitors arranged in a quadrangle. The monitors displayed a moving vertical black and white sinusoidal grating pattern. A blinded observer, unable to see the direction of the moving bars, monitored the tracking behavior of the mouse. Tracking was considered positive when there was a movement of the head (motor response) to the direction of the bars or rotation of the body in the direction concordant with the stimulus. Each eye would be tested separately depending on the direction of rotation of the grating. The staircase method was used to determine the spatial frequency start from 0.15 to 0.40 cycles/deg, the interval is 0.05 cycles/deg. Rotation speed (12°/s) and contrast (100%) were kept constant. Responses were measured before and after treatment by individuals blinded to the group of the animal and the treatment.

(365) Pattern Electroretinogram (pERG)

(366) Mice were anesthetized with ketamine/xylazine (100 mg/kg and 20 mg/kg) and the pupils dilated with one drop of 1% tropicamide ophthalmic solution. The mice were placed on a built-in warming plate (Celeris, Full-Field and Pattern Stimulation for the rodent model), that maintained the body temperature at 37 C and kept under dim red light throughout the procedure. The visual stimuli of a black and white reversing checkerboard pattern with a check size of 1° was displayed on light guide electrode-stimulators placed directly on the ocular surface of both eyes and centered with the pupil. The visual stimuli were presented at 98% contrast and constant mean luminance of 50 cd/m.sup.2, spatial frequency:0.05 cyc/deg; temporal frequency:1 Hz. A total of 300 complete contrast reversals of pERG were repeated twice in each eye and the 600 cycles were segmented and averaged and recorded. The averaged PERGs were analyzed to evaluate the peak to trough N1 to P1 (positive wave) amplitude.

(367) Quantification of Optic Nerve Axons

(368) For quantification of axons, optic nerves were dissected and fixed overnight in Karnovsky's reagent (50% in phosphate buffer). Semi-thin cross-sections of the nerve were taken at 1.0 mm posterior to the globe and stained with 1% p-phenylenediamine (PPD) for evaluation by light microscopy. Optic nerve cross sections were imaged at 60× magnification using a Nikon microscope (Eclipse E800, Nikon, Japan) with the DPController software (Olympus, Japan) and 6-8 non-overlapping photomicrographs were taken to cover the entire area of each optic nerve cross-

section. Using ImageJ (Version 2.0.0-rc-65/1.51u), a 100 μM×100 μM square was placed on each 60× image and all axons within the square (0.01 mm.sup.2) were counted using the threshold and analyze particles function in image J as previously described (Krishnan et al., J Immunol, 2016. 197(12): p. 4626-4638; Mukai et al., PLoS One, 2019. 14(1): p. e0208713; Gao et al., Am J Pathol, 2016. 186(4): p. 985-1005). Damaged axons stain darkly with PPD and are not counted. The average axon counts in the 6-8 images were used to calculate the axon density per square millimeter of optic nerve. Individuals blinded to the experimental groups performed all axon counts.

(369) Quantification of Retinal Ganglion Cells

(370) For ganglion cell counting, images of whole mount retinas were acquired using a 63× oil immersion objective of the Leica TCS SP5 confocal microscope (Leica Microsystems). The retinal whole mount was divided into four quadrants and three to four images (248.53 m by 248.53 m in size) were taken from the midperipheral and peripheral regions of each quadrant, for a total of twelve to sixteen images per retina. were taken from the midperipheral and peripheral regions (4 images per quadrant). The images were obtained as z-stacks (0.5 m) and all Brn3a positive cells in the ganglion cell layer of each image were counted manually as previously described (Gao et al., Am J Pathol, 2016. 186(4): p. 985-1005). Briefly, RGCs were counted using the “Cell Counter” plugin (fiji.sc/Cell_Counter) in Fiji is Just ImageJ software (ImageJ Fiji,version 2.0.0-rc-69/1.52n). Each image was loaded into Fiji and a color counter type was chosen to mark all Brn3a stained RGCs within each image (0.025 mm.sup.2). The average number of RGCs in the 12 to sixteen images were used to calculate the RGC density per square millimeter of retina. Two individuals blinded to the experimental groups performed all RGC counts.

(371) Total RNA Extraction and Sample QC

(372) Total RNA was extracted following the Trizol Reagent User Guide (Thermo Fisher Scientific). 1 ul 10 mg/ml Glycogen was added to the supernatant to increase RNA recovery. RNA was quantified using Qubit 2.0 Fluorometer (Life Technologies, Carlsbad, CA, USA) and RNA integrity was checked with TapeStation (Agilent Technologies, Palo Alto, CA, USA) to see if the concentration met the requirements.

(373) Ultra-Low Input RNA Library Preparation and Multiplexing

(374) RNA samples were quantified using Qubit 2.0 Fluorometer (Life Technologies, Carlsbad, CA, USA) and RNA integrity was checked with 2100 TapeStation (Agilent Technologies, Palo Alto, CA, USA). RNA library preparations, sequencing reactions, and initial bioinformatics analysis were conducted at GENEWIZ, LLC. (South Plainfield, NJ, USA). SMART-Seq v4 Ultra Low Input Kit for Sequencing was used for full-length cDNA synthesis and amplification (Clontech, Mountain View, CA), and Illumina Nextera XT library was used for sequencing library preparation. Briefly, cDNA was fragmented and adaptor was added using Transposase, followed by limited-cycle PCR to enrich and add index to the cDNA fragments. The final library was assessed with Qubit 2.0 Fluorometer and Agilent TapeStation.

(375) Sequencing 2×150 bp PE

(376) The sequencing libraries were multiplexed and clustered on two lanes of a flowcell. After clustering, the flowcell were loaded on the Illumina HiSeq instrument according to manufacturer's instructions. The samples were sequenced using a 2×150 Paired End (PE) configuration. Image analysis and base calling were conducted by the HiSeq Control Software (HCS) on the HiSeq instrument. Raw sequence data (.bcl files) generated from Illumina HiSeq were be converted into fastq files and de-multiplexed using Illumina bcl2fastq v. 2.17 program. One mis-match was allowed for index sequence identification.

(377) RNA-seq Analysis

(378) Paired-end reads were aligned with hisat2 v2.1.0 to the Ensembl GRCm38 primary assembly using splice junctions from the Ensembl release 84 annotation. Paired read counts were quantified using featureCounts v1.6.4 using reads with a MAPQ>=20. Diffentially-expressed genes for each

pairwise comparison were identified with edgeR v3.26, testing only genes with at least 0.1 counts-per-million (CPM) in at least three samples. Gene ontology analysis of differentially-expressed genes was performed with AmiGO v2.5.12. Age-associated sensory perception genes were extracted from the mouse Sensory Perception (GO:0007600) category the Gene Ontology database, including genes that were differentially expressed (q<=0.05) in 12 versus 5 month old mice, excluding genes that were induced by the Control virus alone (q<=0.1).

(379) TABLE-US-00006 TABLE 6 AAV vectors used in Example 16 Vector qPCR Primer for measuring titer Source pAAV-TRE-Oct4 TRE3G Disclosed herein pAAV-TRE-Sox2 TRE3G Disclosed herein pAAV-TRE-Klf4 TRE3G Disclosed herein pAAV-TRE-Oct4-Sox2 TRE3G Disclosed herein pAAV-TRE-OSK TRE3G Disclosed herein pAAV-TRE-d2EGFP TRE3G Disclosed herein pAAV-CMV-rtTAV16 WPRE Disclosed herein pAAV-CAG-tTA hGH Disclosed herein pAAV-sh-Scr-YFP WPRE Plasmid #85741 pAAV-Sh-Tet1-YFP WPRE Plasmid #85742 pAAV-sh-Tet2-YFP WPRE Plasmid #85743

(380) TABLE-US-00007 TABLE 7 Primers SEQ Primer name Sequence ID NO: TRE3G F AACGTATCTACAGTTTACTCCCTATC 53 TRE3G R GGTAGGAAGTGGTACGGAAAG 54 WPRE F CACTGACAATTCCGTGGTGT 55 WPRE R GAGATCCGACTCGTCTGAGG 56 hGH F TGGGAAGACAACCTGTAGGG 57 hGH R TGAAACCCCGTCTCTACCAA 58

(381) TABLE-US-00008 TABLE 8 Primers used for RT-PCR Gene Primer sequence SEQ ID NO: mOct4 F ACA TCG CCA ATC AGC TTG G 59 mOct4 R AGA ACC ATA CTC GAA CCA CAT CC 60 mSox2 F ACA GAT GCA ACC GAT GCA CC 61 mSox2 R TGG AGT TGT ACT GCA GGG CG 62 mKlf4 F GTGCCCCGACTAACCGTTG 63 mKlf4 R GTCGTTGAACTCCTCGGTCT 64 mMyc F ATGCCCCTCAACGTGAACTTC 65 mMyc R CGCAACATAGGATGGAGAGCA 66 mHist1 h2a F GCG ACA ACA AGA AGA CGC GCA T 67 mHist1 h2a R CTG GAT GTT GGG CAG GAC GCC 68 mHist1 h2b F AAG AAG GAC GGC AAG AAG CGC A 69 mHist1 h2b R CGC TCG AAG ATG TCG TTC ACG A 70 mHIST1 H3.1/H3.2 F GAA GAA GCC TCA CCG CTA CCG 71 mHIST1 H3.1/H3.2 R GGT TGG TGT CCT CAA ACA GAC CC 72 mHist1 h4 F AAC ATC CAG GGC ATC ACC AAG C 73 mHist1 h4 R GTT CTC CAG GAA CAC CTT CAG C 74 mLmnb1 F CCG GCC TCA AGG CTC TCT A 75 mLmnb1 R TGC CGC CTC ATA CTC TCG AA 76 mActb F AGT GTG ACG TTG ACA TCC GT 77 mActb R TGC TAG GAG CCA GAG CAG TA 78 mNanog F TCTTCCTGGTCCCCACAGTTT 79 mNanog R GCAAGAATAGTTCTCGGGATGAA 80 mChaf1a R GTG TCT TCC TCA ACT TTC TCC TTG G 81 mChaf1a F CGC GGA CAG CCG CGG CCG TGG ATT GC 82 mChaf1b R GGC TCC TTG CTG TCA TTC ATC TTC CAC 83 mChaf1b F CAC CGC CGT CAG GAT CTG GAA GTT GG 84 mLmnb1 F CCG GCC TCA AGG CTC TCT A 85 mLmnb1 R TGC CGC CTC ATA CTC TCG AA 86 mTet1 F TCAAGCAATGGACCACTGGG 87 mTet1 R TCTCCATGAGCTCCCTGACA 88 mTet2 F ACT CCT GGT GAA CAA AGT CAG A 89 mTet2 R CAT CCC TGA GAG CTC TTG CC 90 mGAPDHF CCA ATG TGT CCG TCG TGG ATC T 91 mGAPDHR GTT GAA GTC GCA GGA GAC AAC C 92 mp16 (Cdkn2a) F ACA TCA AGA CAT CGT GCG ATA TT 93 mp16 (Cdkn2a) R CCA GCG GTA CAC AAA GAC CA 94 mApob F AAG CAC CTC CGA AAG TAC GTG 95 mApob R CTC CAG CTC TAC CTT ACA GTT GA 96 hTet2 F GATAGAACCAACCATGTTGAGGG 97 hTet2 R TGGAGCTTTGTAGCCAGAGGT 98 hActb F CACCATTGGCAATGAGCGGTTC 99 hActb R AGGTCTTTGCGGATGTCCACGT 100

Example 17. Non-Limiting Examples of Sequences

(382) TABLE-US-00009 Nucleotide sequence encoding *Mus Musculus* OCT4 (no

stop codon) (SEQ ID NO: 1):
ATGGCTGGACACCTGGCTTCAGACTTCGCCTTCTCACCCCCACCAGGTGGGGGTG
ATGGGTCAGCAGGGCTGGAGCCGGGCTGGGTGGATCCTCGAACCTGGCTAAGCT
TCCAAGGGCCTCCAGGTGGGCCTGGAATCGGACCAGGCTCAGAGGTATTGGGGA
TCTCCCCATGTCCGCCCGCATACGAGTTCTGCGGAGGGATGGCATACTGTGGACC
TCAGGTTGGACTGGGCCTAGTCCCCCAAGTTGGCGTGGAGACTTTGCAGCCTGAG
GGCCAGGCAGGAGCACGAGTGGAAAGCAACTCAGAGGGAACCTCCTCTGAGCCC
TGTGCCGACCGCCCCAATGCCGTGAAGTTGGAGAAGGTGGAACCAACTCCCGAG
GAGTCCCAGGACATGAAAGCCCTGCAGAAGGAGCTAGAACAGTTTGCCAAGCTG
CTGAAGCAGAAGAGGATCACCTTGGGGTACACCCAGGCCGACGTGGGGCTCACC
CTGGGCGTTCTCTTTGGAAAGGTGTTCAGCCAGACCACCATCTGTCGCTTCGAGG
CCTTGCAGCTCAGCCTTAAGAACATGTGTAAGCTGCGGCCCCTGCTGGAGAAGTG
GGTGGAGGAAGCCGACAACAATGAGAACCTTCAGGAGATATGCAAATCGGAGA
CCCTGGTGCAGGCCCGGAAGAGAAAGCGAACTAGCATTGAGAACCGTGTGAGGT
GGAGTCTGGAGACCATGTTTCTGAAGTGCCCGAAGCCCTCCCTACAGCAGATCAC
TCACATCGCCAATCAGCTTGGGCTAGAGAAGGATGTGGTTCGAGTATGGTTCTGT
AACCGGCGCCAGAAGGGCAAAAGATCAAGTATTGAGTATTCCCAACGAGAAGAG
TATGAGGCTACAGGGACACCTTTCCCAGGGGGGGCTGTATCCTTTCCTCTGCCCC
CAGGTCCCCACTTTGGCACCCCAGGCTATGGAAGCCCCCACTTCACCACACTCTA
CTCAGTCCCTTTTCCTGAGGGCGAGGCCTTTCCCTCTGTTCCCGTCACTGCTCTGG
GCTCTCCCATGCATTCAAAC Amino acid sequence encoding *Mus Musculus* OCT 4 (SEQ ID NO: 2):
MAGHLASDFAFSPPPGGGDGSAGLEPGWVDPRTWLSFQGPPGGPGIGPGSEVLGISP
CPPAYEFCGGMAYCGPQVGLGLVPQVGVETLQPEGQAGARVESNSEGTSSEPCADR
PNAVKLEKVEPTPEESQDMKALQKELEQFAKLLKQKRITLGYTQADVGLTLGVLFG
KVFSQTTICRFEALQLSLKNMCKLRPLLEKWVEEADNNENLQEICKSETLVQARKRK
RTSIENRVRWSLETMELKCPKPSLQQITHIANQLGLEKDVVRVWFCNRRQKGKRSSI
EYSQREEYEATGTPFPGGAVSFPLPPGPHFGTPGYGSPHFTTLYSVPFPEGEAFPSVPV
TALGSPMHSN Nucleotide sequence encoding *Mus Musculus* SOX2 (no stop codon) (SEQ ID NO: 3):
ATGTATAACATGATGGAGACGGAGCTGAAGCCGCCGGGCCCGCAGCAAGCTTCG
GGGGGCGGCGGCGGAGGAGGCAACGCCACGGCGGCGGCGACCGGCGGCAACCA
GAAGAACAGCCCGGACCGCGTCAAGAGGCCCATGAACGCCTTCATGGTATGGTC
CCGGGGGCAGCGGCGTAAGATGGCCCAGGAGAACCCCAAGATGCACAACTCGG
AGATCAGCAAGCGCCTGGGCGCGGAGTGGAAACTTTTGTCCGAGACCGAGAAGC
GGCCGTTCATCGACGAGGCCAAGCGGCTGCGCGCTCTGCACATGAAGGAGCACC
CGGATTATAAATACCGGCCGCGGCGGAAAACCAAGACGCTCATGAAGAAGGATA
AGTACACGCTTCCCGGAGGCTTGCTGGCCCCCGGCGGGAACAGCATGGCGAGCG
GGGTTGGGGTGGGCGCCGGCCTGGGTGCGGGCGTGAACCAGCGCATGGACAGCT
ACGCGCACATGAACGGCTGGAGCAACGGCAGCTACAGCATGATGCAGGAGCAG
CTGGGCTACCCGCAGCACCCGGGCCTCAACGCTCACGGCGCGGCACAGATGCAA
CCGATGCACCGCTACGACGTCAGCGCCCTGCAGTACAACTCCATGACCAGCTCGC
AGACCTACATGAACGGCTCGCCCACCTACAGCATGTCCTACTCGCAGCAGGGCA
CCCCCGGTATGGCGCTGGGCTCCATGGGCTCTGTGGTCAAGTCCGAGGCCAGCTC
CAGCCCCCCCGTGGTTACCTCTTCCTCCCACTCCAGGGCGCCCTGCCAGGCCGGG
GACCTCCGGGACATGATCAGCATGTACCTCCCCGGCGCCGAGGTGCCGGAGCCC
GCTGCGCCCAGTAGACTGCACATGGCCCAGCACTACCAGAGCGGCCCGGTGCCC
GGCACGGCCATTAACGGCACACTGCCCCTGTCGCACATG Amino acid sequence encoding *Mus Musculus* SOX2 (translated) (SEQ ID NO: 4)
MYNMMETELKPPGPQQASGGGGGGGNATAAATGGNQKNSPDRVKRPMNAFMVW

SRGQRRKMAQENPKMHNSEISKRIGAEWKLLSETEKRPFIDEAKRLRALHMKEHPD YKYRPRRKTKTLMKKDKYTLPGGLLAPGGNSMASGVGVGAGLGAGVNQRMDSYA HMNGWSNGSYSMMQEQLGYPQHPGLNAHGAAQMQPMHRYDVSALQYNSMTSSQ TYMNGSPTYSMSYSQQGTPGMALGSMGSVVKSEASSSPPVVTSSSHSRAPCQAGDL RDMISMYLPGAEVPEPAAPSRLHMAQHYQSGPVPGTAINGTLPLSHM Nucleotide sequence encoding *Mus Musculus* KLF4 (no stop codon) (SEQ ID NO: 5): ATGAGGCAGCCACCTGGCGAGTCTGACATGGCTGTCAGCGACGCTCTGCTCCCGT CCTTCTCCACGTTCGCGTCCGGCCCGGCGGGAAGGGAGAAGACACTGCGTCCAG CAGGTGCCCCGACTAACCGTTGGCGTGAGGAACTCTCTCACATGAAGCGACTTCC CCCACTTCCCGGCCGCCCCTACGACCTGGCGGCGACGGTGGCCACAGACCTGGA GAGTGGCGGAGCTGGTGCAGCTTGCAGCAGTAACAACCCGGCCCTCCTAGCCCG GAGGGAGACCGAGGAGTTCAACGACCTCCTGGACCTAGACTTTATCCTTTCCAAC TCGCTAACCCACCAGGAATCGGTGGCCGCCACCGTGACCACCTCGGCGTCAGCTT CATCCTCGTCTTCCCCAGCGAGCAGCGGCCCTGCCAGCGCGCCCTCCACCTGCAG CTTCAGCTATCCGATCCGGGCCGGGGGTGACCCGGGCGTGGCTGCCAGCAACAC AGGTGGAGGGCTCCTCTACAGCCGAGAATCTGCGCCACCTCCCACGGCCCCCTTC AACCTGGCGGACATCAATGACGTGAGCCCCTCGGGCGGCTTCGTGGCTGAGCTC CTGCGGCCGGAGTTGGACCCAGTATACATTCCGCCACAGCAGCCTCAGCCGCCA GGTGGCGGGCTGATGGGCAAGTTTGTGCTGAAGGCGTCTCTGACCACCCCTGGCA GCGAGTACAGCAGCCCTTCGGTCATCAGTGTTAGCAAAGGAAGCCCAGACGGCA GCCACCCCGTGGTAGTGGCGCCCTACAGCGGTGGCCCGCCGCGCATGTGCCCCA AGATTAAGCAAGAGGCGGTCCCGTCCTGCACGGTCAGCCGGTCCCTAGAGGCCC ATTTGAGCGCTGGACCCCAGCTCAGCAACGGCCACCGGCCCAACACACACGACT TCCCCCTGGGGCGGCAGCTCCCCACCAGGACTACCCCTACACTGAGTCCCGAGG AACTGCTGAACAGCAGGGACTGTCACCCTGGCCTGCCTCTTCCCCCAGGATTCCA TCCCCATCCGGGGCCCAACTACCCTCCTTTCCTGCCAGACCAGATGCAGTCACAA GTCCCCTCTCTCCATTATCAAGAGCTCATGCCACCGGGTTCCTGCCTGCCAGAGG AGCCCAAGCCAAAGAGGGGAAGAAGGTCGTGGCCCCGGAAAAGAACAGCCACC CACACTTGIGACTATGCAGGCTGTGGCAAAACCTATACCAAGAGTTCTCATCTCA AGGCACACCTGCGAACTCACACAGGCGAGAAACCTTACCACTGTGACTGGGACG GCTGTGGGTGGAAATTCGCCCGCTCCGATGAACTGACCAGGCACTACCGCAAAC ACACAGGGCACCGGCCCTTTCAGTGCCAGAAGTGCGACAGGGCCTTTTCCAGGT CGGACCACCTTGCCTTACACATGAAGAGGCAC Amino acid sequence encoding *Mus Musculus* KLF4 (translated) (SEQ ID NO: 6): MRQPPGESDMAVSDALLPSFSTFASGPAGREKTLRPAGAPTNRWREELSHMKRLPPL PGRPYDLAATVATDLESGGAGAACSSNNPALLARRETEEFNDLLDLDFILSNSLTHQE SVAATVTTSASASSSSSPASSGPASAPSTCSFSYPIRAGGDPGVAASNTGGGLLYSRES APPPTAPFNLADINDVSPSGGFVAELLRPELDPVYIPPQQPQPPGGGLMGKFVLKASL TTPGSEYSSPSVISVSKGSPDGSHPVVVAPYSGGPPRMCPKIKQEAVPSCTVSRSLEAH LSAGPQLSNGHRPNTHDFPLGRQLPTRTTPTLSPEELLNSRDCHPGLPLPPGFHPHPGP NYPPFLPDQMQSQVPSLHYQELMPPGSCLPEEPKPKRGRRSWPRKRTATHTCDYAG CGKTYTKSSHLKAHLRTHTGEKPYHCDWDGCGWKFARSDELTRHYRKHTGHRPFQ CQKCDRAFSRSDHLALHMKRH TRE3G promoter sequence (non-limiting example of a TRE promoter) (SEQ ID NO: 7): TTTACTCCCTATCAGTGATAGAGAACGTATGAAGAGTTTACTCCCTATCAGTGAT AGAGAACGTATGCAGACTTTACTCCCTATCAGTGATAGAGAACGTATAAGGAGT TTACTCCCTATCAGTGATAGAGAACGTATGACCAGTTTACTCCCTATCAGTGATA GAGAACGTATCTACAGTTTACTCCCTATCAGTGATAGAGAACGTATATCCAGTTT ACTCCCTATCAGTGATAGAGAACGTATAAGCTTTAGGCGTGTACGGTGGGCGCCT ATAAAAGCAGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGCAATTCCACAA

CACTTTTGTCTTATACCAACTTTCCGTACCACTTCCTACCCTCGTAAA SV40-derived terminator sequence (SEQ ID NO: 8): TGCGCGCAGCGGCCGACCATGGCCCAACTTGTTTATTGCAGCTTATAATGGTTAC AAATAAAGCAATAGCATCACAAATTTCACAAATAAAGCATTTTTTTCACTGCATT CTAGTIGTGGTTTGTCCAAACTCATCAATGTATCTTATCATGTCTGGATCTCGGTA CCG T2A sequence (SEQ ID NO: 9): GSGEGRGSLLTCGDVEENPGP Nucleotide sequence encoding rtTA3 (with 2 VP16 domain at 3′ end) (SEQ ID NO: 10): ATGTCTAGGCTGGACAAGAGCAAAGTCATAAACGGAGCTCTGGAATTACTCAAT GGTGTCGGTATCGAAGGCCTGACGACAAGGAAACTCGCTCAAAAGCTGGGAGTT GAGCAGCCTACCCTGTACTGGCACGTGAAGAACAAGCGGGCCCTGCTCGATGCC CTGCCAATCGAGATGCTGGACAGGCATCATACCCACTTCTGCCCCCTGGAAGGCG AGTCATGGCAAGACTTTCTGCGGAACAACGCCAAGTCATACCGCTGTGCTCTCCT CTCACATCGCGACGGGGCTAAAGTGCATCTCGGCACCCGCCCAACAGAGAAACA GTACGAAACCCTGGAAAATCAGCTCGCGTTCCTGTGTCAGCAAGGCTTCTCCCTG GAGAACGCACTGTACGCTCTGTCCGCCGTGGGCCACTTTACACTGGGCTGCGTAT TGGAGGAACAGGAGCATCAAGTAGCAAAAGAGGAAAGAGAGACACCTACCACC GATTCTATGCCCCCACTTCTGAGACAAGCAATTGAGCTGTTCGACCGGCAGGGAG CCGAACCTGCCTTCCTTTTCGGCCTGGAACTAATCATATGTGGCCTGGAGAAACA GCTAAAGTGCGAAAGCGGCGGGCCGACCGACGCCCTTGACGATTTTGACTTAGA CATGCTCCCAGCCGATGCCCTTGACGATTTTGACCTTGACATGCTCCCCGGGTAA Amino acid sequence encoding rtTA3 (SEQ ID NO: 11): MSRLDKSKVINGALELLNGVGIEGLTTRKLAQKLGVEQPTLYWHVKNKRALLDALP IEMLDRHHTHFCPLEGESWQDFLRNNAKSYRCALLSHRDGAKVHLGTRPTEKQYET LENQLAFLCQQGFSLENALYALSAVGHFTLGCVLEEQEHQVAKEERETPTTDSMPPL LRQAIELFDRQGAEPAFLFGLELIICGLEKQLKCESGGPTDALDDFDLDMLPADALDD FDLDMLPG Nucleotide sequence encoding rtTA4 (with 3 VP16 domain at 3′ end) (SEQ ID NO: 12): ATGTCCCGCTTGGATAAGAGCAAGGTAATAAATAGCGCACTCGAACTCCTCAAC GGCGTGGGCATCGAAGGTCTGACTACTCGAAAGCTCGCCCAGAAATTGGGTGTG GAGCAACCTACATTGTATTGGCATGTCAAGAACAAAAGAGCCCTGCTGGACGCT CTTCCTATTGAAATGCTTGACAGGCATCACACTCATTCCTGCCCCCTTGAGGTCG AGAGTTGGCAAGATTTTCTCCGAAACAATGCAAAGTCCTACCGCTGCGCACTTTT GTCCCATAGGGATGGAGCAAAAGTGCACCTGGGAACCAGGCCAACAGAGAAAC AATACGAGACTCTCGAGAACCAGTTGGCTTTCTTGTGCCAACAGGGGTTCTCACT TGAAAATGCCCTTTACGCACTGTCAGCCGTTGGACATTTTACCCTGGGGTGCGTT CTTGAGGAGCAAGAACATCAGGTTGCTAAGGAGGAGCGCGAGACTCCAACCACT GATTCTATGCCACCTTTGCTGAAACAGGCCATTGAACTTTTCGATAGACAGGGTG CTGAACCTGCCTTTCTCTTCGGGTTGGAGCTGATTATTTGTGGTCTCGAAAAACA GCTGAAATGTGAAAGTGGTGGCCCTACTGACGCCCTCGATGATTTCGACCTGGAT ATGCTGCCAGCCGATGCACTTGATGATTTCGATTTGGATATGCTTCCAGCCGACG CACTGGACGACTTCGATTTGGACATGCTTCCCGGTTAA Amino acid sequence encoding rtTA4 (SEQ ID NO: 13): MSRLDKSKVINSALELLNGVGIEGLTTRKLAQKLGVEQPTLYWHVKNKRALLDALPI EMLDRHHTHSCPLEVESWQDFLRNNAKSYRCALLSHRDGAKVHLGTRPTEKQYETL ENQLAFLCQQGFSLENALYALSAVGHFTLGCVLEEQEHQVAKEERETPTTDSMPPLL KQAIELFDRQGAEPAFLFGLELIICGLEKQLKCESGGPTDALDDFDLDMLPADALDDF DLDMLPADALDDFDLDMLPG Nucleotide sequence encoding M2-rtTA (SEQ ID NO: 14): ATGCCTTTGTATCATGCTATTGCTTCCCGTATGGCTTTCATTTTCTCCTCCTTGTAT AAATCCTGGTTGCTGTCTCTTTATGAGGAGTTGTGGCCCGTTGTCAGGCAACGTG

GCGTGGTGTGCACTGTGTTTGCTGACGCAACCCCCACTGGTTGGGGCATTGCCAC CACCTGTCAGCTCCTTTCCGGGACTTTCGCTTTCCCCCTCCCTATTGCCACGGCGG AACTCATCGCCGCCTGCCTTGCCCGCTGCTGGACAGGGGCTCGGCTGTTGGGCAC TGACAATTCCGTGGTGTTGTCGGGGAAATCATCGTCCTTTCCTTGGCTGCTCGCCT GTGTTGCCACCTGGATTCTGCGCGGGACGTCCTTCTGCTACGTCCCTTCGGCCCTC AATCCAGCGGACCTTCCTTCCCGCGGCCTGCTGCCGGCTCTGCGGCCTCTTCCGC GTCTTCGCCTTCGCCCTCAGACGAGTCGGATCTCCCTTTGGGCCGCCTCCCCGCAT CGATACCGTCGACCTCGAGACCTAGAAAAACATGGAGCAATCACAAGTAGCAAT ACAGCAGCTACCAATGCTGATTGTGCCTGGCTAGAAGCACAAGAGGAGGAGGAG GTGGGTTTTCCAGTCACACCTCAGGTACCTTTAAGACCAATGACTTACAAGGCAG CTGTAGATCTTAGCCACTTTTTAAAAGAAAAGGGGGGACTGGAAGGGCTAATTC ACTCCCAACGAAGACAAGATATCCTTGATCTGTGGATCTACCACACACAAGGCTA CTTCCCTGATTGGCAGAACTACACACCAGGGCCAGGGATCAGATATCCACTGAC CTTTGGATGGTGCTACAAGCTAGTACCAGTTGAGCAAGAGAAGGTAGAAGAAGC CAATGAAGGAGAGAACACCCGCTTGTTACACCCTGTGAGCCTGCATGGGATGGA TGACCCGGAGAGAGAAGTATTAGAGTGGAGGTTTGACAGCCGCCTAGCATTTCA TCACATGGCCCGAGAGCTGCATCCGGACTGTACTGGGTCTCTCTGGTTAGACCAG ATCTGA Amino acid sequence encoding M2-rtTA (SEQ ID NO: 15): MPLYHAIASRMAFIFSSLYKSWLLSLYEELWPVVRQRGVVCTVFADATPTGWGIATT CQLLSGTFAFPLPIATAELIAACLARCWTGARLLGTDNSVVLSGKSSSFPWLLACVAT WILRGTSFCYVPSALNPADLPSRGLLPALRPLPRLRLRPQTSRISLWAASPHRYRRPR DLEKHGAITSSNTAATNADCAWLEAQEEEEVGFPVTPQVPLRPMTYKAAVDLSHFL KEKGGLEGLIHSQRRQDILDLWIYHTQGYFPDWQNYTPGPGIRYPLTFGWCYKLVPV EQEKVEEANEGENTRLLHPVSLHGMDDPEREVLEWRFDSRLAFHHMARELHPDCTG SLWLDQI Nucleic acid sequence of pAAV-TRE3G-OSK-SV40pA, TRE-OSK-SV40, or TRE3G-OSK-SV40pA vector (SEQ ID NO: 16): TTATGCAGTGCTGCCATAACCATGAGTGATAACACTGCGGCCAACTTACTTCTGA CAACGATCGGAGGACCGAAGGAGCTAACCGCTTTTTTGCACAACATGGGGGATC ATGTAACTCGCCTTGATCGTTGGGAACCGGAGCTGAATGAAGCCATACCAAACG ACGAGCGTGACACCACGATGCCTGTAGTAATGGTAACAACGTTGCGCAAACTAT TAACTGGCGAACTACTTACTCTAGCTTCCCGGCAACAATTAATAGACTGGATGGA GGCGGATAAAGTTGCAGGACCACTTCTGCGCTCGGCCCTTCCGGCTGGCTGGTTT ATTGCTGATAAATCTGGAGCCGGTGAGCGTGGGTCTCGCGGTATCATTGCAGCAC TGGGGCCAGATGGTAAGCCCTCCCGTATCGTAGTTATCTACACGACGGGGAGTCA GGCAACTATGGATGAACGAAATAGACAGATCGCTGAGATAGGTGCCTCACTGAT TAAGCATTGGTAACTGTCAGACCAAGTTTACTCATATATACTTTAGATTGATTTA AAACTTCATTTTTAATTTAAAAGGATCTAGGTGAAGATCCTTTTTGATAATCTCAT GACCAAAATCCCTTAACGTGAGTTTTCGTTCCACTGAGCGTCAGACCCCGTAGAA AAGATCAAAGGATCTTCTTGAGATCCTTTTTTTCTGCGCGTAATCTGCTGCTTGCA AACAAAAAAACCACCGCTACCAGCGGTGGTTTGTTTGCCGGATCAAGAGCTACC AACTCTTTTTCCGAAGGTAACTGGCTTCAGCAGAGCGCAGATACCAAATACTGTC CTTCTAGTGTAGCCGTAGTTAGGCCACCACTTCAAGAACTCTGTAGCACCGCCTA CATACCTCGCTCTGCTAATCCTGTTACCAGTGGCTGCTGCCAGTGGCGATAAGTC GTGTCTTACCGGGTTGGACTCAAGACGATAGTTACCGGATAAGGCGCAGCGGTC GGGCTGAACGGGGGGTTCGTGCACACAGCCCAGCTTGGAGCGAACGACCTACAC CGAACTGAGATACCTACAGCGTGAGCTATGAGAAAGCGCCACGCTTCCCGAAGG GAGAAAGGCGGACAGGTATCCGGTAAGCGGCAGGGTCGGAACAGGAGAGCGCA CGAGGGAGCTTCCAGGGGGAAACGCCTGGTATCTTTATAGTCCTGTCGGGTTTCG CCACCTCTGACTTGAGCGTCGATTTTTGTGATGCTCGTCAGGGGGGCGGAGCCTA TGGAAAAACGCCAGCAACGCGGCCTTTTTACGGTTCCTGGCCTTTTGCTGGCCTT

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GCGTCGGGCGACCTTTGGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGA GGGAGTGGCCAACTCCATCACTAGGGGTTCCT Nucleic acid sequence of a TRE2 promoter (a non-limiting example of a TRE promoter) (SEQ ID NO: 23): AATTCGTACACGCCTACCTCGACCCATCAAGTGCCACCTGACGTCTCCCTATCAG TGATAGAGAAGTCGACACGTCTCGAGCTCCCTATCAGTGATAGAGAAGGTACGT CTAGAACGTCTCCCTATCAGTGATAGAGAAGTCGACACGTCTCGAGCTCCCTATC AGTGATAGAGAAGGTACGTCTAGAACGTCTCCCTATCAGTGATAGAGAAGTCGA CACGTCTCGAGCTCCCTATCAGTGATAGAGAAGGTACGTCTAGAACGTCTCCCTA TCAGTGATAGAGAAGTCGACACGTCTCGAGCTCCCTATCAGTGATAGAGAAGGT ACCCCCTATATAAGCAGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCC ATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTGGATCG C Nucleic acid sequence of P tight promoter (a non-limiting example of a TRE promoter) (SEQ ID NO: 24): GAGTTTACTCCCTATCAGTGATAGAGAACGTATGTCGAGTTTACTCCCTATCAGT GATAGAGAACGATGTCGAGTTTACTCCCTATCAGTGATAGAGAACGTATGTCGA GTTTACTCCCTATCAGTGATAGAGAACGTATGTCGAGTTTACTCCCTATCAGTGA TAGAGAACGTATGTCGAGTTTATCCCTATCAGTGATAGAGAACGTATGTCGAGTT TACTCCCTATCAGTGATAGAGAACGTATGTCGAGGTAGGCGTGTACGGTGGGAG GCCTATATAAGCAGAGCTCGTTTAGTGAACCGTCAGATCGCC Nucleic acid sequence encoding TetR (SEQ ID NO: 25): ATGGCTAGATTAGATAAAAGTAAAGTGATTAACAGCGCATTAGAGCTGCTTAAT GAGGTCGGAATCGAAGGTTTAACAACCCGTAAACTCGCCCAGAAGCTAGGTGTA GAGCAGCCTACATTGTATTGGCATGTAAAAAATAAGCGGGCTTTGCTCGACGCCT TAGCCATTGAGATGTTAGATAGGCACCATACTCACTTTTGCCCTTTAGAAGGGGA AAGCTGGCAAGATTTTTTACGTAATAACGCTAAAAGTTTTAGATGTGCTTTACTA AGTCATCGCGATGGAGCAAAAGTACATTTAGGTACACGGCCTACAGAAAAACAG TATGAAACTCTCGAAAATCAATTAGCCTTTTTATGCCAACAAGGTTTTTCACTAG AGAATGCATTATATGCACTCAGCGCTGTGGGGCATTTTACTTTAGGTTGCGTATT GGAAGATCAAGAGCATCAAGTCGCTAAAGAAGAAAGGGAAACACCTACTACTG ATAGTATGCCGCCATTATTACGACAAGCTATCGAATTATTTGATCACCAAGGTGC AGAGCCAGCCTTCTTATTCGGCCTTGAATTGATCATATGCGGATTAGAAAAACAA CTTAAATGTGAAAGTGGG Amino acid sequence encoding TetR (SEQ ID NO: 26): MARLDKSKVINSALELLNEVGIEGLTTRKLAQKLGVEQPTLYWHVKNKRALLDALA IEMLDRHHTHFCPLEGESWQDFLRNNAKSFRCALLSHRDGAKVHLGTRPTEKQYET LENQLAFLCQQGFSLENALYALSAVGHFTLGCVLEDQEHQVAKEERETPTTDSMPPL LRQAIELFDHQGAEPAFLFGLELIICGLEKQLKCESG Nucleic acid sequence encoding TetR-Krab (SEQ ID NO: 27) ATGGCTAGATTAGATAAAAGTAAAGTGATTAACAGCGCATTAGAGCTGCTTAAT GAGGTCGGAATCGAAGGTTTAACAACCCGTAAACTCGCCCAGAAGCTAGGTGTA GAGCAGCCTACATTGTATTGGCATGTAAAAAATAAGCGGGCTTTGCTCGACGCCT TAGCCATTGAGATGTTAGATAGGCACCATACTCACTTTTGCCCTTTAGAAGGGGA AAGCTGGCAAGATTTTTTACGTAATAACGCTAAAAGTTTTAGATGTGCTTTACTA AGTCATCGCGATGGAGCAAAAGTACATTTAGGTACACGGCCTACAGAAAAACAG TATGAAACTCTCGAAAATCAATTAGCCTTTTTATGCCAACAAGGTTTTTCACTAG AGAATGCATTATATGCACTCAGCGCTGTGGGGCATTTTACTTTAGGTTGCGTATT GGAAGATCAAGAGCATCAAGTCGCTAAAGAAGAAAGGGAAACACCTACTACTG ATAGTATGCCGCCATTATTACGACAAGCTATCGAATTATTIGATCACCAAGGTGC AGAGCCAGCCTTCTTATTCGGCCTTGAATTGATCATATGCGGATTAGAAAAACAA CTTAAATGTGAAAGTGGGTCGCCAAAAAAGAAGAGAAAGGTCGACGGCGGTGGT GCTTTGTCTCCTCAGCACTCTGCTGTCACTCAAGGAAGTATCATCAAGAACAAGG

AGGGCATGGATGCTAAGTCACTAACTGCCTGGTCCCGGACACTGGTGACCTTCAA
GGATGTATTTGTGGACTTCACCAGGGAGGAGTGGAAGCTGCTGGACACTGCTCA
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CCCTGGCTGGTGGAGAGAGAAATTCACCAAGAGACCCATCCTGATTCAGAGACT
GCATTTGAAATCAAATCATCAGTTTAA Amino acid sequence encoding TetR-KRAB (SEQ ID NO: 28):
MARLDKSKVINSALELLNEVGIEGLTTRKLAQKLGVEQPTLYWHVKNKRALLDALA
IEMLDRHHTHFCPLEGESWQDFLRNNAKSFRCALLSHRDGAKVHLGTRPTEKQYET
LENQLAFLCQQGFSLENALYALSAVGHFTLGCVLEDQEHQVAKEERETPTTDSMPPL
LRQAIELFDHQGAEPAFLFGLELIICGLEKQLKCESGSPKKKRKVDGGGALSPQHSAV
TQGSIIKNKEGMDAKSLTAWSRTLVTFKDVFVDFTREEWKLLDTAQQIVYRNVMLE
NYKNLVSLGYQLTKPDVILRLEKGEEPWLVEREIHQETHPDSETAFEIKSSV Desmin promoter (SEQ ID NO: 29):
ACCTTGCTTCCTAGCTGGGCCTTTCCTTCTCCTCTATAAATACCAGCTCTGGTATT
TCGCCTTGGCAGCTGTTGCTGCTAGGGAGACGGCTGGCTTGACATGCATCTCCTG
ACAAAACACAAACCCGTGGTGTGAGTGGGTGTGGGCGGTGTGAGTAGGGGGATG
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CGGGGGTGCGACTACACGCAGTTGGAAACAGTCGTCAGAAGATTCTGGAAACTA
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CCAATAACCTAAAACCTCTCCTCGGAGAAAATAGGGGGCCTCAAACAAACGAAA
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GCCGGCCGGGGGTCTCCTGTCAGCTCCTTGCCCTGTGAAACCCAGCAGGCCTGCC
TGTCTTCTGTCCTCTTGGGGCTGTCCAGGGGCGCAGGCCTCTTGCGGGGGAGCTG
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TGCAGCTGTCAGGGGAGGGGCGCCGGGGGGTGATGTCAGGAGGGCTACAAATAG
TGCAGACAGCTAAGGGGCTCCGTCACCCATCTTCACATCCACTCCAGCCGGCTGC
CCGCCCGCTGCCTCCTCTGTGCGTCCGCCCAGCCAGCCTCGTCCACGCC Desmin-rtTA4 vector (SEQ ID NO: 30):
TTATGCAGTGCTGCCATAACCATGAGTGATAACACTGCGGCCAACTTACTTCTGA
CAACGATCGGAGGACCGAAGGAGCTAACCGCTTTTTTGCACAACATGGGGGATC
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ACGAGCGTGACACCACGATGCCTGTAGTAATGGTAACAACGTTGCGCAAACTAT
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pAAV2_CMV_rtTA(V16) (SEQ ID NO: 31):

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CCGCTCATGAGACAATAACCCTGATAAATGCTTCAATAATATTGAAAAAGGAAG
AGTATGAGTATTCAACATTTCCGTGTCGCCCTTATTCCCTTTTTTGCGGCATTTTG
CCTTCCTGTTTTTGCTCACCCAGAAACGCTGGTGAAAGTAAAAGATGCTGAAGAT
CAGTTGGGTGCACGAGTGGGTTACATCGAACTGGATCTCAATAGTGGTAAGATCC
TTGAGAGTTTTCGCCCCGAAGAACGTTTTCCAATGATGAGCACTTTTAAAGTTCT
GCTATGTGGCGCGGTATTATCCCGTATTGACGCCGGGCAAGAGCAACTCGGTCGC
CGCATACACTATTCTCAGAATGACTTGGTTGAGTACTCACCAGTCACAGAAAAGC
ATCTTACGGATGGCATGACAGTAAGAGAA VP64, 4 repeats of VP16 (SEQ ID NO: 34) (Non-limiting example of a transactivation domain):
GAGGCCAGCGGTTCCGGACGGGCTGACGCATTGGACGATTTTGATCTGGATATGC
TGGGAAGTGACGCCCTCGATGATTTTGACCTTGACATGCTTGGTTCGGATGCCCT
TGATGACTTTGACCTCGACATGCTCGGCAGTGACGCCCTTGATGATTTCGACCTG
GACATGCTGATTAACTCTAGA P65 (SEQ ID NO: 35) (Non-limiting example of a transactivation domain):
AGCCAGTACCTGCCCGACACCGACGACCGGCACCGGATCGAGGAAAAGCGGAA
GCGGACCTACGAGACATTCAAGAGCATCATGAAGAAGTCCCCCTTCAGCGGCCC
CACCGACCCTAGACCTCCACCTAGAAGAATCGCCGTGCCCAGCAGATCCAGCGC
CAGCGTGCCAAAACCTGCCCCCCAGCCTTACCCCTTCACCAGCAGCCTGAGCACC
ATCAACTACGACGAGTTCCCTACCATGGTGTTCCCCAGCGGCCAGATCTCTCAGG
CCTCTGCTCTGGCTCCAGCCCCTCCTCAGGTGCTGCCTCAGGCTCCTGCTCCTGCA
CCAGCTCCAGCCATGGTGTCTGCACTGGCTCAGGCACCAGCACCCGTGCCTGTGC
TGGCTCCTGGACCTCCACAGGCTGTGGCTCCACCAGCCCCTAAACCTACACAGGC
CGGCGAGGGCACACTGTCTGAAGCTCTGCTGCAGCTGCAGTTCGACGACGAGGA
TCTGGGAGCCCTGCTGGGAAACAGCACCGATCCTGCCGTGTTCACCGACCTGGCC
AGCGTGGACAACAGCGAGTTCCAGCAGCTGCTGAACCAGGGCATCCCTGTGGCC
CCTCACACCACCGAGCCCATGCTGATGGAATACCCCGAGGCCATCACCCGGCTC
GTGACAGGCGCTCAGAGGCCTCCTGATCCAGCTCCTGCCCCTCTGGGAGCACCAG
GCCTGCCTAATGGACTGCTGTCTGGCGACGAGGACTTCAGCTCTATCGC
CGATATGGATTTCTCAGCCTTGCTG RTA (SEQ ID NO: 36) (Non-limiting example of a transactivation domain):
CGGGATTCCAGGGAAGGGATGTTTTTGCCGAAGCCTGAGGCCGGCTCCGCTATTA
GTGACGTGTTTGAGGGCCGCGAGGTGTGCCAGC CAAAACGAA TCCGGCCA
TTTCATCCTCCAGGAAGTCCATGGGCCAACCGCCCACTCCCCGCCAGCCTCGCAC
CAACACCAACCGGTCCAGTACATGAGCCAGTCGGGTCACTGACCCCGGCACCAG TCCC
TCAGCCACTGGATCCAGCGCCCGCAGTGACTCCCGAGGCCAGTCACCTGTTGGA

GGATCCCGATGAAGAGACGAGCCAGGCTGTCAAAGCCCTTCGGGAGATGGCCGA TACTGTGATTCCCCAGAAGGAA GAGGCTGCAA TCTGTGGCCAAA TGGACCTTTCCCA TCCGCCCCCAAGGGGCCA TCTGGA TGAGCT GACAACCACACTTGAGTCCA TGACCGAGGATCTGAACCTGGACTCACCCCTGACCCCGGAATTGAACGAGATTCT GGATACCTTCCTGAACGACGAGTGCCTCTTGCATGCCATGCATATCAGCACAGGA C TGTCCA TCTTCGACACA TCTCTGTTT MPH MS2-P65-HSF1 (SEQ ID NO: 37) (Non-limiting example of a transactivation domain): GCTTCAAACTTTACTCAGTTCGTGCTCGTGGACAATGGTGGGACAGGGGATGTGA CAGTGGCTCCTTCTAATTTCGCTAATGGGGTGGCAGAGTGGATCAGCTCCAACTC ACGGAGCCAGGCCTACAAGGTGACATGCAGCGTCAGGCAGTCTAGTGCCCAGAA GAGAAAGTATACCATCAAGGTGGAGGTCCCCAAAGTGGCTACCCAGACAGTGGG CGGAGTCGAACTGCCTGTCGCCGCTTGGAGGTCCTACCTGAACATGGAGCTCACT ATCCCAATTTTCGCTACCAATTCTGACTGTGAACTCATCGTGAAGGCAATGCAGG GGCTCCTCAAAGACGGTAATCCTATCCCTTCCGCCATCGCCGCTAACTCAGGTAT CTACAGCGCTGGAGGAGGTGGAAGCGGAGGAGGAGGAAGCGGAGGAGGAGGTA GCGGACCTAAGAAAAAGAGGAAGGTGGCGGCCGCTGGATCCCCTTCAGGGCAGA TCAGCAACCAGGCCCTGGCTCTGGCCCCTAGCTCCGCTCCAGTGCTGGCCCAGAC TATGGTGCCCTCTAGTGCTATGGTGCCTCTGGCCCAGCCACCTGCTCCAGCCCCT GTGCTGACCCCAGGACCACCCCAGTCACTGAGCGCTCCAGTGCCCAAGTCTACAC AGGCCGGCGAGGGGACTCTGAGTGAAGCTCTGCTGCACCTGCAGTTCGACGCTG ATGAGGACCTGGGAGCTCTGCTGGGGAACAGCACCGATCCCGGAGTGTTCACAG ATCTGGCCTCCGTGGACAACTCTGAGTTTCAGCAGCTGCTGAATCAGGGCGTGTC CATGTCTCATAGTACAGCCGAACCAATGCTGATGGAGTACCCCGAAGCCATTACC CGGCTGGTGACCGGCAGCCAGCGGCCCCCCGACCCCGCTCCAACTCCCCTGGGA ACCAGCGGCCTGCCTAATGGGCTGTCCGGAGATGAAGACTTCTCAAGCATCGCTG ATATGGACTTTAGTGCCCTGCTGTCACAGATTTCCTCTAGTGGGCAGGGAGGAGG TGGAAGCGGCTTCAGCGTGGACACCAGTGCCCTGCTGGACCTGTTCAGCCCCTCG GTGACCGTGCCCGACATGAGCCTGCCTGACCTTGACAGCAGCCTGGCCAGTATCC AAGAGCTCCTGTCTCCCCAGGAGCCCCCCAGGCCTCCCGAGGCAGAGAACAGCA GCCCGGATTCAGGGAAGCAGCTGGTGCACTACACAGCGCAGCCGCTGTTCCTGC TGGACCCCGGCTCCGTGGACACCGGGAGCAACGACCTGCCGGTGCTGTTTGAGC TGGGAGAGGGCTCCTACTTCTCCGAAGGGGACGGCTTCGCCGAGGACCCCACCA TCTCCCTGCTGACAGGCTCGGAGCCTCCCAAAGCCAAGGACCCCACTGTCTCC OCT4-2A-SOX2-2A-KLF4 (non-limiting example of nucleic acid sequence encoding human OCT4, human SOX2, and human KLF4, each separated by a 2A peptide) (SEQ ID NO: 38): ATGGCGGGACACCTGGCTTCGGATTTCGCCTTCTCGCCCCCTCCAGGTGGTGGAG GTGATGGGCCAGGGGGGCCGGAGCCGGGCTGGGTTGATCCTCGGACCTGGCTAA GCTTCCAAGGCCCTCCTGGAGGGCCAGGAATCGGGCCGGGGGTTGGGCCAGGCT CTGAGGTGTGGGGGATTCCCCCATGCCCCCCGCCGTATGAGTTCTGTGGGGGGAT GGCGTACTGTGGGCCCCAGGTTGGAGTGGGGCTAGTGCCCCAAGGCGGCTTGGA GACCTCTCAGCCTGAGGGCGAAGCAGGAGTCGGGGTGGAGAGCAACTCCGATGG GGCCTCCCCGGAGCCCTGCACCGTCACCCCTGGTGCCGTGAAGCTGGAGAAGGA GAAGCTGGAGCAAAACCCGGAGGAGTCCCAGGACATCAAAGCTCTGCAGAAAG AACTCGAGCAATTTGCCAAGCTCCTGAAGCAGAAGAGGATCACCCTGGGATATA CACAGGCCGATGTGGGGCTCACCCTGGGGGTTCTATTTGGGAAGGTATTCAGCCA AACGACCATCTGCCGCTTTGAGGCTCTGCAGCTTAGCTTCAAGAACATGTGTAAG CTGCGGCCCTTGCTGCAGAAGTGGGTGGAGGAAGCTGACAACAATGAAAATCTT CAGGAGATATGCAAAGCAGAAACCCTCGTGCAGGCCCGAAAGAGAAAGCGAAC

CAGTATCGAGAACCGAGTGAGAGGCAACCTGGAGAATTTGTTCCTGCAGTGCCC
GAAACCCACACTGCAGCAGATCAGCCACATCGCCCAGCAGCTTGGGCTCGAGAA
GGATGTGGTCCGAGTGTGGTTCTGTAACCGGCGCCAGAAGGGCAAGCGATCAAG
CAGCGACTATGCACAACGAGAGGATTTTGAGGCTGCTGGGTCTCCTTTCTCAGGG
GGACCAGTGTCCTTTCCTCTGGCCCCAGGGCCCCATTTTGGTACCCCAGGCTATG
GGAGCCCTCACTTCACTGCACTGTACTCCTCGGTCCCTTTCCCTGAGGGGGAAGC
CTTTCCCCCTGTCTCTGTCACCACTCTGGGCTCTCCCATGCATTCAAACGCTAGCG
GCAGCGGCGCCACGAACTTCTCTCTCTTAAAGCAAGCAGGAGATOTTGAAGAAA
ACCCCGGGCCTGCATGCATGTACAACATGATGGAGACGGAGCTGAAGCCGCCGG
GCCCGCAGCAAACTTCGGGGGGCGGCGGCGGCAACTCCACCGCGGCGGCGGCCG
GCGGCAACCAGAAAAACAGCCCGGACCGCGTCAAGCGGCCCATGAATGCCTTCA
TGGTGTGGTCCCGCGGGCAGCGGCGCAAGATGGCCCAGGAGAACCCCAAGATGC
ACAACTCGGAGATCAGCAAGCGCCTGGGCGCCGAGTGGAAACTTTTGTCGGAGA
CGGAGAAGCGGCCGTTCATCGACGAGGCTAAGCGGCTGCGAGCGCTGCACATGA
AGGAGCACCCGGATTATAAATACCGGCCCCGGCGGAAAACCAAGACGCTCATGA
AGAAGGATAAGTACACGCTGCCCGGCGGGCTGCTGGCCCCCGGCGGCAATAGCA
TGGCGAGCGGGGTCGGGGTGGGCGCCGGCCTGGGCGCGGGCGTGAACCAGCGC
ATGGACAGTTACGCGCACATGAACGGCTGGAGCAACGGCAGCTACAGCATGATG
CAGGACCAGCTGGGCTACCCGCAGCACCCGGGCCTCAATGCGCACGGCGCAGCG
CAGATGCAGCCCATGCACCGCTACGACGTGAGCGCCCTGCAGTACAACTCCATG
ACCAGCTCGCAGACCTACATGAACGGCTCGCCCACCTACAGCATGTCCTACTCGC
AGCAGGGCACCCCTGGCATGGCTCTTGGCTCCATGGGTTCGGTGGTCAAGTCCGA
GGCCAGCTCCAGCCCCCCTGTGGTTACCTCTTCCTCCCACTCCAGGGCGCCCTGC
CAGGCCGGGGACCTCCGGGACATGATCAGCATGTATCTCCCCGGCGCCGAGGTG
CCGGAACCCGCCGCCCCCAGCAGACTTCACATGTCCCAGCACTACCAGAGCGGC
CCGGTGCCCGGCACGGCCATTAACGGCACACTGCCCCTCTCACACATGGCATGCG
GCTCCGGCGAGGGCAGGGGAAGTCTTCTAACATGCGGGGACGTGGAGGAAAATC
CCGGCCCACTCGAGATGGCTGTCAGCGACGCGCTGCTCCCATCTTTCTCCACGTT
CGCGTCTGGCCCGGCGGGAAGGGAGAAGACACTGCGTCAAGCAGGTGCCCCGAA
TAACCGCTGGCGGGAGGAGCTCTCCCACATGAAGCGACTTCCCCCAGTGCTTCCC
GGCCGCCCCTATGACCTGGCGGCGGCGACCGTGGCCACAGACCTGGAGAGCGGC
GGAGCCGGTGCGGCTTGCGGCGGTAGCAACCTGGCGCCCCTACCTCGGAGAGAG
ACCGAGGAGTTCAACGATCTCCTGGACCTGGACTTTATTCTCTCCAATTCGCTGA
CCCATCCTCCGGAGTCAGTGGCCGCCACCGTGTCCTCGTCAGCGTCAGCCTCCTC
TTCGTCGTCGCCGTCGAGCAGCGGCCCTGCCAGCGCGCCCTCCACCTGCAGCTTC
ACCTATCCGATCCGGGCCGGGAACGACCCGGGCGTGGCGCCGGGCGGCACGGGC
GGAGGCCTCCTCTATGGCAGGGAGTCCGCTCCCCCTCCGACGGCTCCCTTCAACC
TGGCGGACATCAACGACGTGAGCCCCTCGGGCGGCTTCGTGGCCGAGCTCCTGC
GGCCAGAATTGGACCCGGTGTACATTCCGCCGCAGCAGCCGCAGCCGCCAGGTG
GCGGGCTGATGGGCAAGTTCGTGCTGAAGGCGTCGCTGAGCGCCCCTGGCAGCG
AGTACGGCAGCCCGTCGGTCATCAGCGTCAGCAAAGGCAGCCCTGACGGCAGCC
ACCCGGTGGTGGTGGCGCCCTACAACGGCGGGCCGCCGCGCACGTGCCCCAAGA
TCAAGCAGGAGGCGGTCTCTTCGTGCACCCACTTGGGCGCTGGACCCCCTCTCAG
CAATGGCCACCGGCCGGCTGCACACGACTTCCCCCTGGGGCGGCAGCTCCCCAG
CAGGACTACCCCGACCCTGGGTCTTGAGGAAGTGCTGAGCAGCAGGGACTGTCA
CCCTGCCCTGCCGCTTCCTCCCGGCTTCCATCCCCACCCGGGGCCCAATTACCCAT
CCTTCCTGCCCGATCAGATGCAGCCGCAAGTCCCGCCGCTCCATTACCAAGAGCT
CATGCCACCCGGTTCCTGCATGCCAGAGGAGCCCAAGCCAAAGAGGGGAAGACG
ATCGTGGCCCCGGAAAAGGACCGCCACCCACACTTGTGATTACGCGGGCTGCGG
CAAAACCTACACAAAGAGTTCCCATCTCAAGGCACACCTGCGAACCCACACAGG

TGAGAAACCTTACCACTGTGACTGGGACGGCTGTGGATGGAAATTCGCCCGCTCA GATGAACTGACCAGGCACTACCGTAAACACACGGGGCACCGCCCGTTCCAGTGC CAAAAATGCGACCGAGCATTTTCCAGGTCGGACCACCTCGCCTTACACATGAAG AGGCATTTT OCT4-2A-SOX2-2A-KLF4 (non-limiting example of an amino acid sequence encoding human OCT4, human SOX2, and human KLF4, each separated by a 2A peptide) (SEQ ID NO: 39): MAGHLASDFAFSPPPGGGGDGPGGPEPGWVDPRTWLSFQGPPGGPGIGPGVGPGSE VWGIPPCPPPYEFCGGMAYCGPQVGVGLVPQGGLETSQPEGEAGVGVESNSDGASP EPCTVTPGAVKLEKEKLEQNPEESQDIKALQKELEQFAKLLKQKRITLGYTQADVGL TLGVLFGKVFSQTTICRFEALQLSFKNMCKLRPLLQKWVEEADNNENLQEICKAETL VQARKRKRTSIENRVRGNLENLFLQCPKPTLQQISHIAQQLGLEKDVVRVWFCNRRQ KGKRSSSDYAQREDFEAAGSPFSGGPVSFPLAPGPHFGTPGYGSPHFTALYSSVPFPE GEAFPPVSVTTLGSPMHSNASGSGATNFSLLKQAGDVEENPGPACMYNMMETELKP PGPQQTSGGGGGNSTAAAAGGNQKNSPDRVKRPMNAFMVWSRGQRRKMAQENPK MHNSEISKRLGAEWKLLSETEKRPFIDEAKRLRALHMKEHPDYKYRPRRKTKTLMK KDKYTLPGGLLAPGGNSMASGVGVGAGLGAGVNQRMDSYAHMNGWSNGSYSMM QDQLGYPQHPGLNAHGAAQMQPMHRYDVSALQYNSMTSSQTYMNGSPTYSMSYS QQGTPGMALGSMGSVVKSEASSSPPVVTSSSHSRAPCQAGDLRDMISMYLPGAEVPE PAAPSRLHMSQHYQSGPVPGTAINGTLPLSHMACGSGEGRGSLLTCGDVEENPGPLE MAVSDALLPSFSTFASGPAGREKTLRQAGAPNNRWREELSHMKRLPPVLPGRPYDL AAATVATDLESGGAGAACGGSNLAPLPRRETEEFNDLLDLDFILSNSLTHPPESVAAT VSSSASASSSSSPSSSGPASAPSTCSFTYPIRAGNDPGVAPGGTGGGLLYGRESAPPPT APFNLADINDVSPSGGFVAELLRPELDPVYIPPQQPQPPGGGLMGKFVLKASLSAPGS EYGSPSVISVSKGSPDGSHPVVVAPYNGGPPRTCPKIKQEAVSSCTHLGAGPPLSNGH RPAAHDFPLGRQLPSRTTPTLGLEEVLSSRDCHPALPLPPGFHPHPGPNYPSFLPDQM QPQVPPLHYQELMPPGSCMPEEPKPKRGRRSWPRKRTATHTCDYAGCGKTYTKSSH LKAHLRTHTGEKPYHCDWDGCGWKFARSDELTRHYRKHTGHRPFQCQKCDRAFSR SDHLALHMKRHF Human OCT4 nucleic acid sequence (non-limiting example of a nucleic acid sequence encoding human OCT4) (SEQ ID NO: 40): ATGGCGGGACACCTGGCTTCGGATTTCGCCTTCTCGCCCCCTCCAGGTGGTGGAG GTGATGGGCCAGGGGGGCCGGAGCCGGGCTGGGTTGATCCTCGGACCTGGCTAA GCTTCCAAGGCCCTCCTGGAGGGCCAGGAATCGGGCCGGGGGTTGGGCCAGGCT CTGAGGTGTGGGGGATTCCCCCATGCCCCCCGCCGTATGAGTTCTGTGGGGGGAT GGCGTACTGTGGGCCCCAGGTTGGAGTGGGGCTAGTGCCCCAAGGCGGCTTGGA GACCTCTCAGCCTGAGGGCGAAGCAGGAGTCGGGGTGGAGAGCAACTCCGATGG GGCCTCCCCGGAGCCCTGCACCGTCACCCCTGGTGCCGTGAAGCTGGAGAAGGA GAAGCTGGAGCAAAACCCGGAGGAGTCCCAGGACATCAAAGCTCTGCAGAAAG AACTCGAGCAATTTGCCAAGCTCCTGAAGCAGAAGAGGATCACCCTGGGATATA CACAGGCCGATGTGGGGCTCACCCTGGGGGTTCTATTTGGGAAGGTATTCAGCCA AACGACCATCTGCCGCTTTGAGGCTCTGCAGCTTAGCTTCAAGAACATGTGTAAG CTGCGGCCCTTGCTGCAGAAGTGGGTGGAGGAAGCTGACAACAATGAAAATCTT CAGGAGATATGCAAAGCAGAAACCCTCGTGCAGGCCCGAAAGAGAAAGCGAAC CAGTATCGAGAACCGAGTGAGAGGCAACCTGGAGAATTTGTTCCTGCAGTGCCC GAAACCCACACTGCAGCAGATCAGCCACATCGCCCAGCAGCTTGGGCTCGAGAA GGATGTGGTCCGAGTGTGGTTCTGTAACCGGCGCCAGAAGGGCAAGCGATCAAG CAGCGACTATGCACAACGAGAGGATTTTGAGGCTGCTGGGTCTCCTTTCTCAGGG GGACCAGTGTCCTTTCCTCTGGCCCCAGGGCCCCATTTTGGTACCCCAGGCTATG GGAGCCCTCACTTCACTGCACTGTACTCCTCGGTCCCTTTCCCTGAGGGGGAAGC CTTTCCCCCTGTCTCTGTCACCACTCTGGGCTCTCCCATGCATTCAAAC Human OCT4 amino acid sequence (non-limiting example of an amino acid sequence encoding human OCT4) (SEQ ID NO: 41): MAGHLASDFAFSPPPGGGGDGPGGPEPGWVDPRTWLSFQGPPGGPGIGPGVGPGSE VWGIPPCPPPYEFCGGMAYCGPQVGVGLVPQGGLETSQPEGEAGVGVESNSDGASP EPCTVTPGAVKLEKEKLEQNPEESQDIKALQKELEQFAKLLKQKRITLGYTQADVGL TLGVLFGKVFSQTTICRFEALQLSFKNMCKLRPLLQKWVEEADNNENLQEICKAETL VQARKRKRTSIENRVRGNLENLFLQCPKPTLQQISHIAQQLGLEKDVVRVWFCNRRQ KGKRSSSDYAQREDFEAAGSPFSGGPVSFPLAPGPHFGTPGYGSPHFTALYSSVPFPE GEAFPPVSVTTLGSPMHSN Human SOX2 nucleic acid sequence (non-limiting example of a nucleic acid sequence encoding human SOX2) (SEQ ID NO: 42): ATGTACAACATGATGGAGACGGAGCTGAAGCCGCCGGGCCCGCAGCAAACTTCG GGGGGCGGCGGCGGCAACTCCACCGCGGCGGCGGCCGGCGGCAACCAGAAAAA CAGCCCGGACCGCGTCAAGCGGCCCATGAATGCCTTCATGGTGTGGTCCCGCGG GCAGCGGCGCAAGATGGCCCAGGAGAACCCCAAGATGCACAACTCGGAGATCA GCAAGCGCCTGGGCGCCGAGTGGAAACTTTTGTCGGAGACGGAGAAGCGGCCGT TCATCGACGAGGCTAAGCGGCTGCGAGCGCTGCACATGAAGGAGCACCCGGATT ATAAATACCGGCCCCGGCGGAAAACCAAGACGCTCATGAAGAAGGATAAGTACA CGCTGCCCGGCGGGCTGCTGGCCCCCGGCGGCAATAGCATGGCGAGCGGGGTCG GGGTGGGCGCCGGCCTGGGCGCGGGCGTGAACCAGCGCATGGACAGTTACGCGC ACATGAACGGCTGGAGCAACGGCAGCTACAGCATGATGCAGGACCAGCTGGGCT ACCCGCAGCACCCGGGCCTCAATGCGCACGGCGCAGCGCAGATGCAGCCCATGC ACCGCTACGACGTGAGCGCCCTGCAGTACAACTCCATGACCAGCTCGCAGACCT ACATGAACGGCTCGCCCACCTACAGCATGTCCTACTCGCAGCAGGGCACCCCTG GCATGGCTCTTGGCTCCATGGGTTCGGTGGTCAAGTCCGAGGCCAGCTCCAGCCC CCCTGTGGTTACCTCTTCCTCCCACTCCAGGGCGCCCTGCCAGGCCGGGGACCTC CGGGACATGATCAGCATGTATCTCCCCGGCGCCGAGGTGCCGGAACCCGCCGCC CCCAGCAGACTTCACATGTCCCAGCACTACCAGAGCGGCCCGGTGCCCGGCACG GCCATTAACGGCACACTGCCCCTCTCACACATG Human SOX2 amino acid sequence (non-limiting example of an amino acid sequence encoding human SOX2) (SEQ ID NO: 43): MYNMMETELKPPGPQQTSGGGGGNSTAAAAGGNQKNSPDRVKRPMNAFMVWSRG QRRKMAQENPKMHNSEISKRLGAEWKLLSETEKRPFIDEAKRLRALHMKEHPDYKY RPRRKTKTLMKKDKYTLPGGLLAPGGNSMASGVGVGAGLGAGVNQRMDSYAHMN GWSNGSYSMMQDQLGYPQHPGLNAHGAAQMQPMHRYDVSALQYNSMTSSQTYM NGSPTYSMSYSQQGTPGMALGSMGSVVKSEASSSPPVVTSSSHSRAPCQAGDLRDMI SMYLPGAEVPEPAAPSRLHMSQHYQSGPVPGTAINGTLPLSHM Human KLF4 (non-limiting example of a nucleotide sequence encoding human KLF4) (SEQ ID NO: 44): ATGGCTGTCAGCGACGCGCTGCTCCCATCTTTCTCCACGTTCGCGTCTGGCCCGG CGGGAAGGGAGAAGACACTGCGTCAAGCAGGTGCCCCGAATAACCGCTGGCGG GAGGAGCTCTCCCACATGAAGCGACTTCCCCCAGTGCTTCCCGGCCGCCCCTATG ACCTGGCGGCGGCGACCGTGGCCACAGACCTGGAGAGCGGCGGAGCCGGTGCG GCTTGCGGCGGTAGCAACCTGGCGCCCCTACCTCGGAGAGAGACCGAGGAGTTC AACGATCTCCTGGACCTGGACTTTATTCTCTCCAATTCGCTGACCCATCCTCCGGA GTCAGTGGCCGCCACCGTGTCCTCGTCAGCGTCAGCCTCCTCTTCGTCGTCGCCG TCGAGCAGCGGCCCTGCCAGCGCGCCCTCCACCTGCAGCTTCACCTATCCGATCC GGGCCGGGAACGACCCGGGCGTGGCGCCGGGCGGCACGGGCGGAGGCCTCCTCT ATGGCAGGGAGTCCGCTCCCCCTCCGACGGCTCCCTTCAACCTGGCGGACATCAA CGACGTGAGCCCCTCGGGCGGCTTCGTGGCCGAGCTCCTGCGGCCAGAATTGGA CCCGGTGTACATTCCGCCGCAGCAGCCGCAGCCGCCAGGTGGCGGGCTGATGGG CAAGTTCGTGCTGAAGGCGTCGCTGAGCGCCCCTGGCAGCGAGTACGGCAGCCC

GTCGGTCATCAGCGTCAGCAAAGGCAGCCCTGACGGCAGCCACCCGGTGGTGGT GGCGCCCTACAACGGCGGGCCGCCGCGCACGTGCCCCAAGATCAAGCAGGAGGC GGTCTCTTCGTGCACCCACTTGGGCGCTGGACCCCCTCTCAGCAATGGCCACCGG CCGGCTGCACACGACTTCCCCCTGGGGCGGCAGCTCCCCAGCAGGACTACCCCG ACCCTGGGTCTTGAGGAAGTGCTGAGCAGCAGGGACTGTCACCCTGCCCTGCCG CTTCCTCCCGGCTTCCATCCCCACCCGGGGCCCAATTACCCATCCTTCCTGCCCGA TCAGATGCAGCCGCAAGTCCCGCCGCTCCATTACCAAGAGCTCATGCCACCCGGT TCCTGCATGCCAGAGGAGCCCAAGCCAAAGAGGGGAAGACGATCGTGGCCCCGG AAAAGGACCGCCACCCACACTTGTGATTACGCGGGCTGCGGCAAAACCTACACA AAGAGTTCCCATCTCAAGGCACACCTGCGAACCCACACAGGTGAGAAACCTTAC CACTGTGACTGGGACGGCTGTGGATGGAAATTCGCCCGCTCAGATGAACTGACC AGGCACTACCGTAAACACACGGGGCACCGCCCGTTCCAGTGCCAAAAATGCGAC CGAGCATTTTCCAGGTCGGACCACCTCGCCTTACACATGAAGAGGCATTTT Human KLF4 (non-limiting example of an amino acid sequence encoding human KLF4) (SEQ ID NO: 45):

MAVSDALLPSFSTFASGPAGREKTLRQAGAPNNRWREELSHMKRLPPVLPGRPYDL AAATVATDLESGGAGAACGGSNLAPLPRRETEEFNDLLDLDFILSNSLTHPPESVAAT VSSSASASSSSSPSSSGPASAPSTCSFTYPIRAGNDPGVAPGGTGGGLLYGRESAPPPT APFNLADINDVSPSGGFVAELLRPELDPVYIPPQQPQPPGGGLMGKFVLKASLSAPGS EYGSPSVISVSKGSPDGSHPVVVAPYNGGPPRTCPKIKQEAVSSCTHLGAGPPLSNGH RPAAHDFPLGRQLPSRTTPTLGLEEVLSSRDCHPALPLPPGFHPHPGPNYPSFLPDQM QPQVPPLHYQELMPPGSCMPEEPKPKRGRRSWPRKRTATHTCDYAGCGKTYTKSSH LKAHLRTHTGEKPYHCDWDGCGWKFARSDELTRHYRKHTGHRPFQCQKCDRAFSR SDHLALHMKRHF Human RCVRN (recoverin) promoter (non-limiting example of a human RCVRN (recoverin) promoter) (SEQ ID NO: 46):

ATTTTAATCTCACTAGGGTTCTGGGAGCACCCCCCCCCACCGCTCCCGCCCTCCA CAAAGCTCCTGGGCCCCTCCTCCCTTCAAGGATTGCGAAGAGCTGGTCGCAAATC CTCCTAAGCCACCAGCATCTCGGTCTTCAGCTCACACCAGCCTTGAGCCCAGCCT GCGGCCAGGGGACCACGCACGTCCCACCCACCCAGCGACTCCCCAGCCGCTGCC CACTCTTCCTCACTCA RSV promoter (non-limiting example of a RSV promoter) (SEQ ID NO: 47):

AATGTAGTCTTATGCAATACTCTTGTAGTCTTGCAACATGGTAACGATGAGTTAG CAACATGCCTTACAAGGAGAGAAAAAGCACCGTGCATGCCGATTGGTGGAAGTA AGGTGGTACGATCGTGCCTTATTAGGAAGGCAACAGACGGGTCTGACATGGATT GGACGAACCACTGAATTGCCGCATTGCAGAGATATTGTATTTAAGTGCCTAGCTC GATACATAAAC CMV promoter (non-limiting example of a CMV promoter) (SEQ ID NO: 48):

CATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATA GCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTG ACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTA ACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGACTATTTACGGTAAACTG CCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGT CAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGAC TTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCG GTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCA AGTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGCACCAAAATCAACGGG ACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCG TGTACGGTGGGAGGTCTATATAAGCAGAGCTGGTTTAGTGAACCGTCAGATCCGC TAGAGATCCGC EFS promoter (non-limiting example of an EFS promoter) (SEQ ID NO: 49):

TCGAGTGGCTCCGGTGCCCGTCAGTGGGCAGAGCGCACATCGCCCACAGTCCCC GAGAAGTTGGGGGGAGGGGTCGGCAATTGAACCGGTGCCTAGAGAAGGTGGCG CGGGGTAAACTGGGAAAGTGATGTCGTGTACTGGCTCCGCCTTTTTCCCGAGGGT GGGGGAGAACCGTATATAAGTGCAGTAGTCGCCGTGAACGTTCTTTTTCGCAACG GGTTTGCCGCCAGAACACAGGTGTCGTGACCGCGG Human GRK1 (rhodopsin kinase) promoter (non-limiting example of a human promoter) (SEQ ID NO: 50): Gggccccagaagcctggggttgtttgtccttctcaggggaaaagtgaggcggccccttggaggaaggggccgg gcagaatgatctaatcggattccaagcagctcaggggattgtctttttctagcaccttcttgccactcctaagcgtcctccgtgacccc ggctgggatttcgcctggtgctgtgtcagccccggtctcccaggggcttcccagtggtccccaggaaccctcgacagggcccggtctct ctcgtccagcaagggcagggacgggccacaggccaagggc Human CRX (cone rod homeobox transcription factor) promoter (non-limiting example of a human CRX promoter) (SEQ ID NO: 51): Gcctgtagccttaatctctcctagcaggggggtttgggggagggaggaggagaaagaaagggcccttatggctga gacacaatgacccagccacaaggagggattaccgggcg Human NRL promoter (neural retina leucine zipper transcription factor enhancer upstream of the human TK terminal promoter) (non-limiting example of a human NRL promoter) (SEQ ID NO: 52): Aggtaggaagtggcctttaactccatagaccctatttaaacagcttcggacaggtttaaacatctccttggataattcct agtatccctgttcccactcctactcagggatgatagctctaagaggtgttaggggattaggctgaaaatgtaggtcacccctcagcca tctcggaactagaatgagtgagagaggagagaggggcagagacacacacattcgcatattaaggtgacgcgtgtggcctcgaacacc gagcgaccctgcagcgacccgcttaa Human red opsin promoter (hred promoter) (SEQ ID NO: 101): Gatccggttccaggcctcggccctaaatagtctccctgggctttcaagagaaccacatgagaaaggaggattcggg ctctgagcagtttcaccacccacccccccagtctgcaaatcctgacccgtgggtccacctgccccaaaggcggacgcaggacagtaga agggaacagagaacacataaacacagagagggccacagcggctcccacagtcaccgccaccttcctggcggggatgggtggggc gtctgagtttggttcccagcaaatccctctgagccgcccttgcgggctcgcctcaggagcaggggagcaagaggtgggaggaggag gtctaagtcccaggcccaattaagagatcaggtagtgtagggtttgggagcttttaaggtgaagaggcccgggctgatcccacaggcc agtataaagcgccgtgaccctcaggtgatgcgccagggccggctgccgtcggggacagggctttccatagc Human rhodopsin promoter (rho promoter) (SEQ ID NO: 102): Agttaatgattaacccgccatgctacttatctacgtagccatgctctaggaagatcggaattcgcccttaagctagcag atcttccccacctagccacctggcaaactgctccttctctcaaaggcccaaacatggcctcccagactgcaacccccaggcagtcagg ccctgtctccacaacctcacagccaccctggacggaatctgcttcttcccacatttgagtcctcctcagcccctgagctcctctgggca gggctgtttctttccatctttgtattcccaggggcctgcaaataaatgtttaatgaacgaacaagagagtgaattccaattccatgcaa caaggattgggctcctgggccctaggctatgtgtctggcaccagaaacggaagctgcaggttgcagcccctgccctcatggagctcct cctgtcagaggagtgtggggactggatgactccagaggtaacttgtgggggaacgaacaggtaaggggctgtgtgacgagatgagagact gggagaataaaccagaaagtctctagctgttccagaggacatagcacagaggcccatggtccctatttcaaacccaggccaaccagact gagctgggaccttgggacagacaagtcatgcagaagttaggggaccttctcctcccttttcctggatggatcctgagtaccttctcctcc ctgacctcaggcttcctcctagtgtcaccttggcccctcttagaagccaattaggccctcagtttctgcagcggggattaatatgatta tgaacacccccaatctcccagatgctgattcagccaggagcttaggagggggaggtcactttataagggtctgggggggtcagaacccag agtcatcccctgaattctgca Mouse cone arrestin promoter (mcar promoter) (SEQ ID NO: 103): Ggttcttcccatttggctacatggtcttttttttaccttttggttcctttggccttttggcttttggcttccagggcttctgga tcccccccaacccctcccatacacatacacatgtgcactcgtgcactcaacccagcacaggataatgttcattcttgacctttccacat acatctggctatgttctctctcttatctacaataaatctcctccactatacttaggagcagttatgttcttcttctttctttcttttt ttttttttcattcagtaacatcatcagaatcccctagctctggcctacctcctcagtaacaatcagctgatccctggccactaatct gtactcactaatctgttttccaaactcttggcccctgagctaattatagcagtgcttcatgccacccaccccaaccctattcttgtt ctctgactcccactaatctacacattcagaggattgtggatataagaggctgggaggccagcttagcaaccagagctggagg Human rhodopsin kinase promoter (hrk promoter) (SEQ ID NO: 104): Gggccccagaagcctggtggttgtttgtccttctcaggggaaaagtgaggcggccccttggaggaaggggccgg gcagaatgatctaatcggattccaagcagctcaggggattgtctttttctagcaccttcttgccactcctaagcgtcctccgtgaccc cggctgggatttagcctggtgctgtgtcagccccggtctcccaggggcttcccagtggtccccaggaaccctcgacagggcccggtct ctctcgtccagcaagggcagggacgggccacaggccaagggc TRE-human OSK-SV40 (SEQ ID NO: 105): TTATGCAGTGCTGCCATAACCATGAGTGATAACACTGCGGCCAACTTA

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GAAGAGGAGAAGGGAAGAAAAACCATCAGATAGGACTTGCAATGAAACTAACC
CAAGACAATCATAATGCAGACAGGAATGTTAAAGGCGTTCAGCAGC

ADDITIONAL EMBODIMENTS

(383) Embodiment 1. A method comprising: inducing in a cell, tissue, organ and/or subject: (i) OCT4 expression; (ii) SOX2 expression; and (iii) KLF4 expression; in the absence of inducing c-MYC expression.

(384) Embodiment 2. The method of embodiment 1, wherein OCT4 expression is induced by administering: (i) a first engineered nucleic acid encoding OCT4 or encoding a Cas9 fusion protein (CRISPR activator) and a guide RNA sequence targeting promoter or enhancer at endogenous

locus of Oct4, optionally wherein the first nucleic acid (e.g., engineered nucleic acid) comprises RNA and/or DNA; (ii) a chemical agent that induces OCT4 expression; (iii) an antibody that induces OCT4 expression; or (iv) an engineered protein encoding OCT4, optionally wherein OCT4 comprises a sequence that is at least 70% identical to SEQ ID NO: 2 or SEQ ID NO: 41.

(385) Embodiment 3. The method of any one of embodiments 1-2, wherein SOX2 expression comprises administering: (v) a second engineered nucleic acid encoding SOX2 encoding a Cas9 fusion protein (CRISPR activator) and a guide RNA sequence targeting promoter or enhancer at endogenous locus of SOX2, wherein the second engineered nucleic acid comprises RNA and/or DNA; (vi) a chemical agent that induces SOX2 expression; (vii) an antibody that induces SOX2 expression; or (viii) an engineered protein encoding SOX2, optionally wherein SOX2 comprises a sequence that is at least 70% identical to SEQ ID NO: 4 or SEQ ID NO: 43.

(386) Embodiment 4. The method of any one of embodiments 1-3, wherein KLF4 expression comprises administering: (ix) a third engineered nucleic acid encoding KLF4 encoding a Cas9 fusion protein (CRISPR activator) and a guide RNA sequence targeting promoter or enhancer at endogenous locus of KLF4, wherein the third nucleic acid (e.g., engineered nucleic acid) comprises RNA and/or DNA; (ix) a chemical agent that induces KLF4 expression; (xi) an antibody that induces KLF4 expression; or (xii) an engineered protein encoding KLF4, optionally wherein KLF4 comprises a sequence that is at least 70% identical to SEQ ID NO: 6 or SEQ ID NO: 45.

(387) Embodiment 5. The method of any one of embodiments 2-4, wherein said first, second, third engineered nucleic acids, or a combination thereof are present on an expression vector or are not present on an expression vector, optionally wherein the first, second, third engineered nucleic acids are mRNA or plasmid DNA.

(388) Embodiment 6. The method of embodiment 5, wherein two or three of said first, second and third engineered nucleic acids are present in the same expression vector.

(389) Embodiment 7. The method of any one of embodiments 1-5, wherein said first, second and third engineered nucleic acids are present in separate expression vectors.

(390) Embodiment 8. The method of any one of embodiments 5-7, wherein said expression vector(s) include an inducible promoter operably linked to the first, second, third engineered nucleic acids, or a combination thereof, optionally wherein said method further comprises administering an inducing agent.

(391) Embodiment 9. The method of embodiment 8 wherein said promoter comprises a tetracycline response element (TRE).

(392) Embodiment 10. The method of embodiment 9, wherein administration of the inducing agent comprises administering a protein or a fourth engineered nucleic acid encoding the inducing agent, optionally wherein the fourth engineered nucleic acid is introduced simultaneously as the first, second, and third engineered nucleic acids.

(393) Embodiment 11. The method of embodiment 10, wherein the fourth engineered nucleic acid is present on a separate expression vector from the first, second, and third engineered nucleic acids.

(394) Embodiment 12. The method of embodiment 10, wherein the fourth engineered nucleic acid is present on the same expression vector with at least one of the first, second, and third engineered nucleic acids.

(395) Embodiment 13. The method of any one of embodiments 9-12, wherein the inducing agent is capable of inducing expression of the first, second, third engineered nucleic acids, or a combination thereof from the inducible promoter in the presence of a tetracycline and the method further comprises administering tetracycline and/or removing tetracycline, optionally wherein the tetracycline is doxycycline.

(396) Embodiment 14. The method of embodiment 13, wherein the inducing agent is reverse tetracycline-controlled transactivator (rtTA).

(397) Embodiment 15. The method of embodiment 14, wherein the rtTA is M2-rtTA or rtTA3.

(398) Embodiment 16. The method of embodiment 15, wherein the M2-rtTA comprises an amino

acid sequence that is at least 70% identical to SEQ ID NO: 15 or the rtTA3 comprises an amino acid sequence that is at least 70% identical to SEQ ID NO: 11.
(399) Embodiment 17. The method of any one of embodiments 9-12, wherein the inducing agent is capable of inducing expression of the first, second, third engineered nucleic acids, or a combination thereof from the inducible promoter in the absence of a tetracycline, optionally, wherein the tetracycline is doxycycline.
(400) Embodiment 18. The method of embodiment 17, wherein the inducing agent is a temperature, a chemical, a pH, a nucleic acid, a protein, optionally wherein the protein is a tetracycline-controlled transactivator (tTA).
(401) Embodiment 19. The method of any one of embodiments embodiment 11 or 13-18, wherein the first, second, and third engineered nucleic acids are present in a first expression vector and the fourth engineered nucleic acid is present in a second expression vector.
(402) Embodiment 20. The method of any one of embodiments 9-19, wherein the promoter is a TRE3G, a TRE2 promoter, or a P tight promoter, optionally, wherein the promoter comprises a engineered nucleic acid sequence that is at least 70% identical to SEQ ID NO: 7, optionally, wherein the promoter comprises a engineered nucleic acid sequence that is at least 70% identical to SEQ ID NO: 23, and optionally wherein the promoter comprises a sequence that is at least 70% identical to SEQ ID NO: 24.
(403) Embodiment 21. The method of any one of embodiments 1-7 or 10-20, wherein said expression vector(s) comprise a constitutive promoter operably linked to the first, second, third, fourth engineered nucleic acids, or any combination thereof.
(404) Embodiment 22. The method of embodiment 21, wherein the constitutive promoter is operably linked to the fourth engineered nucleic acid but not to the first, second, or third engineered nucleic acids, optionally wherein the constitutive promoter is CP1, CMV, EF1 alpha, SV40, PGK1, Ubc, human beta actin, CAG, Ac5, polyhedrin, TEF1, GDS, CaM3 5S, Ubi, H1, and U6 promoter, or a tissue-specific promoter.
(405) Embodiment 23. The method of embodiment 19-22, wherein the first expression vector comprises the sequence provided in SEQ ID NO: 16, optionally wherein the second expression vector comprises the sequence provided in SEQ ID NO: 31 or SEQ ID NO: 32.
(406) Embodiment 24. The method of any one of embodiments 2-23, wherein at least one of (i)-(xii) is delivered in a viral vector or is delivered without a viral vector, wherein the viral vector is selected from the group consisting of a lentivirus, a retrovirus, an adenovirus, alphavirus, vaccinia virus, and an adeno-associated virus (AAV) vector, optionally wherein delivery without a viral vector comprises administration of a naked nucleic acid, electroporation, use of a nanoparticle, or use of liposomes.
(407) Embodiment 25. The method of any one of embodiments 19-24, wherein the first expression vector is a first viral vector, and the second expression vector is a viral vector, optionally wherein the first and second viral vectors are AAV vectors.
(408) Embodiment 26. The method of any one of embodiments 1-25 wherein at least one engineered nucleic acid comprises an SV40-derived sequence including a sequence that is at least 70% identical to SEQ ID NO: 8.
(409) Embodiment 27. The methods of any one of embodiments 1-26, wherein OCT4, KLF4, or SOX2 is a mammalian protein.
(410) Embodiment 28. The method of any one of embodiments 1-27, wherein the cell or tissue is in a subject, wherein the subject has a condition, is suspected of having a condition, or at risk for a condition, optionally wherein the condition is selected from the group consisting of ocular disease, aging, cancer, musculoskeletal disease, age-related disease, a disease affecting a non-human animal and neurodegenerative disease.
(411) Embodiment 29. The method of any one of embodiments 1-28, wherein the method further comprises regulating: cellular reprogramming, tissue repair, tissue survival, tissue regeneration,

tissue growth, tissue function, organ regeneration, organ survival, organ function, disease, or any combination thereof, optionally wherein regulating comprises inducing cellular reprogramming, reversing aging, improving tissue function, improving organ function, tissue repair, tissue survival, tissue regeneration, tissue growth, promoting angiogenesis, treating a disease, reducing scar formation, reducing the appearance of aging, promoting organ regeneration, promoting organ survival, altering the taste and quality of agricultural products derived from animals, treating a disease, or any combination thereof, ex vivo or in vitro and optionally wherein treating a disease comprises inducing expression of OCT4, KLF4, and/or SOX2 prior to the onset of disease or wherein treating a disease a disease comprises inducing expression of OCT4, KLF4, and/or SOX2 after the onset of disease.
(412) Embodiment 30. The method of embodiment 29, wherein the cell or tissue is from eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine, optionally wherein the tissue is damaged or the tissue may be considered healthy but suboptimal for performance or survival in current or future conditions.
(413) Embodiment 31. The method of any one of embodiments 1-30, wherein the engineered nucleic acid further comprises a self-cleaving peptide, optionally wherein the self-cleaving peptide is a 2A peptide that is at least 70% identical to SEQ ID NO: 9.
(414) Embodiment 32. The method of any one of embodiments 1-31, wherein the engineered nucleic acid further comprises inverted terminal repeats (ITRs) flanking the first nucleic acid, the second nucleic acid, the third nucleic acid, or a combination thereof, optionally, wherein the distance between the ITRs is 4.7 kb or less.
(415) Embodiment 33. An expression vector comprising: (i) a first engineered nucleic acid encoding OCT4; (ii) a second engineered nucleic acid encoding SOX2; and (iii) a third engineered nucleic acid encoding KLF4; in the absence of an engineered nucleic acid capable of expressing c-MYC.
(416) Embodiment 34. The expression vector of embodiment 33, wherein the OCT4 protein comprises a sequence that is at least 70% identical to SEQ ID NO: 2 or SEQ ID NO: 41.
(417) Embodiment 35. The expression vector of any one of embodiments 33-34, wherein the SOX2 protein comprises a sequence that is at least 70% identical to SEQ ID NO: 4 or SEQ ID NO: 43.
(418) Embodiment 36. The expression vector of any one of embodiments 33-35, wherein the KLF4 protein comprises a sequence that is at least 70% identical to SEQ ID NO: 6 or SEQ ID NO: 45.
(419) Embodiment 37. The expression vector of any one of embodiments 33-36, further comprising an inducible promoter operably linked to the first, second, third engineered nucleic acids, or any combination thereof.
(420) Embodiment 38. The expression vector of embodiment 37, wherein an inducing agent is capable of inducing expression of the first, second, third engineered nucleic acids, or any combination thereof from the inducible promoter in the presence of a tetracycline, optionally wherein the tetracycline is doxycycline.
(421) Embodiment 39. The expression vector of embodiment 38, wherein the inducing agent is reverse tetracycline-controlled transactivator (rtTA).
(422) Embodiment 40. The expression vector of embodiment 39, wherein the rtTA is M2-rtTA or rtTA3.
(423) Embodiment 41. The expression vector of embodiment 40, wherein the M2-rtTA comprises an amino acid sequence that is at least 70% identical to SEQ ID NO: 15 or the rtTA3 comprises an amino acid sequence that is at least 70% identical to SEQ ID NO: 11.
(424) Embodiment 42. The expression vector of any one of embodiments 38-41, wherein the inducing agent is capable of inducing expression of the first, second, third engineered nucleic acids, or any combination thereof from the inducible promoter in the absence of a tetracycline, optionally,

wherein the tetracycline is doxycycline.
(425) Embodiment 43. The expression vector of embodiment 42, wherein the inducing agent is a tetracycline-controlled transactivator (tTA).
(426) Embodiment 44. The expression vector of any one of embodiments 37-43, wherein the inducible promoter comprises a tetracycline-responsive element (TRE), optionally, wherein the promoter is a TRE3G promoter comprising a engineered nucleic acid sequence that is at least 70% identical to SEQ ID NO: 7, optionally, wherein the promoter comprises a engineered nucleic acid sequence that is at least 70% identical to SEQ ID NO: 23, and optionally wherein the promoter comprises a sequence that is at least 70% identical to SEQ ID NO: 24.
(427) Embodiment 45. The expression vector of any one of embodiments 33-36, wherein said expression vector(s) comprise a constitutive promoter operably linked to the first, second, third engineered nucleic acids, or a combination thereof.
(428) Embodiment 46. The expression vector of any one of embodiments 33-44, wherein the expression vector comprises the sequence provided in SEQ ID NO: 16.
(429) Embodiment 47. The expression vector of any one of embodiments 33-46, wherein the expression vector is a viral vector, wherein the viral vector is selected from the group consisting of a lentivirus, alphavirus, vaccinia virus, a herpes virus, a retrovirus, an adenovirus, and an adeno-associated virus (AAV) vector.
(430) Embodiment 48. The expression vector of any one of embodiments 33-47, wherein at least one engineered nucleic acid comprises an SV40-derived sequence including a sequence that is at least 70% identical to SEQ ID NO: 8.
(431) Embodiment 49. The expression vectors of any one of embodiments 33-48, wherein OCT4, KLF4, or SOX2 is a mammalian protein.
(432) Embodiment 50. The expression vector of any one of embodiments 33-49, wherein the expression vector further comprises a self-cleaving peptide, optionally wherein the self-cleaving peptide is 2A peptide, optionally wherein the 2A peptide comprises a sequence that is at least 70% identical to SEQ ID NO: 9.
(433) Embodiment 51. The expression vector of any one of embodiments 37-44 and 46-50, wherein the expression vector comprises one inducible promoter.
(434) Embodiment 52. The expression vector of any one of embodiments 45-50, wherein the expression vector comprises one constitutive promoter.
(435) Embodiment 53. The expression vector of any one of embodiments 33-52, wherein the engineered nucleic acid further comprises inverted terminal repeats (ITRs) flanking the first nucleic acid, the second nucleic acid, the third nucleic acid, or a combination thereof.
(436) Embodiment 54. The expression vector of embodiment 32, wherein the distance between the ITRs is 4.7 kb or less.
(437) Embodiment 55. A recombinant virus comprising the expression vector of any one of embodiments 47-54, optionally wherein the recombinant virus is a retrovirus, an adenovirus, an AAV, alphavirus, vaccinia virus, a herpes virus, or a lentivirus.
(438) Embodiment 56. An engineered cell produced by any one of the methods of embodiments 1-32, 63-66, 70-75, 81, and 85-87, optionally wherein the engineered cell comprises the expression vector of any one of embodiments 33-54.
(439) Embodiment 57. A composition comprising the, expression vector of any one of embodiments 33-54, the recombinant virus of embodiment 55, the engineered cell of embodiment 56, a chemical agent that is capable of inducing OCT4, KLF4, and/or SOX2 expression, an engineered protein selected from the group consisting of OCT4, KLF4, and/or SOX2, an antibody capable of inducing expression of OCT4, KLF4, and/or SOX2, optionally wherein the composition comprises a pharmaceutically acceptable carrier.
(440) Embodiment 58. The composition of embodiment 57, further comprising a second expression vector encoding an inducing agent, a second protein encoding an inducing agent, or a second

recombinant virus encoding an inducing agent, optionally wherein the second expression vector is an AAV vector and/or the second recombinant virus is an AAV.
(441) Embodiment 59. The composition of embodiment 58, wherein the inducing agent is reverse tetracycline transactivator (rtTA) or tetracycline transactivator (tTA).
(442) Embodiment 60. The composition of any one of embodiments 58-59, wherein the inducing agent is encoded by a viral vector, optionally, wherein the viral vector is selected from the group consisting of a lentiviral vector, an adenoviral vector, an adeno-associated viral vector, and a retroviral vector.
(443) Embodiment 61. The composition of embodiment 60, wherein the viral vector encoding the inducing agent comprises a sequence set forth in SEQ ID NO: 31 or SEQ ID NO: 32.
(444) Embodiment 62. A kit comprising the expression vector of any one of embodiments 33-54, recombinant virus of embodiment 55, the engineered cell of embodiment 56, a chemical agent that is capable of inducing OCT4, KLF4, and/or SOX2 expression, an engineered protein selected from the group consisting of OCT4, KLF4, and/or SOX2, an antibody capable of inducing expression of OCT4, KLF4, and/or SOX2, or the composition of any one of embodiments 56-61.
(445) Embodiment 63. A method of producing an engineered cell comprising the method of any one of embodiments 1-32, thereby producing the engineered cell.
(446) Embodiment 64. The method of embodiment 63, wherein the engineered cell is an induced pluripotent stem cell.
(447) Embodiment 65. The method of any one of embodiments 63-64, wherein the engineered cell is the cell of embodiment 56.
(448) Embodiment 66. A method of producing an engineered cell, comprising the method of any one of embodiments 1-32 and 63-65, wherein the engineered cell is produced ex vivo.
(449) Embodiment 67. The method of any one of embodiments 63-66, further comprising generating an engineered tissue or engineered organ.
(450) Embodiment 68. The method of any one of embodiments 66-67, further comprising administering the engineered cell, engineered tissue, and/or engineered organ to a subject in need thereof, optionally wherein the cell, tissue, and/or organ is from eye, ear, nose, mouth including gum and roots of teeth, bone, lung, breast, udder, pancreas, stomach, oesophagus, muscle including cardiac muscle, liver, blood vessel, skin including hair, heart, brain, nerve tissue, kidney, testis, prostate, penis, cloaca, fin, ovary, or intestine cell.
(451) Embodiment 69. The method of any one of embodiments 63-68, wherein the method further comprises treating a disease, optionally wherein the disease is selected from the group consisting of acute injuries, neurodegenerative diseases, chronic diseases, proliferative diseases, ocular disease, cardiovascular diseases, genetic diseases, inflammatory diseases, autoimmunue diseases, neurological diseases, hematological diseases, painful conditions, psychiatric disorders, metabolic disorders, chronic diseases, cancers, aging, age-related diseases, and diseases affecting any tissue in a subject, optionally wherein the disease is an ocular disease.
(452) Embodiment 70. A method comprising: (i) activating OCT4; (ii) activating SOX2; and (iii) activating KLF4;
in a cell, tissue, organ, and/or subject and in the absence of activating c-Myc.
(453) Embodiment 71. The method of embodiment 71, wherein the activating in any one of (i)-(iii) comprises administering an antibody, protein, nucleic acid, or chemical agent.
(454) Embodiment 72. The method of any one of embodiments 72, wherein the nucleic acid, antibody, protein, and/or chemical agent replaces OCT4, SOX2, and/or KLF4.
(455) Embodiment 73. The method of embodiment 72, wherein the replacing comprises promoting cellular reprogramming.
(456) Embodiment 74. The method of any one of embodiments 70-73, wherein activating of any one of (i)-(iii) comprises replacing OCT4, SOX2, and/or KLF4, selected from the group consisting of an antibody, a protein, a nucleic acid, and a chemical agent.

(457) Embodiment 75. The method of embodiment 74, wherein the replacing of OCT4, SOX2, and/or KLF4 comprises administering a nucleic acid and/or protein encoding Tet1, NR5A-2, Sal14, E-cadherin, NKX3-1, NANOG, and/or Tet2.
(458) Embodiment 76. The method of any one of embodiments 1-32 and 70-75, wherein the subject is healthy.
(459) Embodiment 77. The method of any one of embodiments 1-32 and 70-76, wherein the subject is a pediatric subject.
(460) Embodiment 78. The method of any one of embodiments 1-32 and 70-76, wherein the subject is an adult subject.
(461) Embodiment 79. The method of any one of embodiments 28-32 and 70-78, wherein the subject has, is suspected of having, or at risk for glaucoma.
(462) Embodiment 80. The method of any one of embodiments 28-32 and 70-79, wherein the subject has, is suspected of having, or at risk for age-related decline in visual acuity, and/or retinal function.
(463) Embodiment 81. A method comprising administering a nucleic acid and/or protein encoding Tet1 or Tet2 to a cell, tissue, organ, and/or subject.
(464) Embodiment 82. The method of embodiment 81, wherein the subject has a disease.
(465) Embodiment 83. The method of embodiment 82, wherein the disease is selected from acute injuries, neurodegenerative diseases, chronic diseases, proliferative diseases, ocular disease, cardiovascular diseases, genetic diseases, inflammatory diseases, autoimmunue diseases, neurological diseases, hematological diseases, painful conditions, psychiatric disorders, metabolic disorders, chronic diseases, cancers, aging, age-related diseases, and diseases affecting any tissue in a subject.
(466) Embodiment 84. The method of embodiment 83, wherein the disease is an ocular disease.
(467) Embodiment 85. The method of any one of embodiments 1-32 and 63-84, further comprising activating an enhancer of reprogramming in the cell, tissue, organ and/or subject.
(468) Embodiment 86. The method of any one of embodiments 1-32 and 63-85, further comprising inhibiting a barrier of reprogramming in the cell, tissue, organ and/or subject.
(469) Embodiment 87. The method of embodiment 86, wherein the barrier of reprogramming is a DNA methyltransferase (DNMT) in the cell, tissue, organ and/or subject.
(470) Embodiment 88. A method comprising: inducing in a subject: (i) OCT4 expression; (ii) SOX2 expression; and (iii) KLF4 expression; in the absence of inducing c-MYC expression, wherein the subject has been treated with a chemotherapy drug.
(471) Embodiment 89. The method of embodiment 89, wherein the chemotherapy drug is vincristine (VCS).
(472) Embodiment 90. A method comprising inducing in a cell, tissue, organ, and/or subject: (i) OCT4 expression; (ii) SOX2 expression; and (iii) KLF4 expression;
wherein OCT4, SOX2, and KLF4 is encoded by a nucleic acid and expression of OCT4, SOX2, and/or KLF4 is induced from a single promoter.
(473) Embodiment 91. A method comprising: inducing in a cell, tissue, organ and/or subject: (i) OCT4 expression; (ii) SOX2 expression; (iii) KLF4 expression; or (iv) any combination of (i)-(iii), in the absence of inducing c-MYC expression.
(474) Embodiment 92. The method of embodiment 91, wherein the combination of (i)-(iii) comprises (i) and (ii); (i) and (iii); (ii) and (iii); or (i), (ii), and (iii).
(475) Embodiment 93. An expression vector comprising: (i) a first engineered nucleic acid encoding OCT4; (ii) a second engineered nucleic acid encoding SOX2; (iii) a third engineered nucleic acid encoding KLF4; or (iv) any combination of (i)-(iii),
in the absence of an engineered nucleic acid capable of inducing c-MYC expression.
(476) Embodiment 94. The expression vector of embodiment 93, wherein the combination of (i)-(iii) comprises (i) and (ii); (i) and (iii); (ii) and (iii); or (i), (ii), and (iii).

(477) Embodiment 95. A recombinant virus comprising the expression vector of any one of embodiments 47-54 and 93-94, optionally wherein the recombinant virus is a retrovirus, an adenovirus, an AAV, alphavirus, vaccinia virus, a herpes virus, or a lentivirus.
(478) Embodiment 96. An engineered cell produced by any one of the methods of embodiments 1-32, 63-66, 70-75, 81, 85-87, and 91-92, optionally wherein the engineered cell comprises the expression vector of any one of embodiments 33-54 and 93-94.
(479) Embodiment 97. A composition comprising the expression vector of any one of embodiments 33-54 and 93-94, the recombinant virus of embodiment 55 or embodiment 95, the engineered cell of embodiment 56 or 96, a chemical agent that is capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, an engineered protein selected from the group consisting of OCT4; KLF4; SOX2; or any combination thereof, an antibody capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, optionally wherein the composition comprises a pharmaceutically acceptable carrier.
(480) Embodiment 98. A kit comprising the expression vector of any one of embodiments 33-54 and 93-94, recombinant virus of embodiment 55 or 95, the engineered cell of embodiment 56 or 96, a chemical agent that is capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, an engineered protein selected from the group consisting of OCT4; KLF4; SOX2; or any combination thereof, an antibody capable of inducing expression of OCT4; KLF4; SOX2; or any combination thereof, or the composition of any one of embodiments 56-61 or 97.
(481) Embodiment 99. A method of producing an engineered cell comprising the method of any one of embodiments 1-32 and 91-92, thereby producing the engineered cell.
(482) Embodiment 100. A method of producing an engineered cell, comprising the method of any one of embodiments 1-32, 63-65, 91-92, and 99, wherein the engineered cell is produced in vivo.
(483) Embodiment 101. A method of producing an engineered cell, comprising the method of any one of embodiments 1-32, 63-65, 91-92, and 99, wherein the engineered cell is produced ex vivo.
(484) Embodiment 102. A method comprising: (i) activating OCT4; (ii) activating SOX2; (iii) activating KLF4; or (iv) any combination of (i)-(iii),
in a cell, tissue, organ, subject, or any combination thereof, and in the absence of activating c-Myc above endogenous levels.
(485) Embodiment 103. The method of embodiment 102, wherein the combination of (i)-(iii) comprises (i) and (ii); (i) and (iii); (ii) and (iii); or (i), (ii), and (iii).
(486) Embodiment 104. A method comprising: inducing in a subject: (i) OCT4 expression; (ii) SOX2 expression; (iii) KLF4 expression; or (iv) any combination of (i)-(iii),
in the absence of inducing c-MYC expression, wherein the subject has been treated with a chemotherapy drug.
(487) Embodiment 105. The method of embodiment 104, wherein the combination of (i)-(iii) comprises (i) and (ii); (i) and (iii); (ii) and (iii); or (i), (ii), and (iii).
(488) Embodiment 106. A method comprising inducing in a cell, tissue, organ, subject, or any combination thereof: (i) OCT4 expression; (ii) SOX2 expression; (iii) KLF4 expression; or (iv) any combination of (i)-(iii),
wherein OCT4, SOX2, KLF4, or any combination thereof is encoded by a nucleic acid and expression of OCT4, SOX2, KLF4, or any combination thereof is induced from a single promoter.
(489) Embodiment 107. The method of embodiment 106, wherein the combination of (i)-(iii) comprises (i) and (ii); (i) and (iii); (ii) and (iii); or (i), (ii), and (iii).
(490) Embodiment 108. The method of any one of embodiments 1-32, 68-92, or 102-107 wherein the subject is a human.
(491) Embodiment 109. The method of any one of embodiments 1-32, 68-92, or 102-108, wherein the method does not induce teratoma formation.
(492) Embodiment 110. The method of any one of embodiments 1-32, 68-92, or 102-109, wherein the method does not induce tumor formation or tumor growth.

(493) Embodiment 111. The method of embodiment 110, wherein the method reduces tumor formation or tumor growth.
(494) Embodiment 112. The method of any one of embodiments 1-32, 68-92, or 102-111, wherein the method increases visual acuity in the subject.
(495) Embodiment 113. The method of any one of embodiments 1-32, 68-92, or 102-112, wherein the method does not induce cancer.
(496) Embodiment 114. The method of any one of embodiments 1-32, 68-92, or 102-113, wherein the method does not induce glaucoma.
(497) Embodiment 115. The method of any one of embodiments 1-32, 68-92, or 102-114, wherein the method reverses the epigenetic clock of the cell, the tissue, the organ, the subject, or any combination thereof.
(498) Embodiment 116. The method of embodiment 115, wherein the epigenetic clock is determined using a DNA methylation-based (DNAm) age estimator.
(499) Embodiment 117. The method of any one of embodiments 1-32, 68-92, or 102-116, wherein the method alters the expression of one or more genes associated with ageing.
(500) Embodiment 118. The method of embodiment 117, wherein the method reduces expression of one or more genes associated with ageing.
(501) Embodiment 119. The method of embodiment 118, wherein the method reduces expression of 0610040J01Rik, 1700080N15Rik, 2900064F13Rik, 4833417C18Rik, 4921522P10Rik, 4930447C04Rik, 4930488N15Rik, Ace, Ackr1, Acot10, Acvr1, Adamts17, Adra1b, AI504432, Best3, Boc, Cadm3, Cand2, Cc19, Cd14, Cd36, Cfh, Chrm3, Chrna4, Cntn4, Cracr2b, Cryaa, CT573017.2, Cyp26a1, Cyp27a1, D330050G23Rik, D930007P13Rik, Ddo, Dgkg, Dlk2, Dnajal-ps, Drd2, Dsel, Dytn, Ecscr, Edn1, Ednrb, Efemp1, Elfn2, Epha10, Ephx1, Erbb4, Fam20a, Fbxw21, Ffar4, Flt4, Fmod, Foxp4, Fzd7, Gabrd, Galntl5, Galntl8, Gfra2, Ggt1, Gm10416, Gm14964, Gm17634, Gm2065, Gm32352, Gm33172, Gm34280, Gm35853, Gm36298, Gm36356, Gm36937, Gm3898, Gm42303, Gm42484, Gm42537, Gm42743, Gm43151, Gm43843, Gm44545, Gm44722, Gm45516, Gm45532, Gm47494, Gm47982, Gm47989, Gm48398, Gm48495, Gm48593, Gm48958, Gm49089, Gm49326, Gm49331, Gm49760, Gm5796, Gm6374, Gm7276, Gm8237, Gm9796, Gm9954, Gpr75, Gprc5c, Grid2ip, Gsg112, Hapln4, Hcn3, Hcn4, Hhat1, Hs6st2, Htr3a, Il1rap, Il1rap12, Inka1, Kbtbd12, Kcnj11, Kcnk4, Kdelc2, Klhl33, Lamc3, Lilra5, Lman11, Lrfn2, Lrrc38, Lrrn4c1, Ltc4s, Mansc1, Mir344c, Msr1, Mycbpap, Myoc, Ngfr, Nipal2, Olfr1372-ps1, Otop3, P2rx5, P2ry12, P4ha2, Pcdha12, Pcdha2, Pcdhac2, Pcdhb18, Pcdhb5, Pcsk2os1, Pcsk6, Perp, Pkp1, Plxna4, Prickle2, Qsox1, Rapgef4os2, Rbp4, Rcn3, Sec1415, Sel113, Serpinh1, Sgpp2, Shisa6, Siah3, Siglech, Slc12a4, Slc24a2, Slc2a5, Slc4a4, Slitrk3, Smagp, Smoc2, Speer4b, Spon2, Sstr2, Sstr3, St3gal3, Stc1, Stc2, Syndig1, Syt10, Thsd7a, Tlr8, Tmem132a, Tmem132d, Tmem200a, Tmem44, Trpc4, Trpv4, Unc5b, Vgf, Vmn1r90, Vwc21, Wfikkn2, Wnt11, Wnt6, Zeb2os, Zfp608, Zfp976, or any combination thereof.
(502) Embodiment 120. The method of embodiment 119, wherein the gene is a sensory gene.
(503) Embodiment 121. The method of any one of embodiments 118-120, wherein the gene is Ace, Kcnk4, Lamc3, Edn1, Syt10, Ngfr, Gprc5c, Cd36, Chrna4, Ednrb, Drd2, or a combination thereof.
(504) Embodiment 122. The method of embodiment 117, wherein the method increases expression of one or more genes associated with ageing.
(505) Embodiment 123. The method of any one of embodiments 1-32, 68-92, 102-122, wherein the method increases expression of 1700031P21Rik, 1810053B23Rik, 2900045020Rik, 2900060B14Rik, 4921504E06Rik, 4930402F11Rik, 4930453C13Rik, 4930455B14Rik, 4930500H12Rik, 4930549P19Rik, 4930555B11Rik, 4930556J02Rik, 4932442E05Rik, 4933431K23Rik, 4933438K21Rik, 6720475M21Rik, 9830132P13Rik, A430010J10Rik, A530064D06Rik, A530065N20Rik, Abcb5, Abhd17c, AC116759.2, AC131705.1, AC166779.3, Acot12, Adig, Akr1c1, Ankrd1, Asb15, Atp2c2, AU018091, AW822073, Btn110, C130093G08Rik, C730027H18Rik, Ccdc162, Chil6, Col26a1, Corin, Crls1, Cybrd1, Cyp2d12, Cyp7a1,

D830005E20Rik, Dlx3, Dnah14, Dsc3, Dthd1, Eid2, Eps811, EU599041, Fam90ala, Fancf, Fau-ps2, Fezf1, Gja5, Gm10248, Gm10513, Gm10635, Gm10638, Gm10718, Gm10722, Gm10800, Gm10801, Gm11228, Gm11251, Gm11264, Gm11337, Gm11368, Gm11485, Gm11693, Gm12793, Gm13050, Gm13066, Gm13323, Gm13339, Gm13346, Gm13857, Gm14387, Gm14770, Gm15638, Gm16072, Gm16161, Gm16181, Gm17200, Gm17791, Gm18025, Gm18757, Gm18795, Gm18848, Gm19719, Gm20121, Gm20356, Gm2093, Gm21738, Gm21940, Gm22933, Gm24000, Gm24119, Gm25394, Gm26555, Gm27047, Gm28262, Gm28530, Gm29295, Gm29825, Gm29844, Gm3081, Gm32051, Gm32122, Gm33056, Gm33680, Gm34354, Gm34643, Gm3551, Gm36660, Gm36948, Gm37052, Gm37142, Gm37262, Gm37535, Gm37569, Gm37589, Gm37647, Gm37648, Gm37762, Gm38058, Gm38069, Gm38137, Gm38218, Gm39139, Gm42535, Gm42680, Gm42895, Gm42994, Gm43027, Gm43158, Gm43288, Gm43366, Gm44044, Gm44081, Gm44187, Gm44280, Gm44535, Gm45338, Gm45644, Gm45740, Gm46555, Gm46565, Gm4742, Gm47485, Gm47853, Gm47992, Gm48225, Gm48314, Gm48383, Gm48673, Gm48804, Gm48832, Gm4994, Gm5487, Gm5724, Gm595, Gm6012, Gm6024, Gm7669, Gm7730, Gm8043, Gm8953, Gm9348, Gm9369, Gm9495, H2a12a, Ido2, Igfbp1, Kif7, Klhl31, Lrrc31, Mc5r, Mgam, Msh4, Muc12, Mug1, Mybl2, Myh15, Nek10, Neurod6, Nrlh5, Olfr1042, Olfr1043, Olfr1082, Olfr1090, Olfr1124, Olfr1167, Olfr1205, Olfr1206, Olfr1223, Olfr1263, Olfr1264, Olfr1269, Olfr127, Olfr1291-ps1, Olfr1406, Olfr1469, Olfr215, Olfr273, Olfr328, Olfr355, Olfr372, Olfr390, Olfr427, Olfr456, Olfr466, Olfr481, Olfr522, Olfr6, Olfr601, Olfr603, Olfr706, Olfr727, Olfr728, Olfr741, Olfr801, Olfr812, Olfr816, Olfr822, Olfr860, Olfr890, Olfr923, Olfr943, Otog1, Pi15, Pkhd1, Pkhd111, Platr6, Pou3f4, Prr9, Pvalb, Rhag, Sav1, Serpinb9b, Skint1, Skint3, Skint5, Slc10a5, Slc6a4, Smok2a, Tcaf3, Tomm20l, Trcg1, Trdn, Ugt1a6a, Usp17la, Vmn1r178, Vmn1r179, Vmn1r33, Vmn1r74, Vmn1r87, Vmn2r102, Vmn2r113, Vmn2r17, Vmn2r52, Vmn2r66, Vmn2r68, Vmn2r76, Vmn2r78, Wnt16, or any combination thereof.

(506) Embodiment 124. The method of embodiment 123, wherein the method increases expression of Olfr816, Olfr812, Olfr1264, Olfr727, Olfr923, Olfr1090, Olfr328, Olfr1124, Olfr522, Olfr1082, Olfr1206, Olfr1167, Olfr706, Olfr6, Pou3f4, Olfr603, Olfr127, Olfr1263, Olfr1269, Olfr1205, Olfr390, Olfr601, Olfr860, Olfr215, Olfr741, Olfr1469, Olfr355, Olfr481, Olfr456, Olfr1042, Olfr728, Olfr372, Olfr801, Olfr1223, Olfr822, Otog1, Olfr943, Olfr1406, Olfr273, Olfr466, Olfr1043, Olfr427, Olfr890, Rbp4, or any combination thereof.

(507) Embodiment 125. A method of reprogramming comprising rejuvenating the epigenetic clock of a cell, tissue, organ, subject, or any combination thereof.

(508) Embodiment 126. The method of embodiment 125, wherein rejuvenating the epigenetic clock of a cell, tissue, organ, subject, or any combination thereof comprises introducing, activating, and/or expressing OCT4, KLF4, SOX2, or any combination thereof.

(509) Embodiment 127. The method of any one of embodiments 126, wherein the epigenetic clock of a cell, tissue, organ, subject, or any combination thereof is rejuvenated to that of a young cell, tissue, organ, subject, or any combination thereof.

(510) Embodiment 128. The method of any one of embodiments 125-127, wherein rejuvenating the epigenetic clock comprises altering expression of one or more genes associated with ageing in the cell, tissue, organ, subject, or the combination thereof.

(511) Embodiment 129. The method of embodiment 128, wherein the method comprises reducing expression of one or more genes associated with ageing.

(512) Embodiment 130. The method of embodiment 129, wherein the method comprises reducing expression of 0610040J01Rik, 1700080N15Rik, 2900064F13Rik, 4833417C18Rik, 4921522P10Rik, 4930447C04Rik, 4930488N15Rik, Ace, Ackr1, Acot10, Acvr1, Adamts17, Adra1b, AI504432, Best3, Boc, Cadm3, Cand2, Cc19, Cd14, Cd36, Cfh, Chrm3, Chrna4, Cntn4, Cracr2b, Cryaa, CT573017.2, Cyp26a1, Cyp27a1, D330050G23Rik, D930007P13Rik, Ddo, Dgkg, Dlk2, Dnajal-ps, Drd2, Dsel, Dytn, Ecscr, Edn1, Ednrb, Efemp1, Elfn2, Epha10, Ephx1, Erbb4,

Fam20a, Fbxw21, Ffar4, Flt4, Fmod, Foxp4, Fzd7, Gabrd, Galntl5, Galntl8, Gfra2, Ggt1, Gm10416, Gm14964, Gm17634, Gm2065, Gm32352, Gm33172, Gm34280, Gm35853, Gm36298, Gm36356, Gm36937, Gm3898, Gm42303, Gm42484, Gm42537, Gm42743, Gm43151, Gm43843, Gm44545, Gm44722, Gm45516, Gm45532, Gm47494, Gm47982, Gm47989, Gm48398, Gm48495, Gm48593, Gm48958, Gm49089, Gm49326, Gm49331, Gm49760, Gm5796, Gm6374, Gm7276, Gm8237, Gm9796, Gm9954, Gpr75, Gprc5c, Grid2ip, Gsg112, Hapln4, Hcn3, Hcn4, Hhat1, Hs6st2, Htr3a, Il1rap, Il1rap12, Inka1, Kbtbd12, Kcnj11, Kcnk4, Kdelc2, Klhl33, Lamc3, Lilra5, Lman11, Lrfn2, Lrrc38, Lrrn4c1, Ltc4s, Mansc1, Mir344c, Msr1, Mycbpap, Myoc, Ngfr, Nipal2, Olfr1372-ps1, Otop3, P2rx5, P2ry12, P4ha2, Pcdha12, Pcdha2, Pcdhac2, Pcdhb18, Pcdhb5, Pcsk2os1, Pcsk6, Perp, Pkp1, Plxna4, Prickle2, Qsox1, Rapgef4os2, Rbp4, Rcn3, Sec1415, Sel113, Serpinh1, Sgpp2, Shisa6, Siah3, Siglech, Slc12a4, Slc24a2, Slc2a5, Slc4a4, Slitrk3, Smagp, Smoc2, Speer4b, Spon2, Sstr2, Sstr3, St3gal3, Stc1, Stc2, Syndig1, Syt10, Thsd7a, Tlr8, Tmem132a, Tmem132d, Tmem200a, Tmem44, Trpc4, Trpv4, Unc5b, Vgf, Vmn1r90, Vwc21, Wfikkn2, Wnt11, Wnt6, Zeb2os, Zfp608, Zfp976, or any combination thereof.

(513) Embodiment 131. The method of embodiment 128-130, wherein the one or more genes is one or more sensory genes.

(514) Embodiment 132. The method of any one of embodiments 128-131, wherein the gene is Ace, Kcnk4, Lamc3, Edn1, Syt10, Ngfr, Gprc5c, Cd36, Chrna4, Ednrb, Drd2, or a combination thereof.

(515) Embodiment 133. The method of any one of embodiments 128-132, wherein the method comprises increasing expression of one or more genes associated with ageing.

(516) Embodiment 134. The method of embodiment 133, wherein the method increases expression of 1700031P21Rik, 1810053B23Rik, 2900045020Rik, 2900060B14Rik, 4921504E06Rik, 4930402F11Rik, 4930453C13Rik, 4930455B14Rik, 4930500H12Rik, 4930549P19Rik, 4930555B11Rik, 4930556J02Rik, 4932442E05Rik, 4933431K23Rik, 4933438K21Rik, 6720475M21Rik, 9830132P13Rik, A430010J10Rik, A530064D06Rik, A530065N20Rik, Abcb5, Abhd17c, AC116759.2, AC131705.1, AC166779.3, Acot12, Adig, Akr1c1, Ankrd1, Asb15, Atp2c2, AU018091, AW822073, Btn110, C130093G08Rik, C730027H18Rik, Ccdc162, Chil6, Col26a1, Corin, Crls1, Cybrdl, Cyp2d12, Cyp7a1, D830005E20Rik, Dlx3, Dnah14, Dsc3, Dthd1, Eid2, Eps811, EU599041, Fam90ala, Fancf, Fau-ps2, Fezf1, Gja5, Gm10248, Gm10513, Gm10635, Gm10638, Gm10718, Gm10722, Gm10800, Gm10801, Gm11228, Gm11251, Gm11264, Gm11337, Gm11368, Gm11485, Gm11693, Gm12793, Gm13050, Gm13066, Gm13323, Gm13339, Gm13346, Gm13857, Gm14387, Gm14770, Gm15638, Gm16072, Gm16161, Gm16181, Gm17200, Gm17791, Gm18025, Gm18757, Gm18795, Gm18848, Gm19719, Gm20121, Gm20356, Gm2093, Gm21738, Gm21940, Gm22933, Gm24000, Gm24119, Gm25394, Gm26555, Gm27047, Gm28262, Gm28530, Gm29295, Gm29825, Gm29844, Gm3081, Gm32051, Gm32122, Gm33056, Gm33680, Gm34354, Gm34643, Gm3551, Gm36660, Gm36948, Gm37052, Gm37142, Gm37262, Gm37535, Gm37569, Gm37589, Gm37647, Gm37648, Gm37762, Gm38058, Gm38069, Gm38137, Gm38218, Gm39139, Gm42535, Gm42680, Gm42895, Gm42994, Gm43027, Gm43158, Gm43288, Gm43366, Gm44044, Gm44081, Gm44187, Gm44280, Gm44535, Gm45338, Gm45644, Gm45740, Gm46555, Gm46565, Gm4742, Gm47485, Gm47853, Gm47992, Gm48225, Gm48314, Gm48383, Gm48673, Gm48804, Gm48832, Gm4994, Gm5487, Gm5724, Gm595, Gm6012, Gm6024, Gm7669, Gm7730, Gm8043, Gm8953, Gm9348, Gm9369, Gm9495, H2a12a, Ido2, Igfbp1, Kif7, Klhl31, Lrrc31, Mc5r, Mgam, Msh4, Muc12, Mug1, Mybl2, Myh15, Nek10, Neurod6, Nrlh5, Olfr1042, Olfr1043, Olfr1082, Olfr1090, Olfr1124, Olfr1167, Olfr1205, Olfr1206, Olfr1223, Olfr1263, Olfr1264, Olfr1269, Olfr127, Olfr1291-ps1, Olfr1406, Olfr1469, Olfr215, Olfr273, Olfr328, Olfr355, Olfr372, Olfr390, Olfr427, Olfr456, Olfr466, Olfr481, Olfr522, Olfr6, Olfr601, Olfr603, Olfr706, Olfr727, Olfr728, Olfr741, Olfr801, Olfr812, Olfr816, Olfr822, Olfr860, Olfr890, Olfr923, Olfr943, Otog1, Pi15, Pkhd1, Pkhd111, Platr6, Pou3f4, Prr9, Pvalb, Rhag, Sav1, Serpinb9b, Skint1, Skint3, Skint5, Slc10a5, Slc6a4, Smok2a, Tcaf3, Tomm20l, Trcg1, Trdn, Ugt1a6a, Usp17la, Vmn1r178, Vmn1r179,

Vmn1r33, Vmn1r74, Vmn1r87, Vmn2r102, Vmn2r113, Vmn2r17, Vmn2r52, Vmn2r66, Vmn2r68, Vmn2r76, Vmn2r78, Wnt16, or any combination thereof.

(517) Embodiment 135. The method of any one of embodiments 133-134, wherein the method comprises increasing expression of Olfr816, Olfr812, Olfr1264, Olfr727, Olfr923, Olfr1090, Olfr328, Olfr1124, Olfr522, Olfr1082, Olfr1206, Olfr1167, Olfr706, Olfr6, Pou3f4, Olfr603, Olfr127, Olfr1263, Olfr1269, Olfr1205, Olfr390, Olfr601, Olfr860, Olfr215, Olfr741, Olfr1469, Olfr355, Olfr481, Olfr456, Olfr1042, Olfr728, Olfr372, Olfr801, Olfr1223, Olfr822, Otog1, Olfr943, Olfr1406, Olfr273, Olfr466, Olfr1043, Olfr427, Olfr890, Rbp4, or any combination thereof.

(518) Embodiment 136. A method of reprogramming comprising altering the expression of one or more genes associated with ageing.

(519) Embodiment 137. The method of embodiment 136, comprising increasing expression of OCT4, KLF4, SOX2, or any combination thereof.

(520) Embodiment 138. The method of any one of embodiments 136-137, wherein the method rejuvenates the epigenetic clock of a cell, tissue, organ, subject, or any combination thereof.

(521) Embodiment 139. The method of any one of embodiments embodiment 136-138, wherein the method comprises reducing expression of one or more genes associated with ageing.

(522) Embodiment 140. The method of embodiment 139, wherein the method reduces expression of 0610040J01Rik, 1700080N15Rik, 2900064F13Rik, 4833417C18Rik, 4921522P10Rik, 4930447C04Rik, 4930488N15Rik, Ace, Ackr1, Acot10, Acvr1, Adamts17, Adra1b, AI504432, Best3, Boc, Cadm3, Cand2, Cc19, Cd14, Cd36, Cfh, Chrm3, Chrna4, Cntn4, Cracr2b, Cryaa, CT573017.2, Cyp26a1, Cyp27a1, D330050G23Rik, D930007P13Rik, Ddo, Dgkg, Dlk2, Dnajal-ps, Drd2, Dsel, Dytn, Ecscr, Edn1, Ednrb, Efemp1, Elfn2, Epha10, Ephx1, Erbb4, Fam20a, Fbxw21, Ffar4, Flt4, Fmod, Foxp4, Fzd7, Gabrd, Galntl5, Galntl8, Gfra2, Ggt1, Gm10416, Gm14964, Gm17634, Gm2065, Gm32352, Gm33172, Gm34280, Gm35853, Gm36298, Gm36356, Gm36937, Gm3898, Gm42303, Gm42484, Gm42537, Gm42743, Gm43151, Gm43843, Gm44545, Gm44722, Gm45516, Gm45532, Gm47494, Gm47982, Gm47989, Gm48398, Gm48495, Gm48593, Gm48958, Gm49089, Gm49326, Gm49331, Gm49760, Gm5796, Gm6374, Gm7276, Gm8237, Gm9796, Gm9954, Gpr75, Gprc5c, Grid2ip, Gsg112, Hapln4, Hcn3, Hcn4, Hhat1, Hs6st2, Htr3a, Il1rap, Il1rap12, Inka1, Kbtbd12, Kcnj11, Kcnk4, Kdelc2, Klhl33, Lamc3, Lilra5, Lman11, Lrfn2, Lrrc38, Lrrn4c1, Ltc4s, Mansc1, Mir344c, Msr1, Mycbpap, Myoc, Ngfr, Nipal2, Olfr1372-ps1, Otop3, P2rx5, P2ry12, P4ha2, Pcdha12, Pcdha2, Pcdhac2, Pcdhb18, Pcdhb5, Pcsk2os1, Pcsk6, Perp, Pkp1, Plxna4, Prickle2, Qsox1, Rapgef4os2, Rbp4, Rcn3, Sec1415, Sel113, Serpinh1, Sgpp2, Shisa6, Siah3, Siglech, Slc12a4, Slc24a2, Slc2a5, Slc4a4, Slitrk3, Smagp, Smoc2, Speer4b, Spon2, Sstr2, Sstr3, St3gal3, Stc1, Stc2, Syndig1, Syt10, Thsd7a, Tlr8, Tmem132a, Tmem132d, Tmem200a, Tmem44, Trpc4, Trpv4, Unc5b, Vgf, Vmn1r90, Vwc21, Wfikkn2, Wnt11, Wnt6, Zeb2os, Zfp608, Zfp976, or any combination thereof.

(523) Embodiment 141. The method of any one of embodiments 136-140, wherein the one or more genes is one or more sensory genes.

(524) Embodiment 142. The method of any one of embodiments 136-140, wherein the gene is Ace, Kcnk4, Lamc3, Edn1, Syt10, Ngfr, Gprc5c, Cd36, Chrna4, Ednrb, Drd2, or a combination thereof.

(525) Embodiment 143. The method of any one of embodiments embodiment 136-142, wherein the method comprises increasing expression of one or more genes associated with ageing.

(526) Embodiment 144. The method of embodiment 143, wherein the method comprises increasing expression of 1700031P21Rik, 1810053B23Rik, 2900045020Rik, 2900060B14Rik, 4921504E06Rik, 4930402F11Rik, 4930453C13Rik, 4930455B14Rik, 4930500H12Rik, 4930549P19Rik, 4930555B11Rik, 4930556J02Rik, 4932442E05Rik, 4933431K23Rik, 4933438K21Rik, 6720475M21Rik, 9830132P13Rik, A430010J10Rik, A530064D06Rik, A530065N20Rik, Abcb5, Abhd17c, AC116759.2, AC131705.1, AC166779.3, Acot12, Adig, Akr1c1, Ankrd1, Asb15, Atp2c2, AU018091, AW822073, Btn110, C130093G08Rik,

C730027H18Rik, Ccdc162, Chil6, Col26a1, Corin, Crls1, Cybrd1, Cyp2d12, Cyp7a1, D830005E20Rik, Dlx3, Dnah14, Dsc3, Dthd1, Eid2, Eps811, EU599041, Fam90ala, Fancf, Fau-ps2, Fezf1, Gja5, Gm10248, Gm10513, Gm10635, Gm10638, Gm10718, Gm10722, Gm10800, Gm10801, Gm11228, Gm11251, Gm11264, Gm11337, Gm11368, Gm11485, Gm11693, Gm12793, Gm13050, Gm13066, Gm13323, Gm13339, Gm13346, Gm13857, Gm14387, Gm14770, Gm15638, Gm16072, Gm16161, Gm16181, Gm17200, Gm17791, Gm18025, Gm18757, Gm18795, Gm18848, Gm19719, Gm20121, Gm20356, Gm2093, Gm21738, Gm21940, Gm22933, Gm24000, Gm24119, Gm25394, Gm26555, Gm27047, Gm28262, Gm28530, Gm29295, Gm29825, Gm29844, Gm3081, Gm32051, Gm32122, Gm33056, Gm33680, Gm34354, Gm34643, Gm3551, Gm36660, Gm36948, Gm37052, Gm37142, Gm37262, Gm37535, Gm37569, Gm37589, Gm37647, Gm37648, Gm37762, Gm38058, Gm38069, Gm38137, Gm38218, Gm39139, Gm42535, Gm42680, Gm42895, Gm42994, Gm43027, Gm43158, Gm43288, Gm43366, Gm44044, Gm44081, Gm44187, Gm44280, Gm44535, Gm45338, Gm45644, Gm45740, Gm46555, Gm46565, Gm4742, Gm47485, Gm47853, Gm47992, Gm48225, Gm48314, Gm48383, Gm48673, Gm48804, Gm48832, Gm4994, Gm5487, Gm5724, Gm595, Gm6012, Gm6024, Gm7669, Gm7730, Gm8043, Gm8953, Gm9348, Gm9369, Gm9495, H2a12a, Ido2, Igfbp1, Kif7, Klhl31, Lrrc31, Mc5r, Mgam, Msh4, Mucl2, Mug1, Mybl2, Myh15, Nek10, Neurod6, Nrlh5, Olfr1042, Olfr1043, Olfr1082, Olfr1090, Olfr1124, Olfr1167, Olfr1205, Olfr1206, Olfr1223, Olfr1263, Olfr1264, Olfr1269, Olfr127, Olfr1291-ps1, Olfr1406, Olfr1469, Olfr215, Olfr273, Olfr328, Olfr355, Olfr372, Olfr390, Olfr427, Olfr456, Olfr466, Olfr481, Olfr522, Olfr6, Olfr601, Olfr603, Olfr706, Olfr727, Olfr728, Olfr741, Olfr801, Olfr812, Olfr816, Olfr822, Olfr860, Olfr890, Olfr923, Olfr943, Otog1, Pi15, Pkhd1, Pkhd111, Platr6, Pou3f4, Prr9, Pvalb, Rhag, Sav1, Serpinb9b, Skint1, Skint3, Skint5, Slc10a5, Slc6a4, Smok2a, Tcaf3, Tomm20l, Trcg1, Trdn, Ugt1a6a, Usp17la, Vmn1r178, Vmn1r179, Vmn1r33, Vmn1r74, Vmn1r87, Vmn2r102, Vmn2r113, Vmn2r17, Vmn2r52, Vmn2r66, Vmn2r68, Vmn2r76, Vmn2r78, Wnt16, or any combination thereof.

(527) Embodiment 145. The method of embodiment 144, wherein the method comprises increasing expression of Olfr816, Olfr812, Olfr1264, Olfr727, Olfr923, Olfr1090, Olfr328, Olfr1124, Olfr522, Olfr1082, Olfr1206, Olfr1167, Olfr706, Olfr6, Pou3f4, Olfr603, Olfr127, Olfr1263, Olfr1269, Olfr1205, Olfr390, Olfr601, Olfr860, Olfr215, Olfr741, Olfr1469, Olfr355, Olfr481, Olfr456, Olfr1042, Olfr728, Olfr372, Olfr801, Olfr1223, Olfr822, Otog1, Olfr943, Olfr1406, Olfr273, Olfr466, Olfr1043, Olfr427, Olfr890, Rbp4, or any combination thereof.

(528) Embodiment 146. A method comprising resetting the transcriptional profile of old cells in vitro.

(529) Embodiment 147. A method comprising resetting the transcriptional profile of old cells in vivo.

(530) Embodiment 148. A method comprising inducing in a subject: (i) OCT4 expression; (ii) SOX2 expression; and/or (iii) KLF4 expression;

in the absence of inducing c-MYC expression, wherein the subject has, is at risk for, or is suspected of having a condition that increases the DNA methylation-based age of a cell, of a tissue, and/or of an organ within the subject, as compared to a control cell, a control tissue, and/or of a control organ of a control subject that does not have the condition.

(531) Embodiment 149. The method of embodiment 148, wherein the method reduces the DNA methylation-based age of the cell, the tissue, the organ, and/or the subject.

(532) Embodiment 150. A method of transdifferentiation comprising inducing in one type of cell: (i) OCT4 expression; (ii) SOX2 expression; (iii) KLF4 expression; and (iv) expression of a lineage determining factor,

wherein (i)-(iii) are expressed from a single vector, thereby transdifferentiating the cell into another cell type.

(533) Embodiment 151. A method of transdifferentiation comprising inducing in a cell: (i) OCT4

expression; (ii) SOX2 expression; and (iii) KLF4 expression; and
reducing expression of a lineage determining factor, wherein (i)-(iii) are expressed from a single vector.

EQUIVALENTS AND SCOPE

(534) In the claims articles such as "a," "an," and "the" may mean one or more than one unless indicated to the contrary or otherwise evident from the context. Claims or descriptions that include "or" between one or more members of a group are considered satisfied if one, more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process unless indicated to the contrary or otherwise evident from the context. The disclosure includes embodiments in which exactly one member of the group is present in, employed in, or otherwise relevant to a given product or process. The disclosure includes embodiments in which more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process.

(535) Furthermore, the disclosure encompasses all variations, combinations, and permutations in which one or more limitations, elements, clauses, and descriptive terms from one or more of the listed claims is introduced into another claim. For example, any claim that is dependent on another claim can be modified to include one or more limitations found in any other claim that is dependent on the same base claim. Where elements are presented as lists, e.g., in Markush group format, each subgroup of the elements is also disclosed, and any element(s) can be removed from the group. It should it be understood that, in general, where the disclosure, or aspects described herein, is/are referred to as comprising particular elements and/or features, certain embodiments described herein or aspects described herein consist, or consist essentially of, such elements and/or features. For purposes of simplicity, those embodiments have not been specifically set forth in haec verba herein. It is also noted that the terms "comprising" and "containing" are intended to be open and permits the inclusion of additional elements or steps. Where ranges are given, endpoints are included. Furthermore, unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or sub-range within the stated ranges in different embodiments described herein, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.

(536) This application refers to various issued patents, published patent applications, journal articles, and other publications, all of which are incorporated herein by reference. If there is a conflict between any of the incorporated references and the instant specification, the specification shall control. In addition, any particular embodiment of the present disclosure that falls within the prior art may be explicitly excluded from any one or more of the claims. Because such embodiments are deemed to be known to one of ordinary skill in the art, they may be excluded even if the exclusion is not set forth explicitly herein. Any particular embodiment described herein can be excluded from any claim, for any reason, whether or not related to the existence of prior art.

(537) Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation many equivalents to the specific embodiments described herein. The scope of the present embodiments described herein is not intended to be limited to the above Description, but rather is as set forth in the appended claims. Those of ordinary skill in the art will appreciate that various changes and modifications to this description may be made without departing from the spirit or scope of the present disclosure, as defined in the following claims.

Claims

1. A method of increasing survival and/or regeneration of neuronal cells, the method comprising: contacting the neuronal cells with one or more polynucleotides encoding OCT4, SOX2, and KLF4, but not c-Myc, operably linked to one or more promoters, or an expression vector comprising the one or more polynucleotides, wherein the method maintains the neuronal cells as neuronal cells

without loss of cellular identity.
2. The method of claim 1, wherein the one or more promoters are inducible promoters.
3. The method of claim 2, wherein the one or more inducible promoters comprise: a mifepristone-responsive promoter, or a coumermycin-responsive promoter.
4. The method of claim 2, wherein the one or more inducible promoters comprise one or more tetracycline-responsive elements (TREs).
5. The method of claim 4, wherein the method comprises introducing into the neuronal cells a polynucleotide encoding a reverse tetracycline-controlled transactivator (rtTA).
6. The method of claim 5, wherein the expression vector comprises the polynucleotide encoding the rtTA.
7. The method of claim 2, wherein the inducible promoter is a TRE3G promoter.
8. The method of claim 2, wherein the method comprises administering an inducing agent for the one or more inducible promoters.
9. The method of claim 8, wherein the inducing agent is a tetracycline-class antibiotic.
10. The method of claim 8, wherein the inducing agent is tetracycline.
11. The method of claim 8, wherein the inducing agent is doxycycline.
12. The method of claim 1, wherein the method further comprises measuring expression of one or more of Esrrb, Nanog, Lin28, TRA-1-3/TRA-1-81/TRA-2-54, SSEA1, or SSEA4.
13. The method of claim 1, wherein the method further comprises measuring DNA methylation-based age (DNAmAge) in the neuronal cells.
14. The method of claim 1, wherein the method further comprises measuring expression of Nanog.
15. The method of claim 1, wherein the expression vector is an adeno-associated virus (AAV) vector.
16. The method of claim 1, wherein the expression vector is a lentiviral vector.
17. The method of claim 1, wherein the expression vector comprises a polynucleotide sequence encoding a self-cleaving peptide.
18. The method of claim 1, wherein: i) OCT4 comprises an amino acid sequence having at least 90% identity to SEQ ID NO: 41; ii) SOX2 comprises an amino acid sequence having at least 90% identity to SEQ ID NO: 43; and iii) KLF4 comprises an amino acid sequence having at least 90% identity to SEQ ID NO: 45.
19. The method of claim 1, wherein: i) OCT4 comprises an amino acid sequence having at least 95% identity to SEQ ID NO: 41; ii) SOX2 comprises an amino acid sequence having at least 95% identity to SEQ ID NO: 43; and iii) KLF4 comprises an amino acid sequence having at least 95% identity to SEQ ID NO: 45.
20. The method of claim 1, wherein: i) OCT4 comprises the amino acid sequence of SEQ ID NO: 41; ii) SOX2 comprises the amino acid sequence of SEQ ID NO: 43; and iii) KLF4 comprises the amino acid sequence of SEQ ID NO: 45.
21. The method of claim 1, wherein the expression vector does not encode other transcription factors besides OCT4, SOX2, and KLF4.
22. The method of claim 1, wherein after the contacting step, the neuronal cells do not express at least one stem cell marker.
23. The method of claim 22, wherein the at least one stem cell marker is Esrrb, Nanog, Lin28, TRA-1-60/TRA-1-81/TRA-2-54, SSEA1, SSEA4, or any combination thereof.